

Efficient Extraction from Effectful E-Graphs



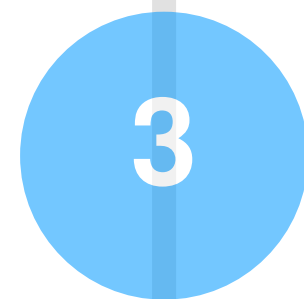
Oliver Flatt, Anjali Pal, Yihong Zhang, Ryan Tjoa, Kirsten Graham, Alex Fischman,
Chandrakana Nandi, Eli Rosenthal, Zachary Tatlock, Haobin Ni



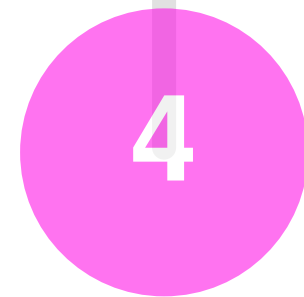
Motivate Effectful E-Graphs



Example Effectful E-Graph



Effect-Safe Extraction



It Works!



Motivate Effectful E-Graphs



Motivate Effective E-Graphs

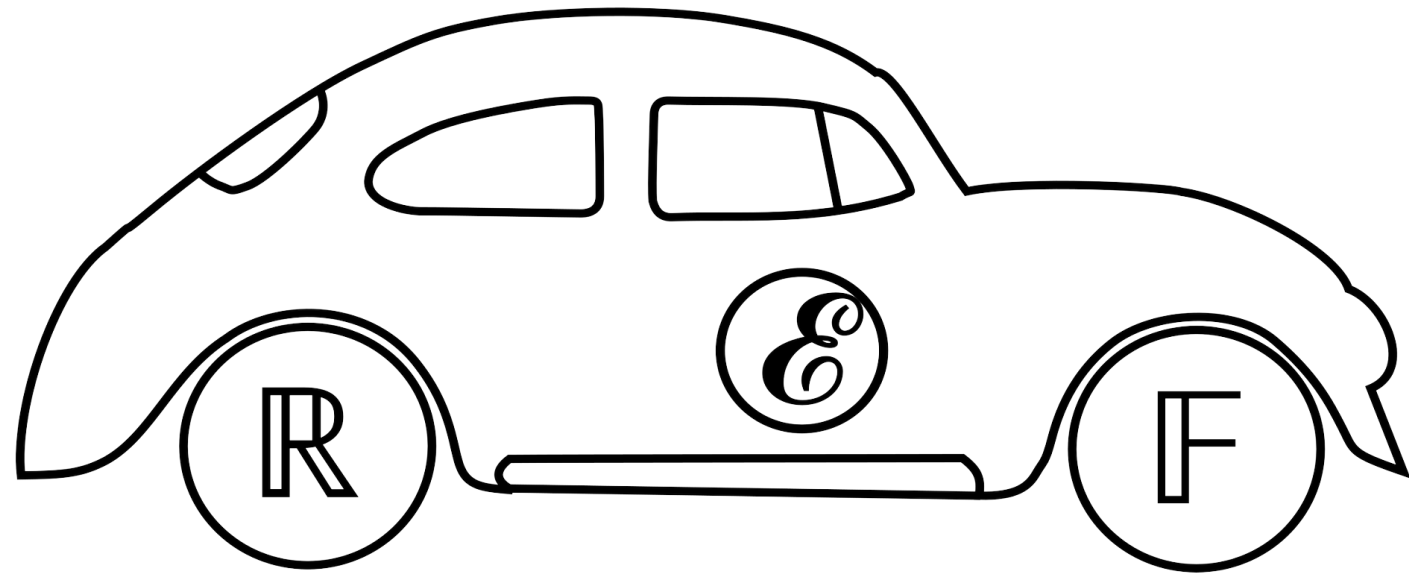
E-Graphs are Super Useful!



Motivate Effectful E-Graphs

E-Graphs are Super Useful!

Floating point



1

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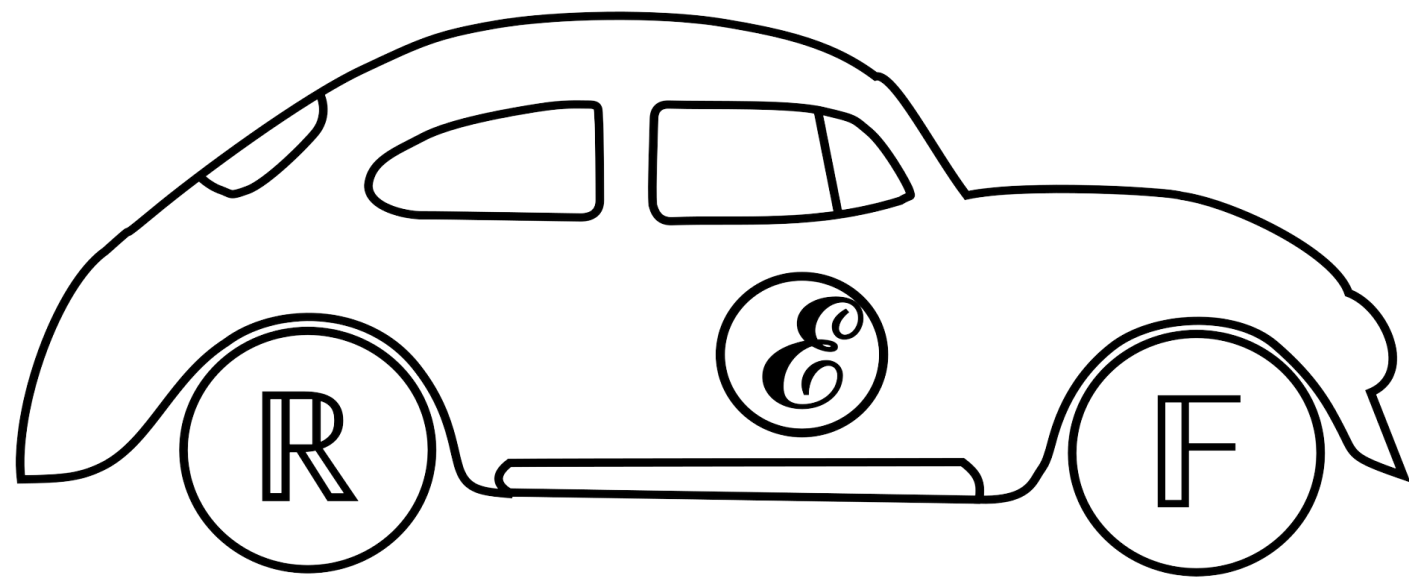
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Motivate Effectful E-Graphs

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Floating point



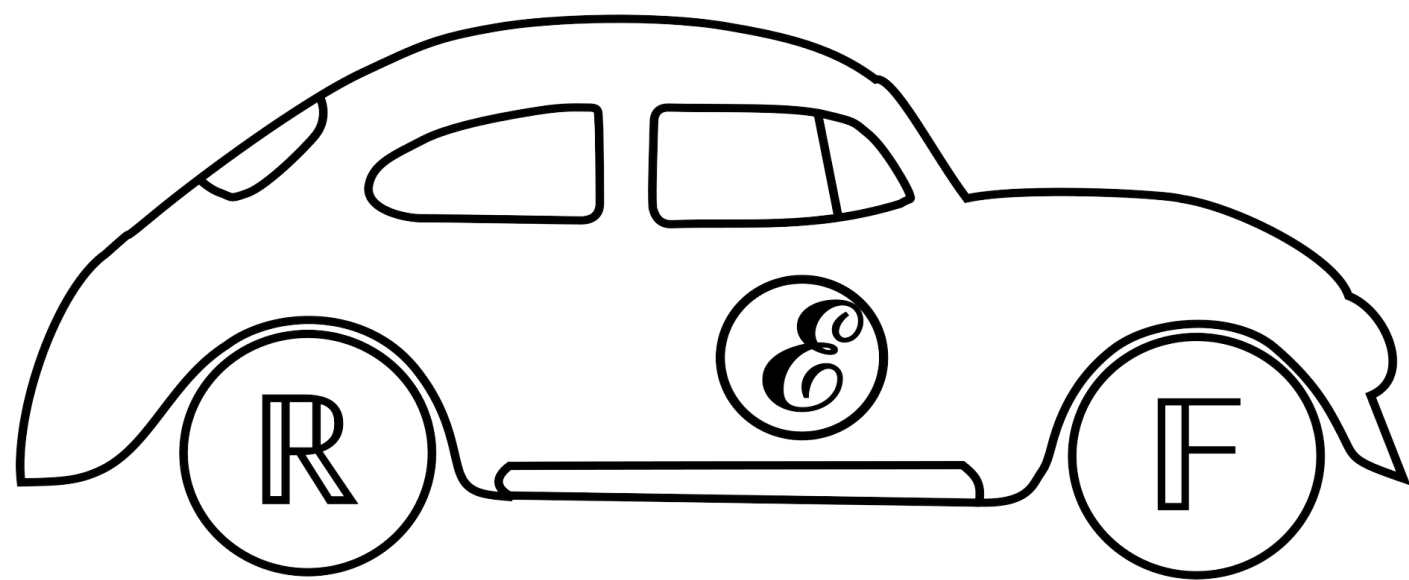
Python Simulations





E-Graphs are Super Useful!

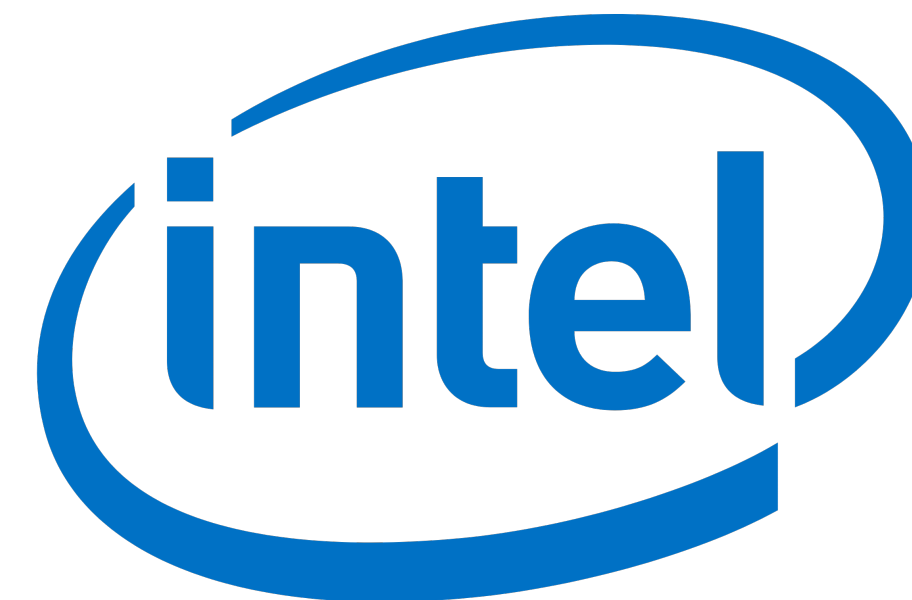
Floating point



Python Simulations



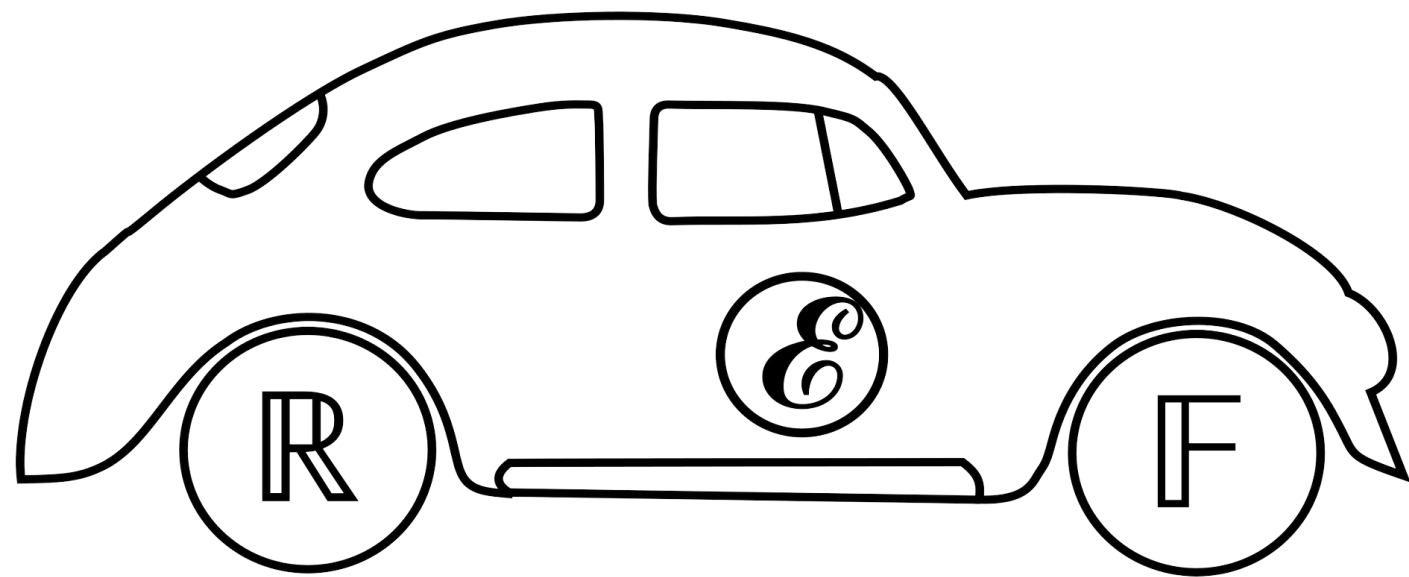
RTL Optimization





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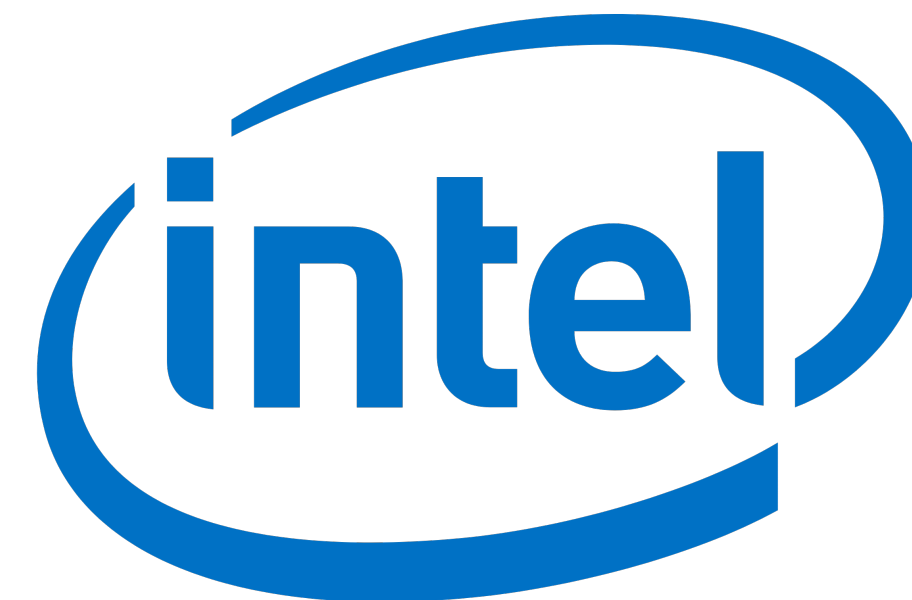
Floating point



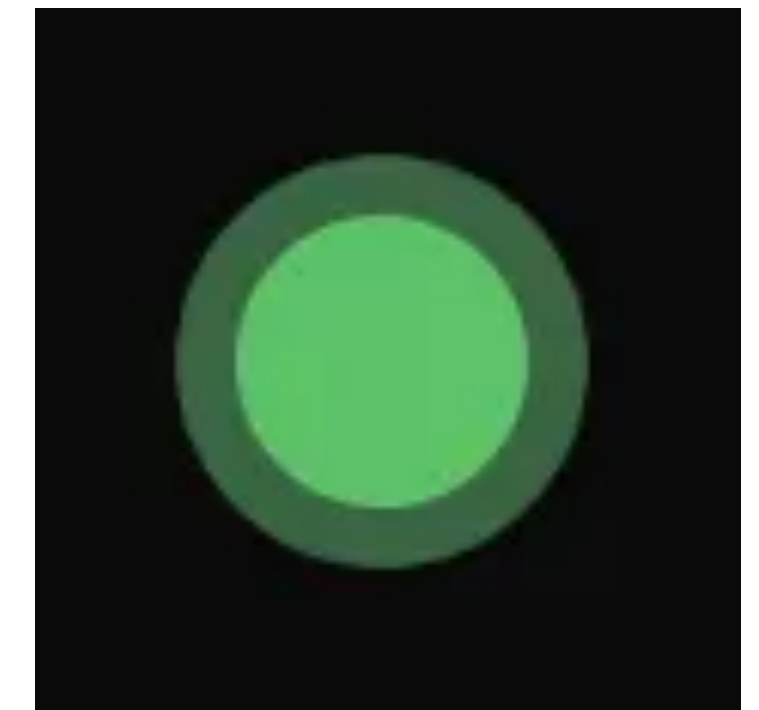
Python Simulations



RTL Optimization



ML Kernels



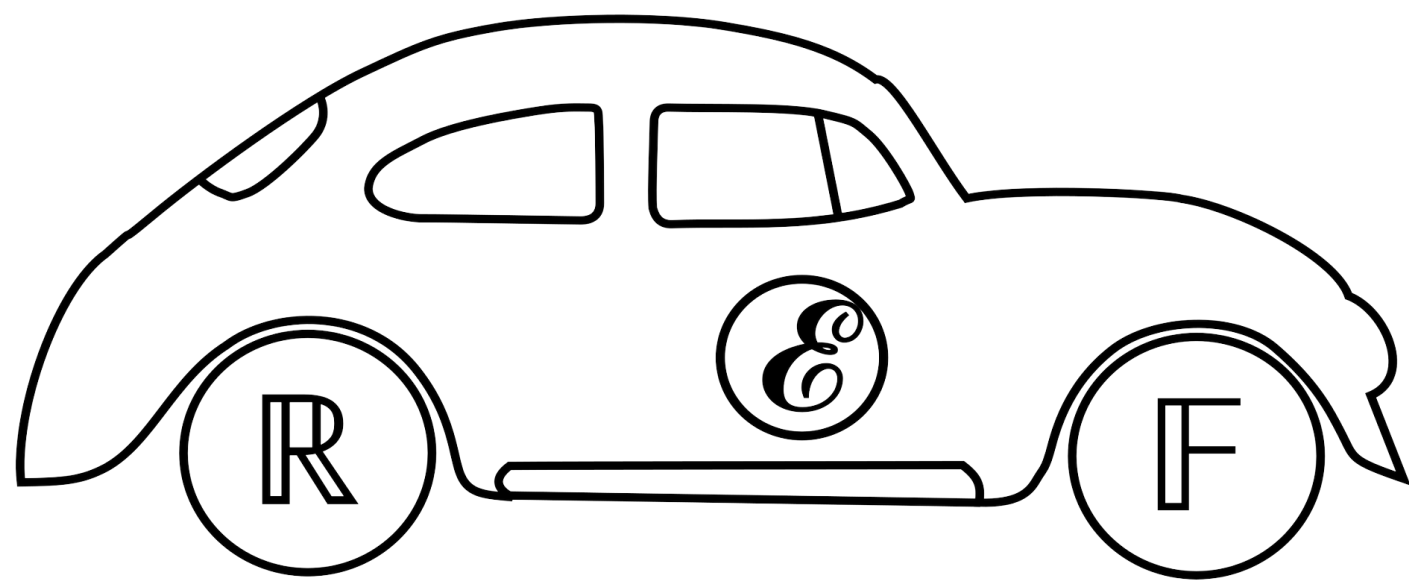
Luminal



Motivate Effective E-Graphs

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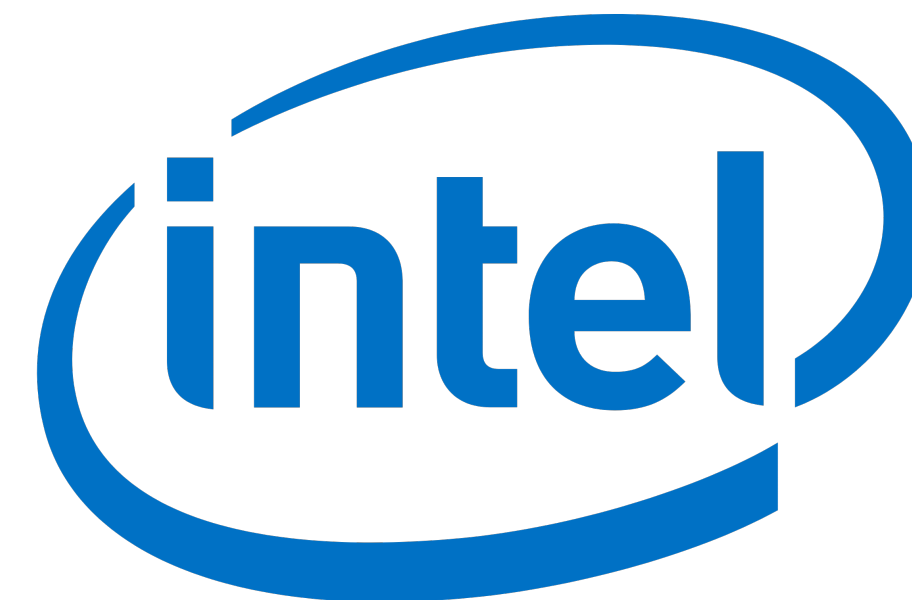
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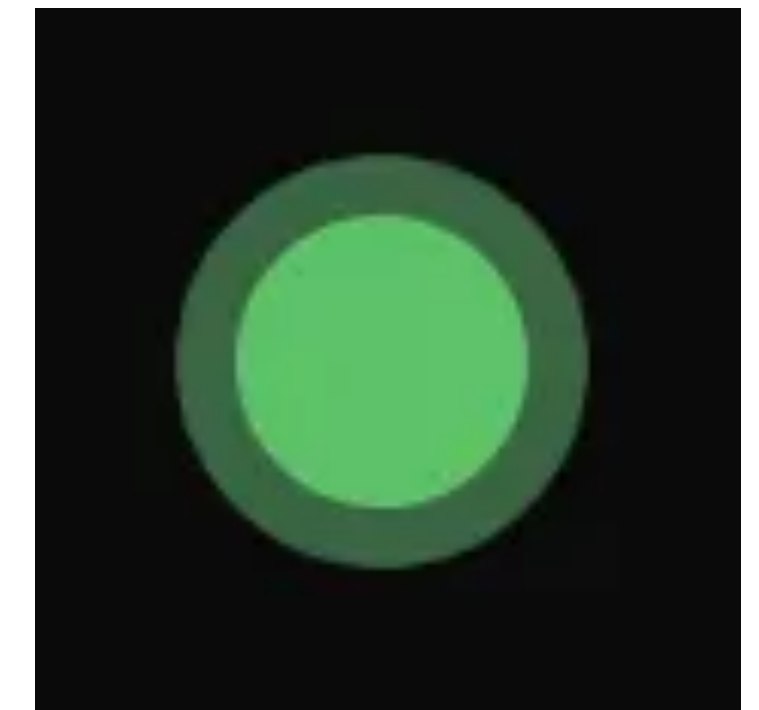
Python Simulations



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ML Kernels



Luminal

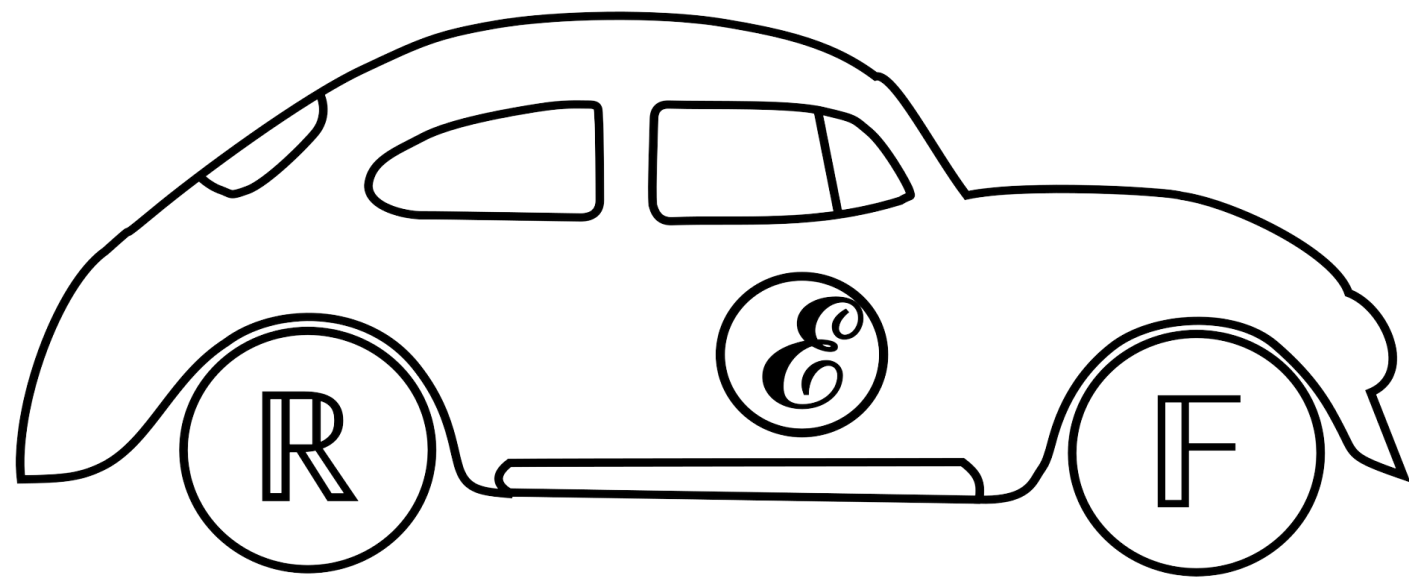
SQL



RisingWave

E-Graphs are Super Useful!

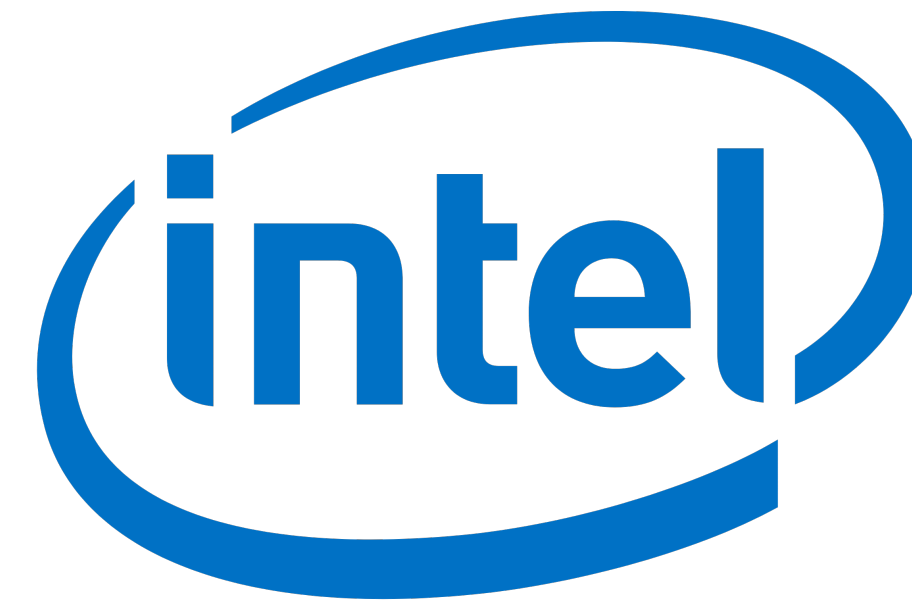
Floating point



Python Simulations



RTL Optimization



ML Kernels



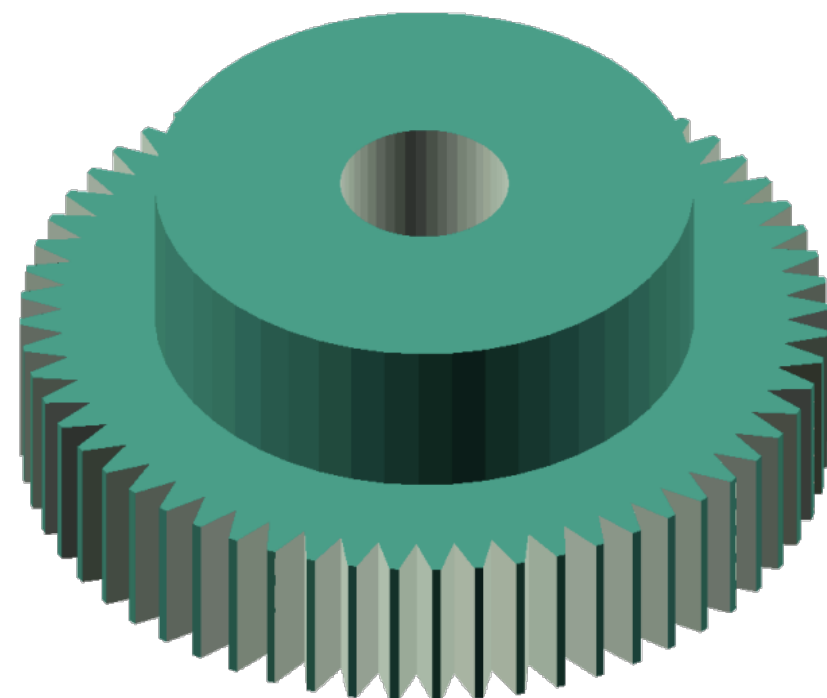
Luminal

SQL



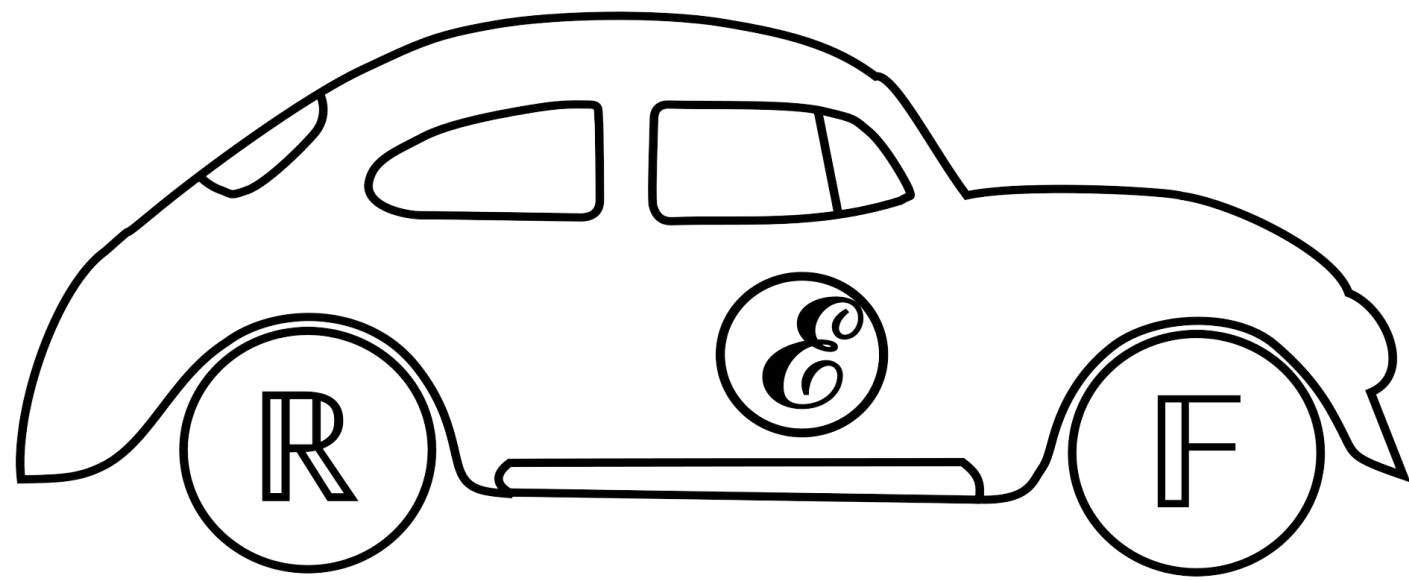
RisingWave

CAD



E-Graphs are Super Useful!

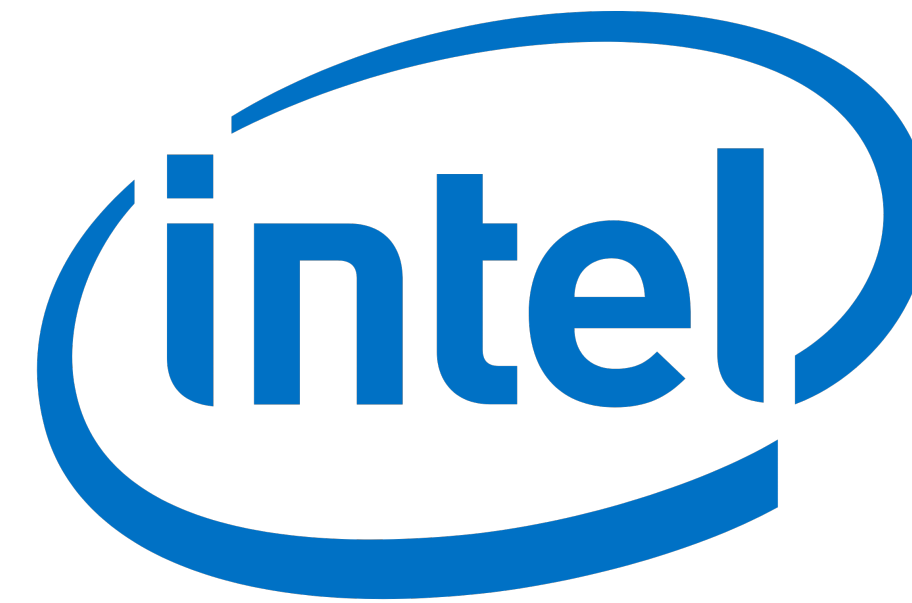
Floating point



Python Simulations



RTL Optimization



ML Kernels



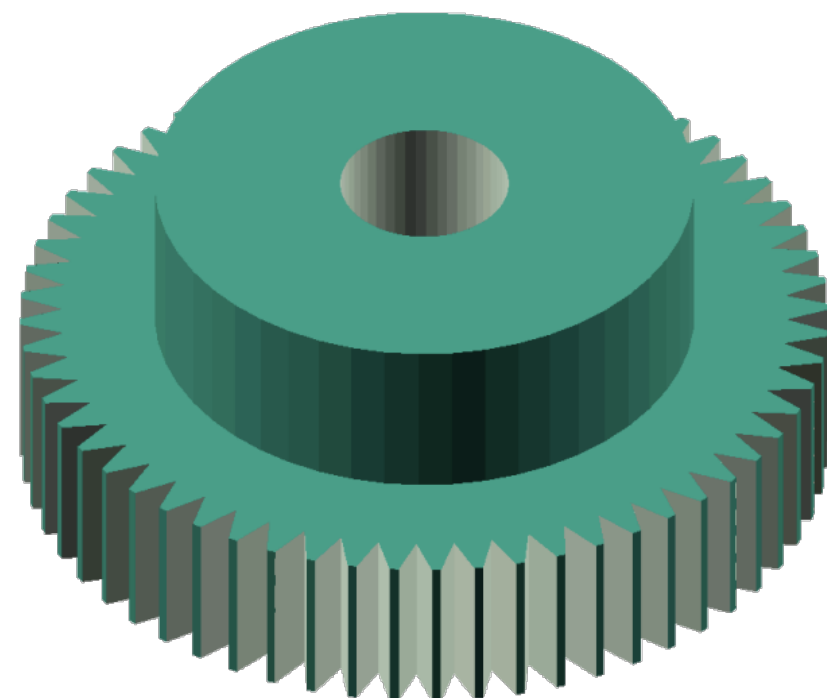
Luminal

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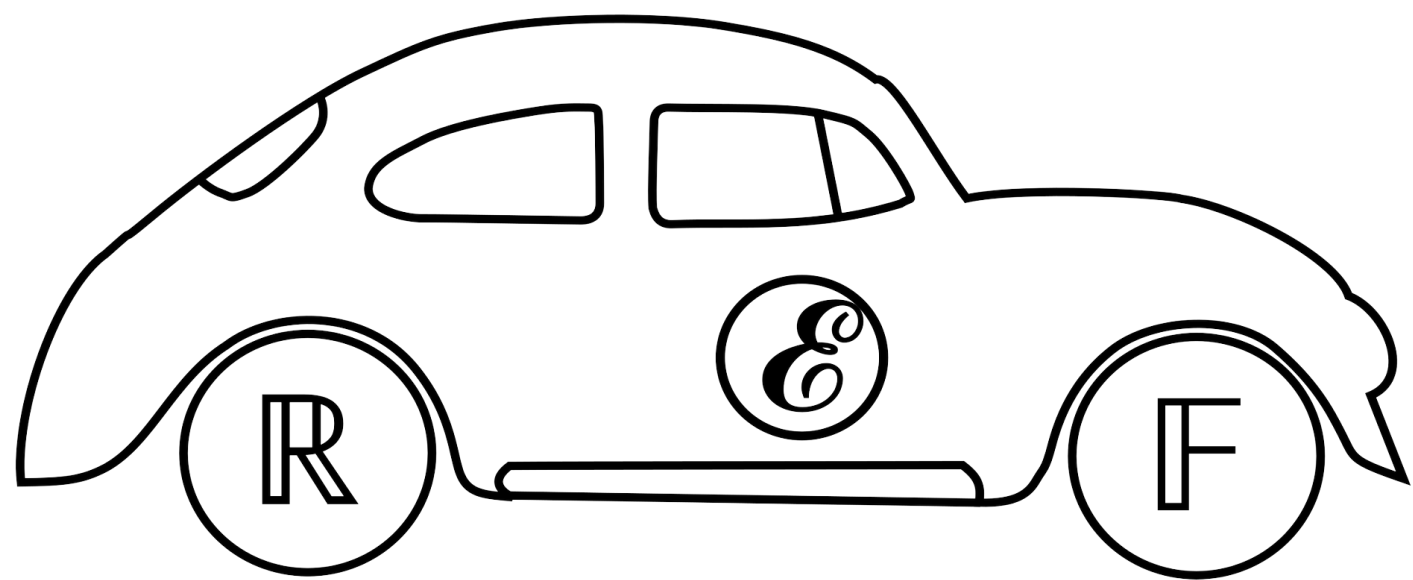


WASM



E-Graphs are Super Useful!

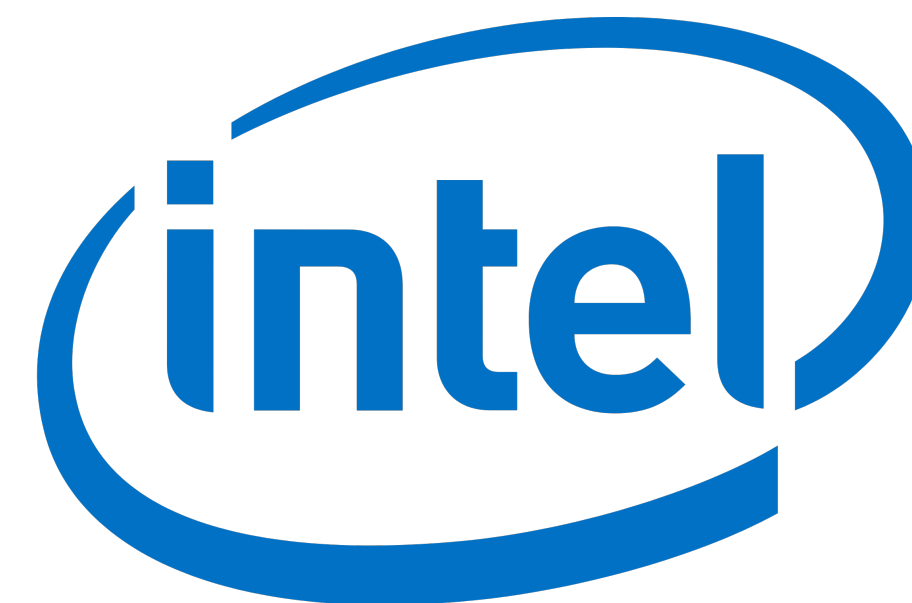
Floating point



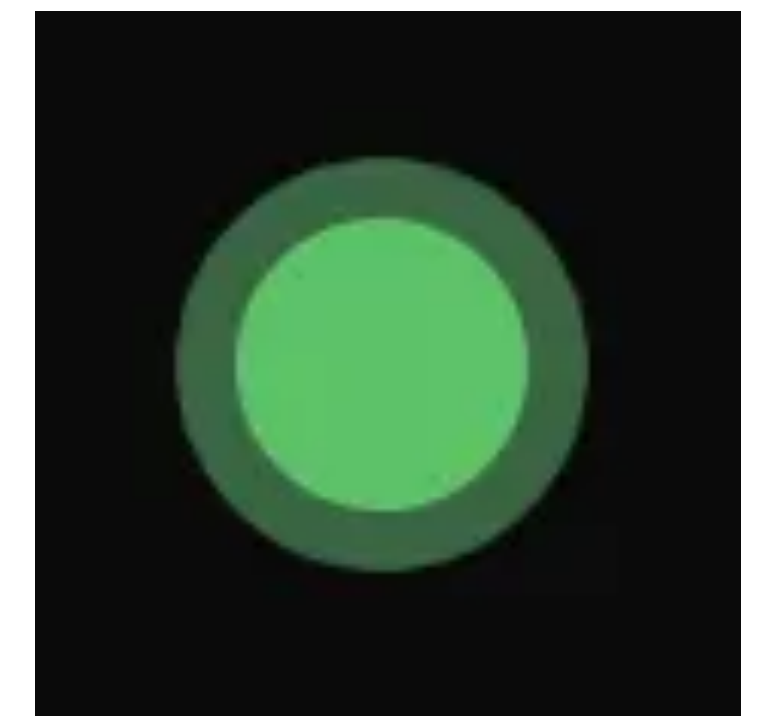
Python Simulations



RTL Optimization



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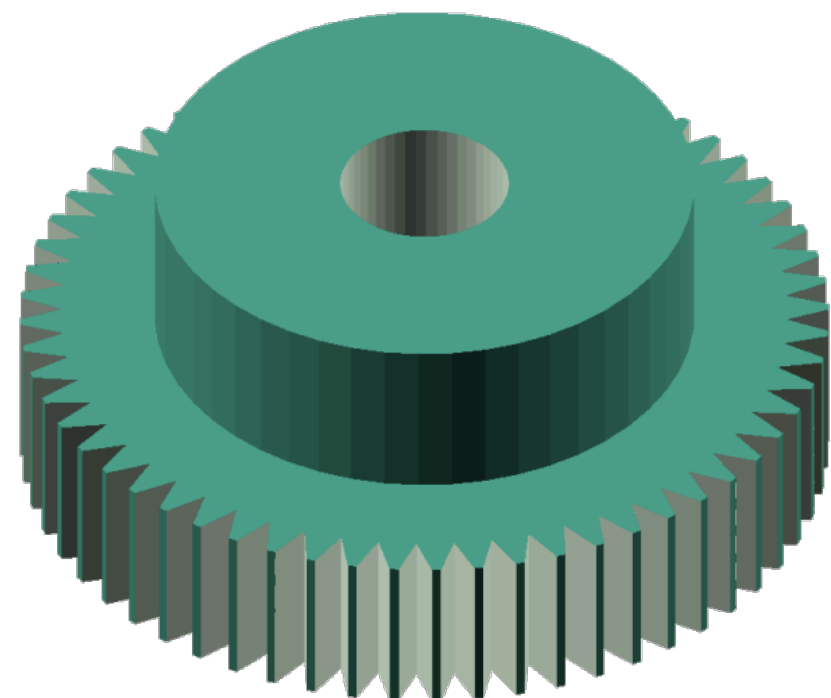


SQL

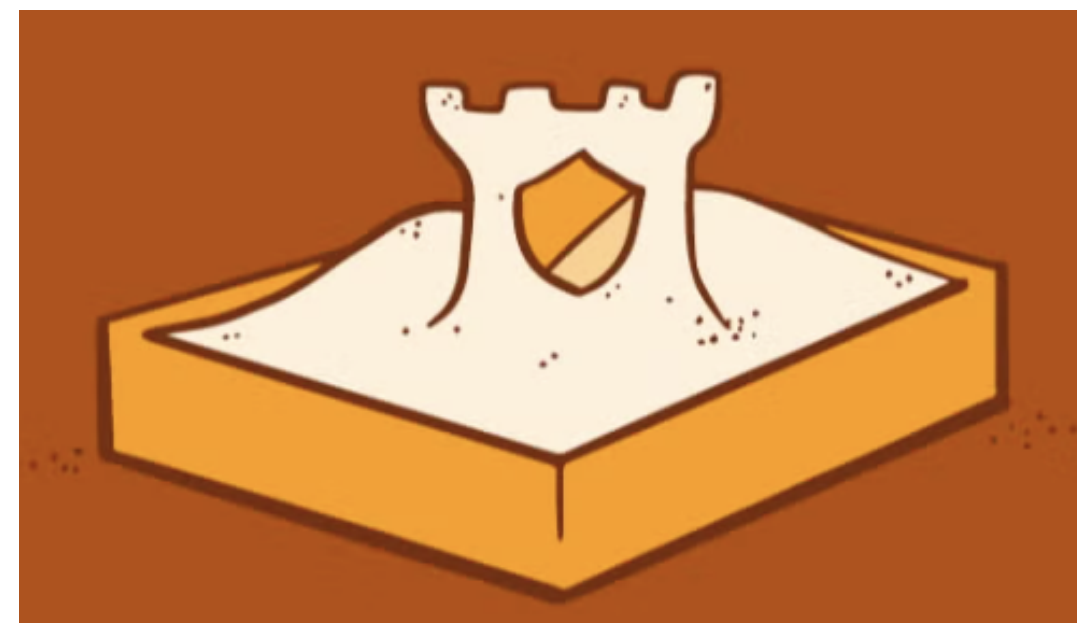


RisingWave

CAD



WASM

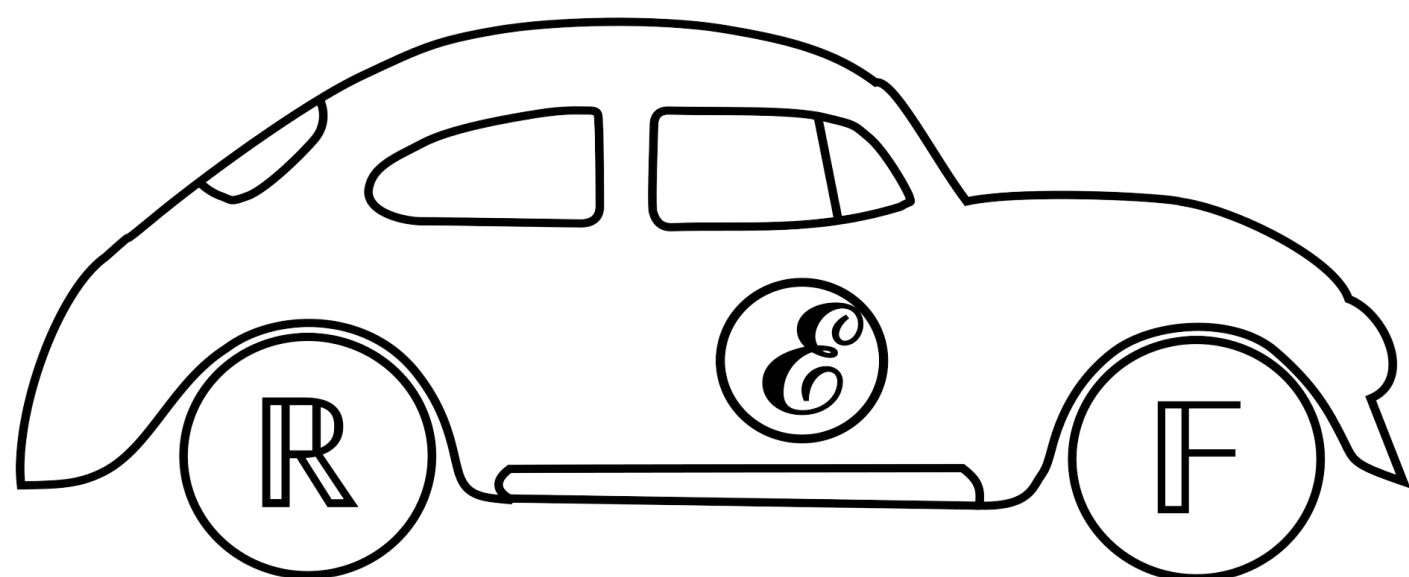


Luminal
User DSL Optimizations



E-Graphs are Super Useful!

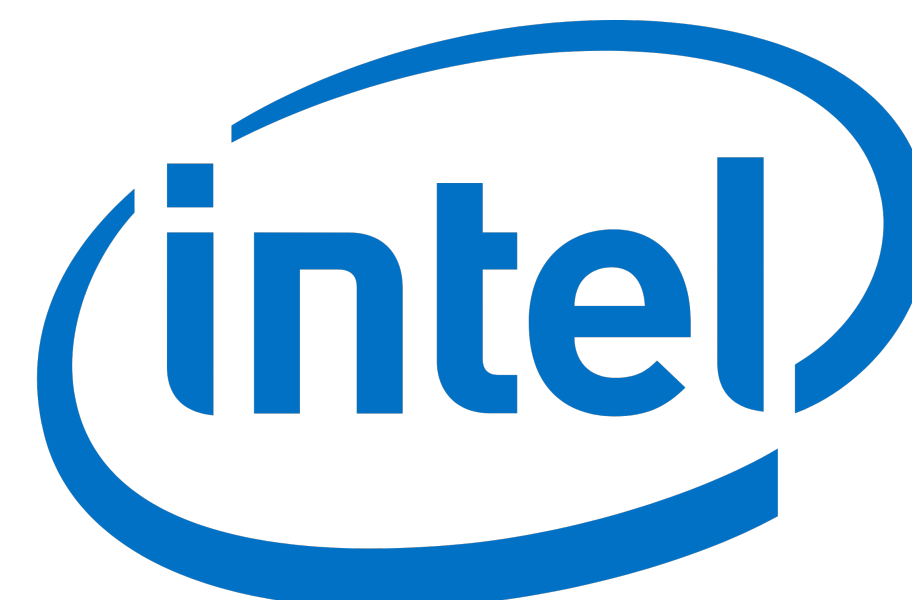
Floating point



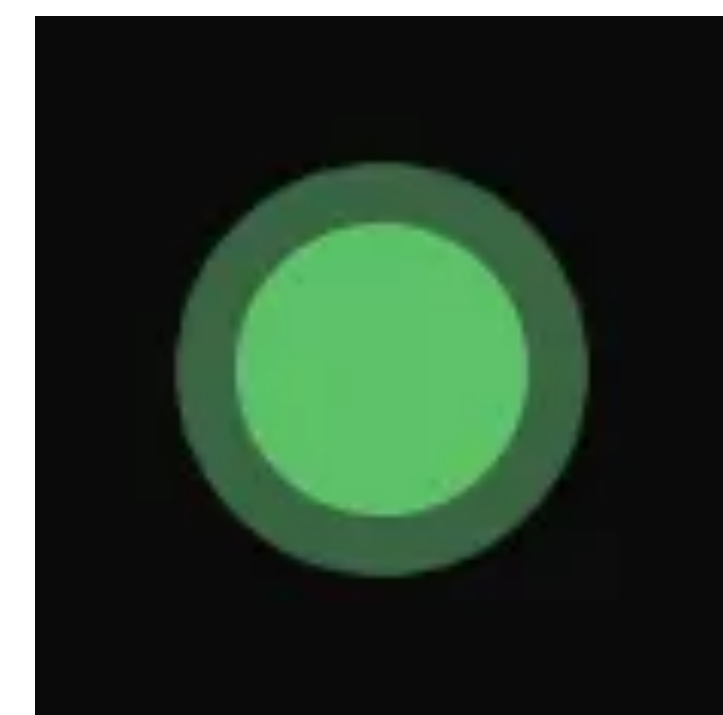
Python Simulations



RTL Optimization



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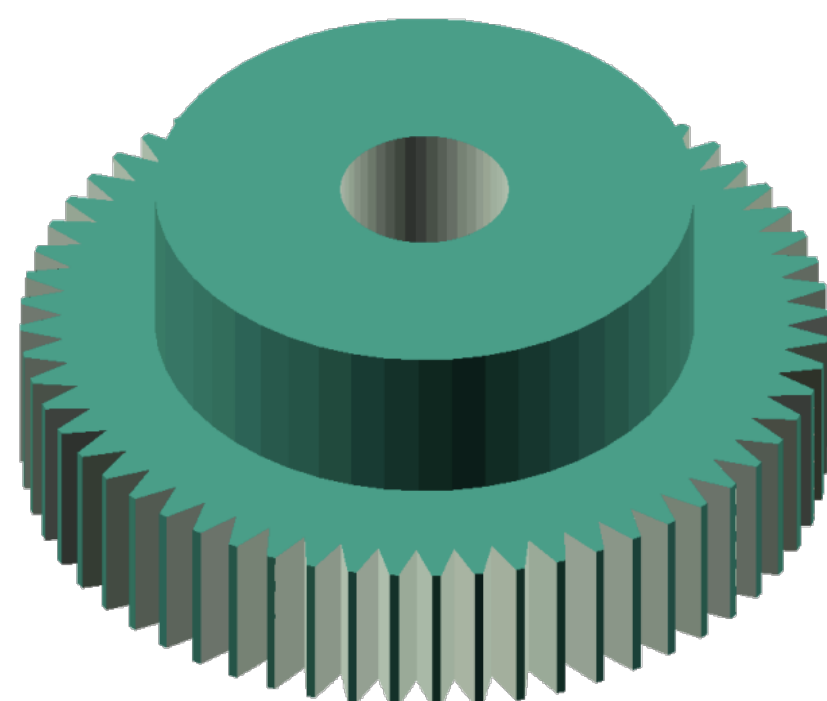


SQL

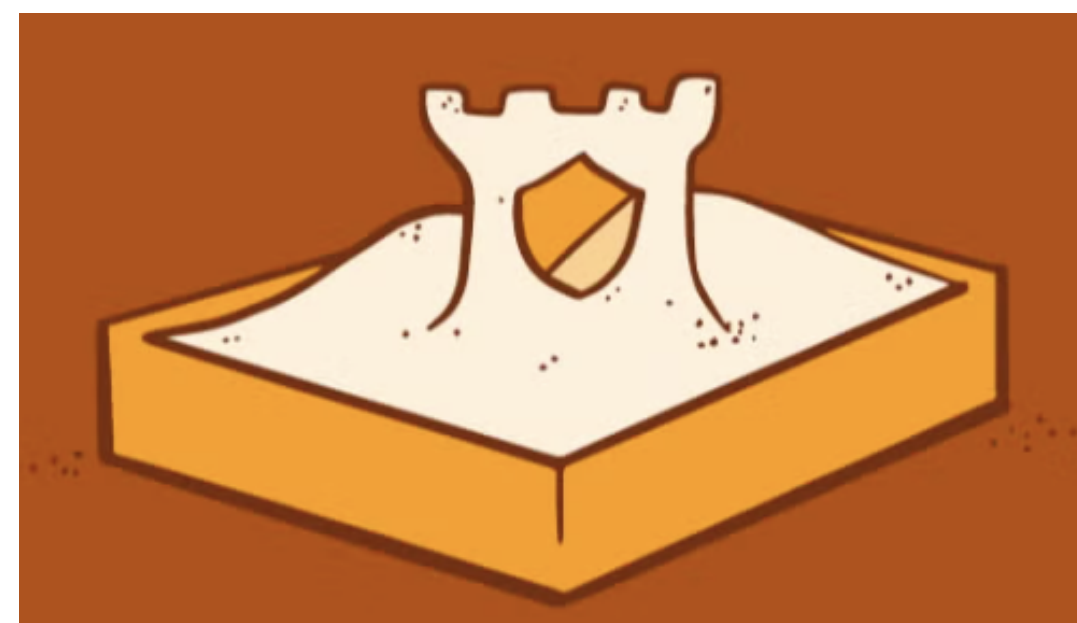


RisingWave

CAD



WASM



User DSL Optimizations



...and more!

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Motivate Effective E-Graphs

E-Graphs Explore a Big Space

Input Program

$(a + 1) - 1$

E-Graphs Explore a Big Space

Input Program

$(a + 1) - 1$

Optimization Rules

$(\alpha + \beta) - \gamma \rightarrow \alpha + (\beta - \gamma)$

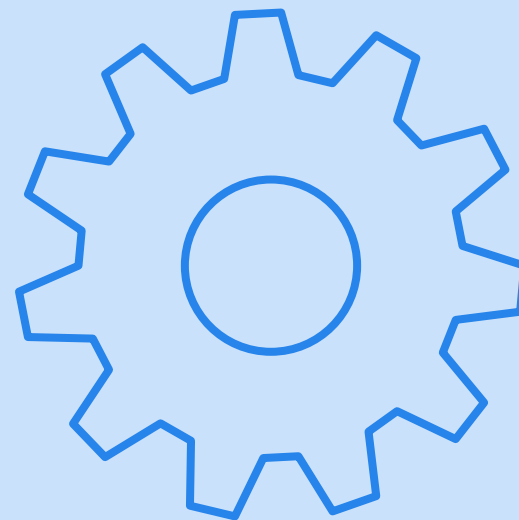
$\alpha - \alpha \rightarrow 0$

E-Graphs Explore a Big Space

Input Program

$(a + 1) - 1$

E-Graph



Optimization Rules

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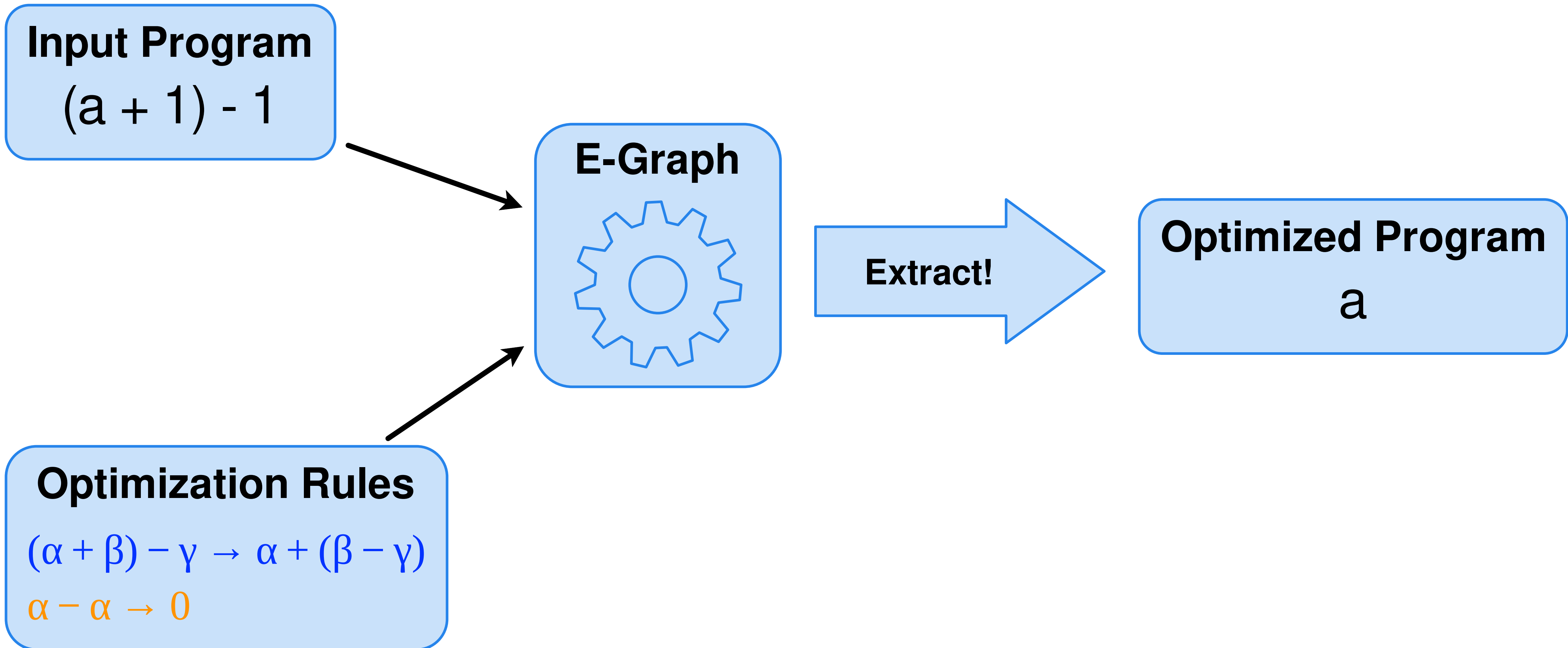
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E-Graphs Explore a Big Space



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Motivate Effective E-Graphs

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$$(\alpha + \beta) - \gamma \rightarrow \alpha + (\beta - \gamma) \quad (a + 1)$$

$$\alpha - \alpha \rightarrow 0$$

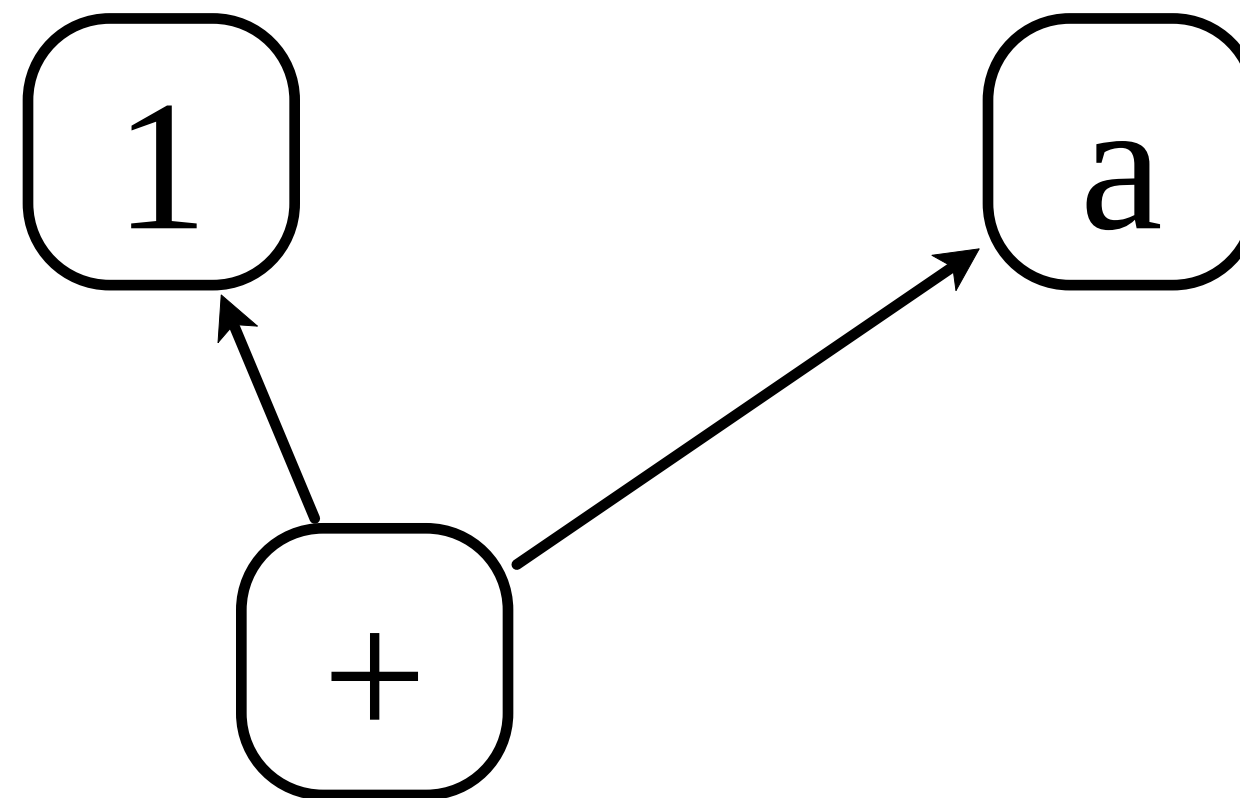
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data flow



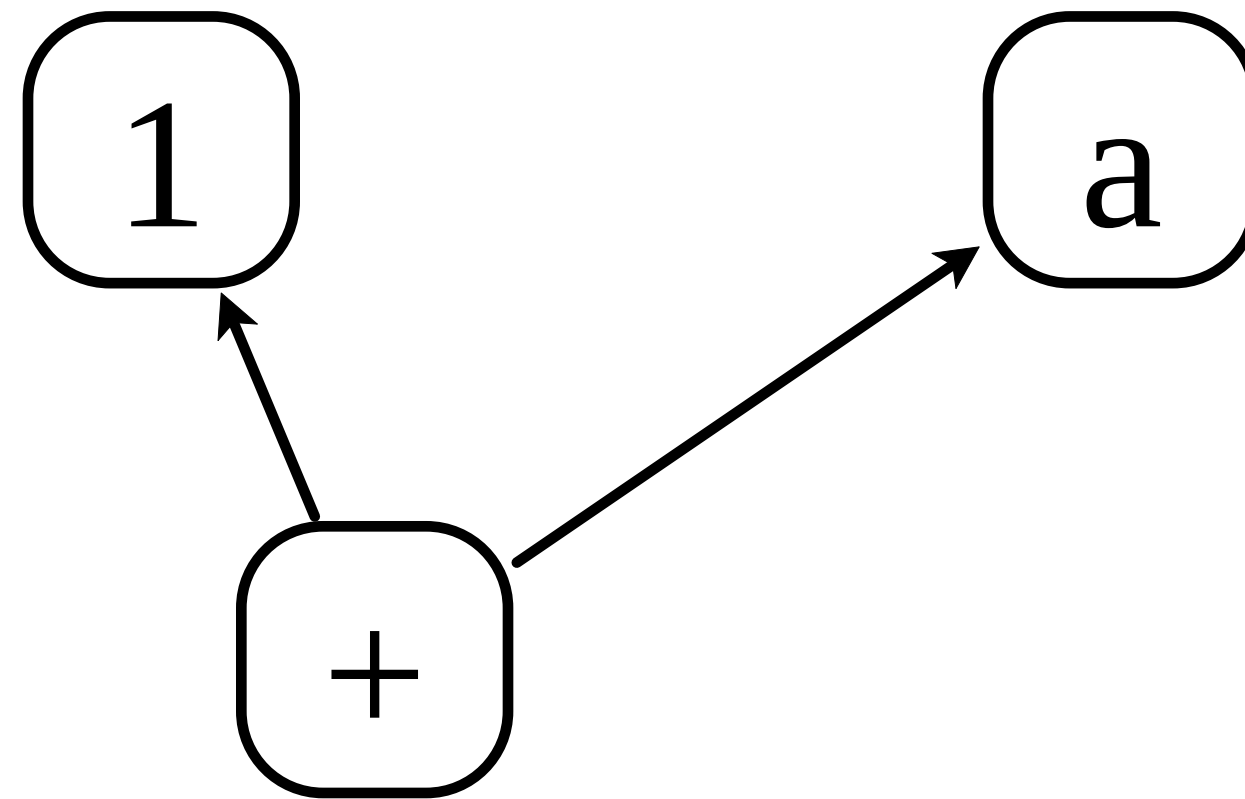
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data flow



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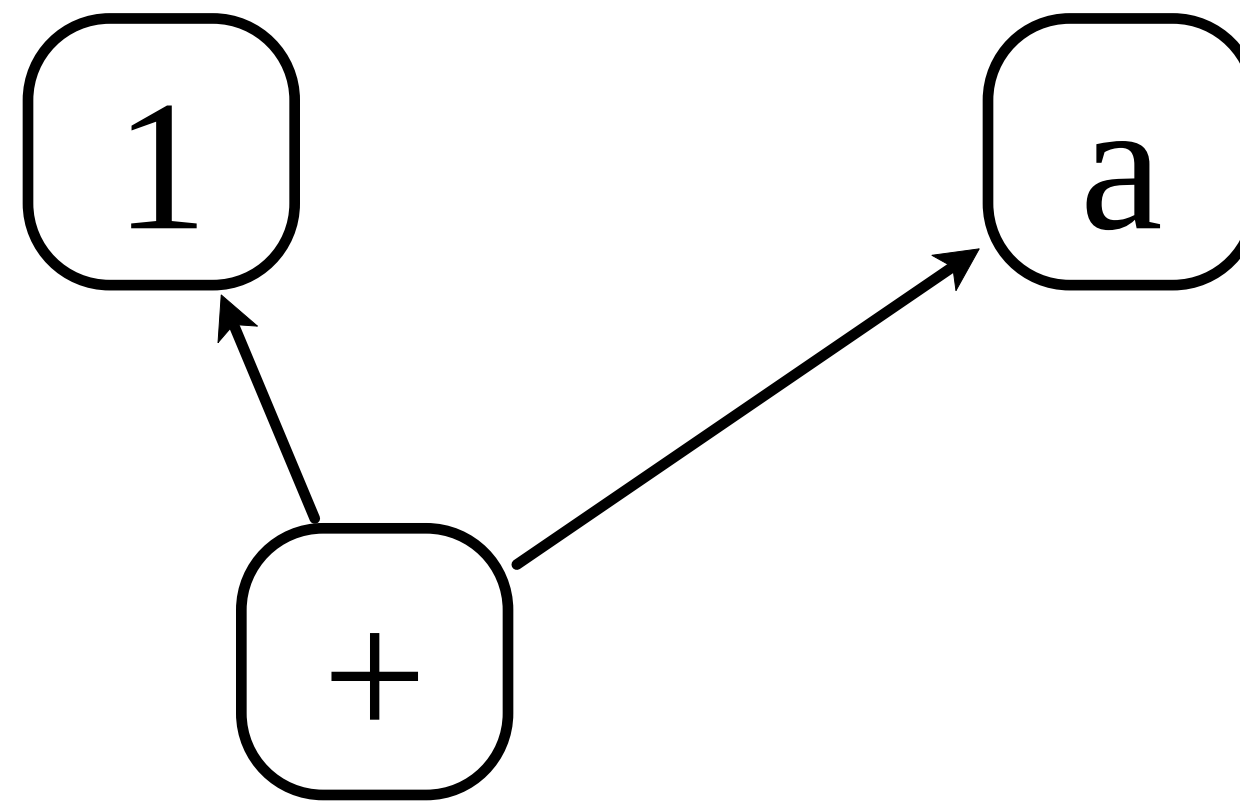
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$(a + 1) - 1$

data flow



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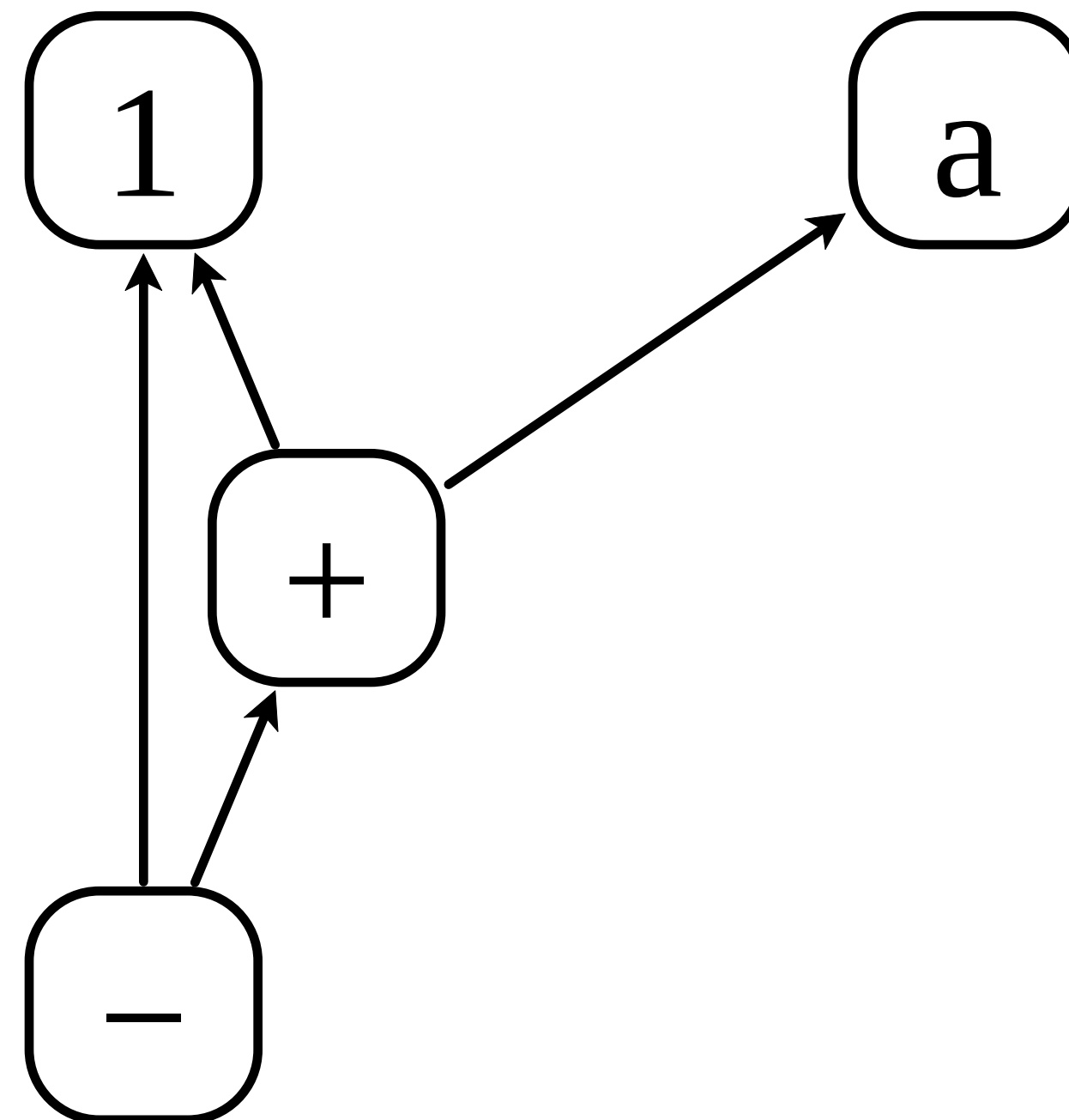
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data flow



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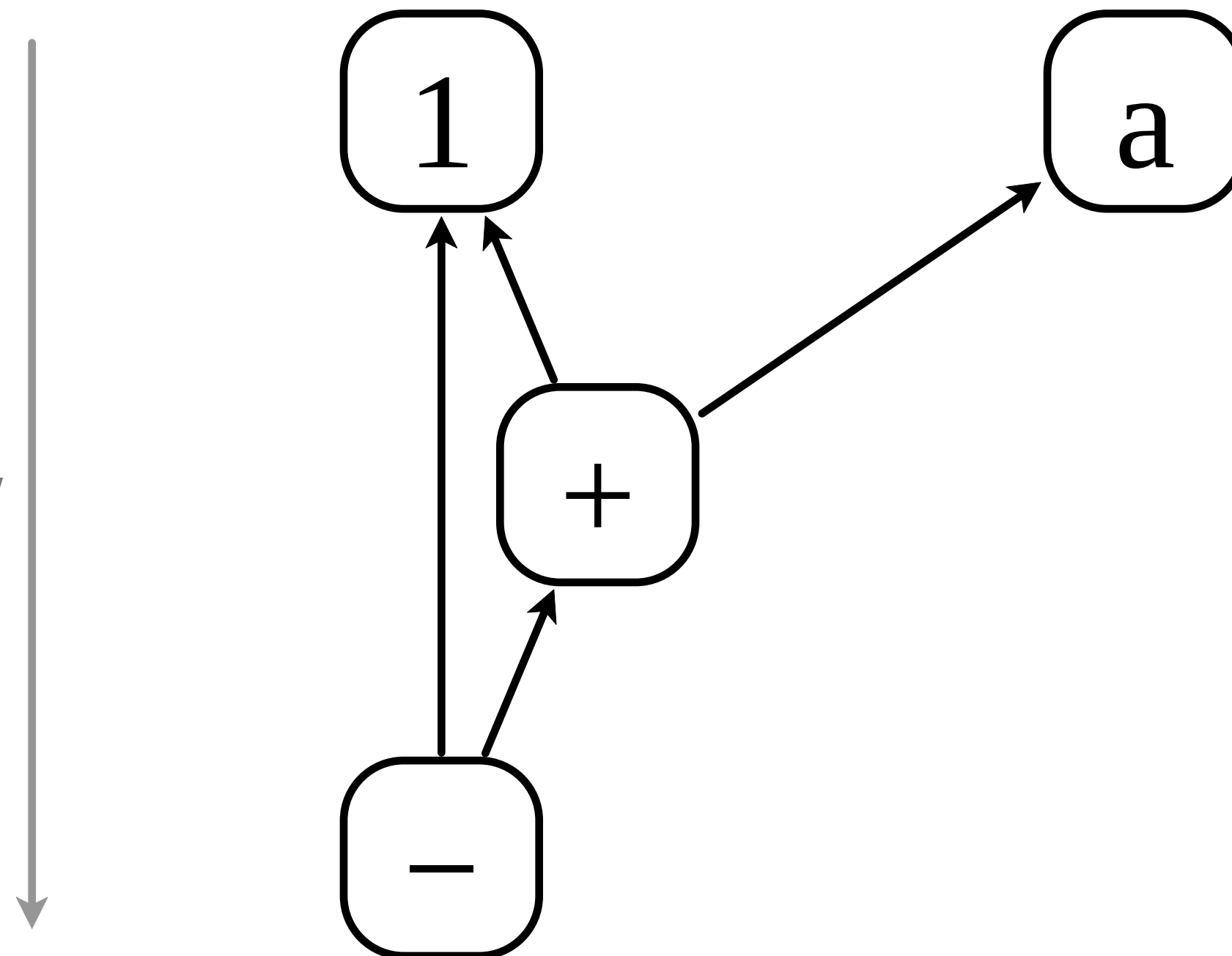
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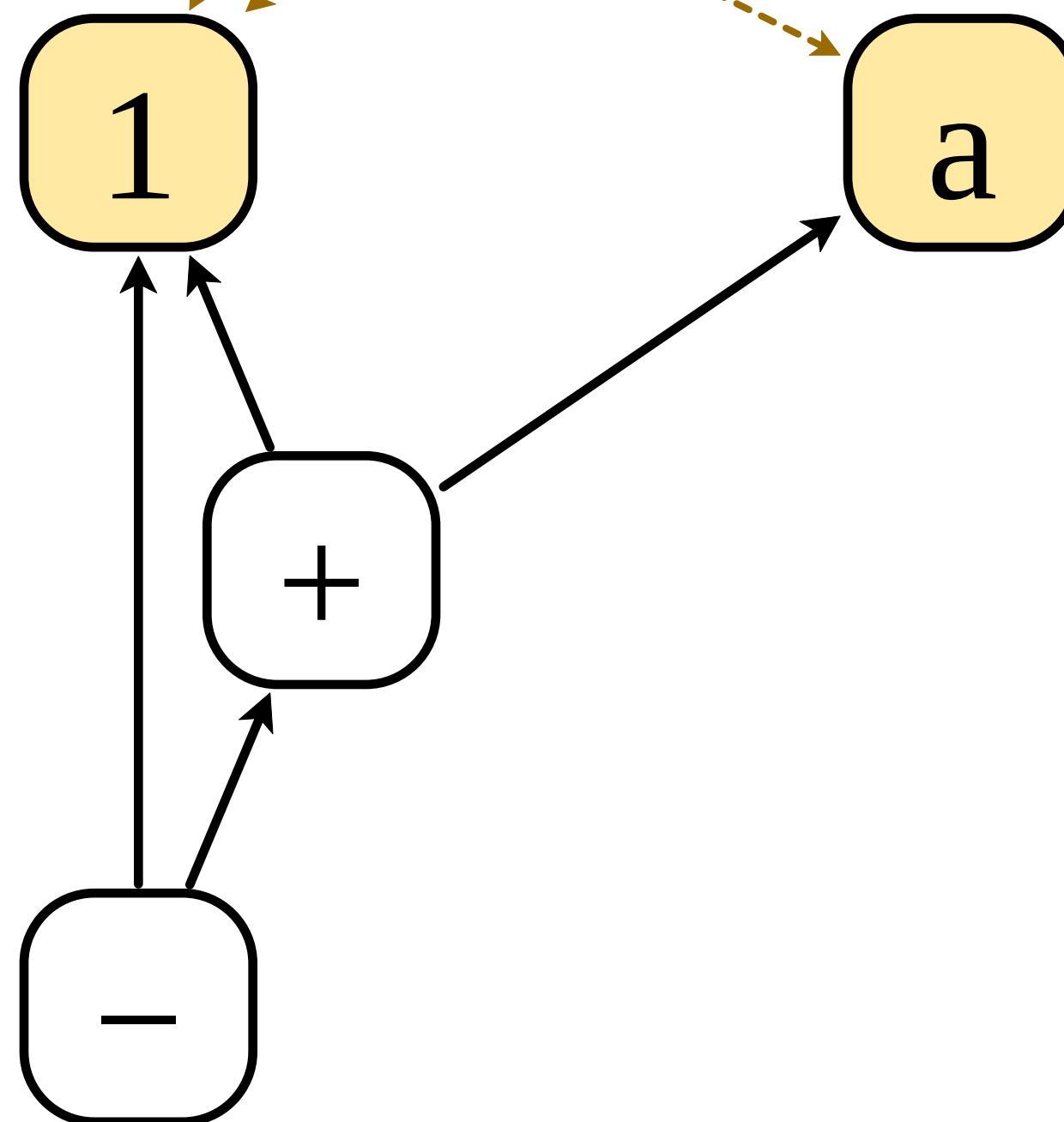
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data flow

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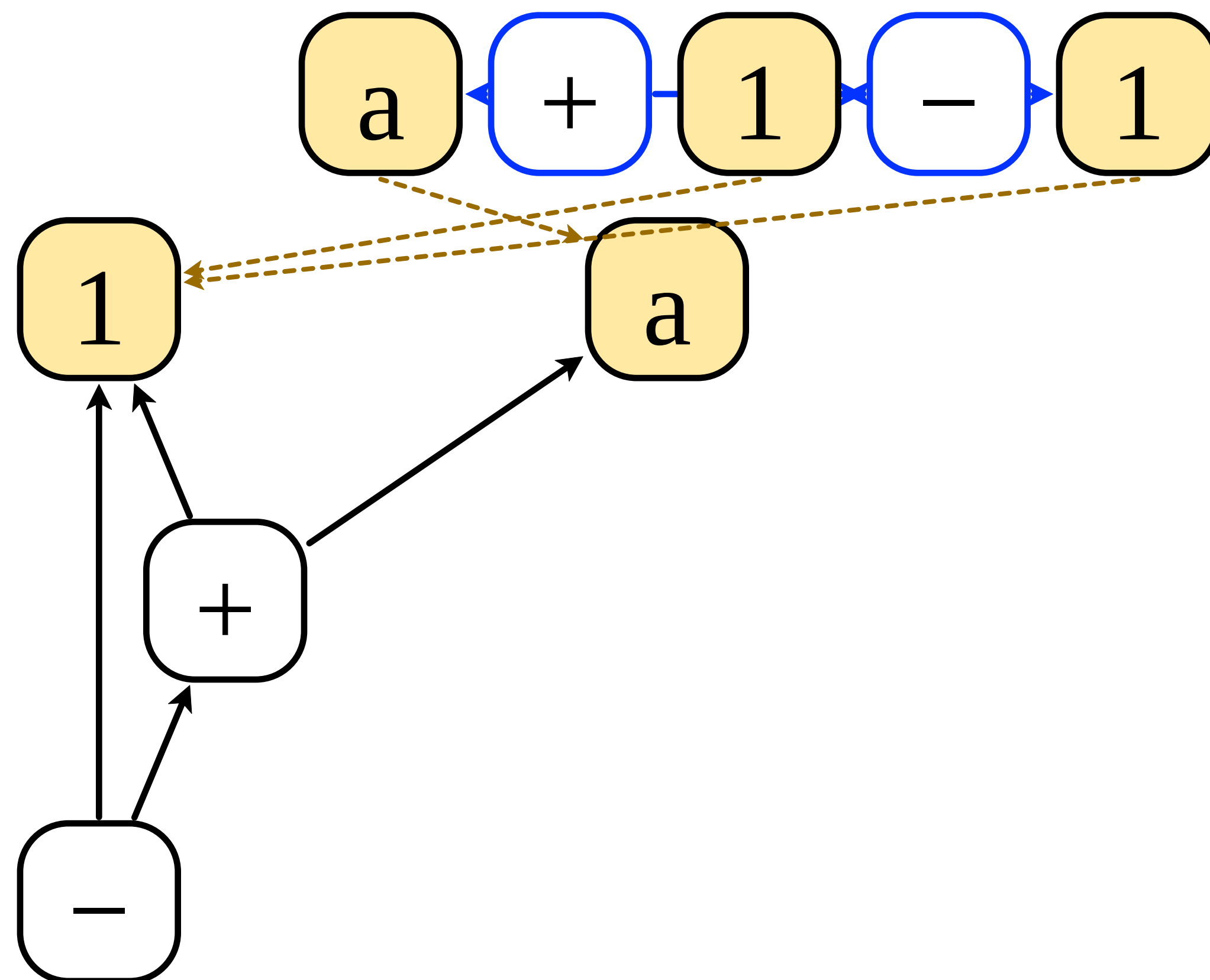
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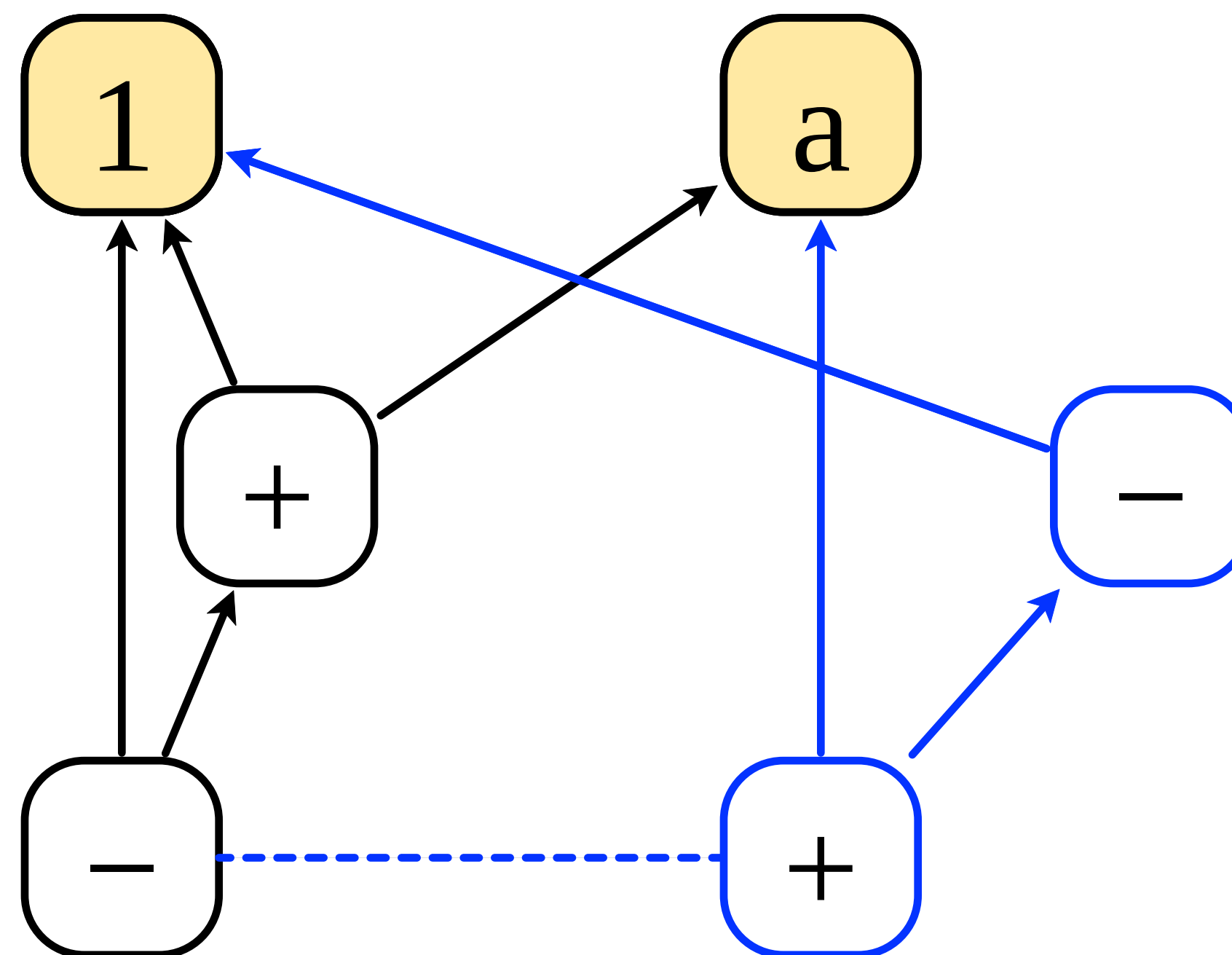
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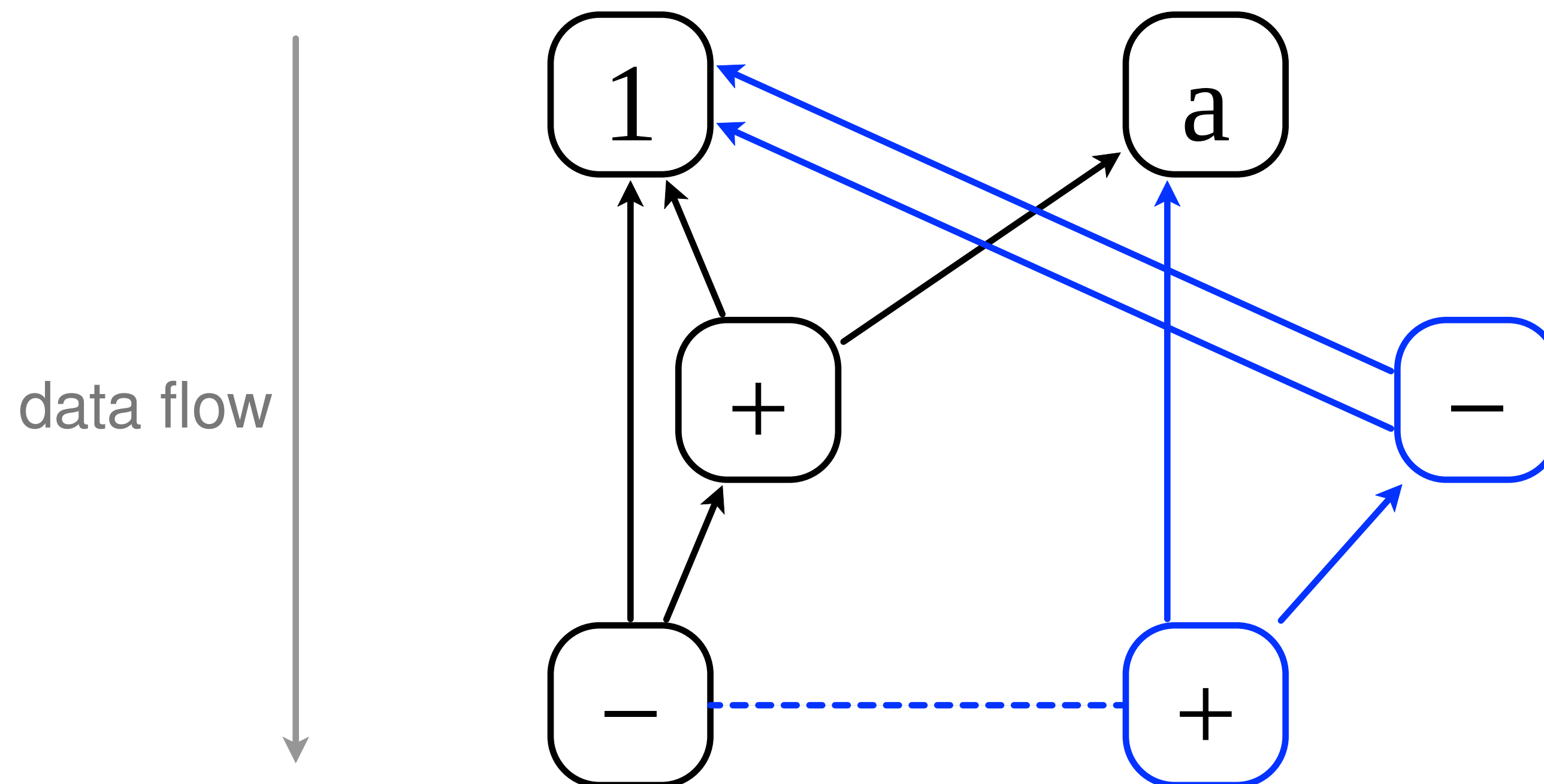


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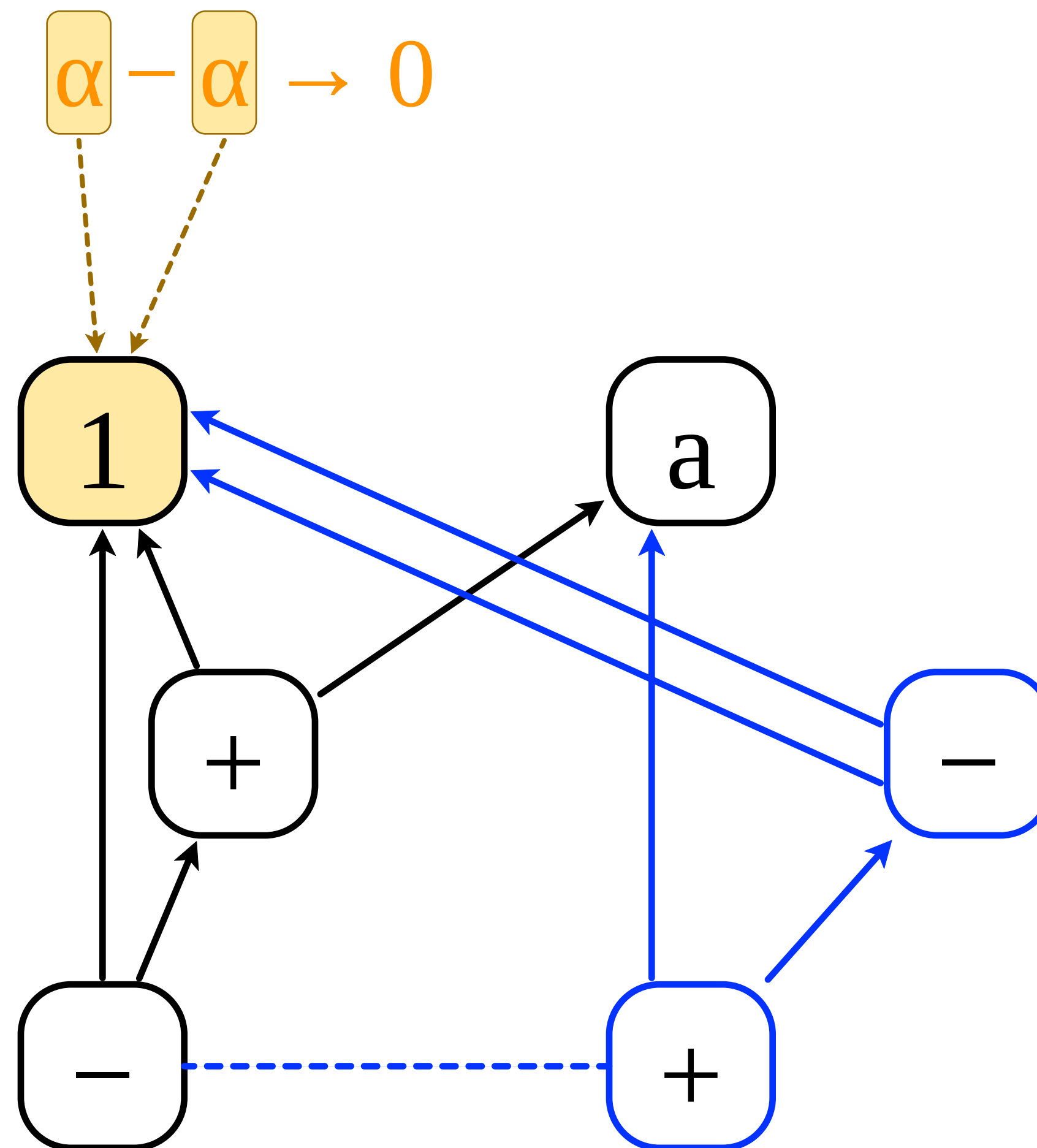


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data flow

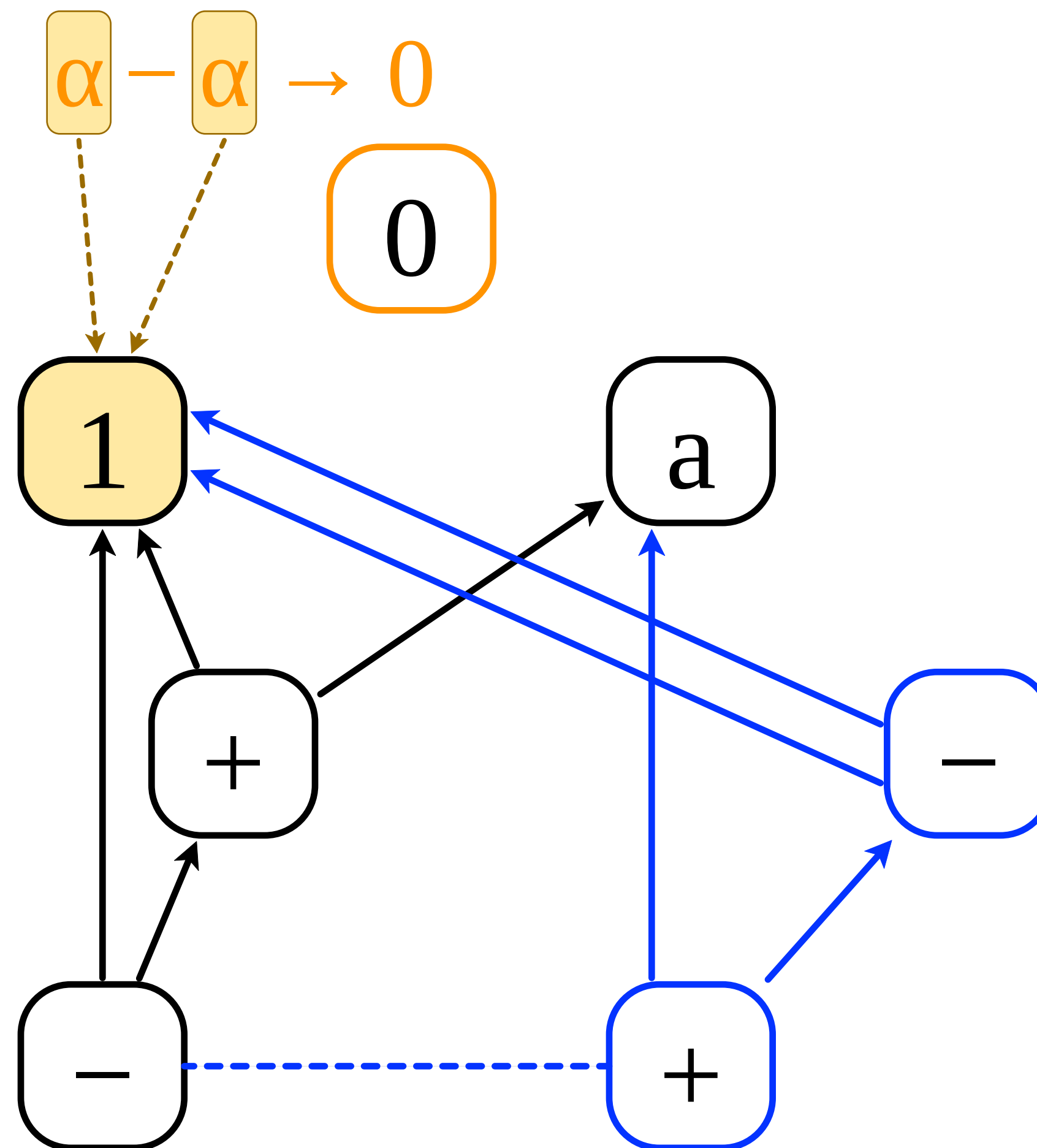


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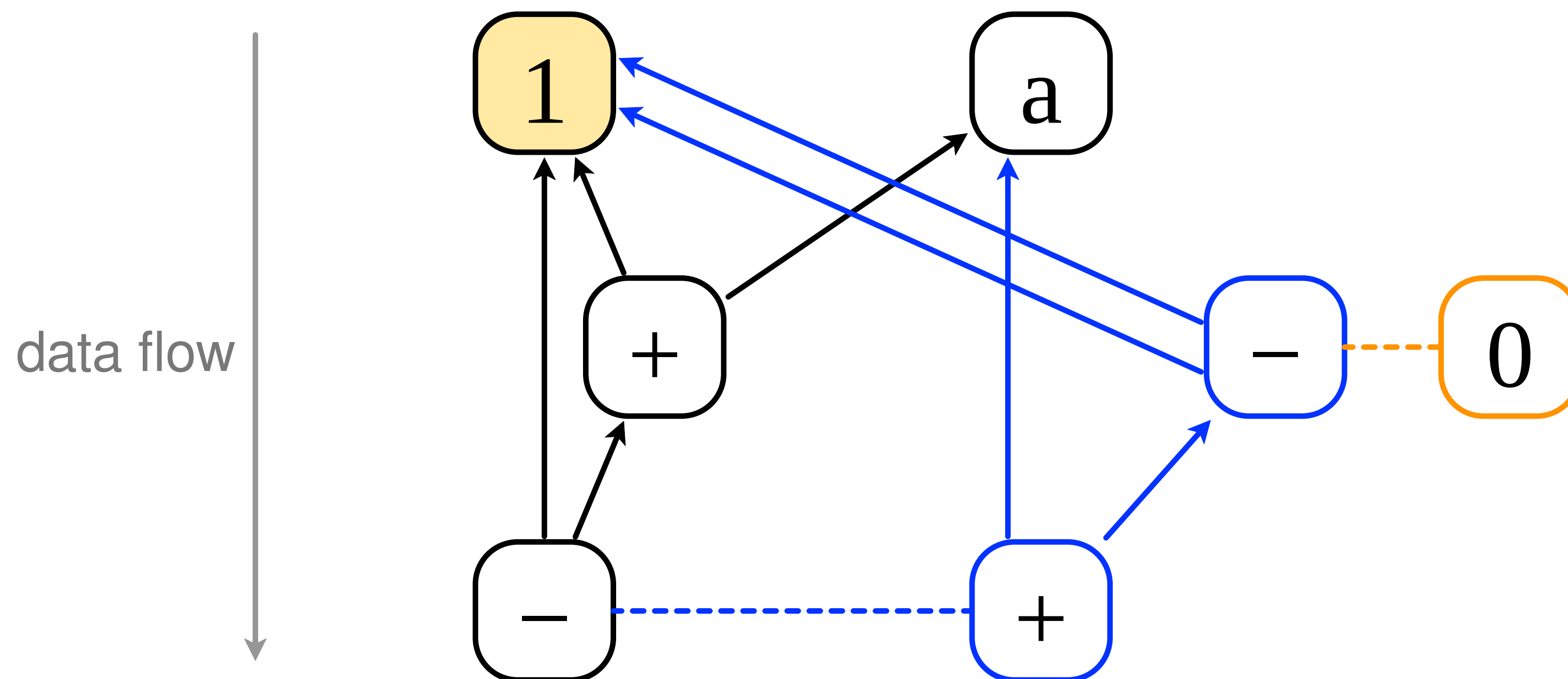


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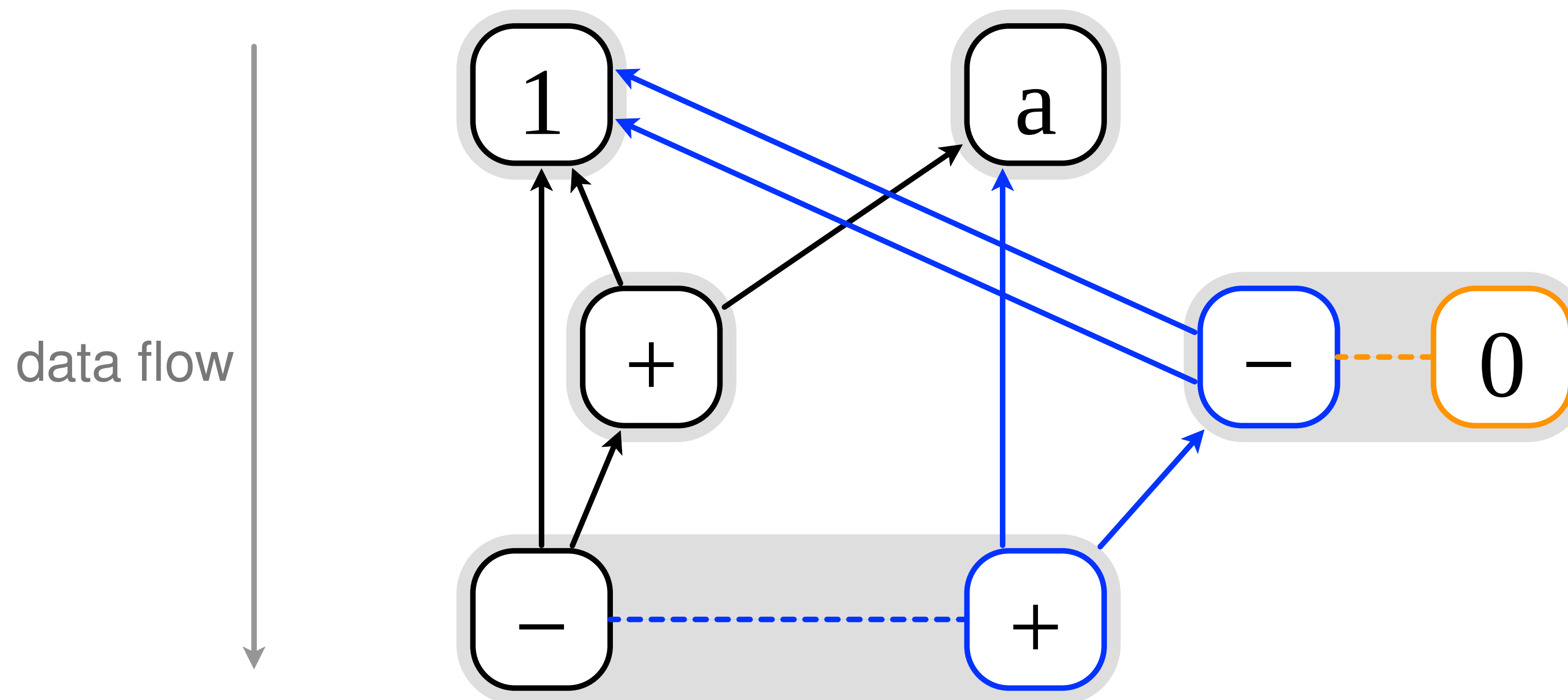


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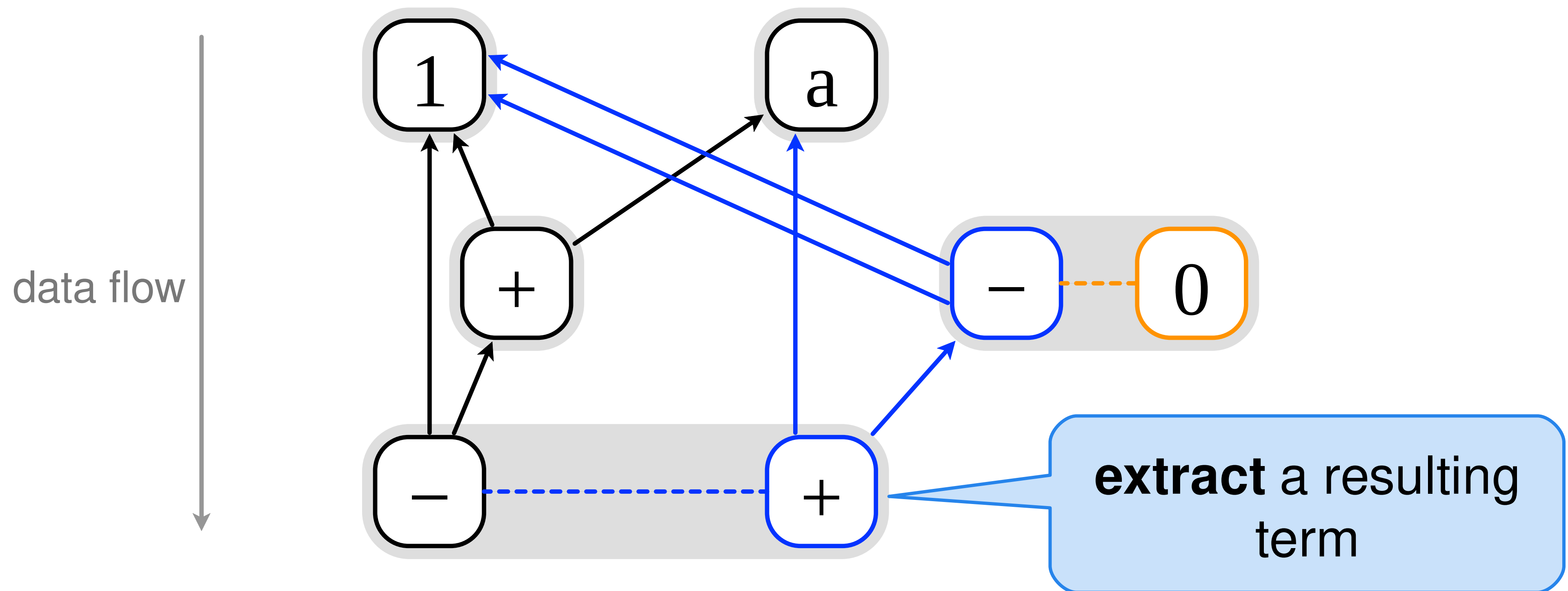


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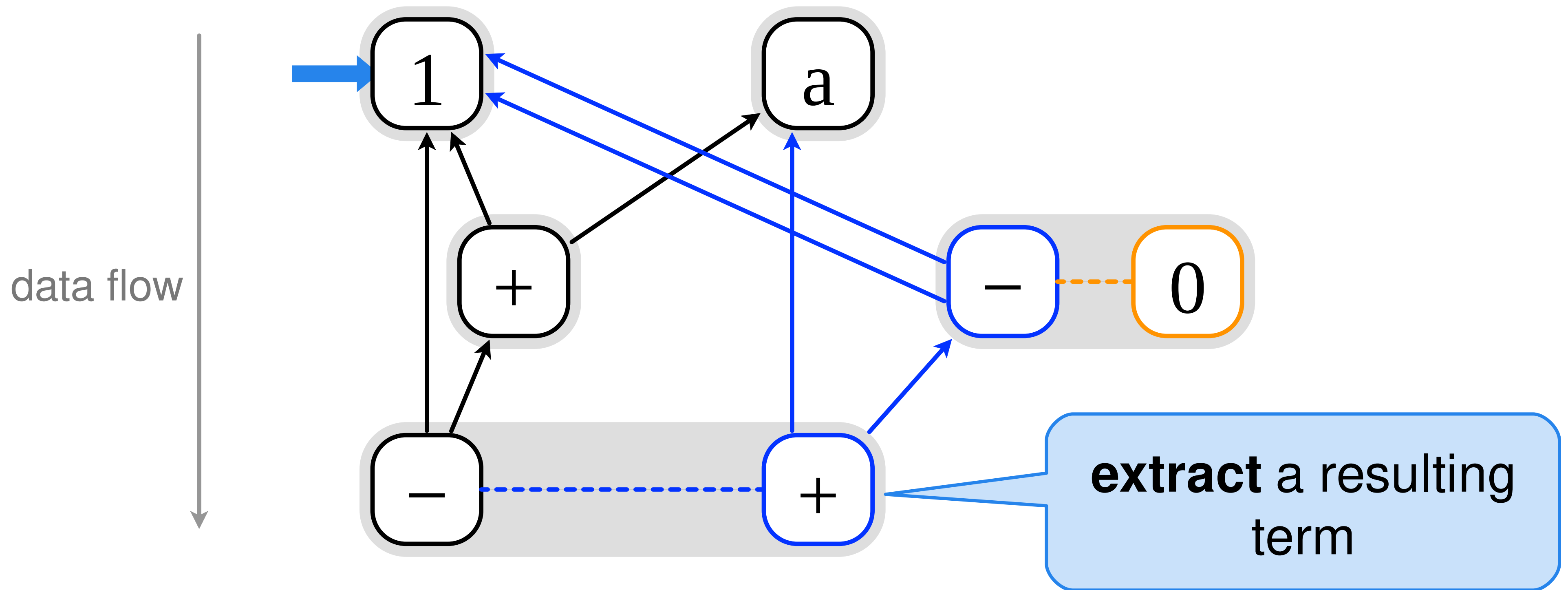


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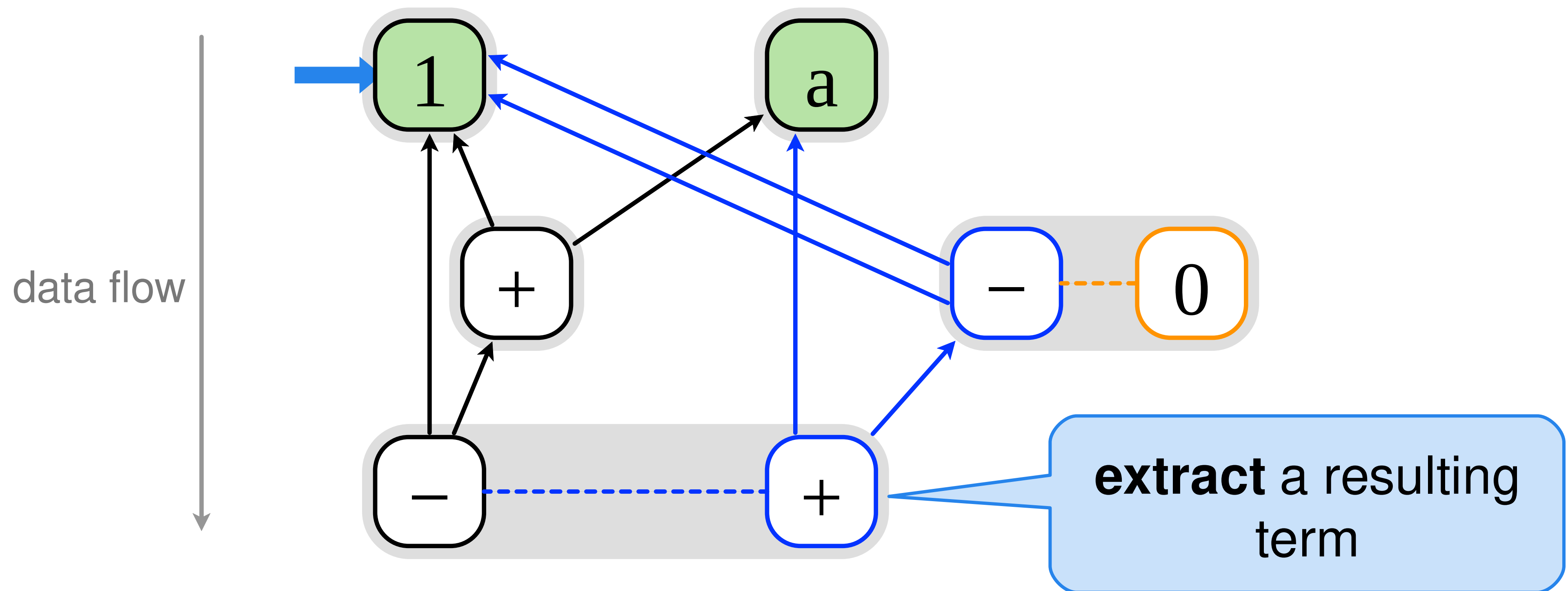


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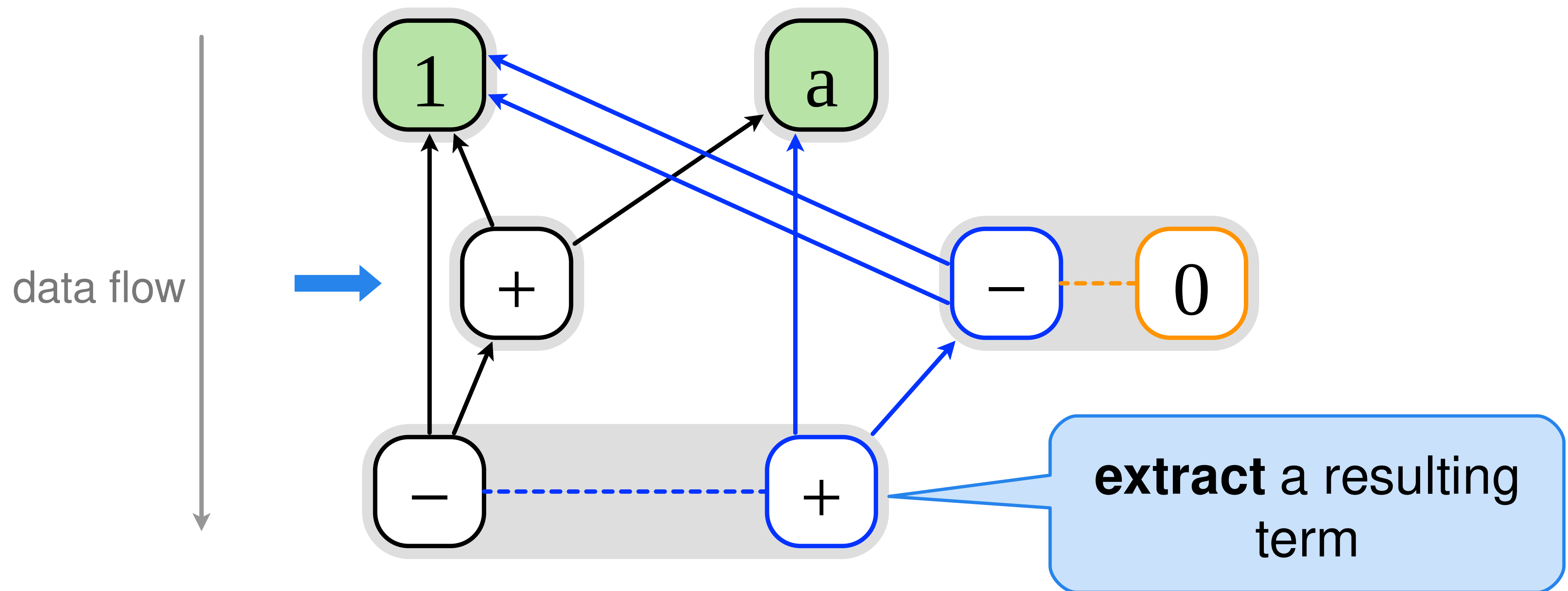


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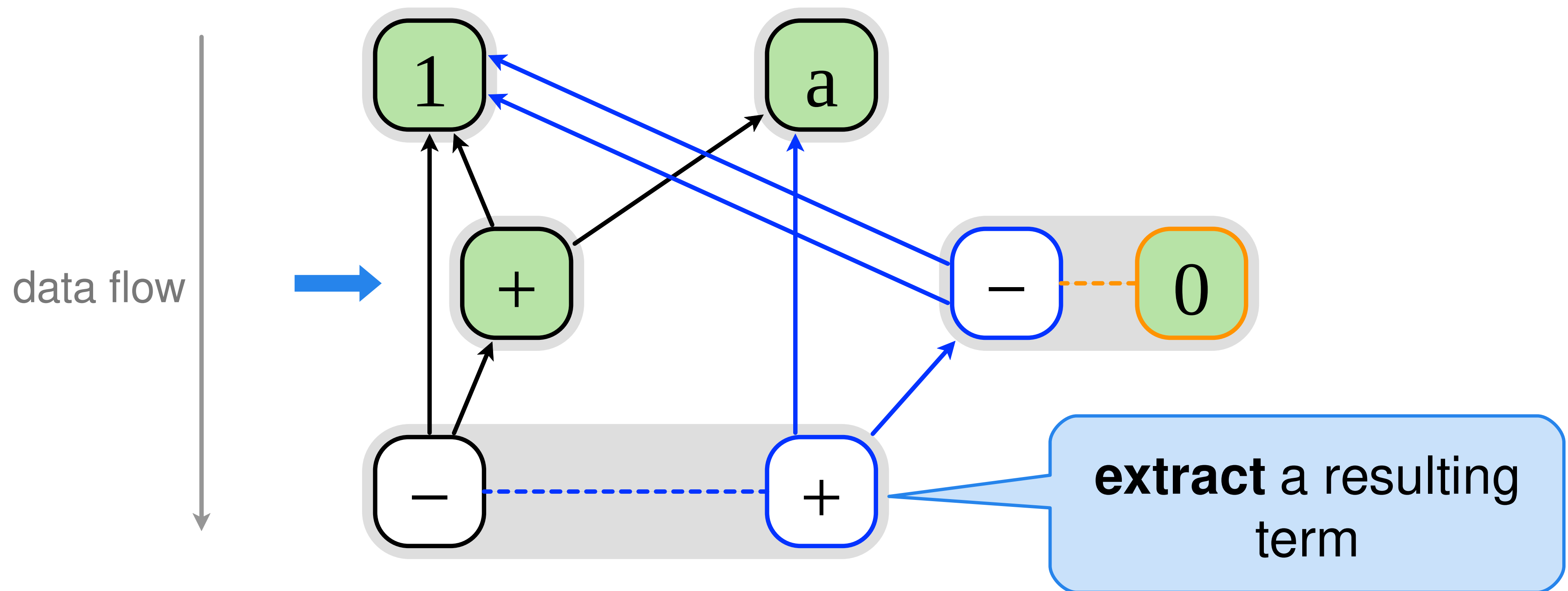


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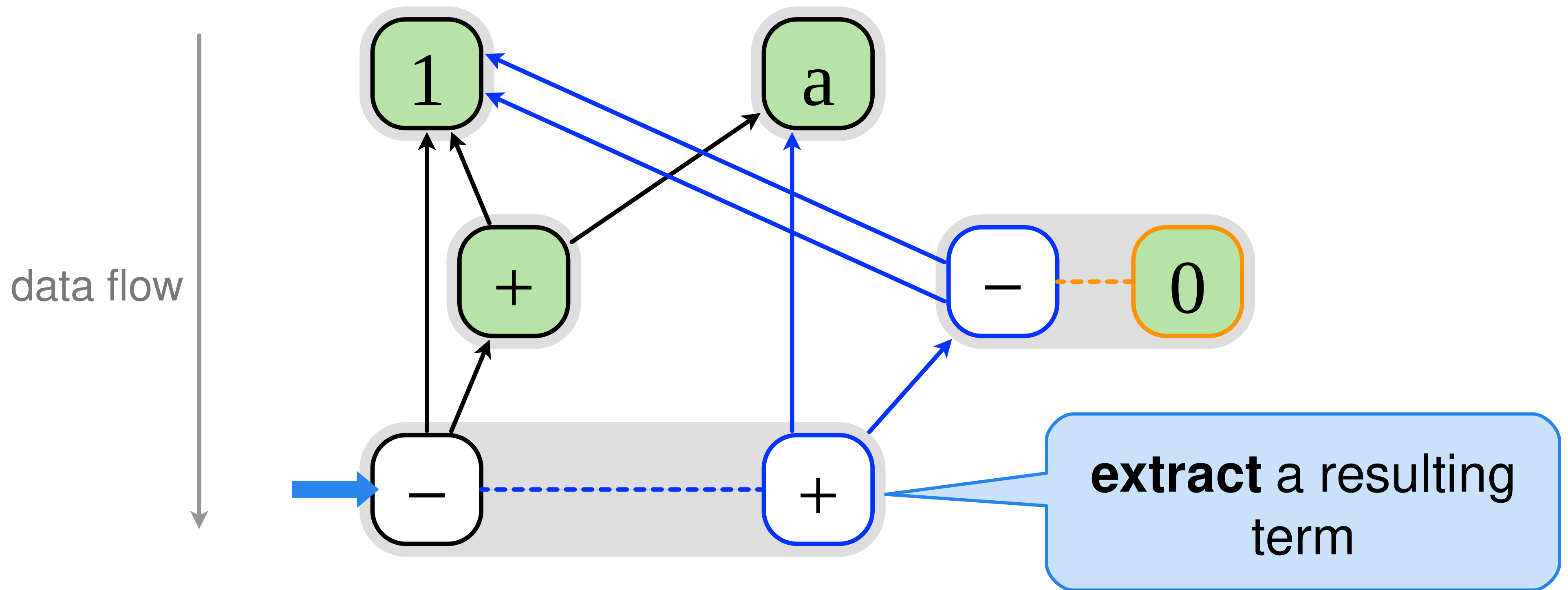


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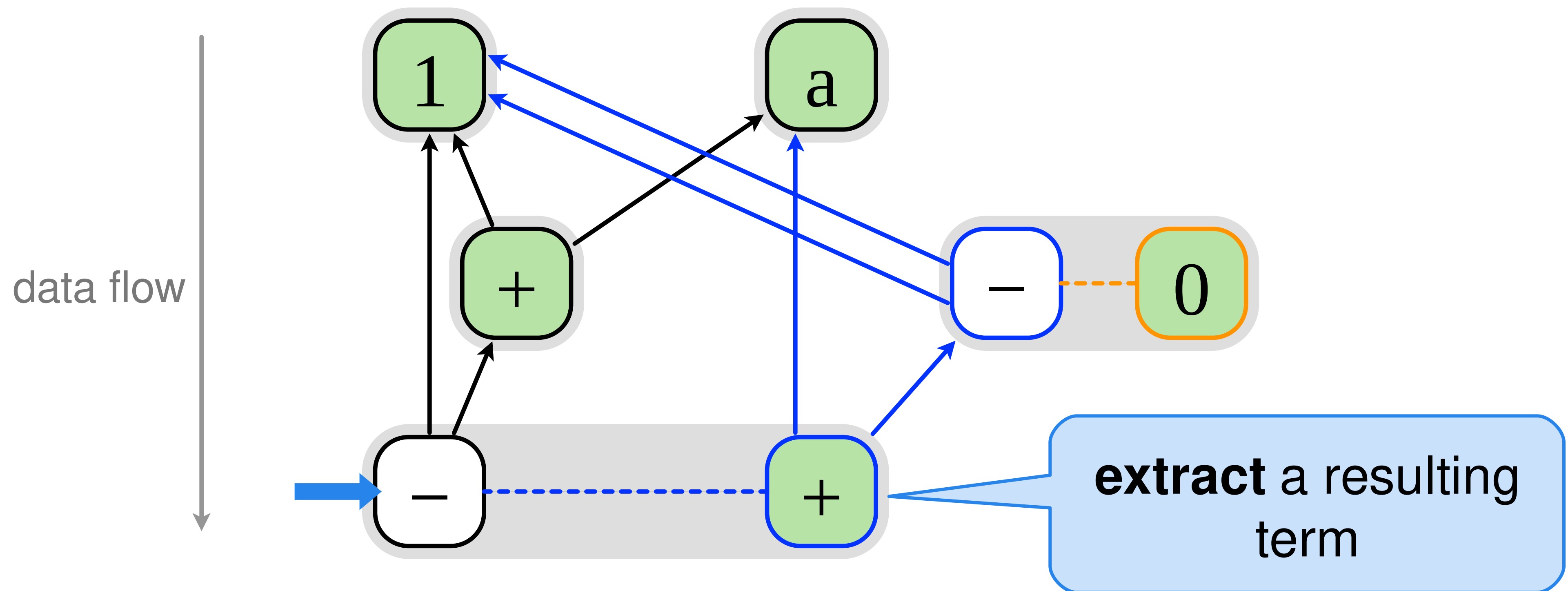


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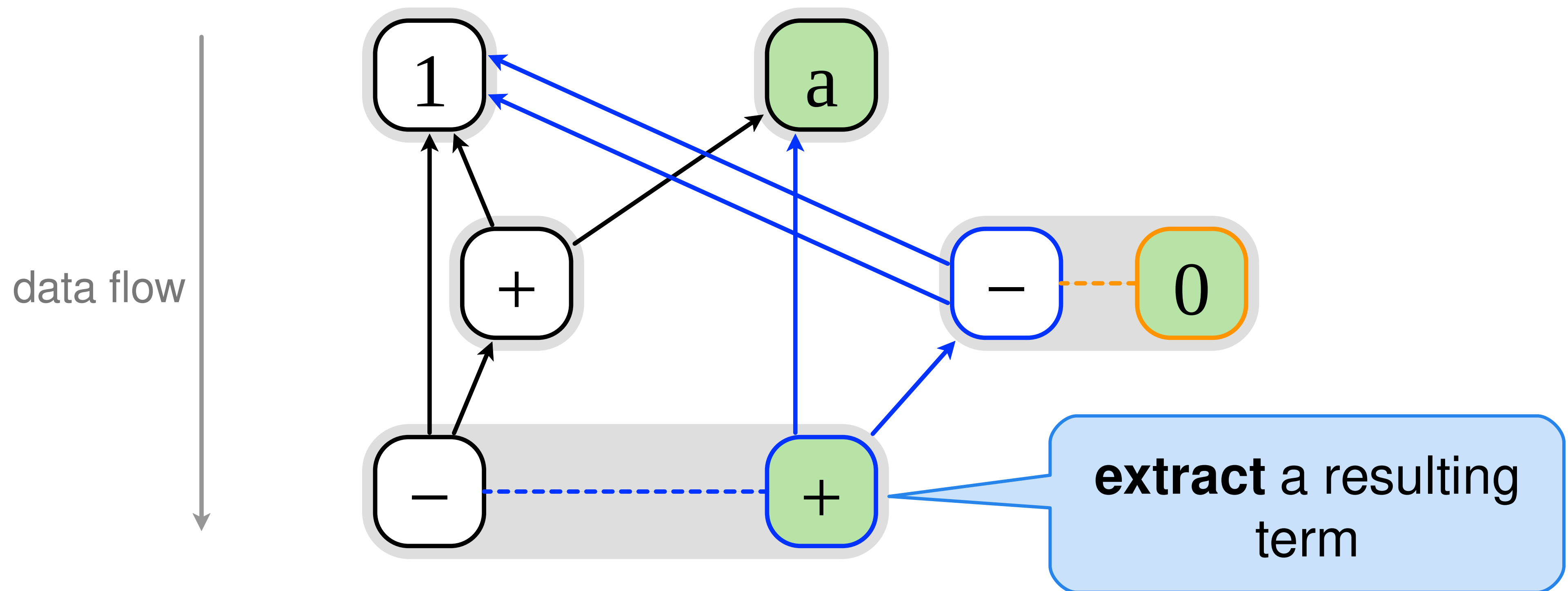


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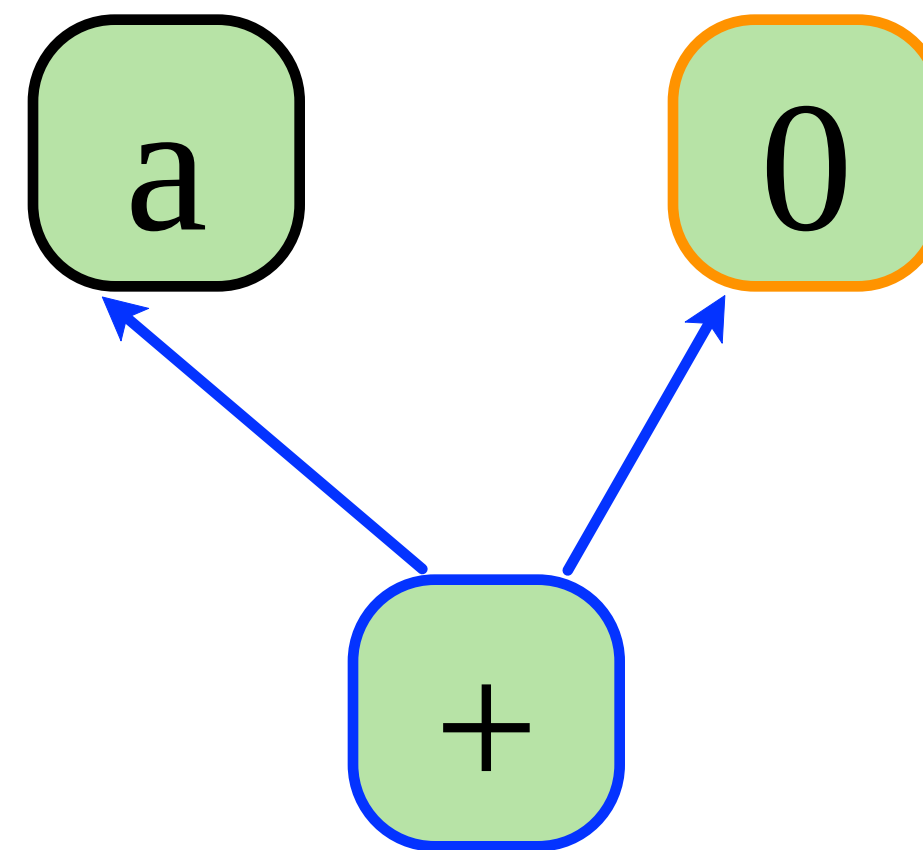


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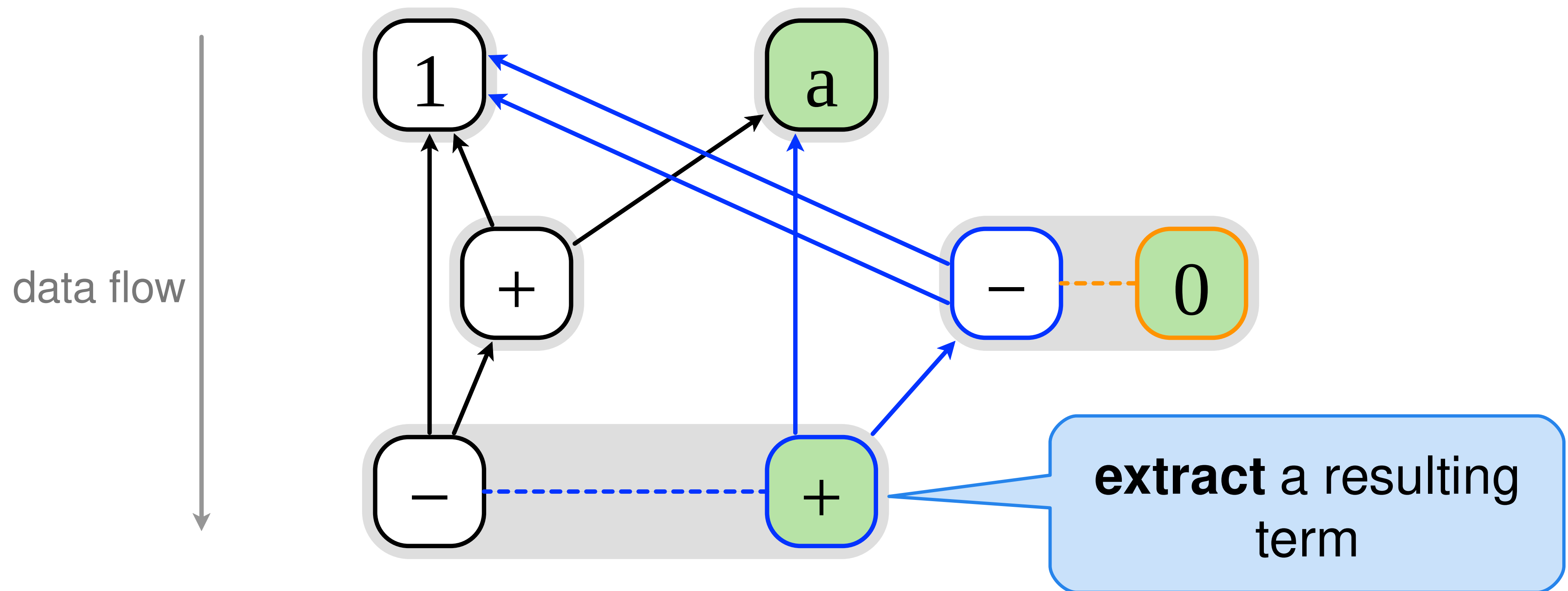


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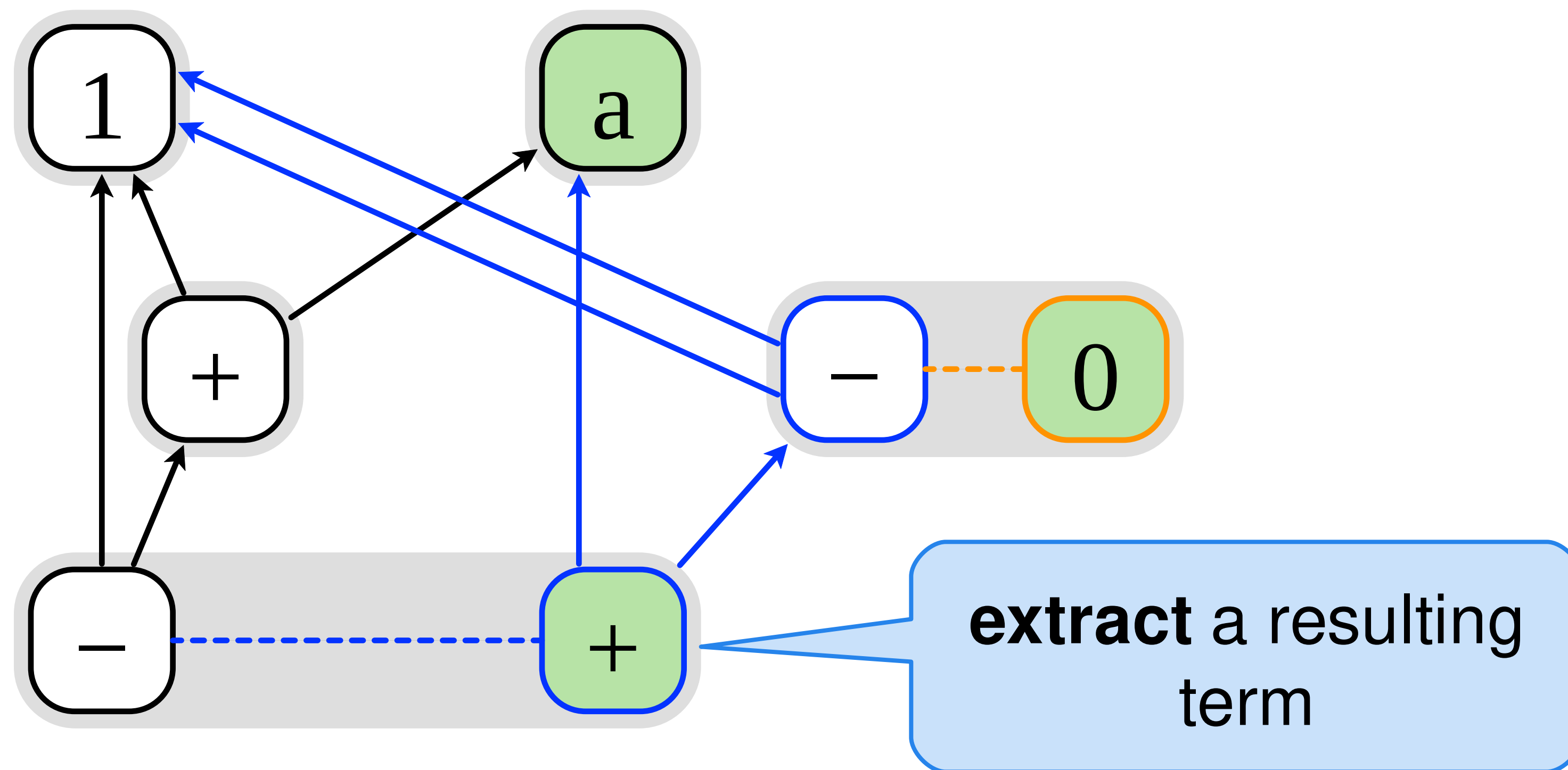
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data flow



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E-Graphs Explore a Big Space

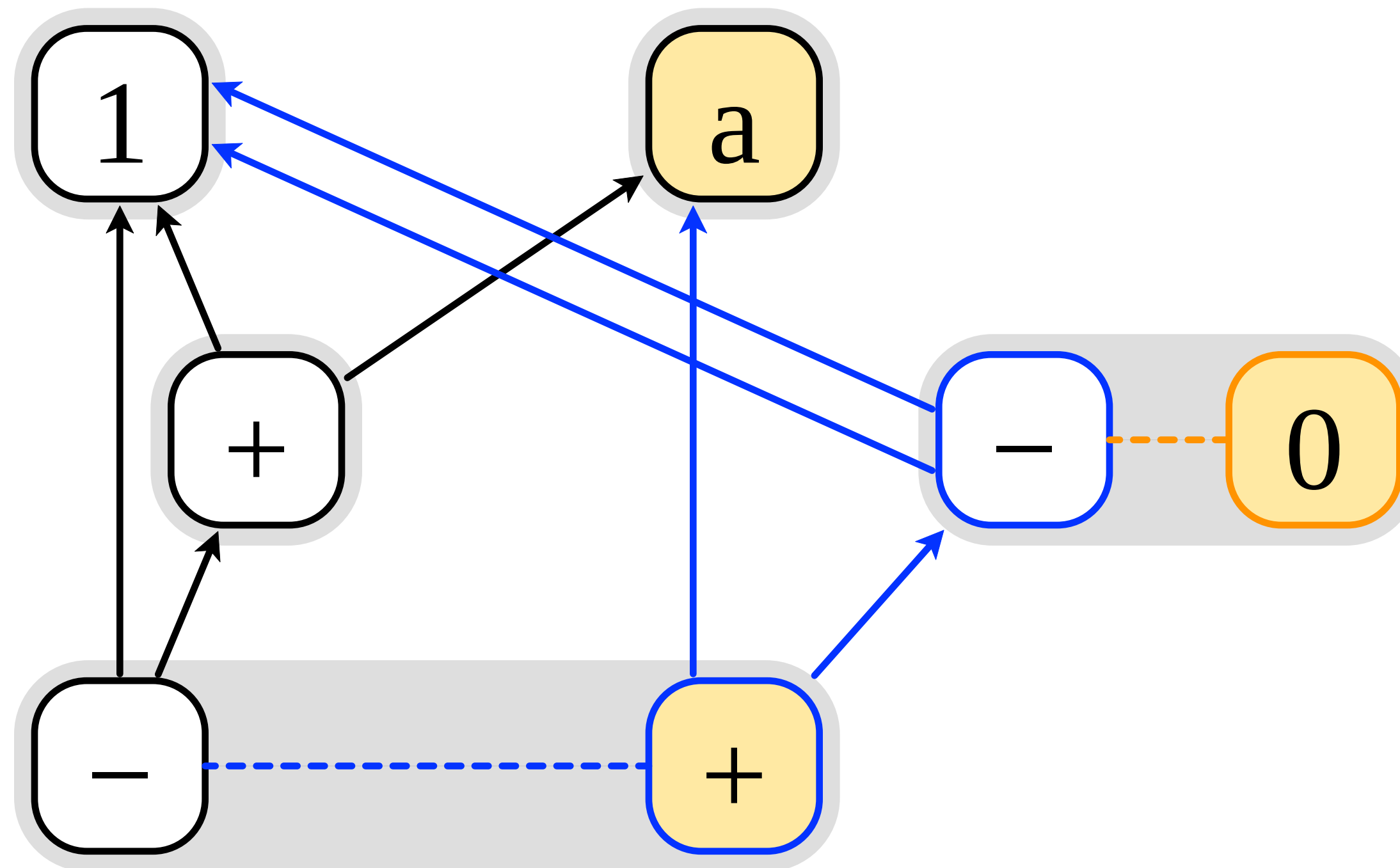
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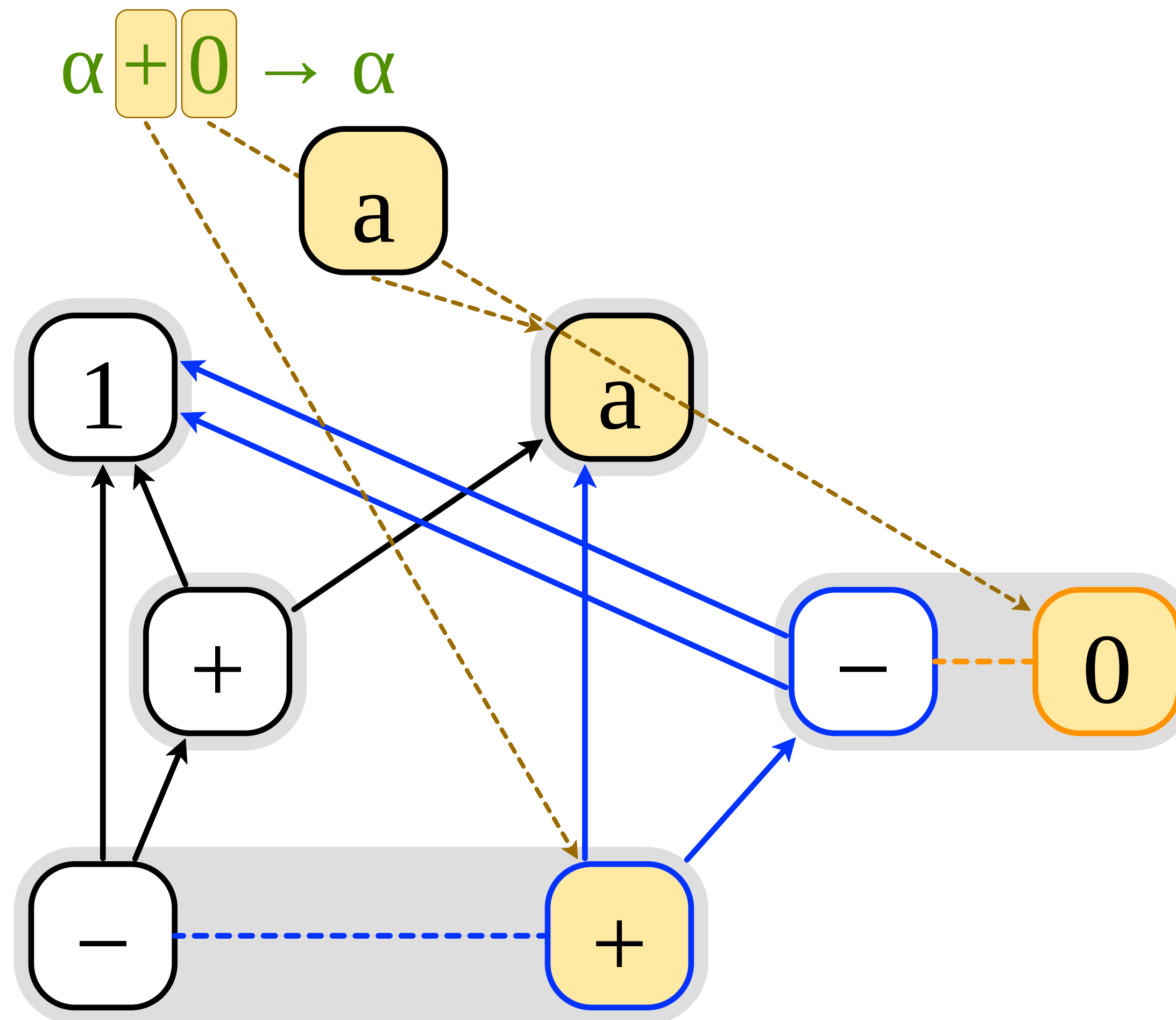
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E-Graphs Explore a Big Space

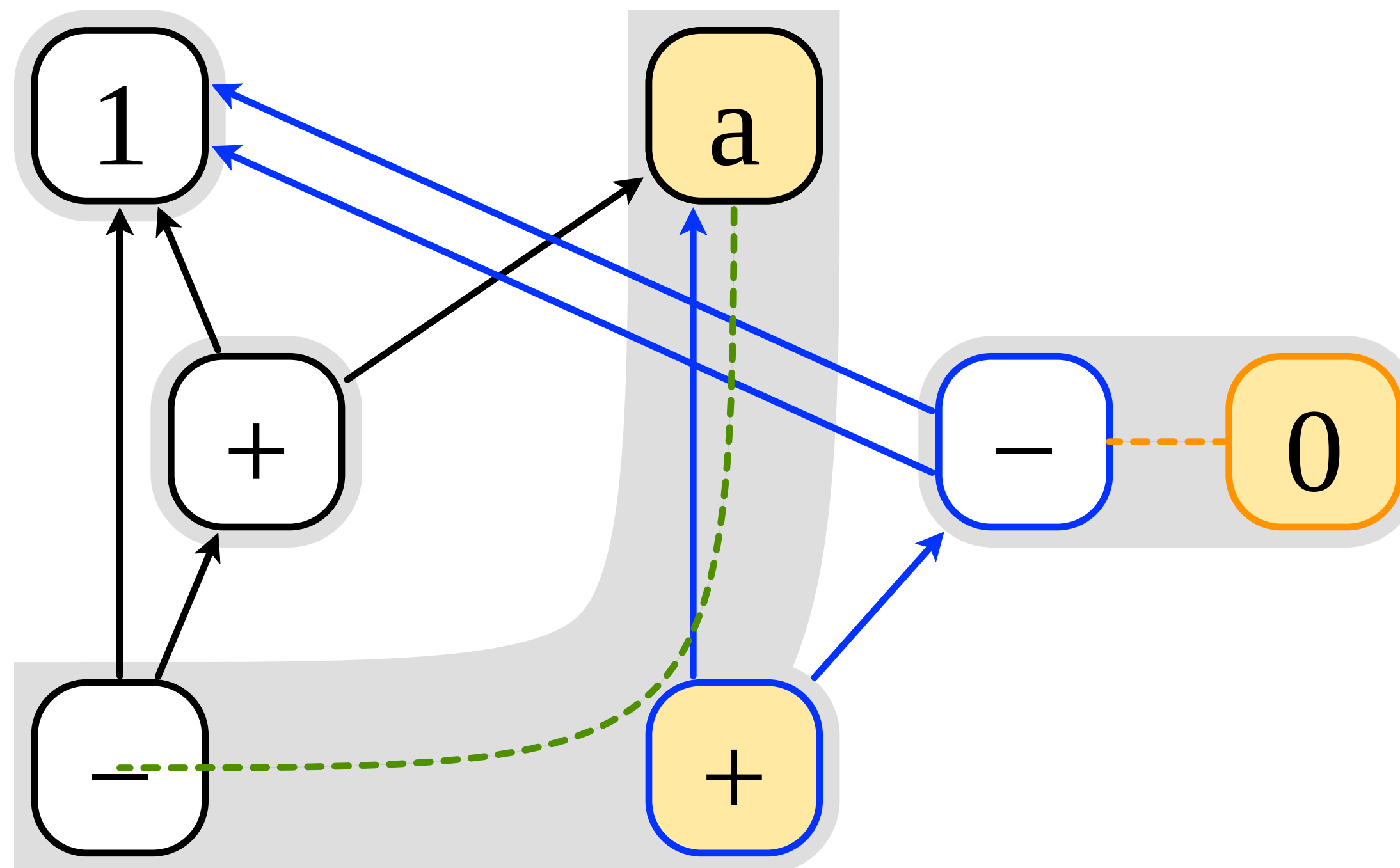
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data flow



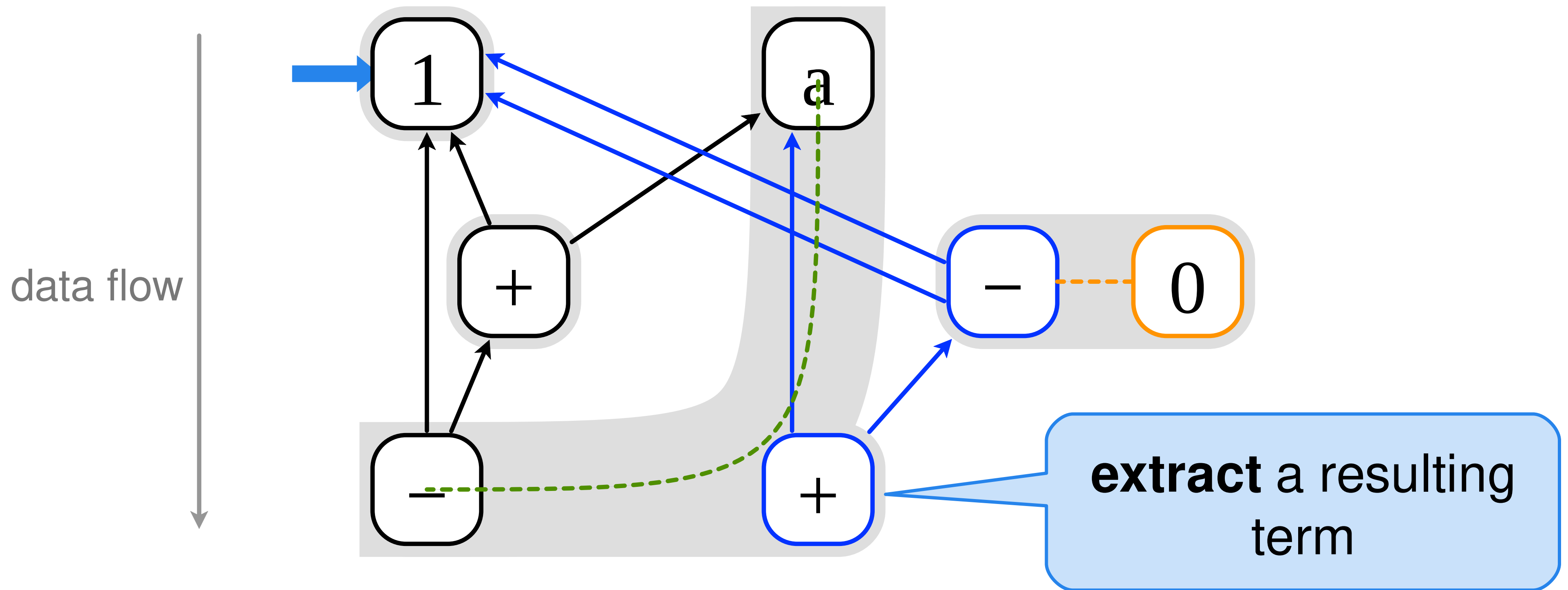
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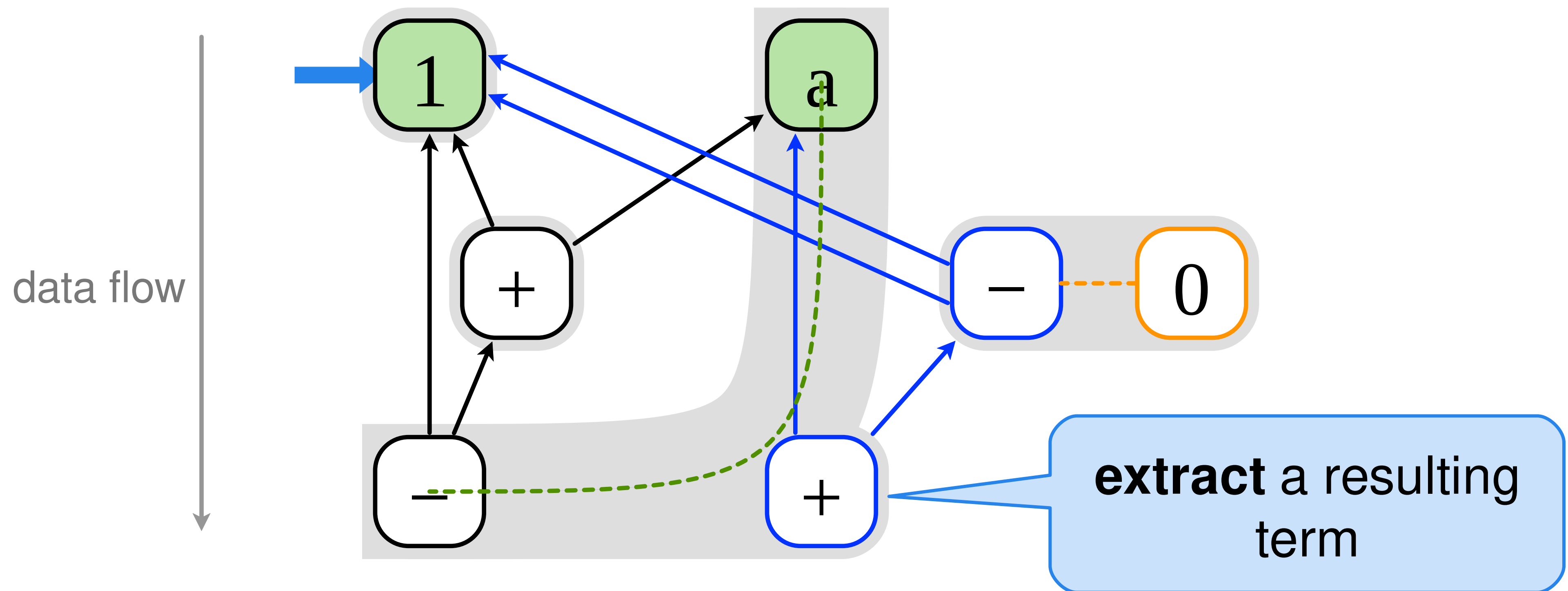
$$\alpha + 0 \rightarrow \alpha$$



$$(a + 1) - 1$$

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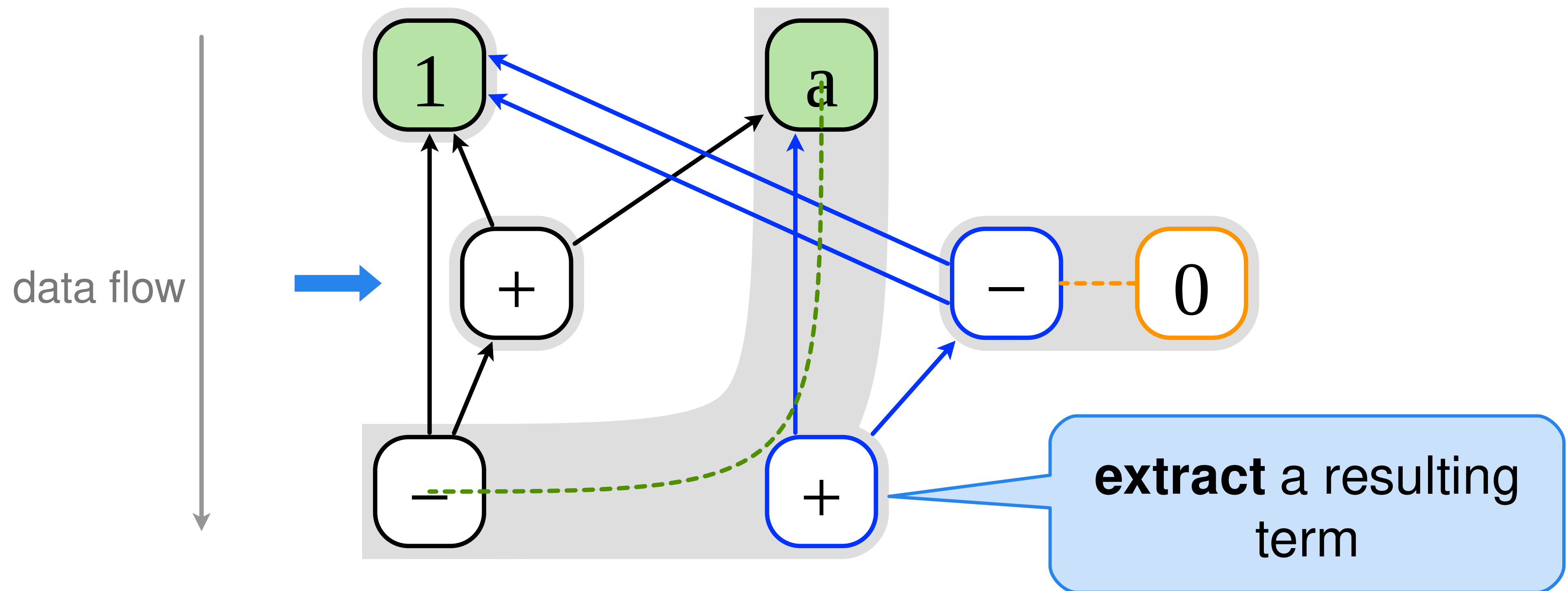
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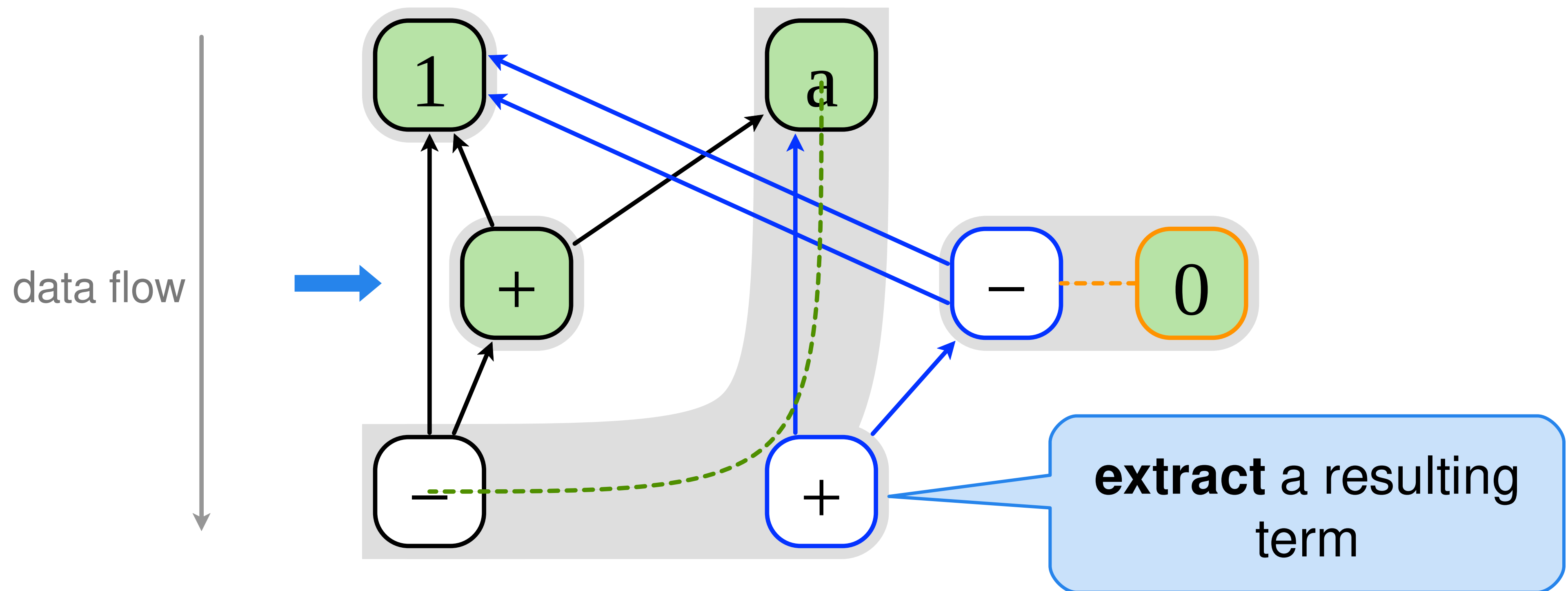
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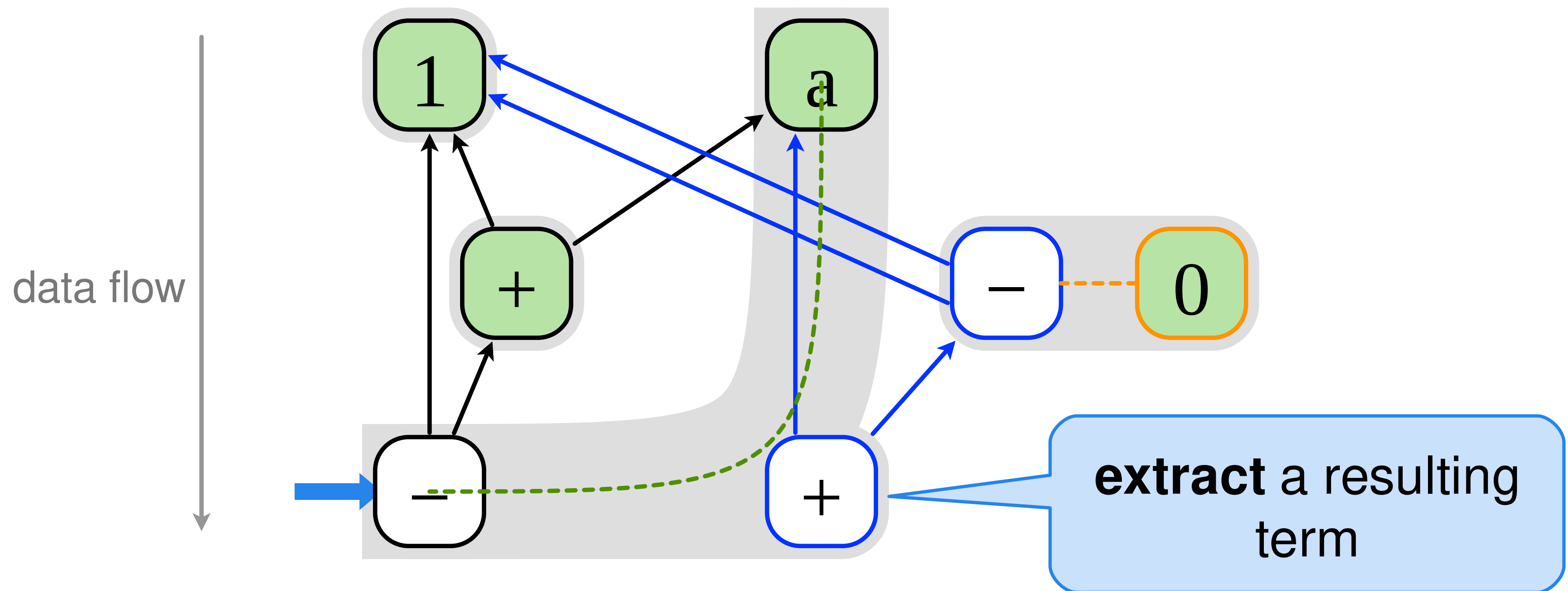
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$$\alpha + 0 \rightarrow \alpha$$



E-Graphs Explore a Big Space

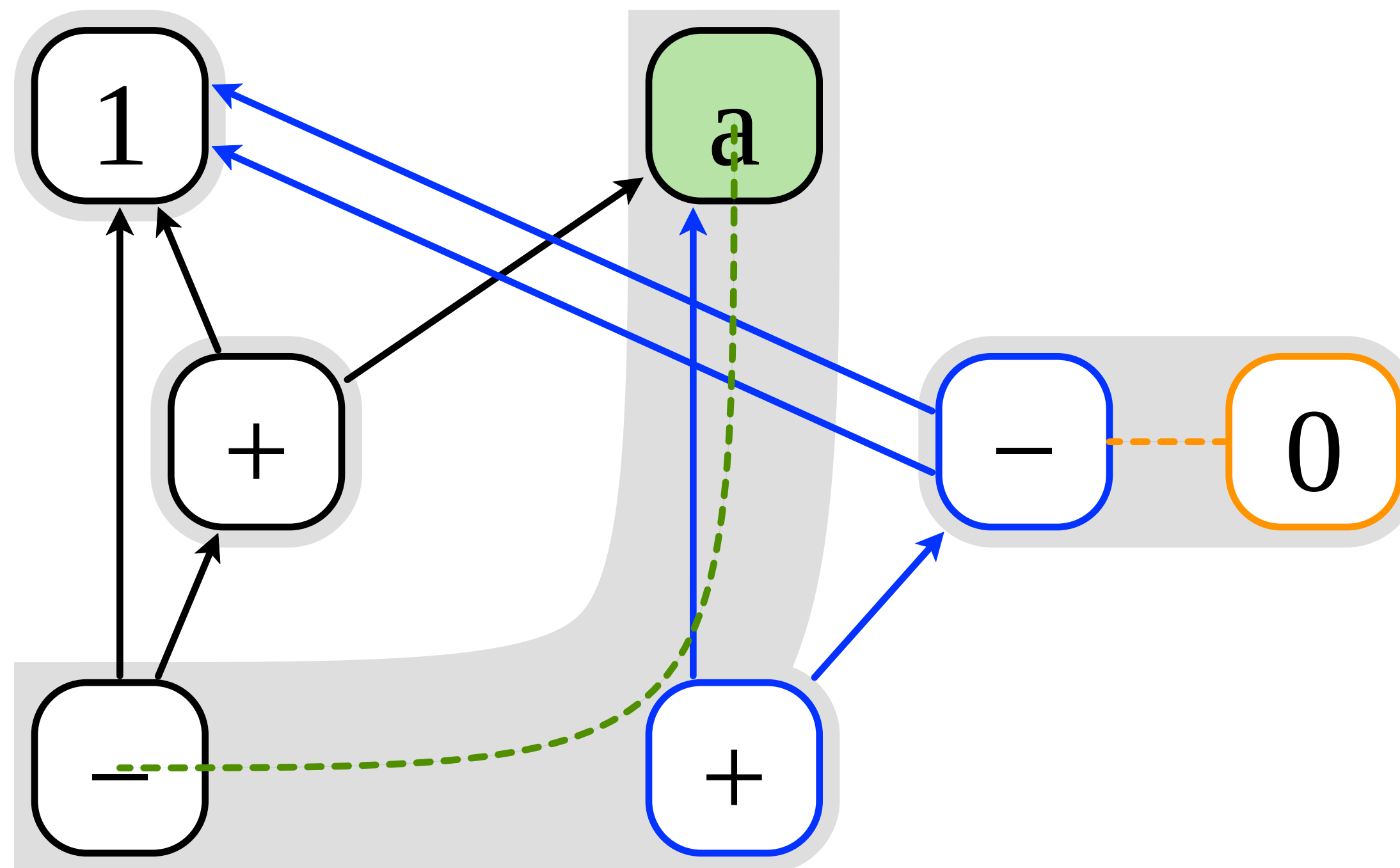
$$(a + 1) - 1$$

$$(\alpha + \beta) - \gamma \rightarrow \alpha + (\beta - \gamma)$$

$$\alpha - \alpha \rightarrow 0$$

$$\alpha + 0 \rightarrow \alpha$$

data flow



1

2

3

4

Motivate Effectful E-Graphs

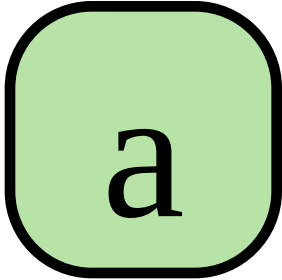
E-Graphs Explore a Big Space

$$(a + 1) - 1$$

$$(\alpha + \beta) - \gamma \rightarrow \alpha + (\beta - \gamma)$$

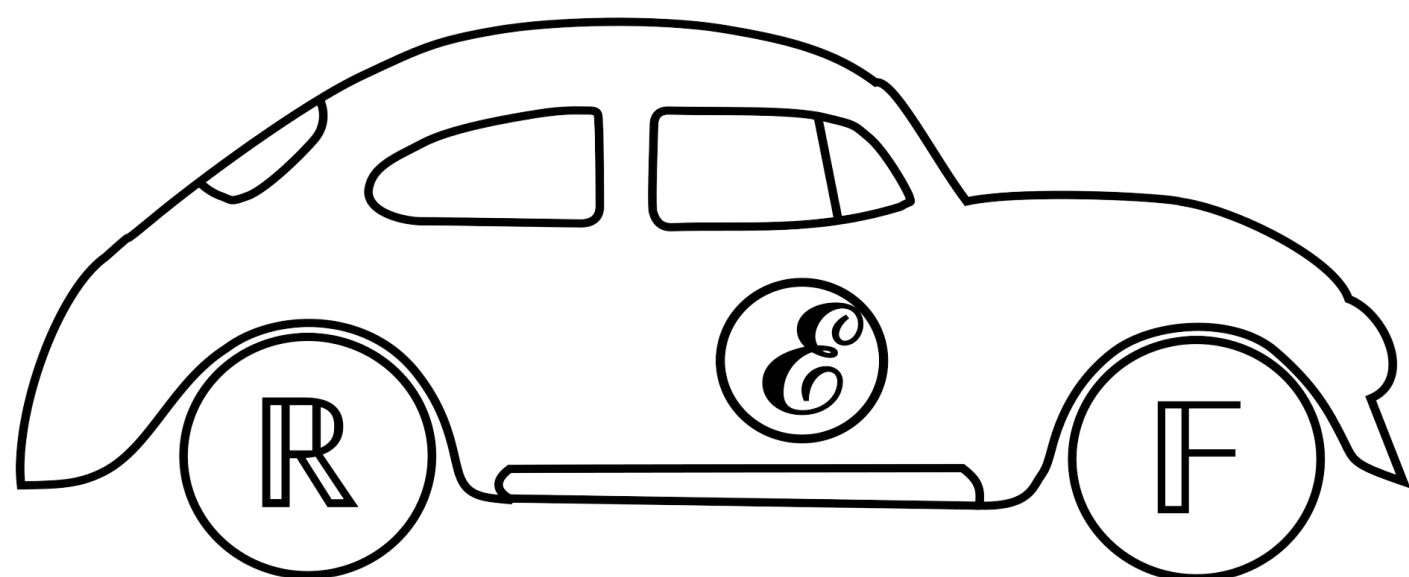
$$\alpha - \alpha \rightarrow 0$$

$$\alpha + 0 \rightarrow \alpha$$

a

E-Graphs are Super Useful!

Floating point

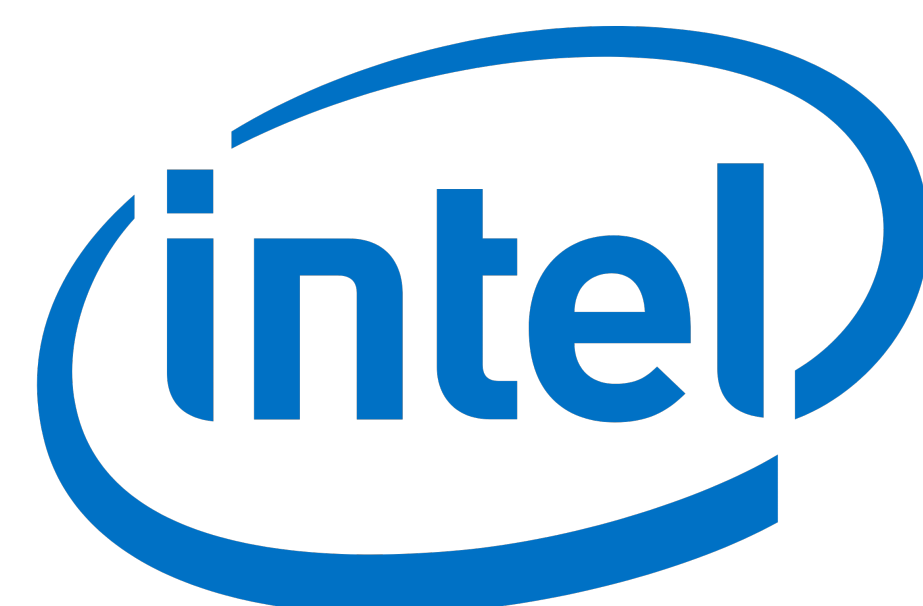


(Herbie)

Python Simulations



RTL Optimization



ML Kernels



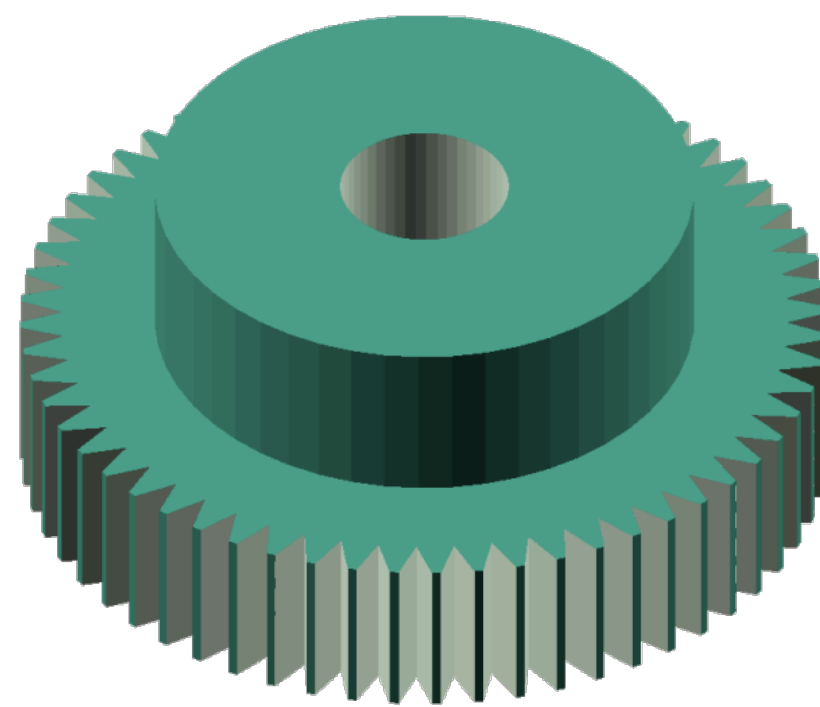
Luminal

SQL



RisingWave

CAD



(Szalinski)

WASM



(Cranelift)

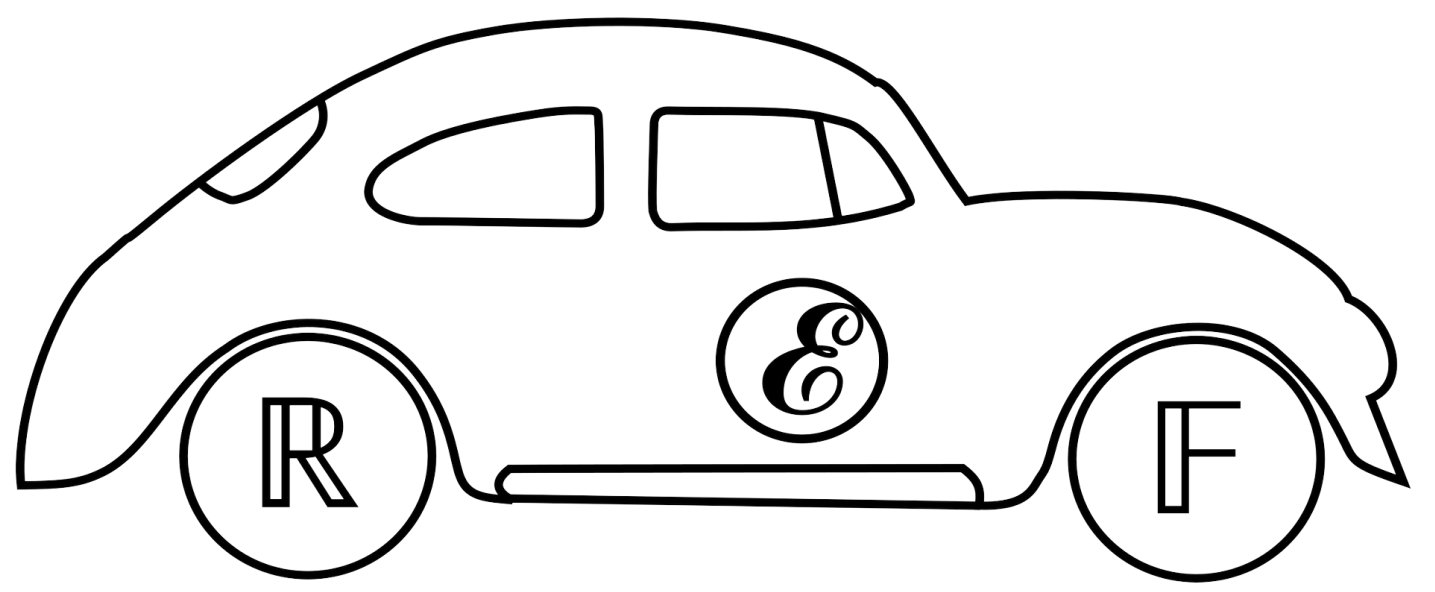
User DSL Optimizations



...and more!

E-Graphs are Super Useful!

Floating point

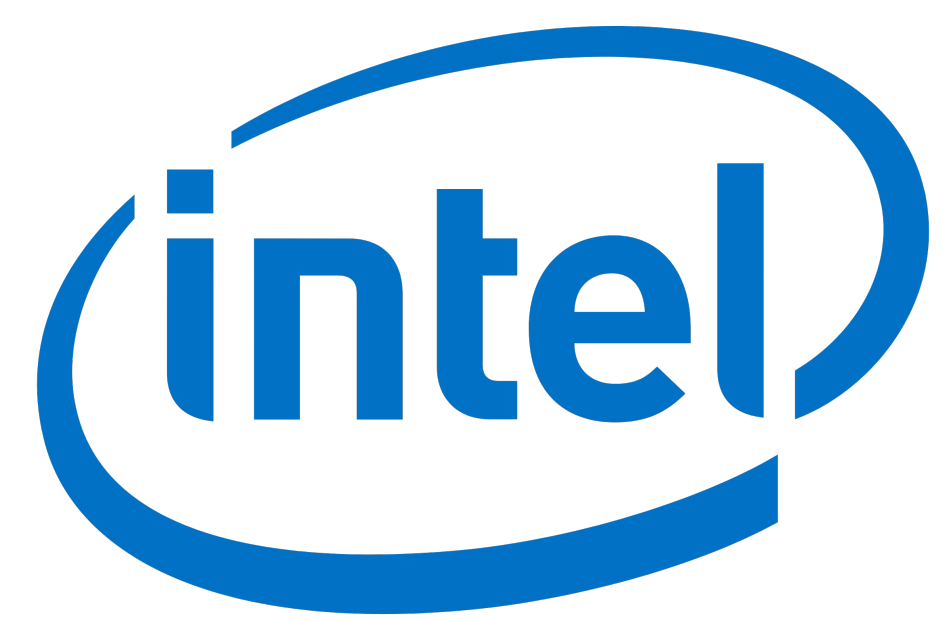


(Herbie)

Python Simulations



RTL Optimization



ML Kernels

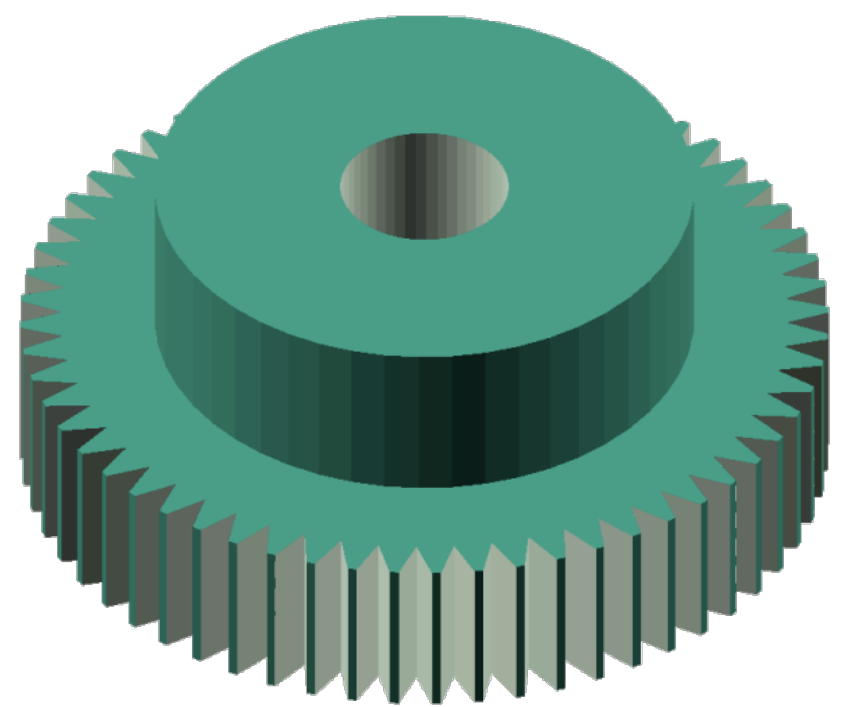


SQL



RisingWave

CAD

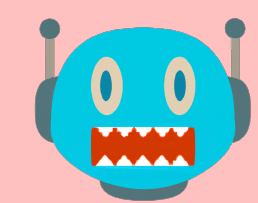


(Szalinski)

WASM



(Cranelift)



need side effects!

L Optimizations



MLIR

...and more!

1

Motivate Effectful E-Graphs

2

Example Effectful E-Graph

3

Effect-Safe Extraction

4

It Works!



Example Effectful E-Graph

1

2

3

4

Example Effectful E-Graph

Update

```
fun foo(a, x):  
  v1 = update(a, x)  
  v2 = update(a, x+1)  
  return v1
```


1

2

3

4

Update

```
fun foo(a, x):  
  v1 = update(a, x)  
  v2 = update(a, x+1)  
  return v1
```

Write **x** to address **a**,
return the previous value at **a**

Update

```
fun foo(a, x,  $\omega$ ):  
  ( $v_1$ ,  $\omega_1$ ) = update(a, x,  $\omega$ )  
  ( $v_2$ ,  $\omega_2$ ) = update(a, x+1,  $\omega_1$ )  
  return ( $v_1$ ,  $\omega_2$ )
```

ω_1 is the state of memory
after the update

1

2

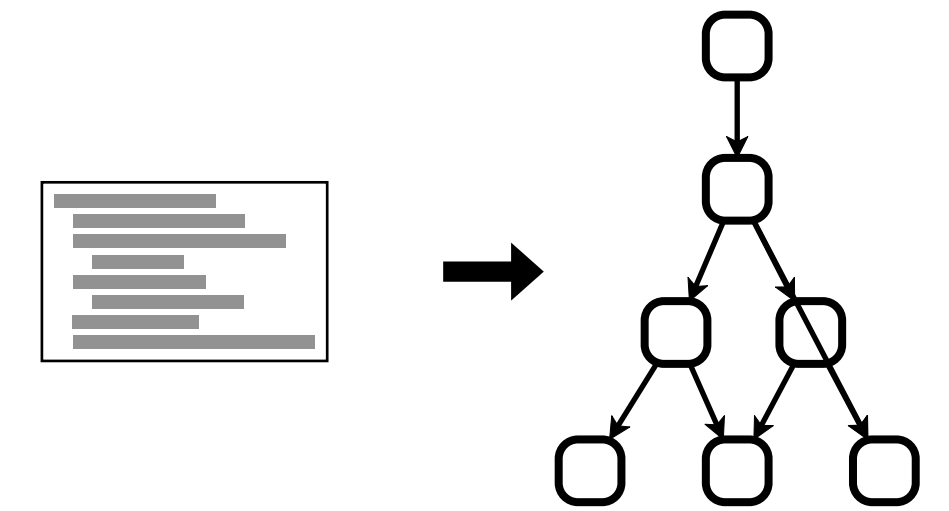
3

4

Example Effectful E-Graph

The Effectful E-Graph

```
fun foo(a, x,  $\omega$ ):  
  ( $v_1$ ,  $\omega_1$ ) = update(a, x,  $\omega$ )  
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  return ( $v_1$ ,  $\omega_2$ )
```



1

2

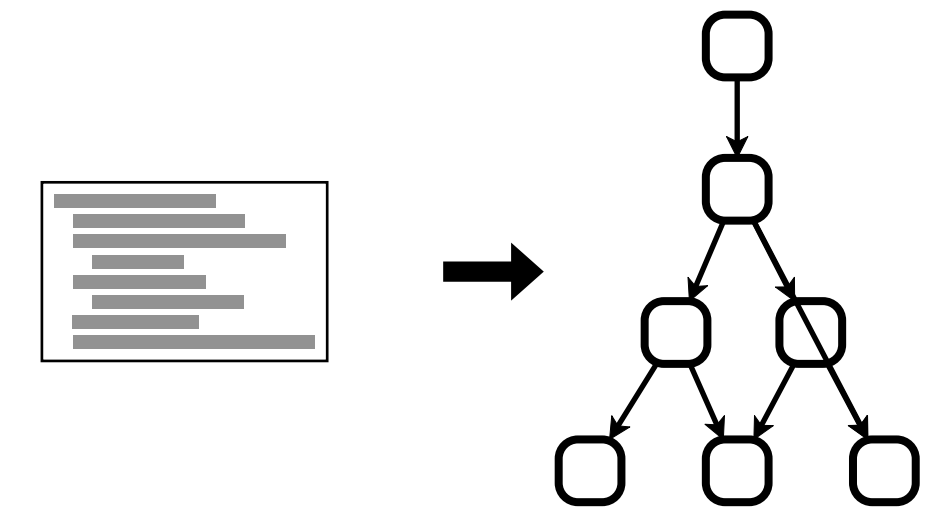
3

4

Example Effectful E-Graph

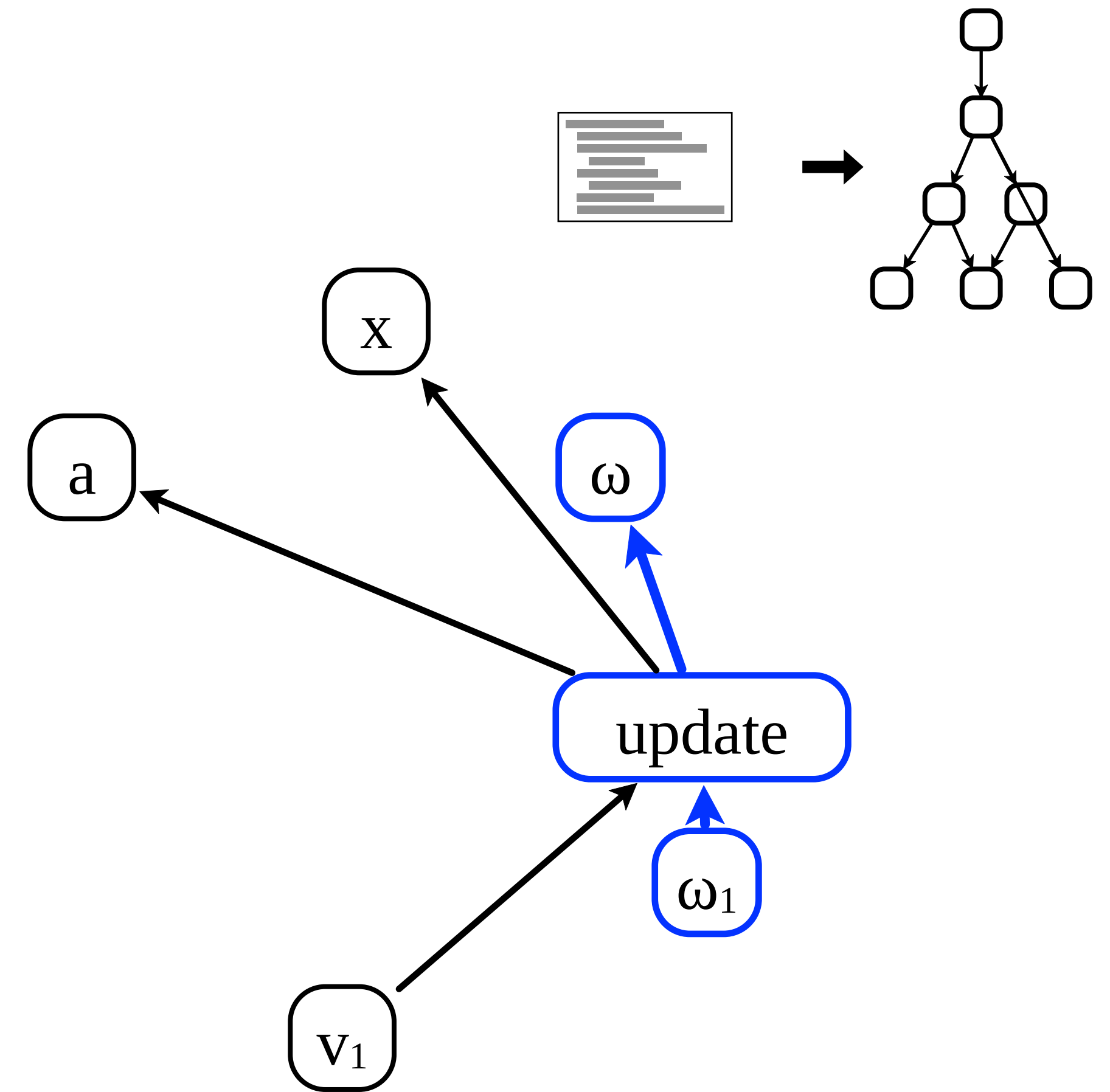
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fun foo(a, x,  $\omega$ ):  
  ( $v_1$ ,  $\omega_1$ ) = update(a, x,  $\omega$ )  
  ( $v_2$ ,  $\omega_2$ ) = update(a, x+1,  $\omega_1$ )  
  return ( $v_1$ ,  $\omega_2$ )
```



The Effectful E-Graph

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fun foo(a, x,  $\omega$ ):  
  ( $v_1$ ,  $\omega_1$ ) = update(a, x,  $\omega$ )  
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  return ( $v_1$ ,  $\omega_2$ )
```



1

2

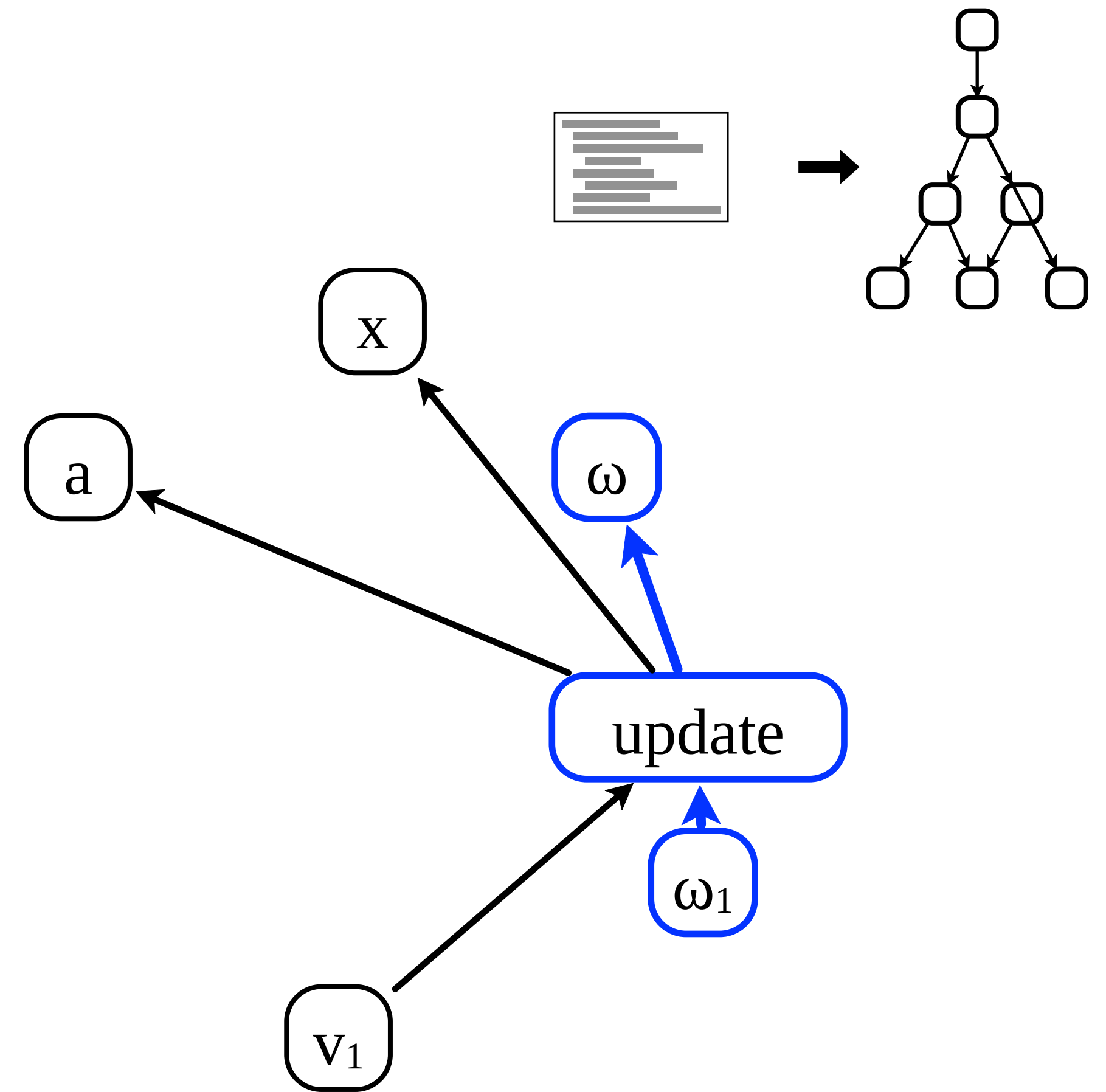
3

4

Example Effectful E-Graph

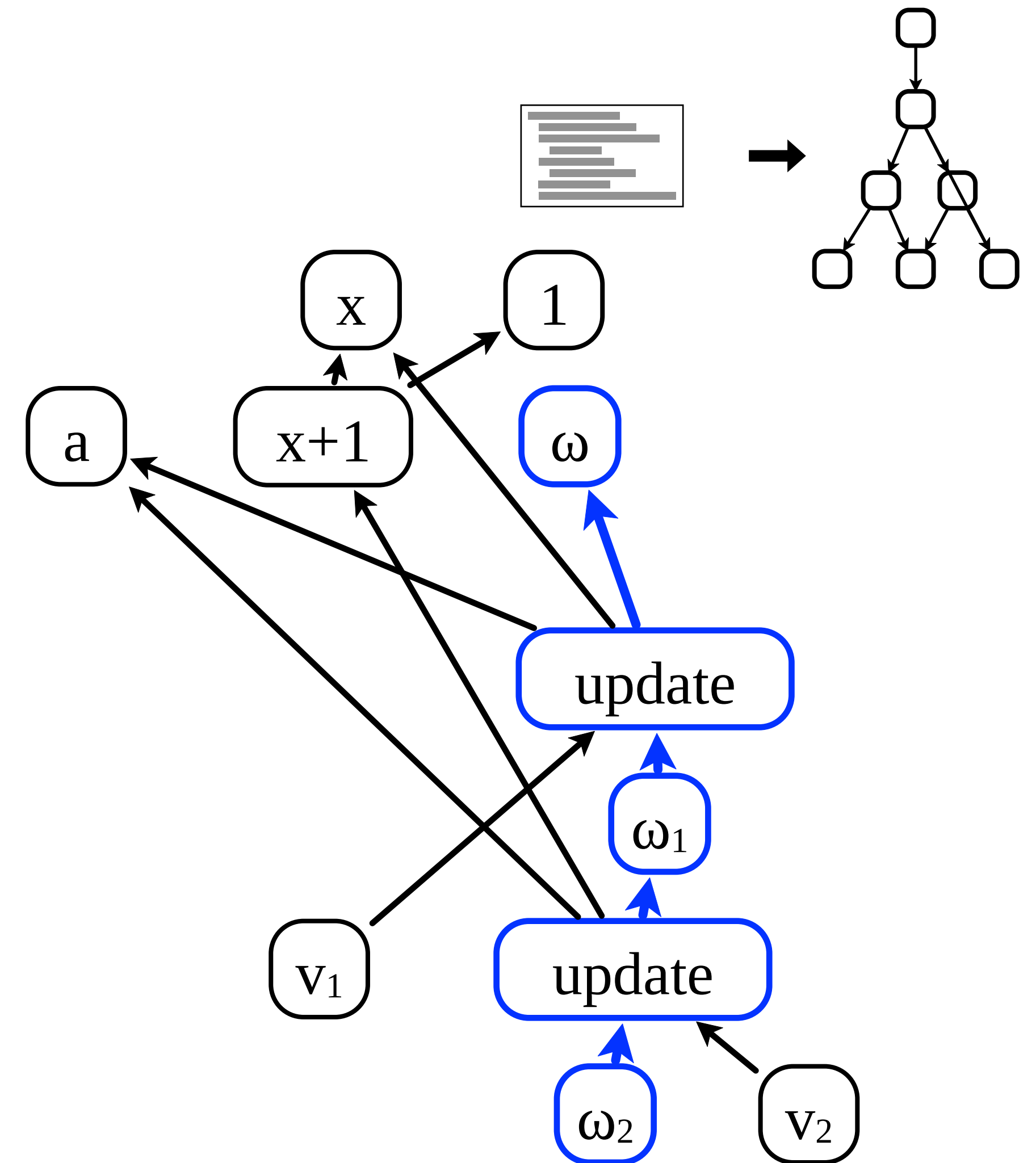
The Effectful E-Graph

```
fun foo(a, x,  $\omega$ ):  
  ( $v_1$ ,  $\omega_1$ ) = update(a, x,  $\omega$ )  
  ( $v_2$ ,  $\omega_2$ ) = update(a, x+1,  $\omega_1$ )  
  return ( $v_1$ ,  $\omega_2$ )
```



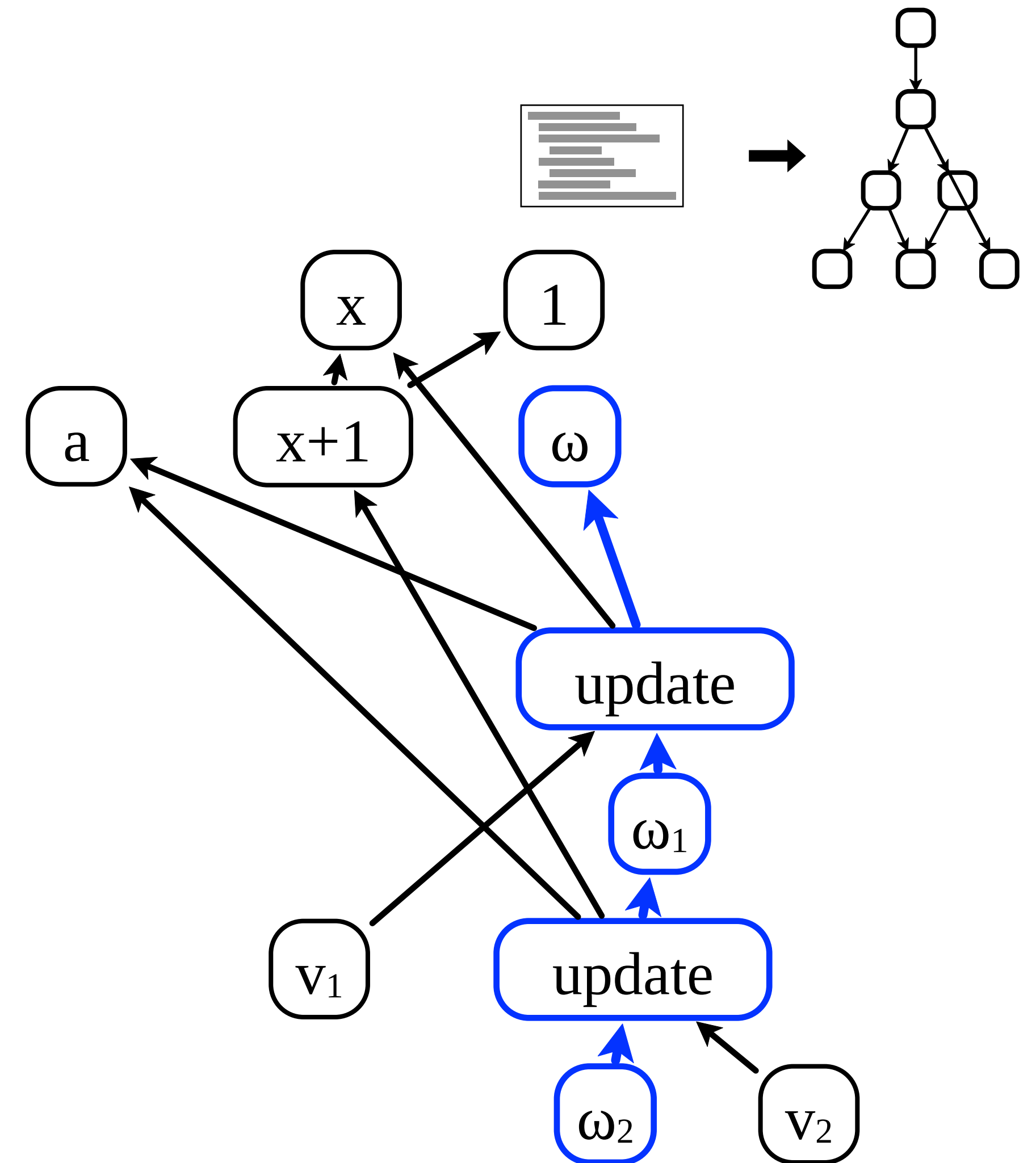
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  ( $v_2$ ,  $\omega_2$ ) = update(a, x+1,  $\omega_1$ )  
  return ( $v_1$ ,  $\omega_2$ )
```



The Effectful E-Graph

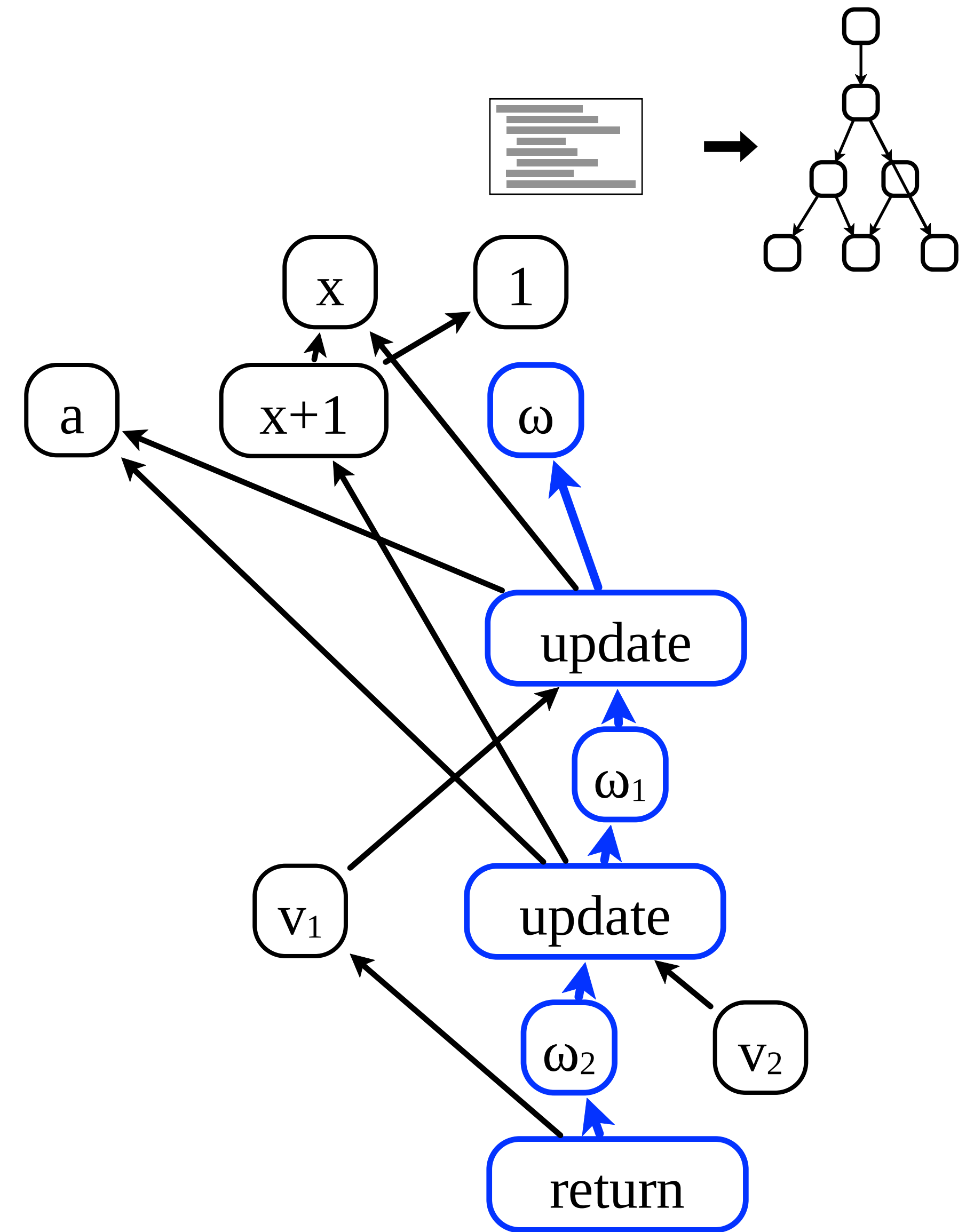
```
fun foo(a, x,  $\omega$ ):  
  ( $v_1$ ,  $\omega_1$ ) = update(a, x,  $\omega$ )  
  ( $v_2$ ,  $\omega_2$ ) = update(a, x+1,  $\omega_1$ )  
  return ( $v_1$ ,  $\omega_2$ )
```



The Effectful E-Graph

```

fun foo(a, x, ω):
  (v1, ω1) = update(a, x, ω)
  (v2, ω2) = update(a, x+1, ω1)
  return (v1, ω2)
  
```



1

2

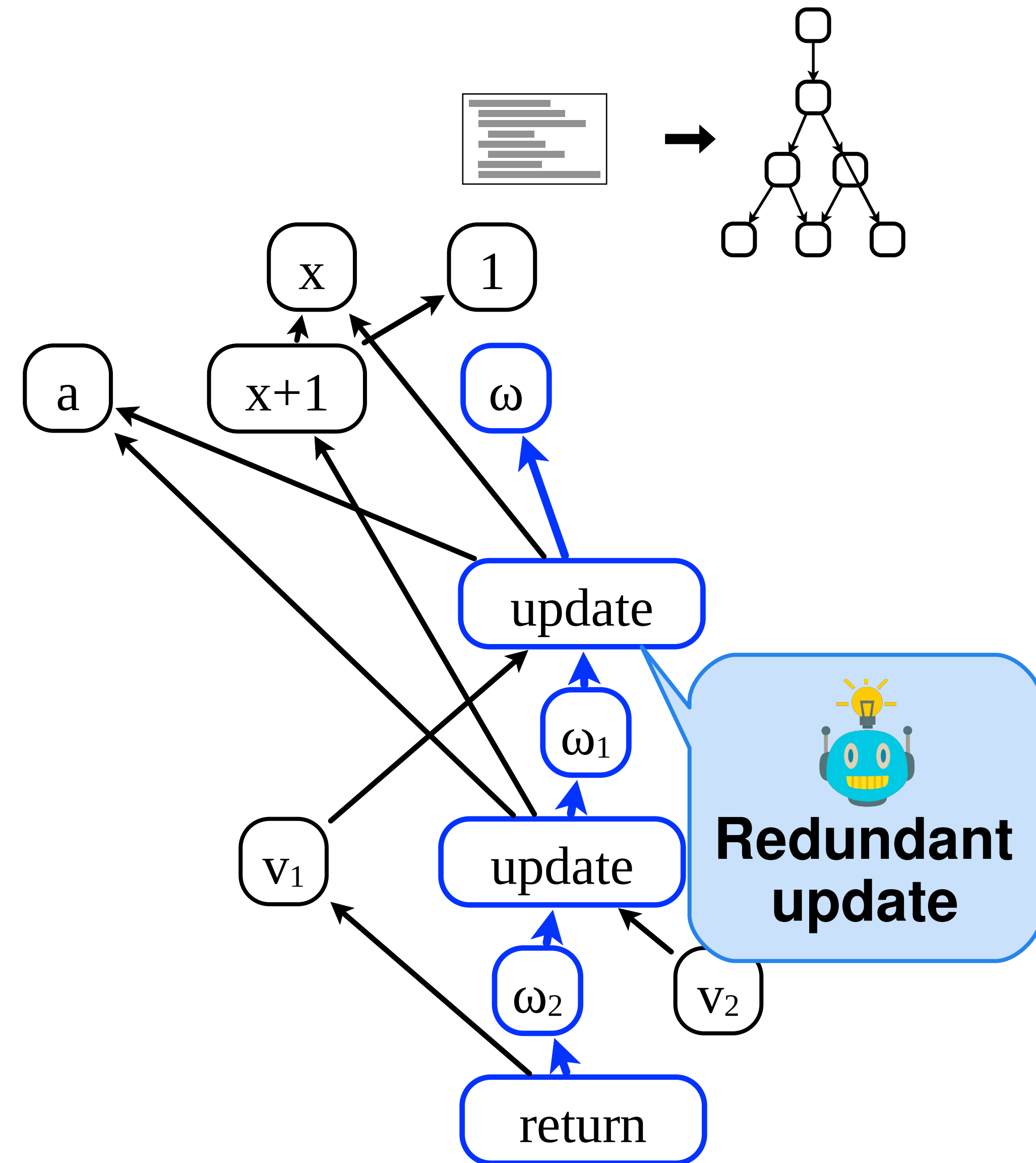
3

4

Example Effectful E-Graph

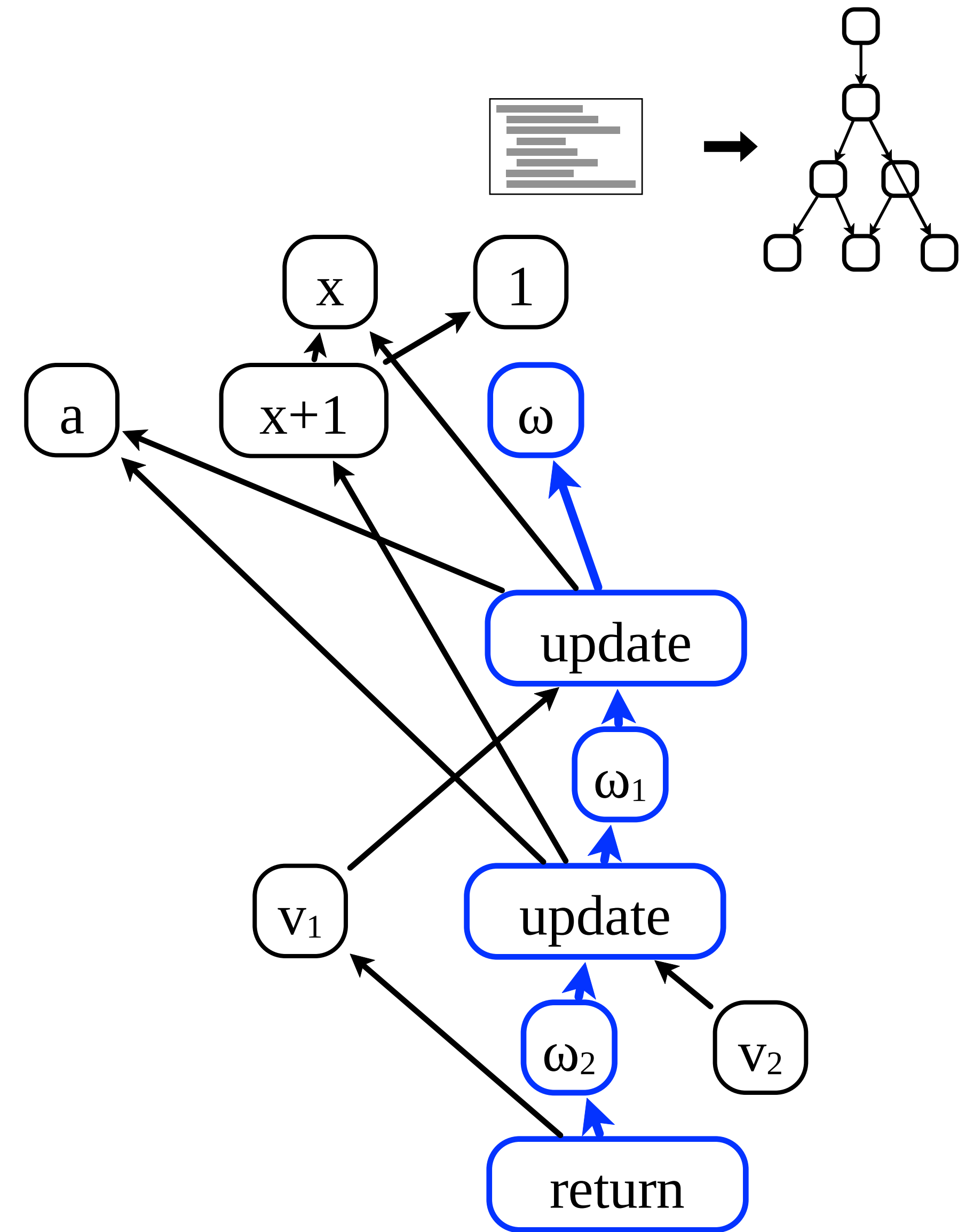
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  return ( $v_1$ ,  $\omega_2$ )
```



Dead Store Elimination

```
fun foo(a, x, ω):  
  (v1, ω1) = update(a, x, ω)  
  (v2, ω2) = update(a, x+1, ω1)  
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```



Dead Store Elimination

```
fun foo(a, x, ω):
```

```
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```

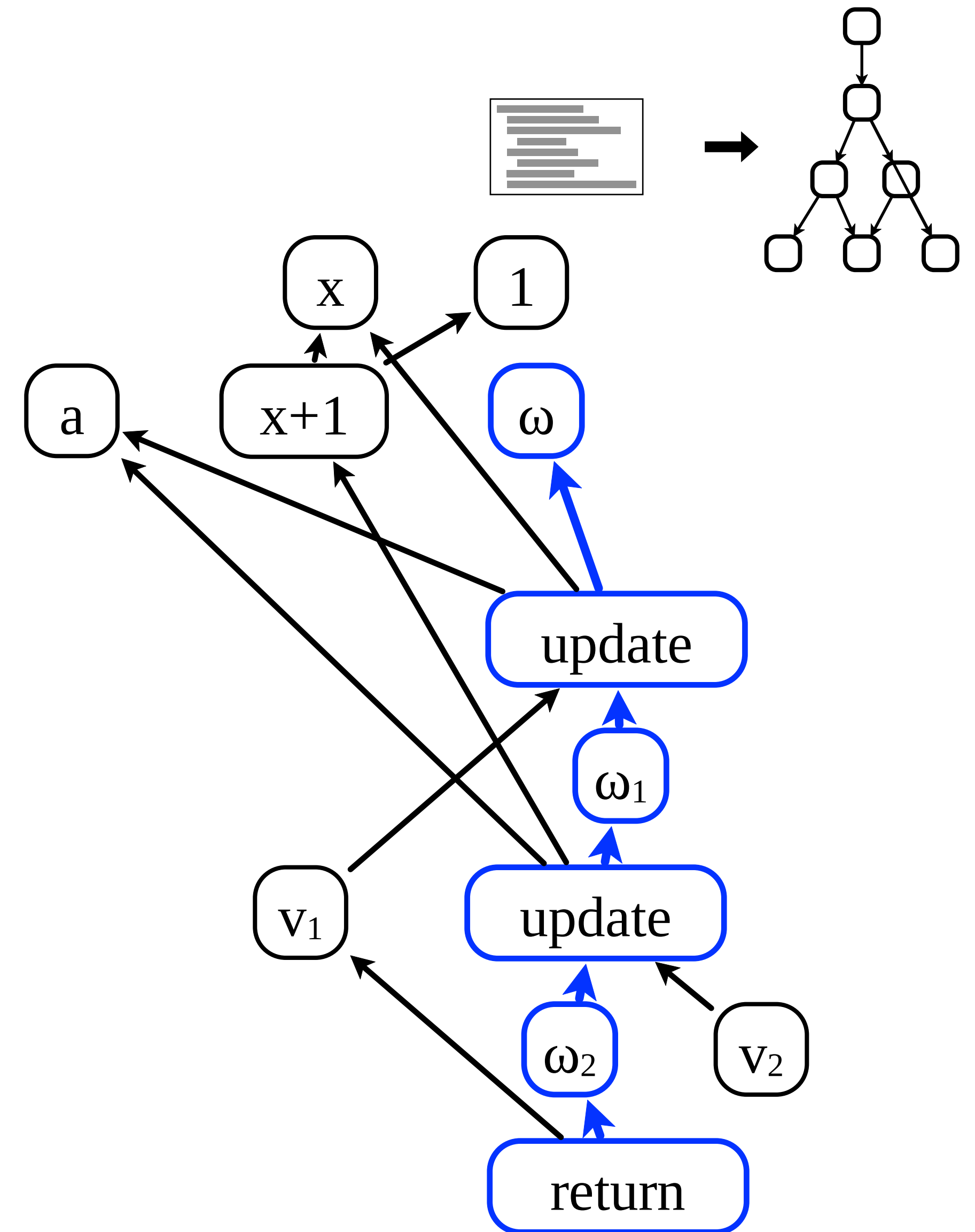
```
  (v2, ω2) = update(a, x+1, ω1)
```

```
  return (v1, ω2)
```

$$\begin{array}{l} (v_1, \omega_1) = \text{update}(a, x, \omega) \\ (v_2, \omega_2) = \text{update}(a, x+1, \omega_1) \end{array}$$

$$(v_3, \omega_3) = \text{update}(a, x+1, \omega)$$

$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



Dead Store Elimination

```
fun foo(a, x, ω):
```

```
  (v1, ω1) = update(a, x, ω)
```

```
  (v2, ω2) = update(a, x+1, ω1)
```

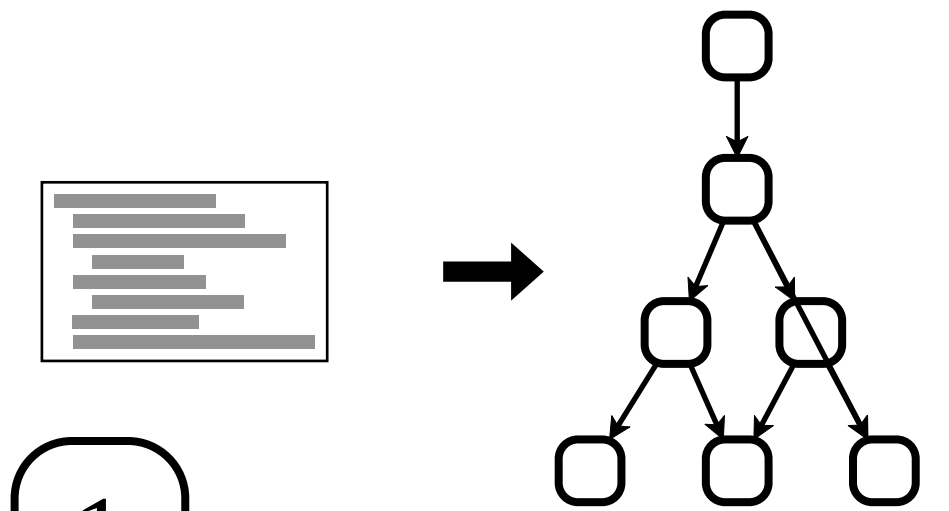
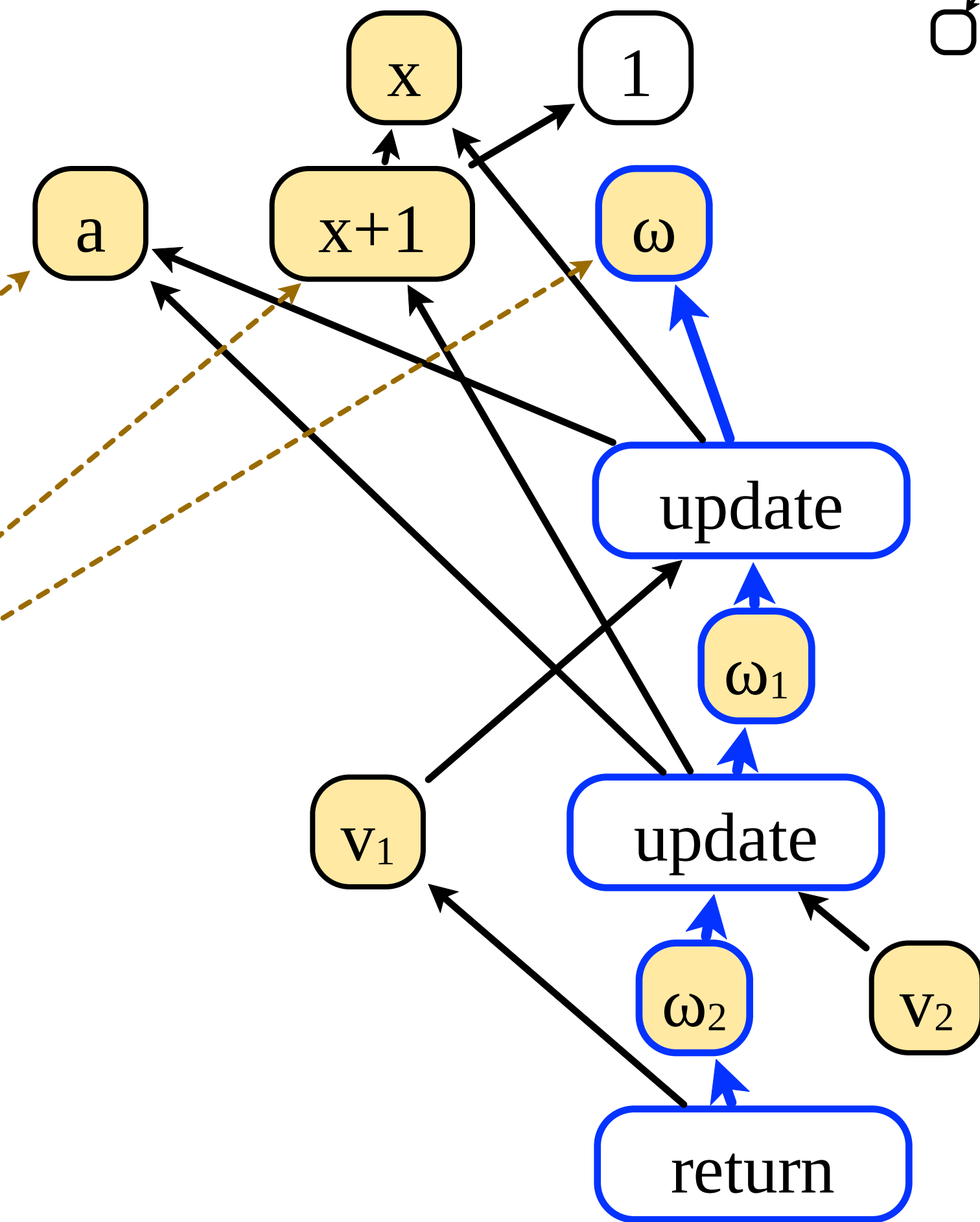
```
  return (v1, ω2)
```

$$(v_1, \omega_1) = \text{update}(a, x, \omega)$$

$$(v_2, \omega_2) = \text{update}(a, x+1, \omega_1)$$

$$(v_3, \omega_3) = \text{update}(a, x+1, \omega)$$

$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



Dead Store Elimination

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```
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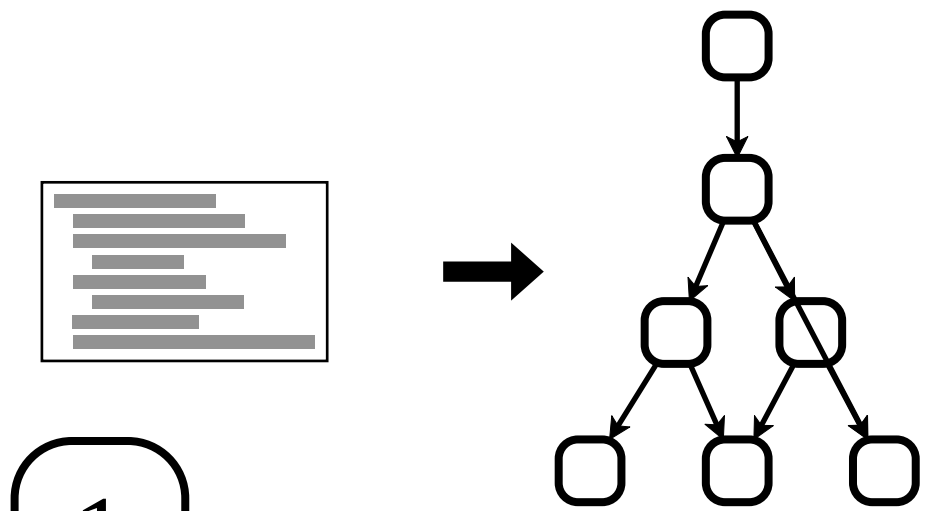
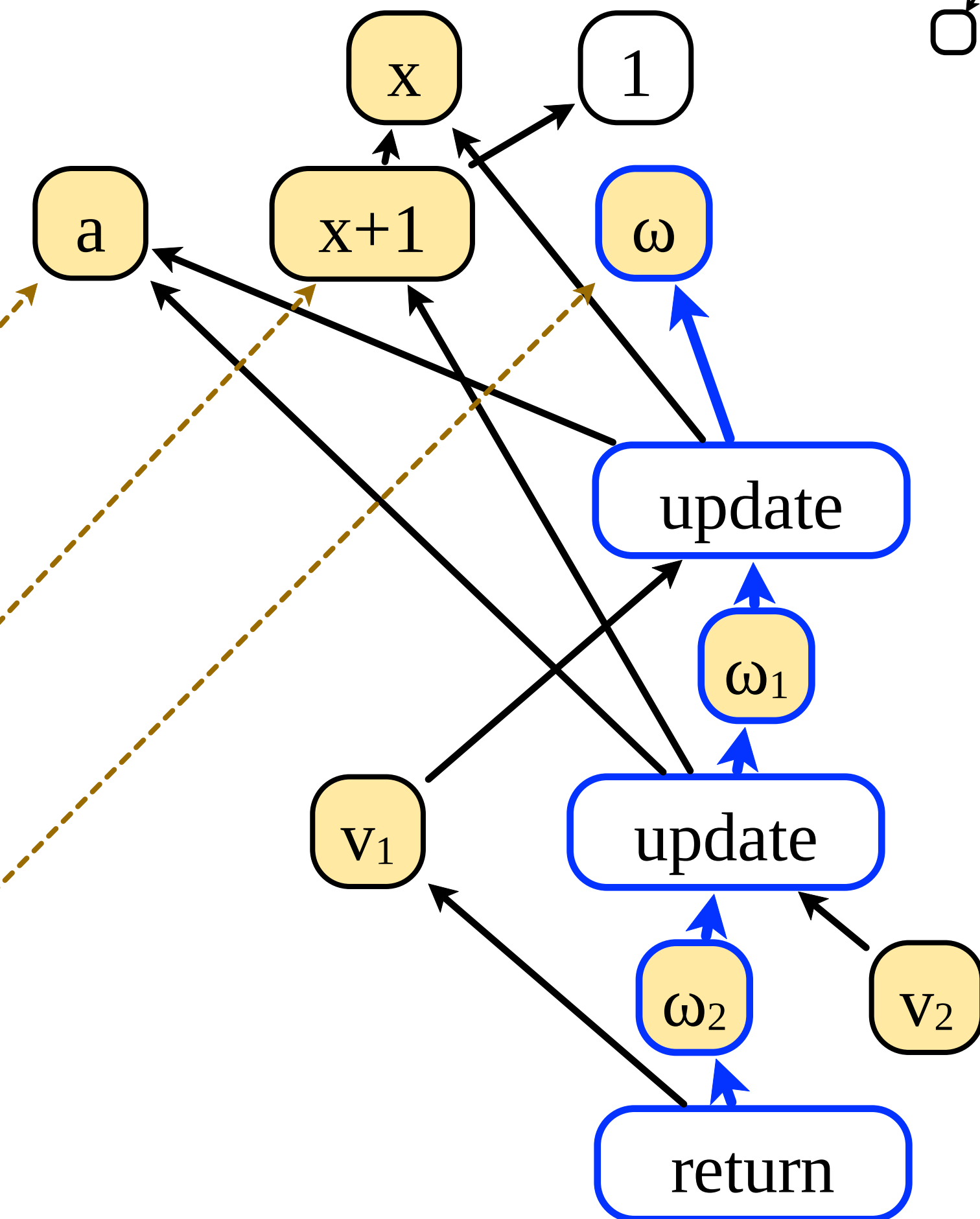
```
  (v2, ω2) = update(a, x+1, ω1)
```

```
  return (v1, ω2)
```

```
(v1, ω1) = update(a, x, ω)
(v2, ω2) = update(a, x+1, ω1)
```

```
(v3, ω3) = update(a, x+1, ω)
```

$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



Dead Store Elimination

```
fun foo(a, x, ω):
```

```
  (v1, ω1) = update(a, x, ω)
```

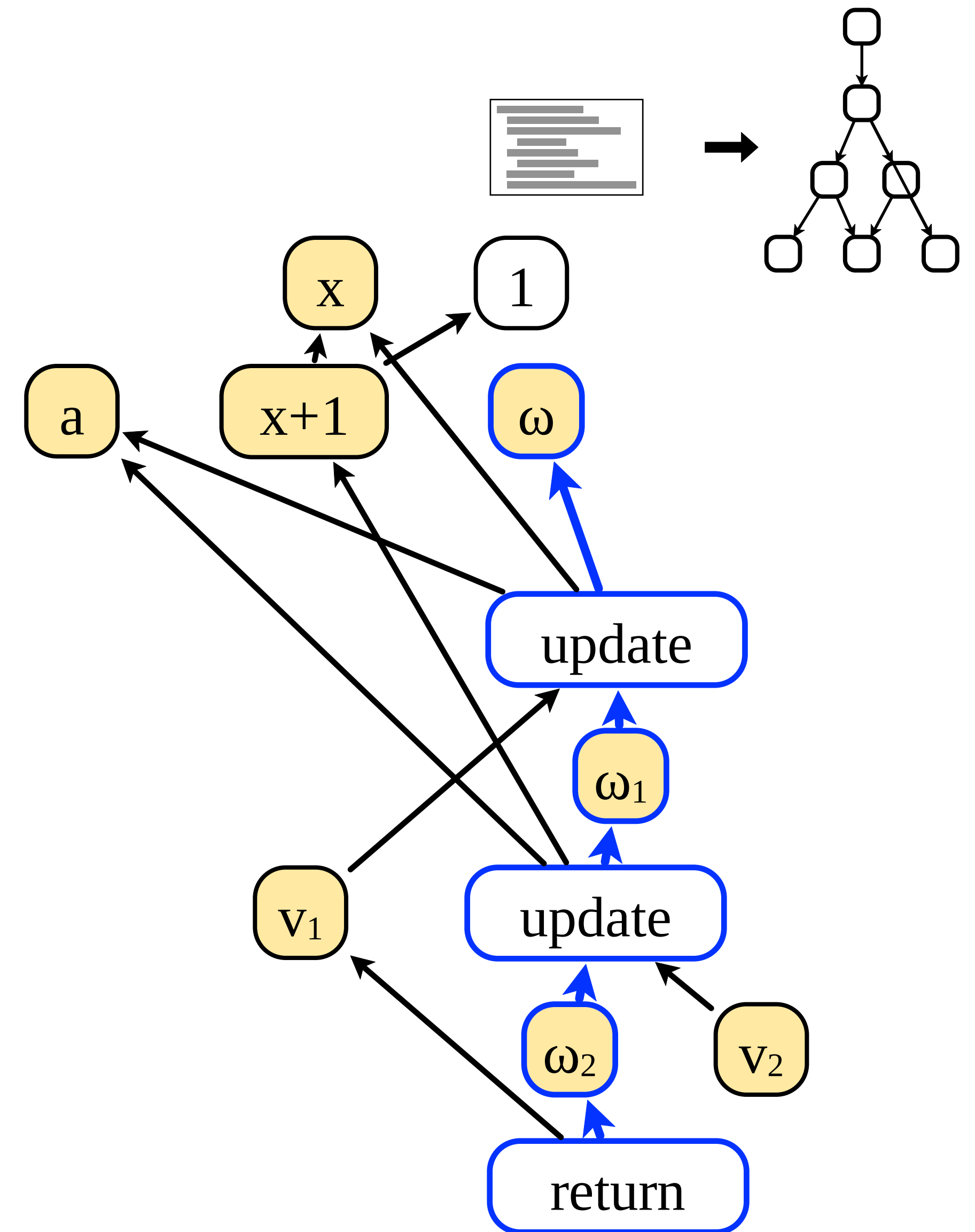
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  (v2, ω2) = update(a, x+1, ω1)
```

```
  return (v1, ω2)
```

$$\begin{array}{l} (v_1, \omega_1) = \text{update}(a, x, \omega) \\ (v_2, \omega_2) = \text{update}(a, x+1, \omega_1) \end{array}$$

$$((v_3), \omega_3) = \text{update}(a, x+1, \omega)$$

$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



Dead Store Elimination

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  (v1, ω1) = update(a, x, ω)
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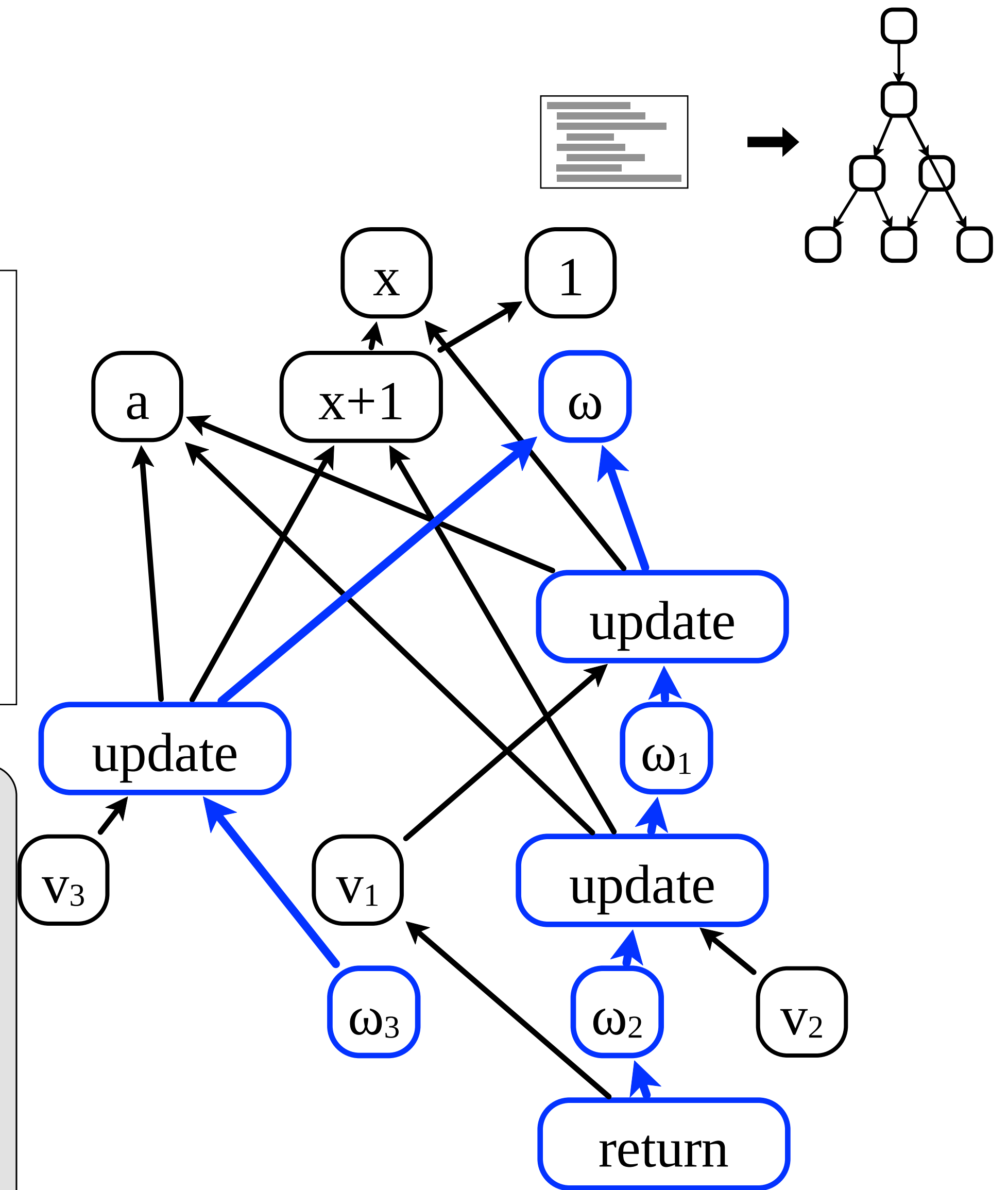
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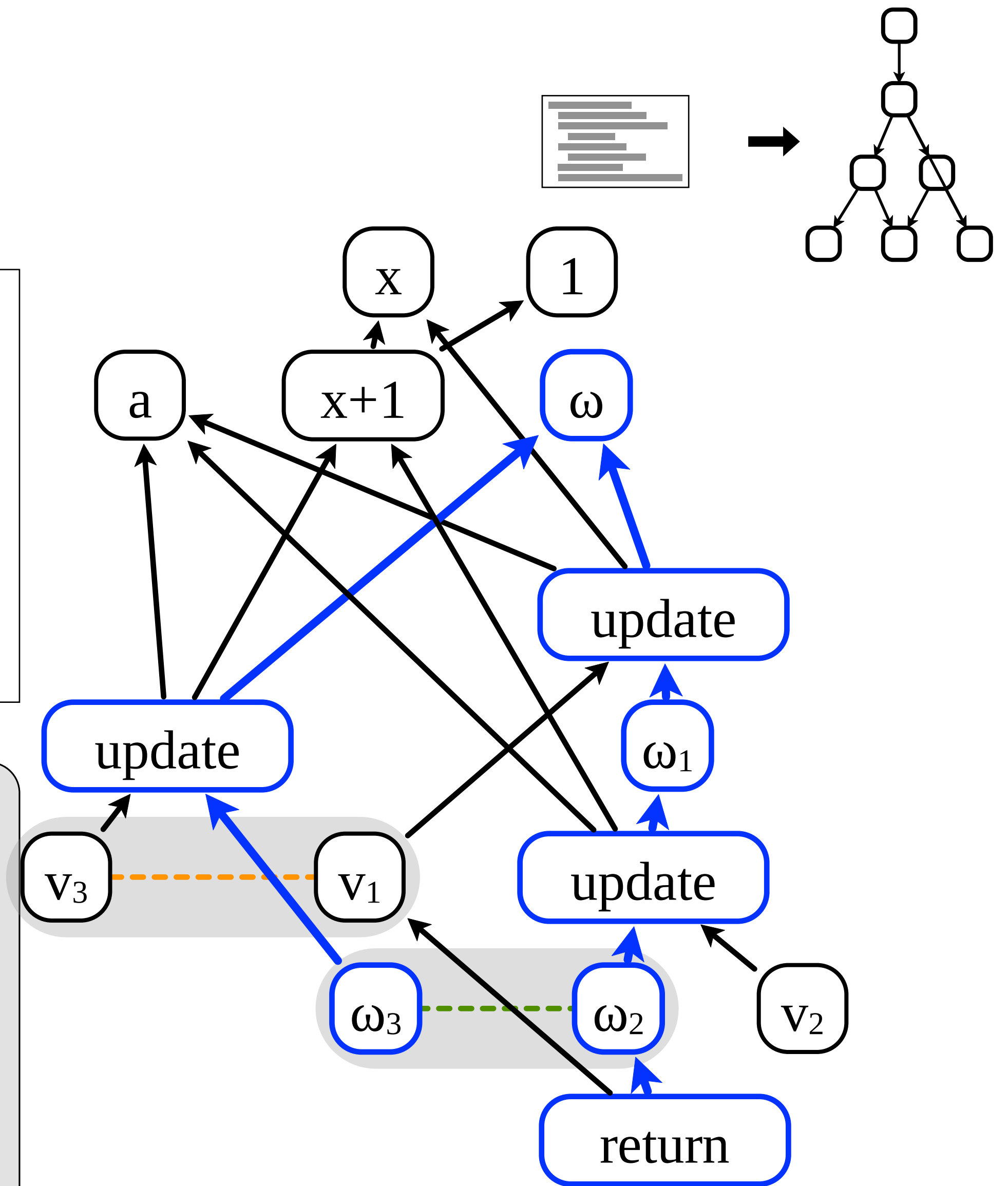
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$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



Extraction Goes Wrong

```
fun foo(a, x, ω):
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  (v1, ω1) = update(a, x, ω)
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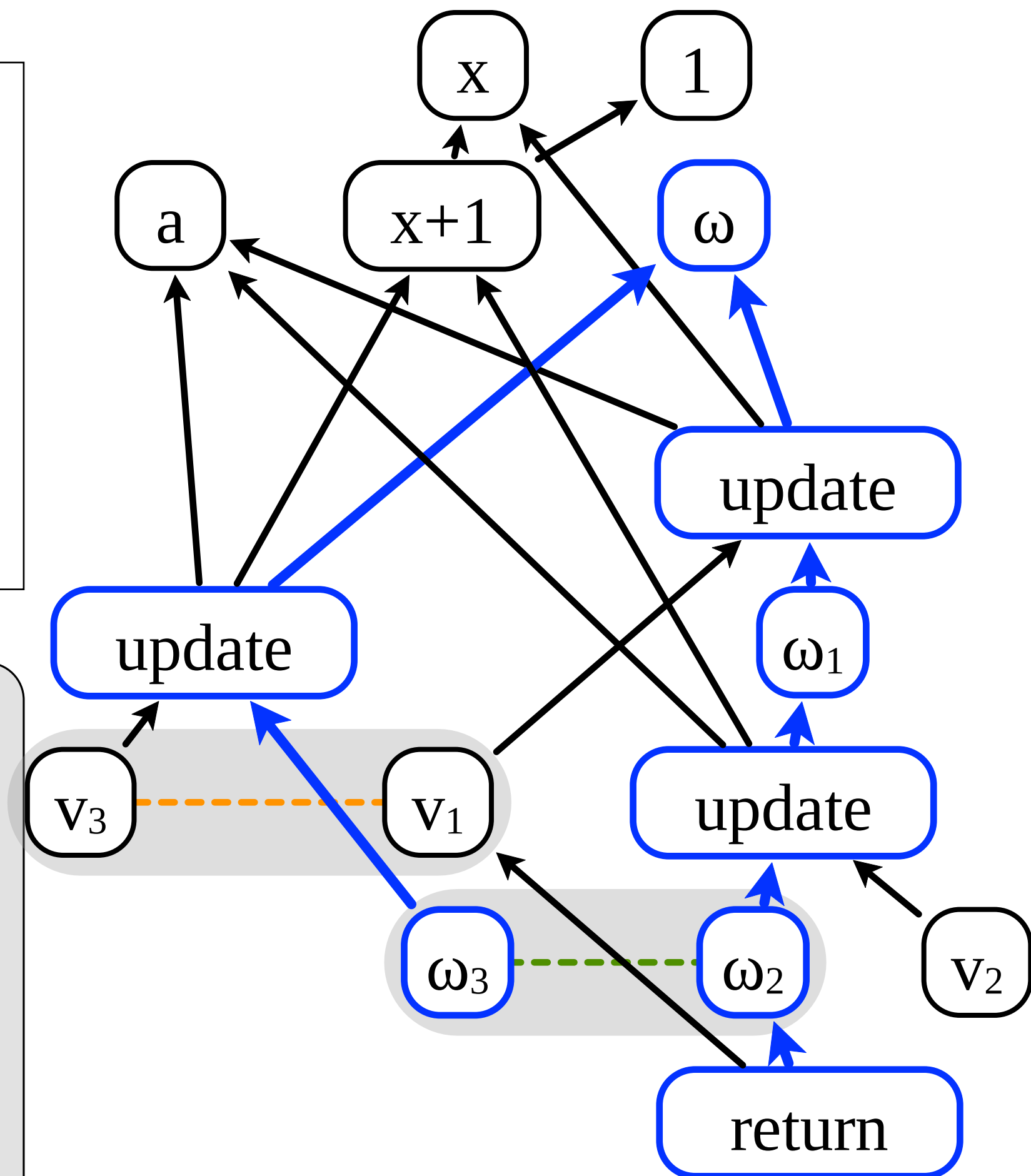
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Extraction Goes Wrong

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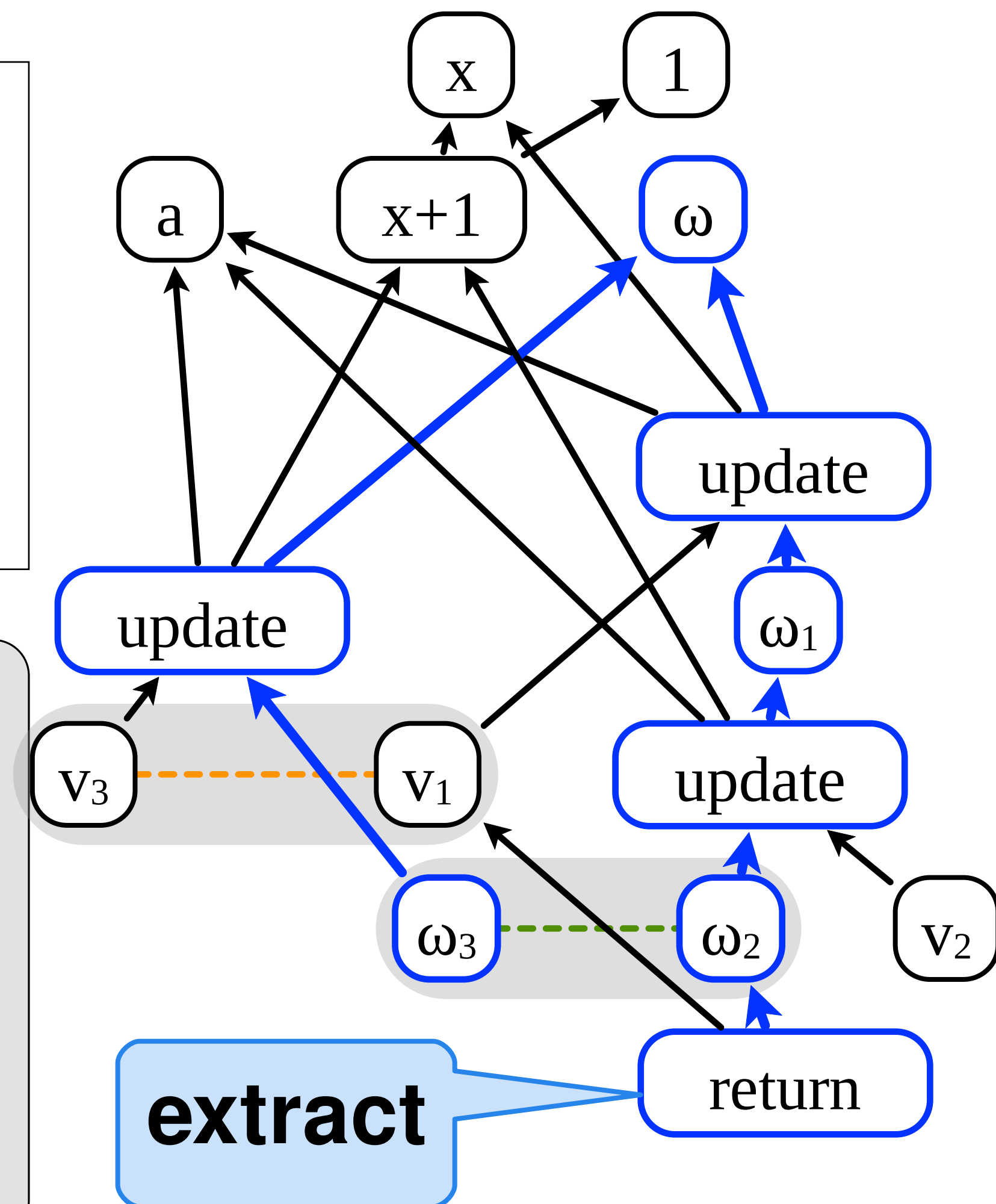
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Extraction Goes Wrong

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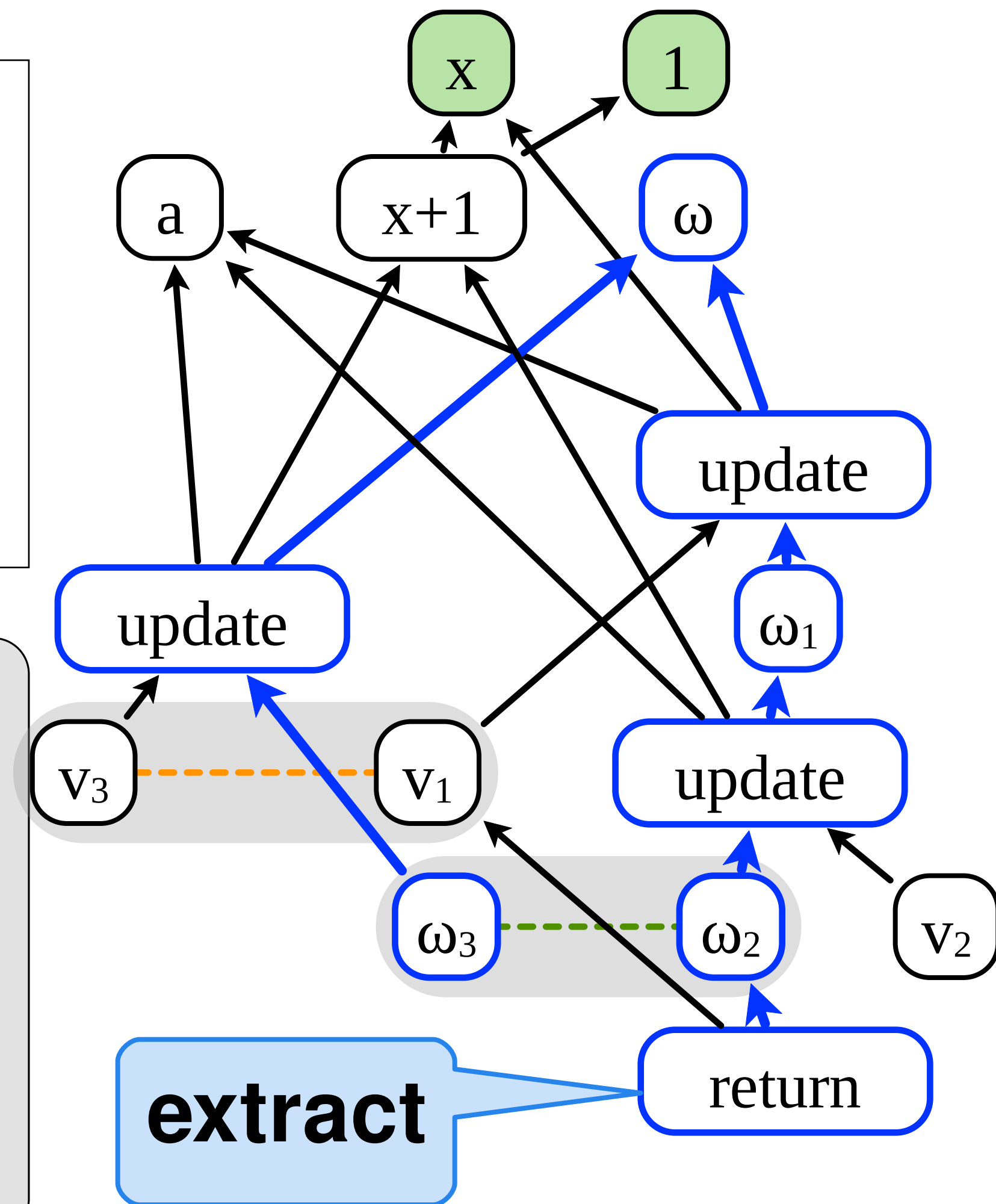
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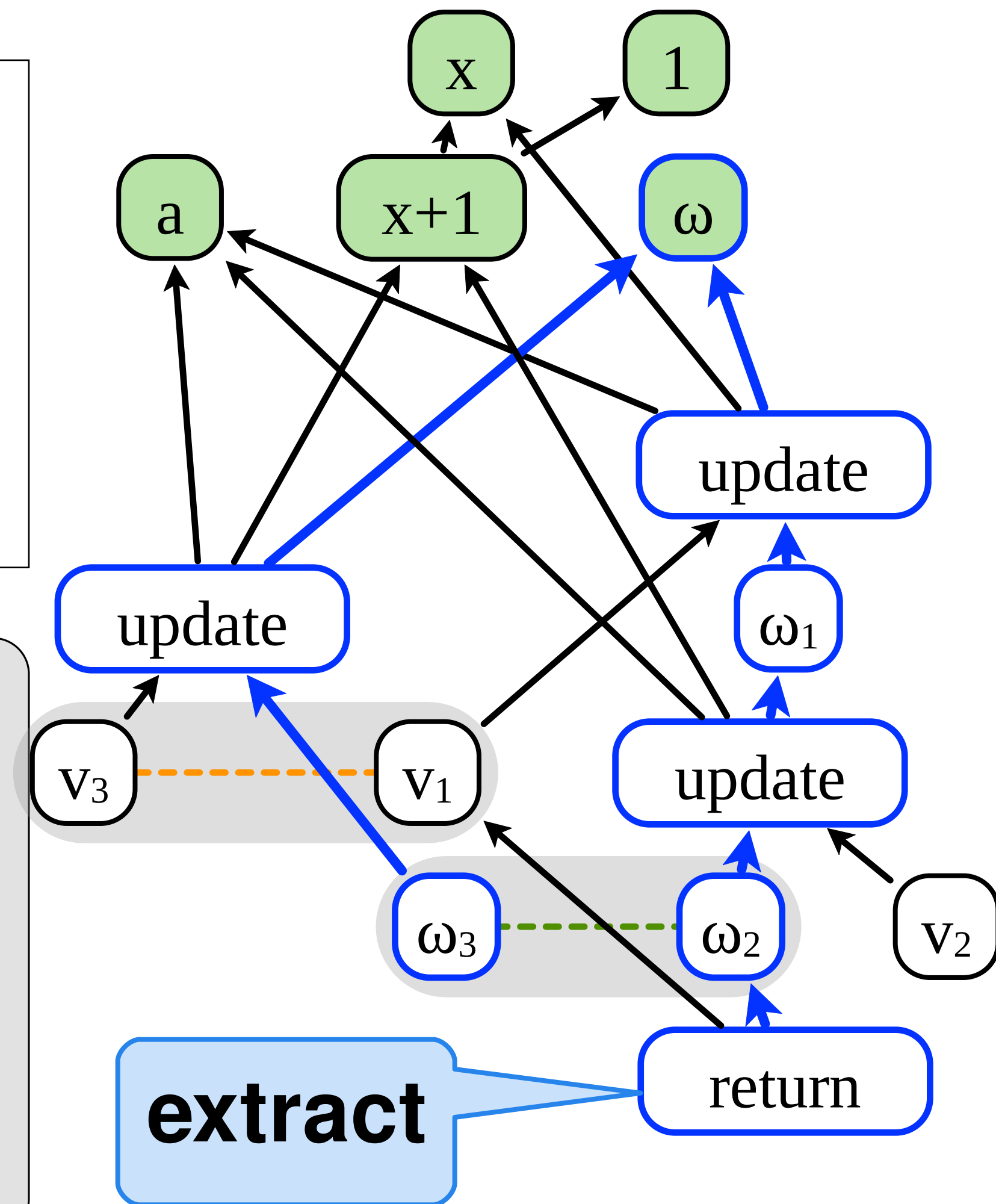
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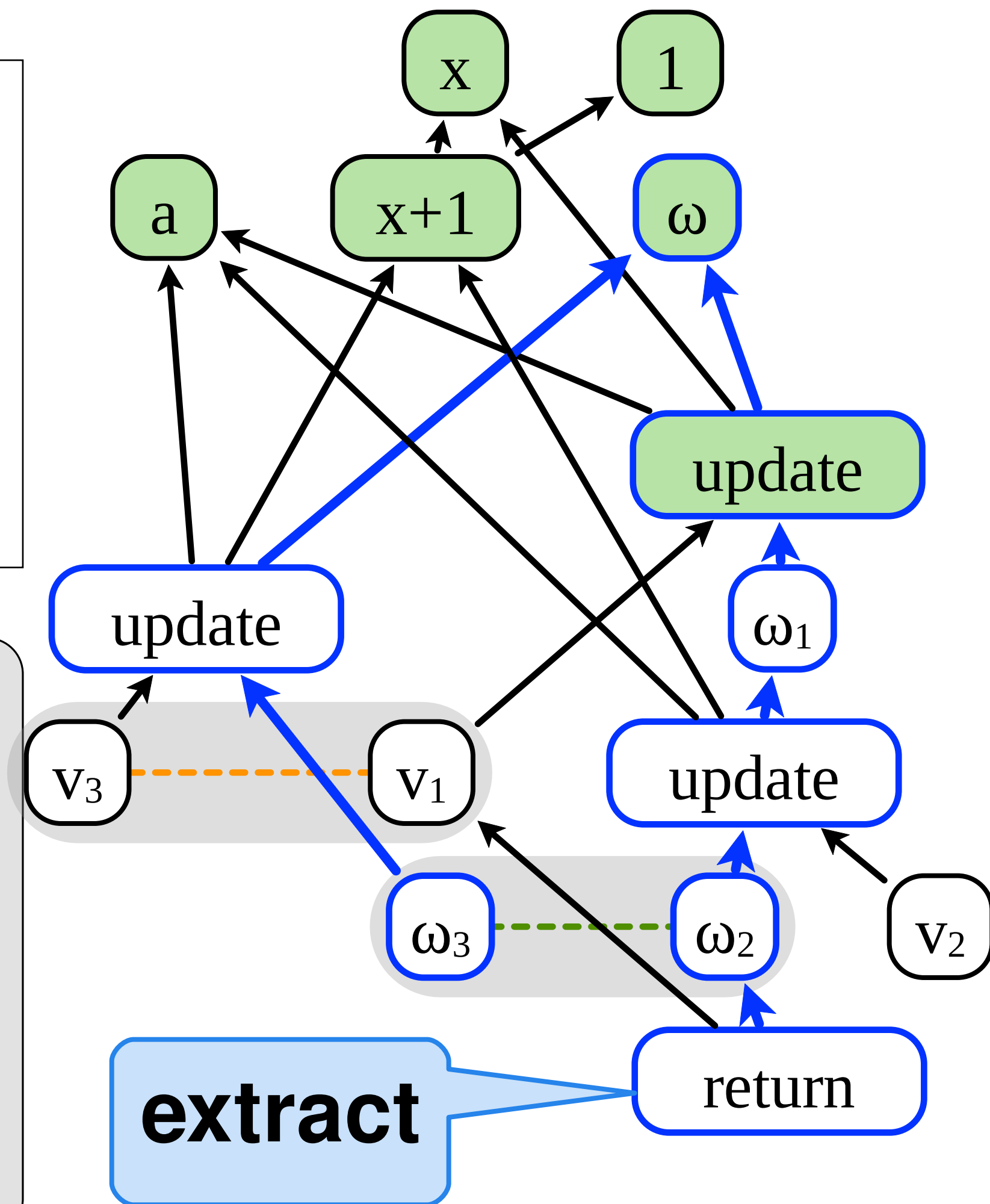
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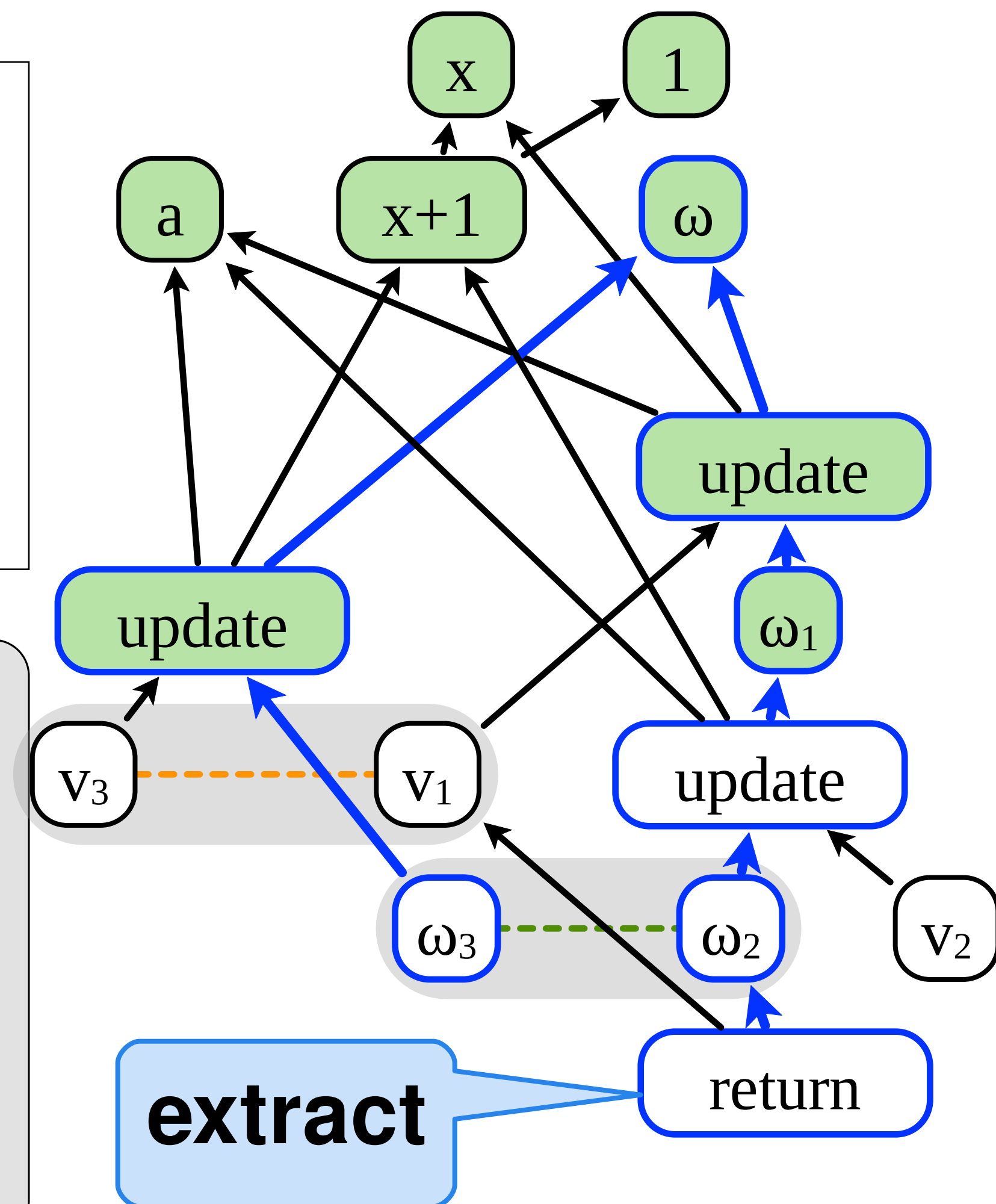
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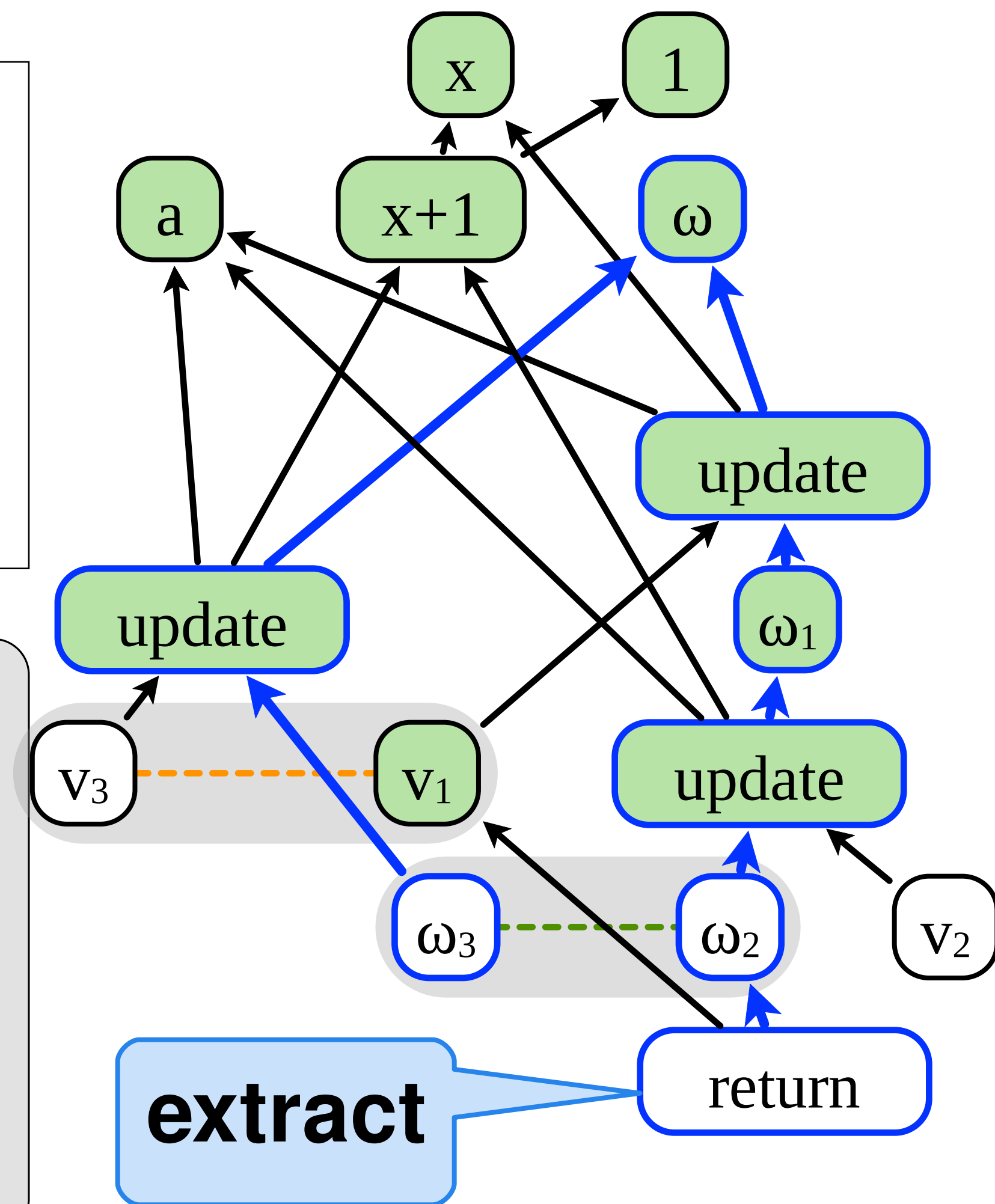
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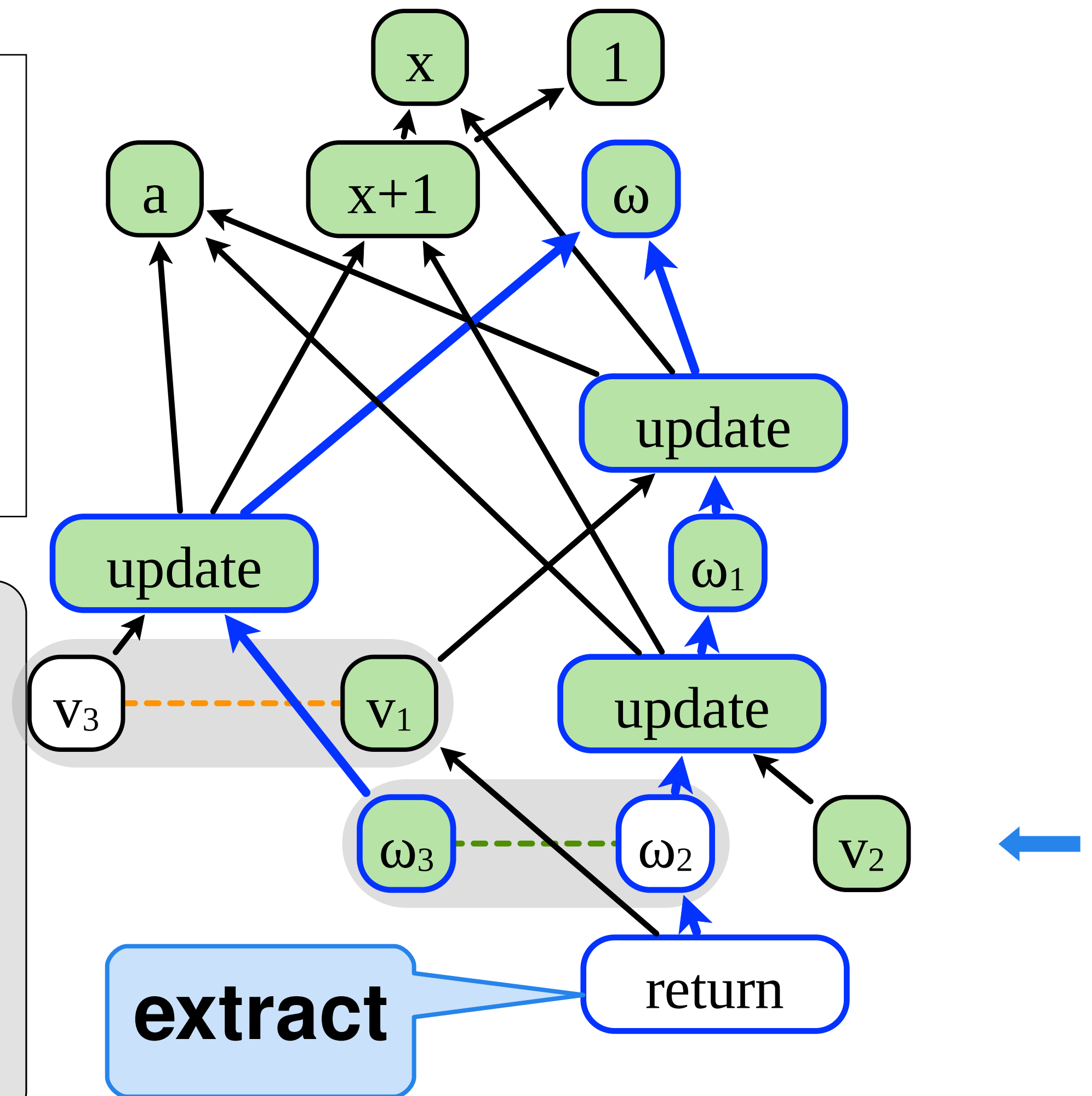
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```
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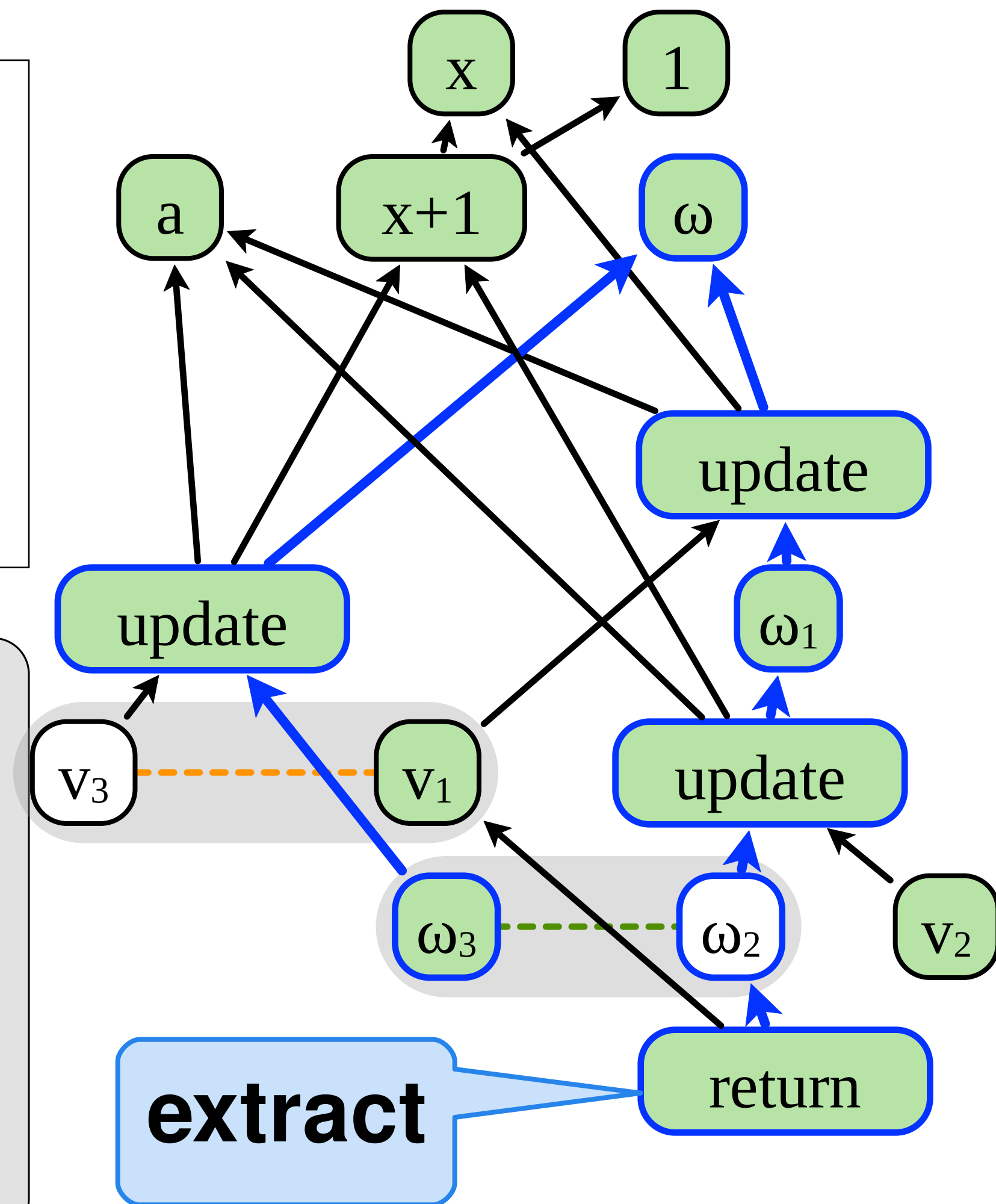
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$v_1 \equiv v_3 \quad \omega_2 \equiv \omega_3$



Extraction Goes Wrong

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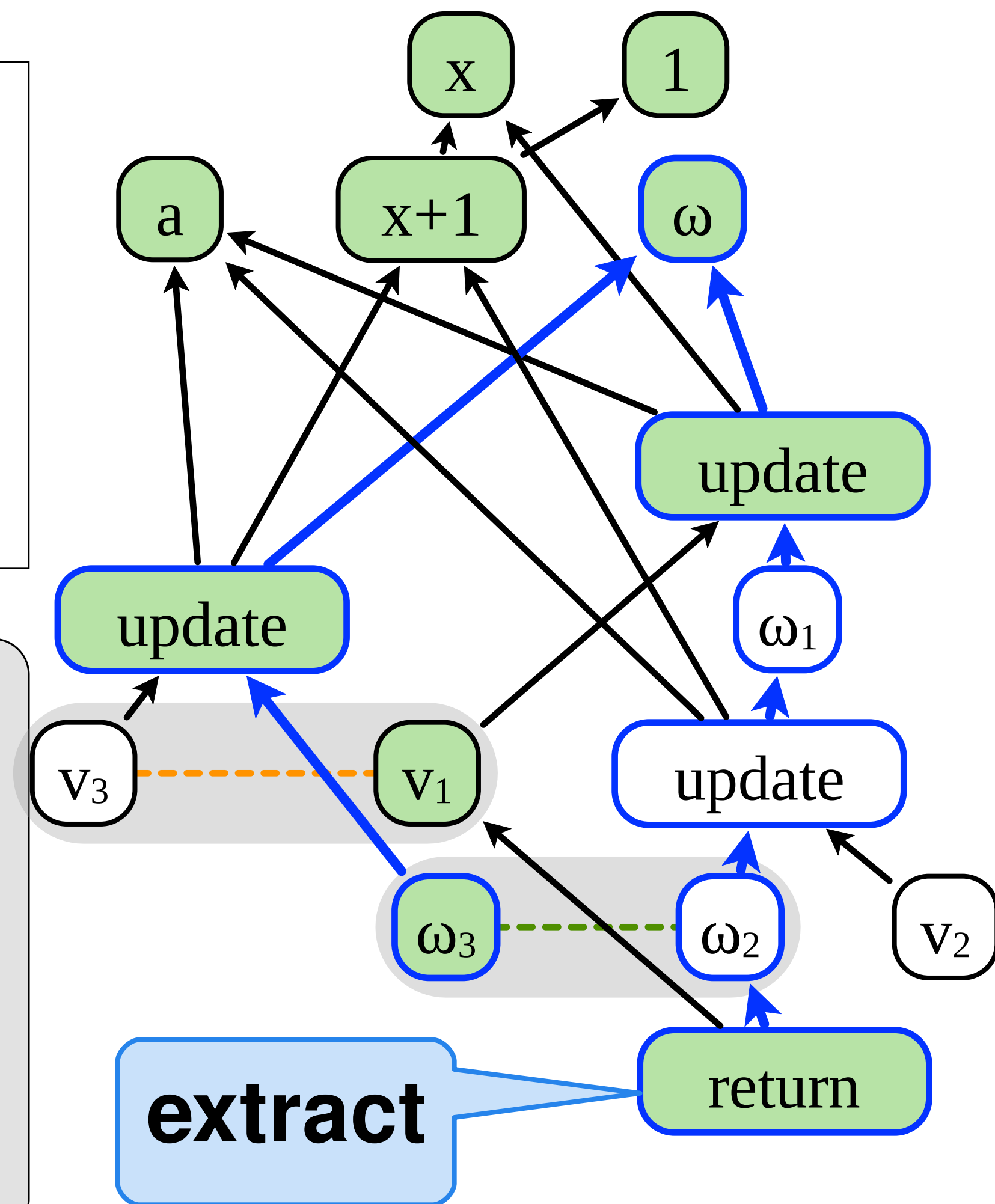
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Extraction Goes Wrong

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fun foo(a, x, ω):
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  (v1, ω1) = update(a, x, ω)
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```

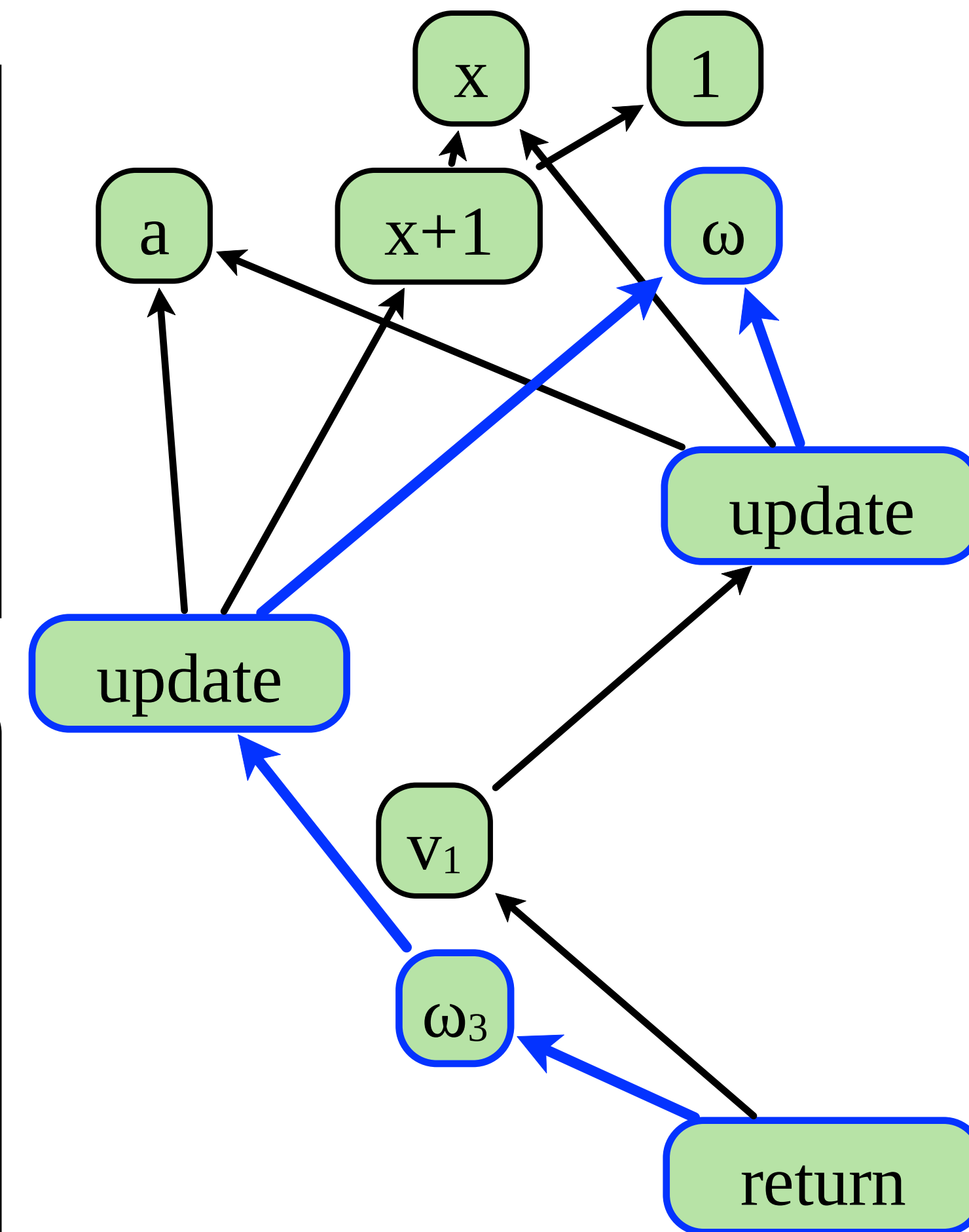
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$$(v_1, \omega_1) = \text{update}(a, x, \omega)$$

$$(v_2, \omega_2) = \text{update}(a, x+1, \omega_1)$$

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$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



Extraction Goes Wrong

```
fun foo(a, x, ω):
```

```
  (v1, ω1) = update(a, x, ω)
```

```
  (v2, ω2) = update(a, x+1, ω1)
```

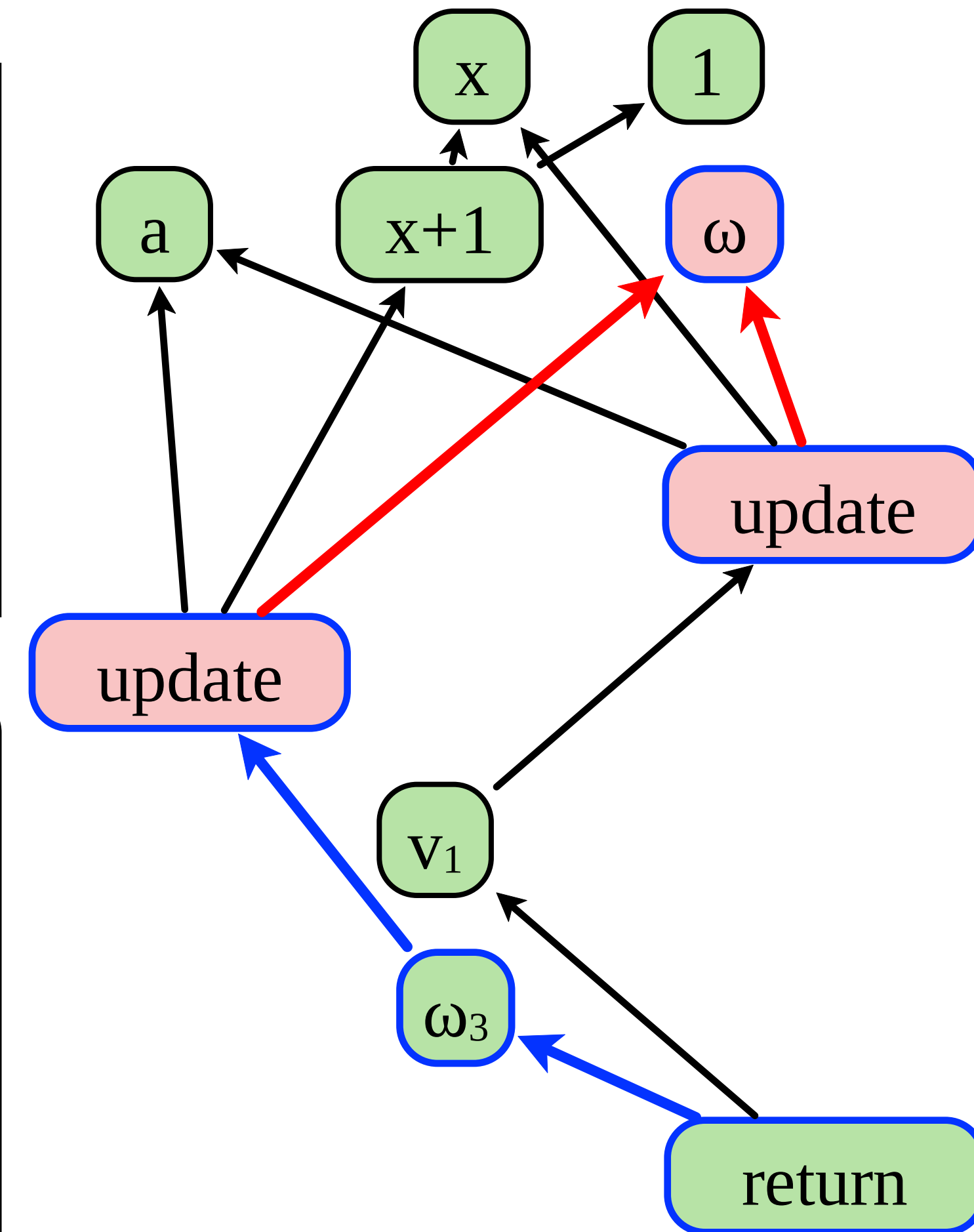
```
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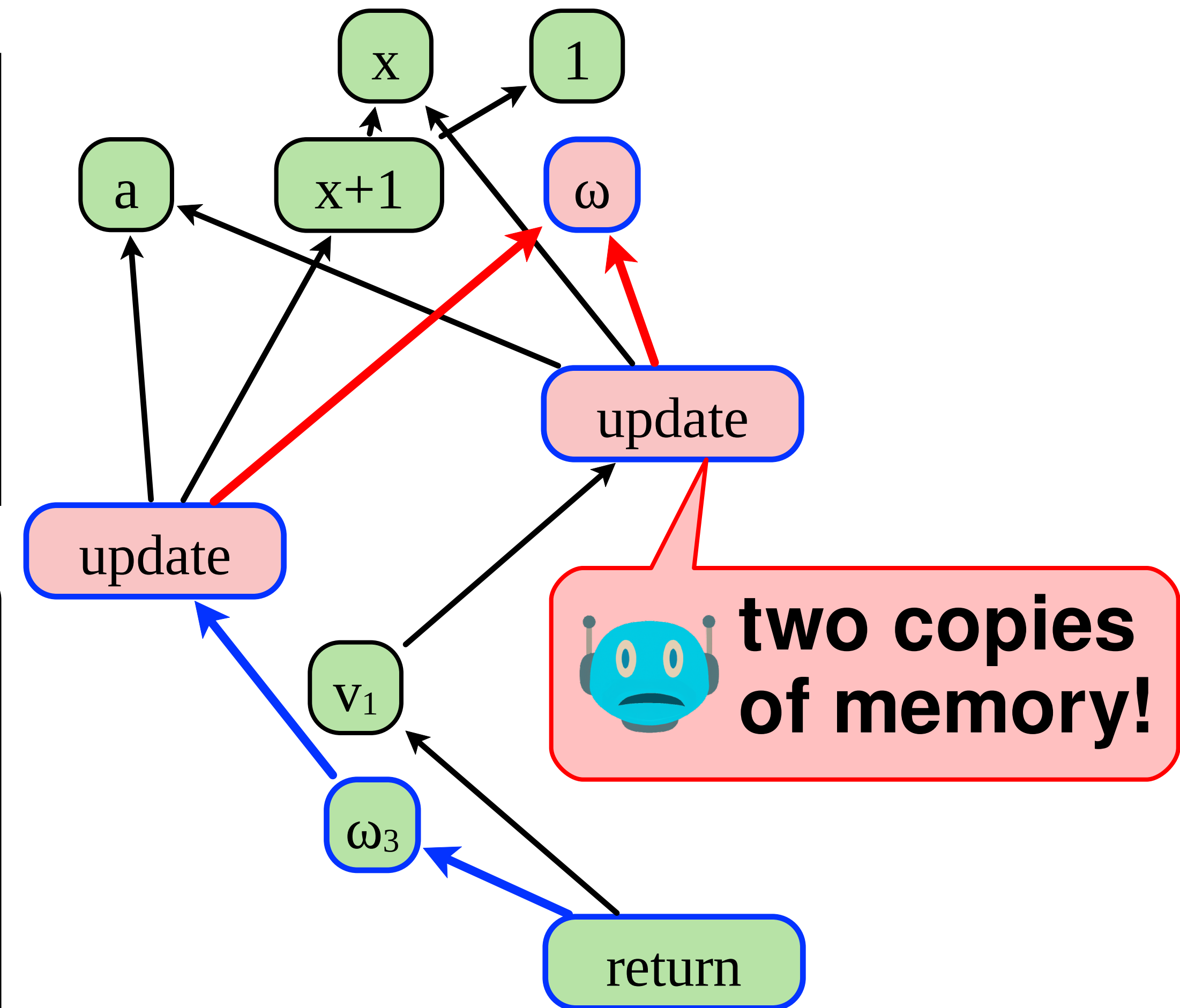
```
  (v1, ω1) = update(a, x, ω)
```

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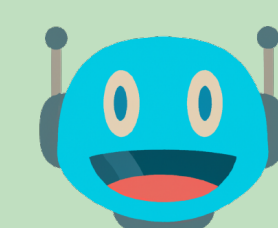
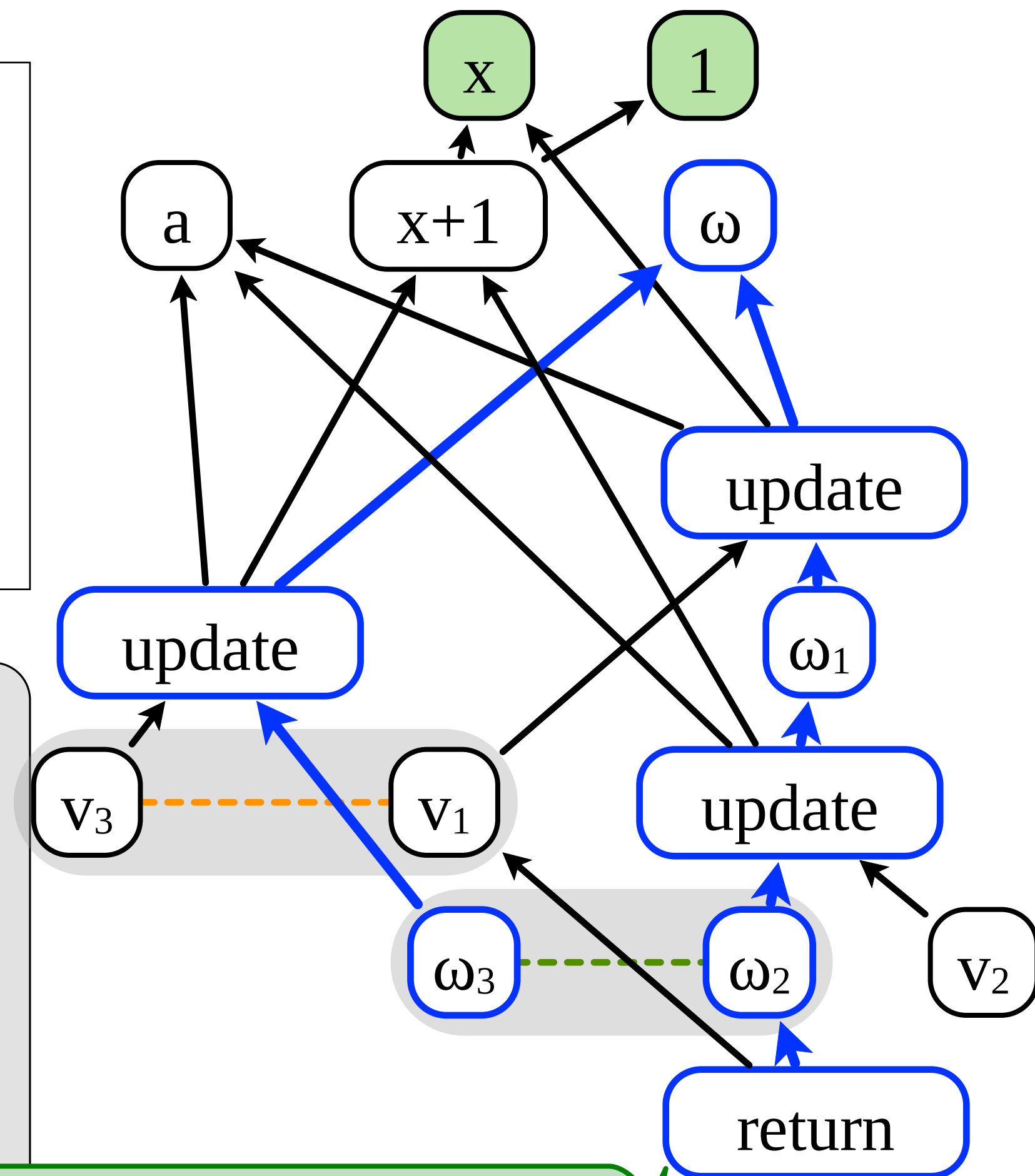
```
  (v2, ω2) = update(a, x+1, ω1)
```

```
  return (v1, ω2)
```

```
(v1, ω1) = update(a, x, ω)  
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```

```
(v3, ω3) = update(a, x+1, ω)
```

$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



**extract
effect-safely**

Extraction Goes Wrong

```
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```

```
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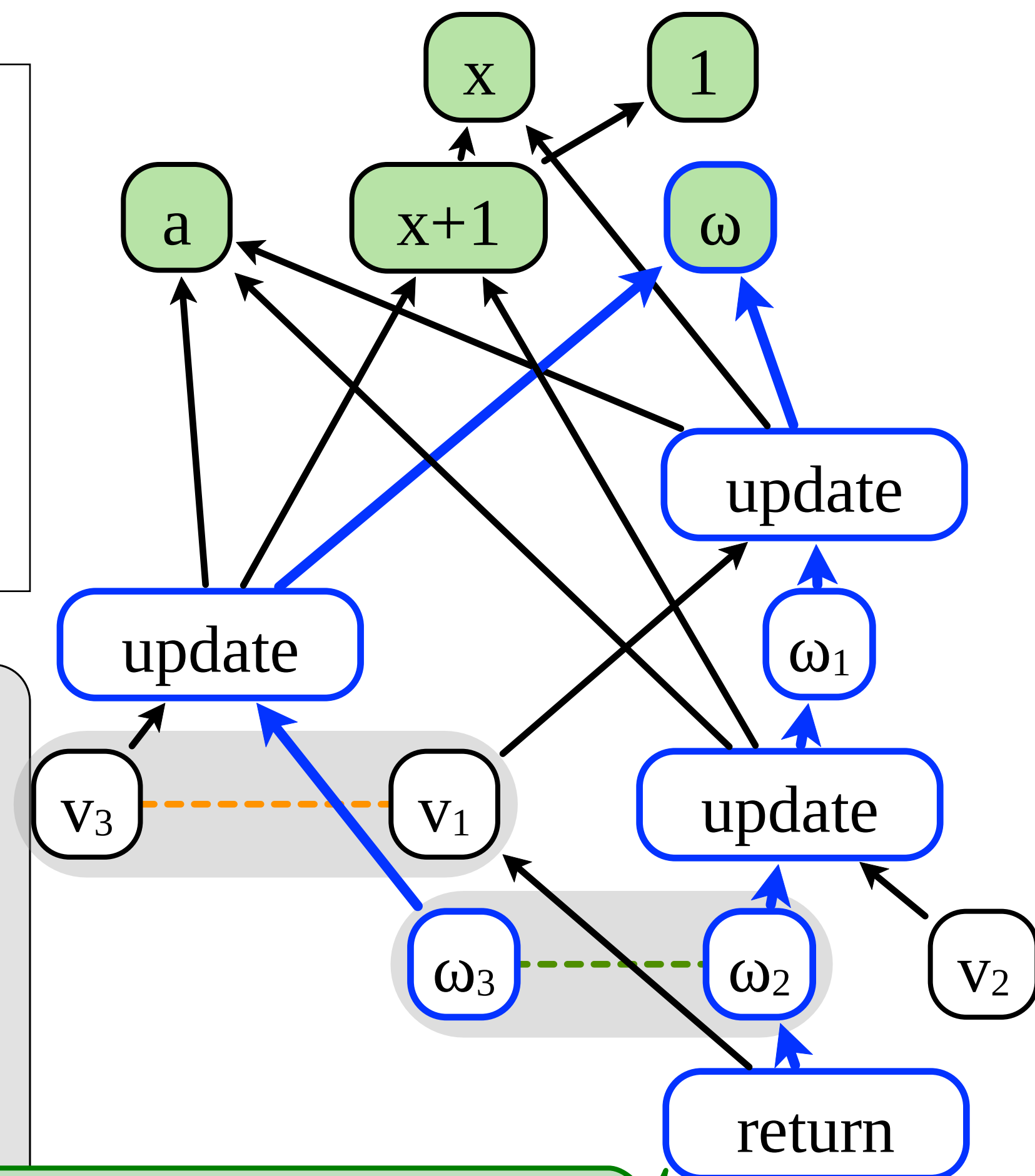
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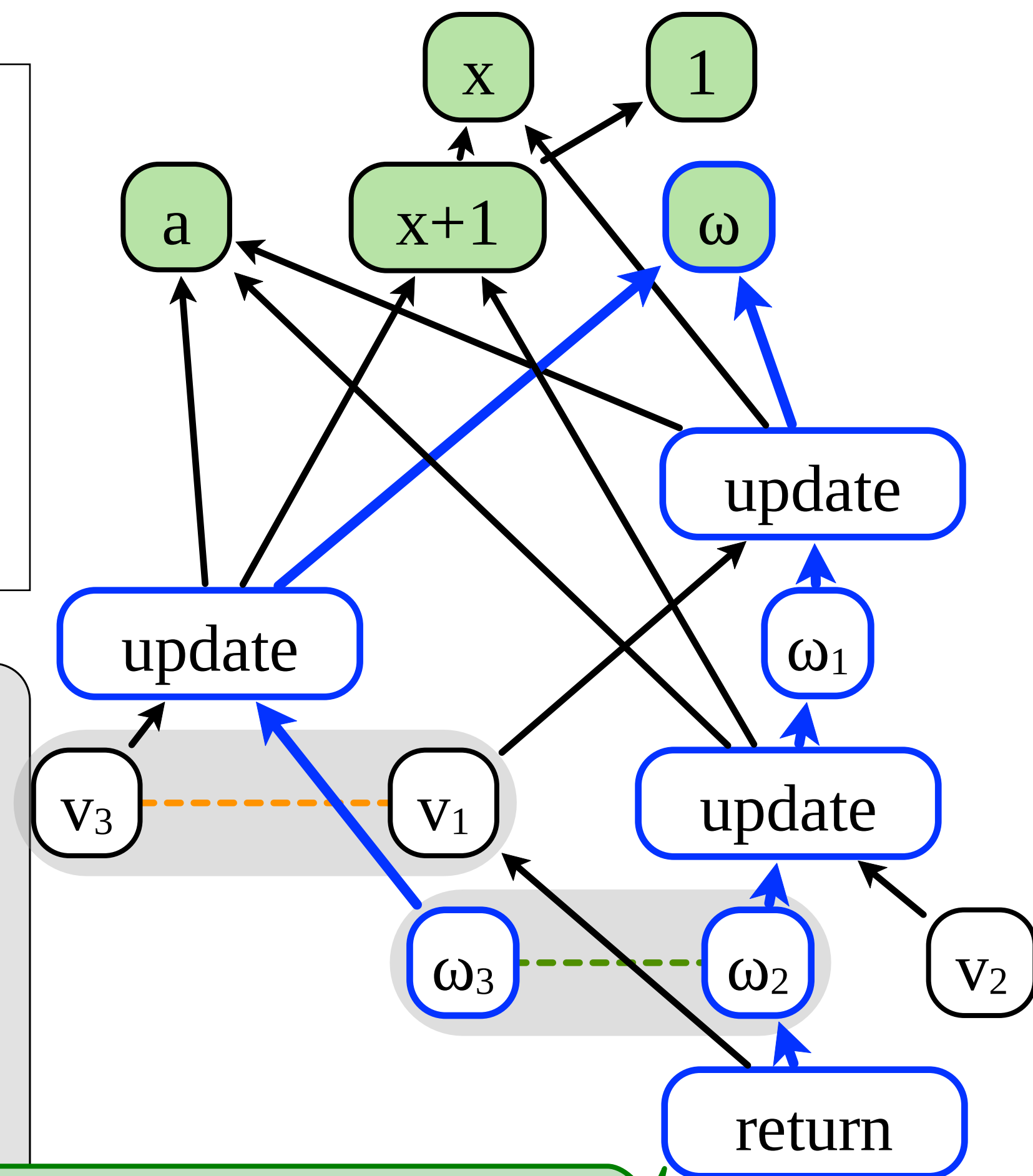
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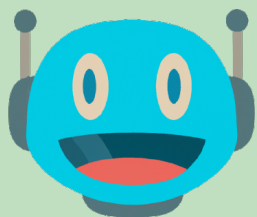
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 **extract
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Extraction Goes Wrong

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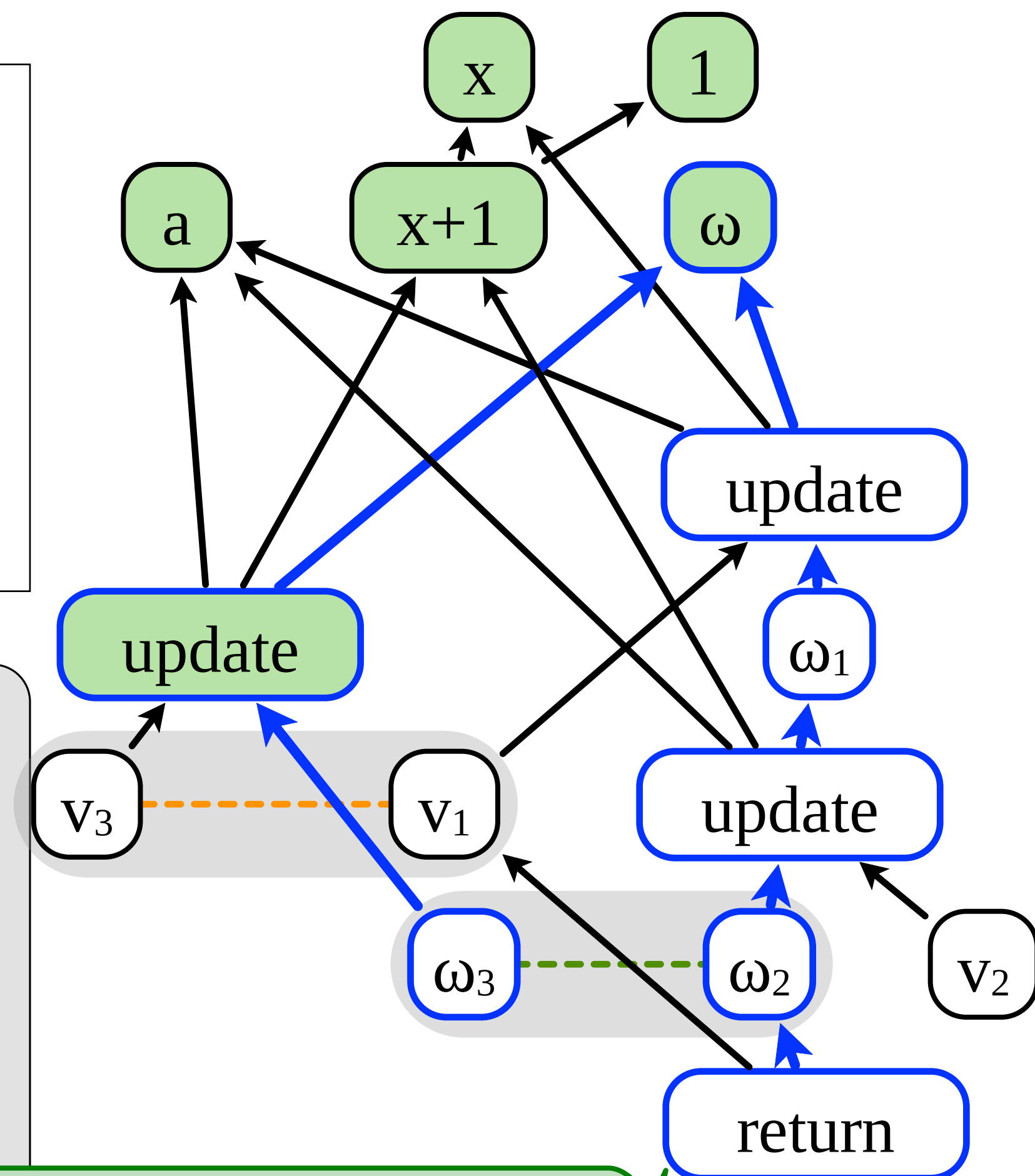
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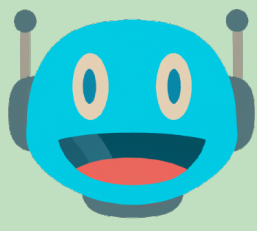
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 **extract
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Extraction Goes Wrong

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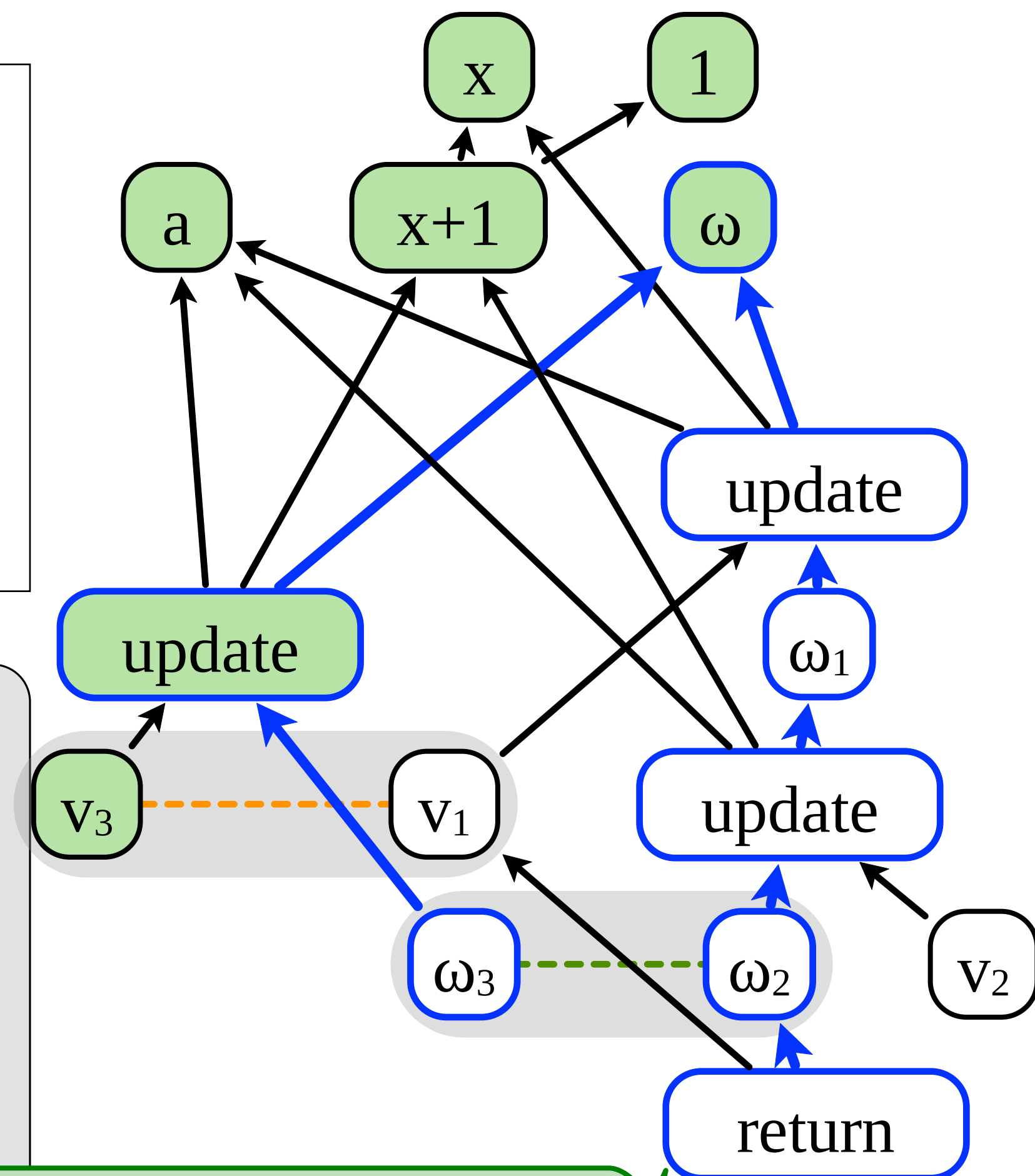
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Extraction Goes Wrong

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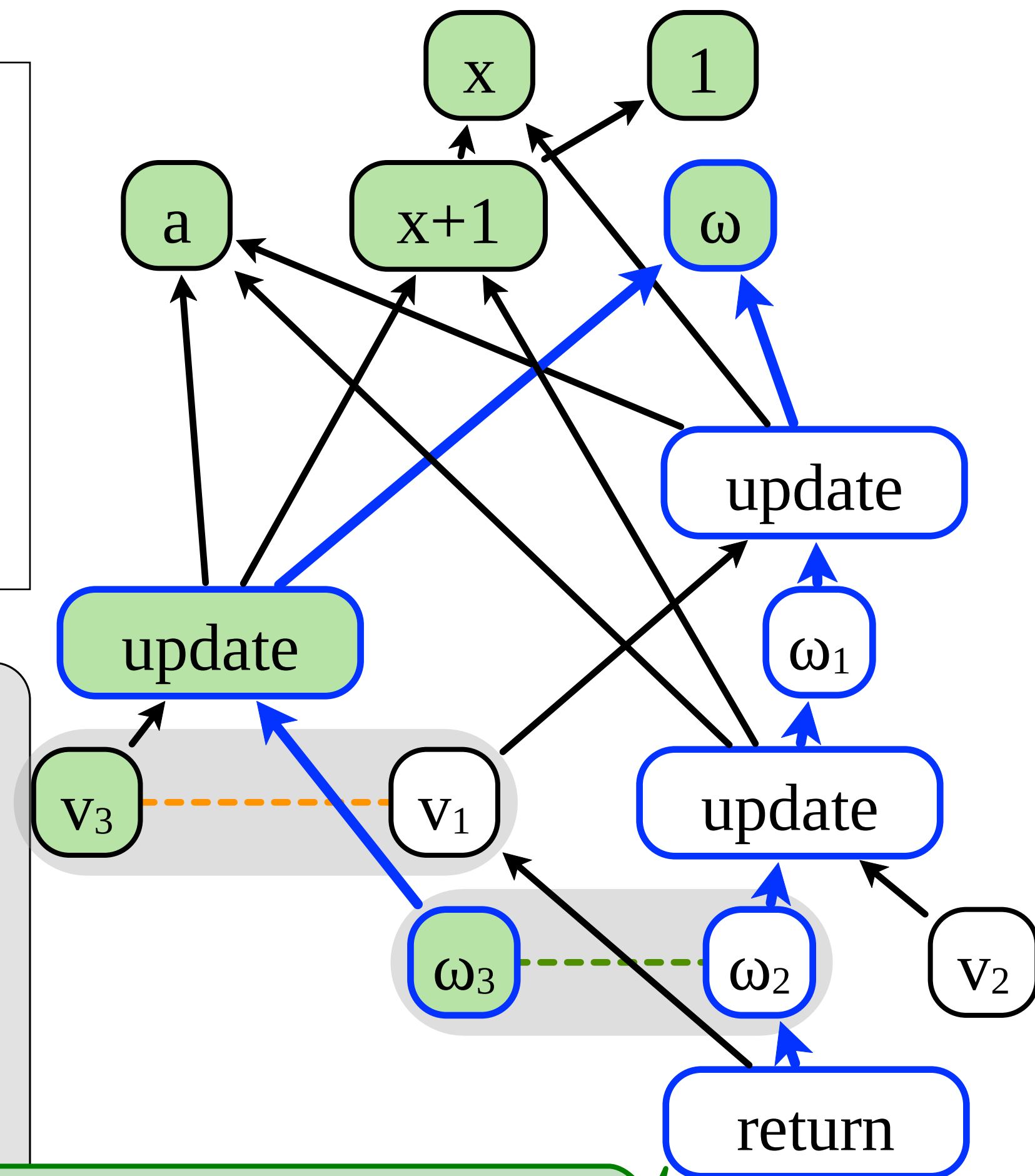
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$$\frac{\begin{array}{l} (v_1, \omega_1) = \text{update}(a, x, \omega) \\ (v_2, \omega_2) = \text{update}(a, x+1, \omega_1) \end{array}}{(v_3, \omega_3) = \text{update}(a, x+1, \omega)}$$

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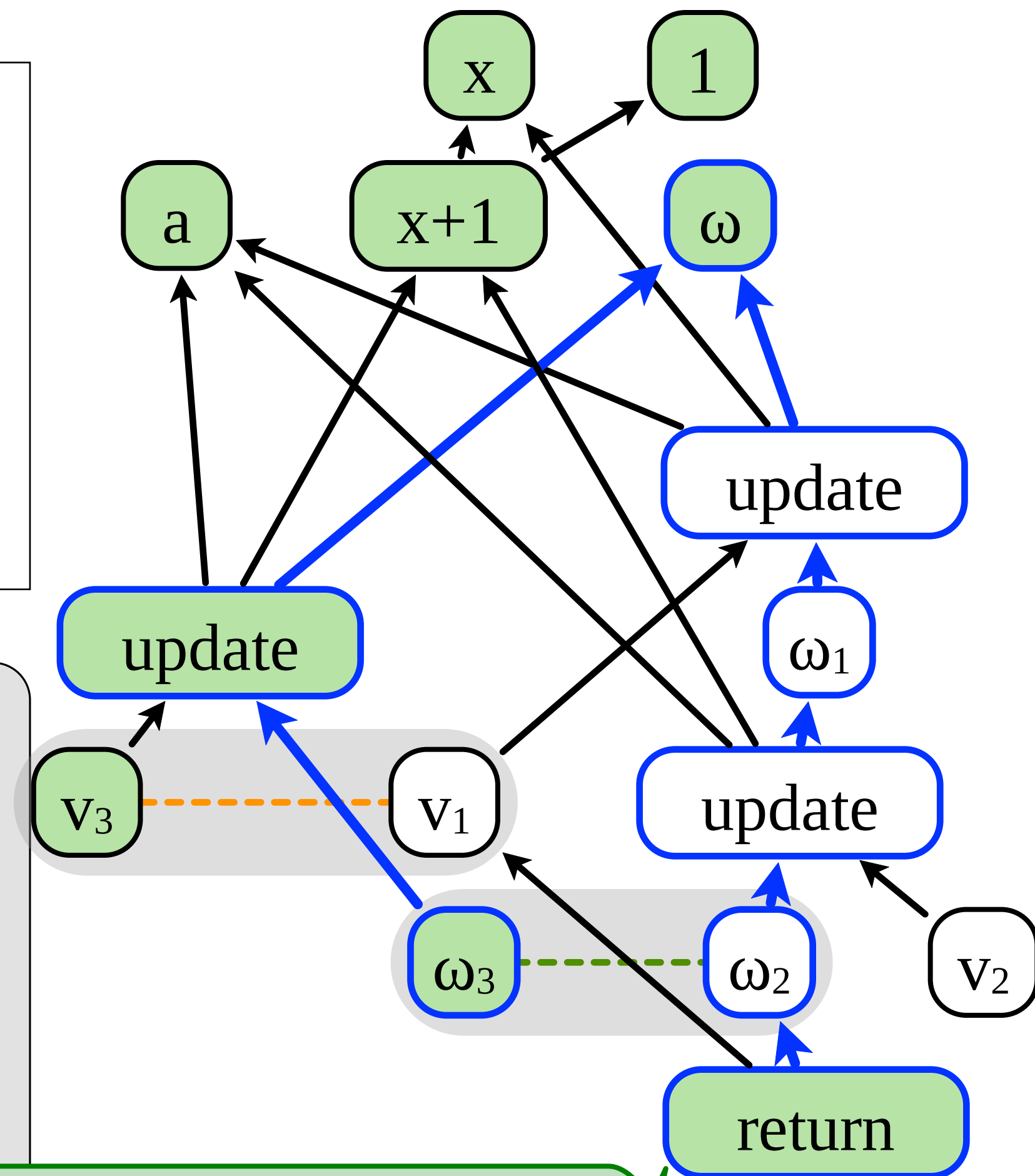
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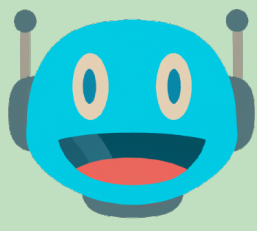
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 **extract
effect-safely**

Extraction Goes Wrong

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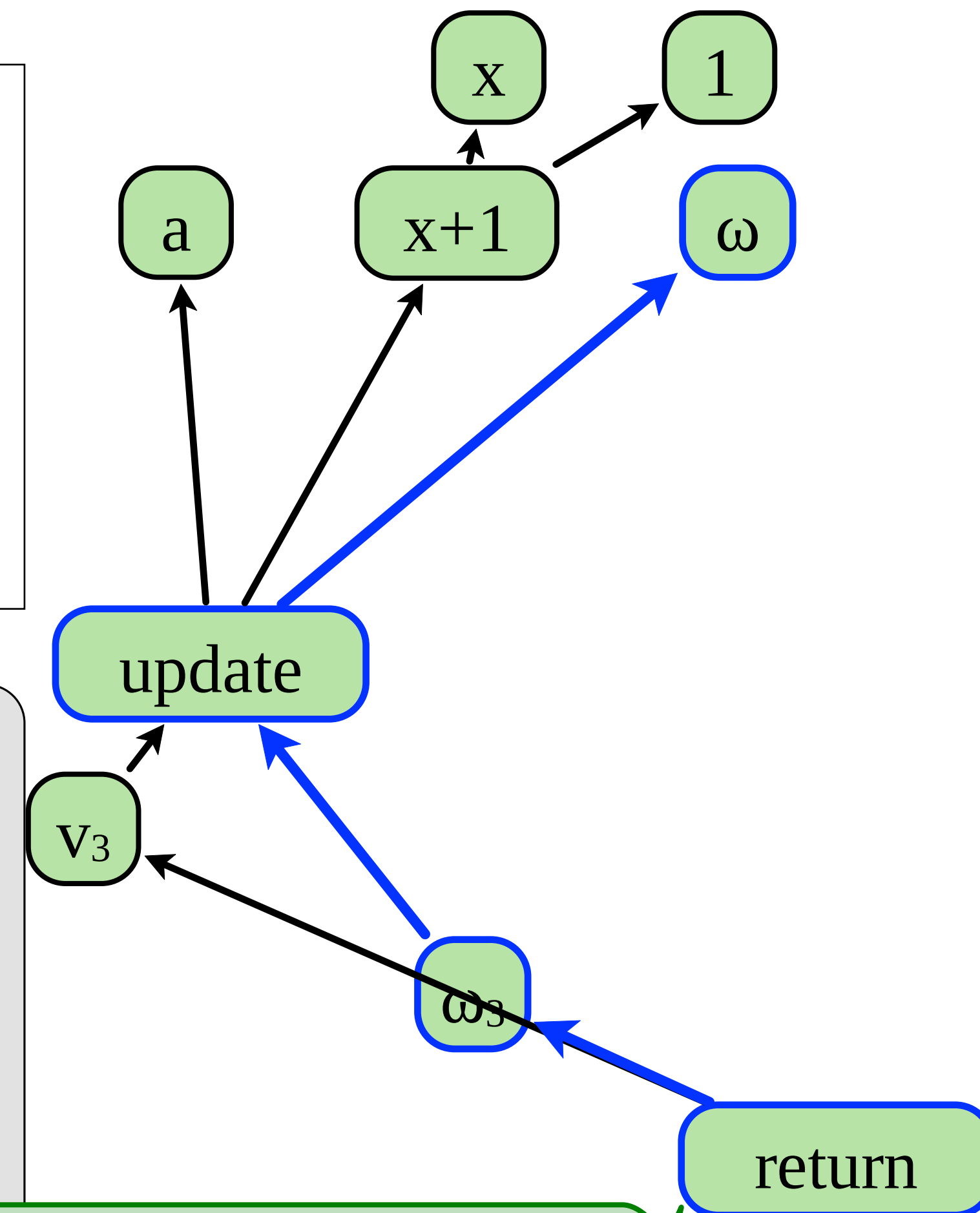
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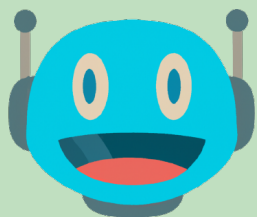
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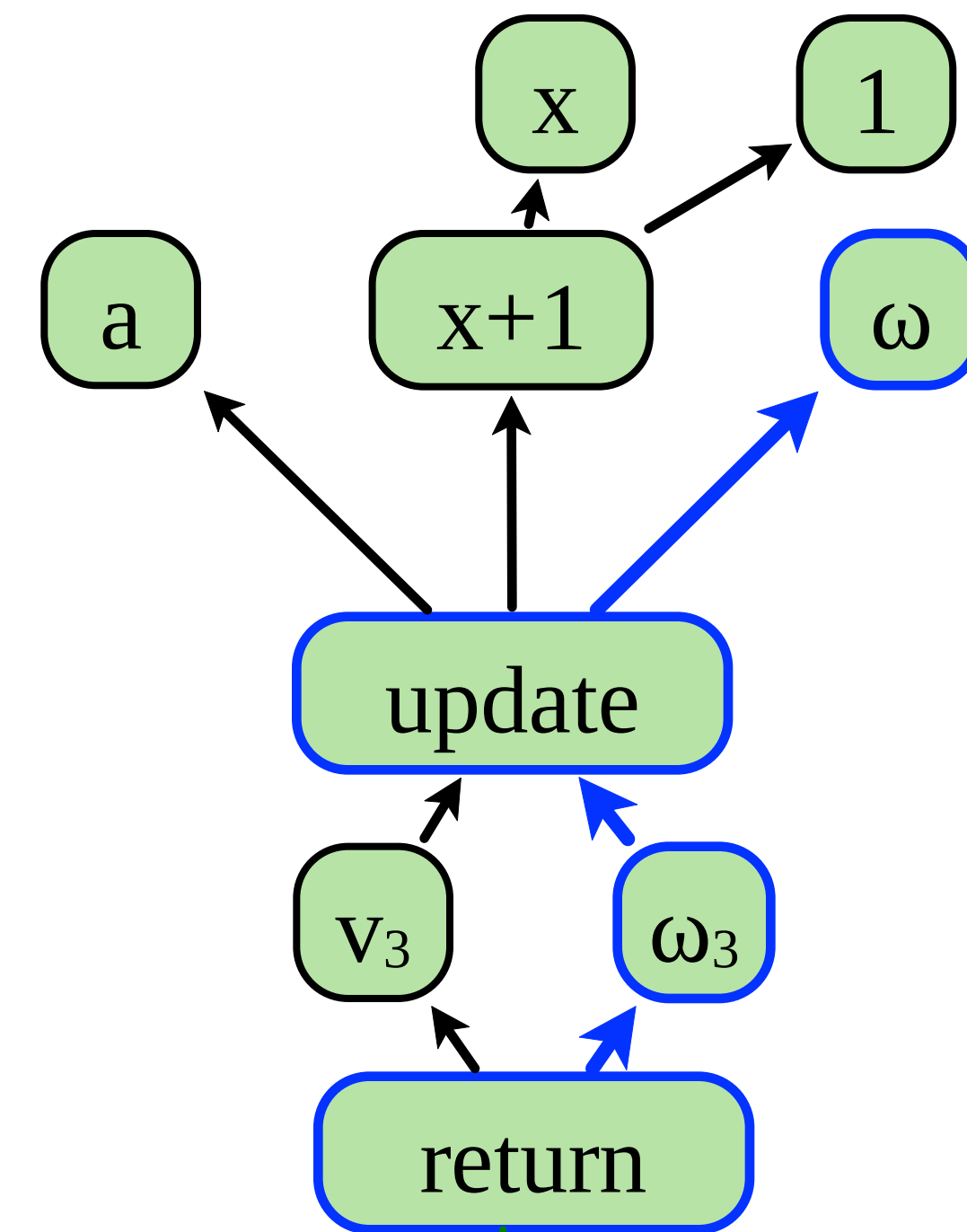
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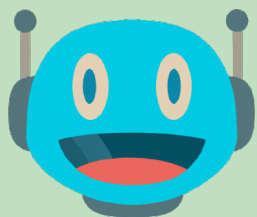
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 **extract
effect-safely**

1

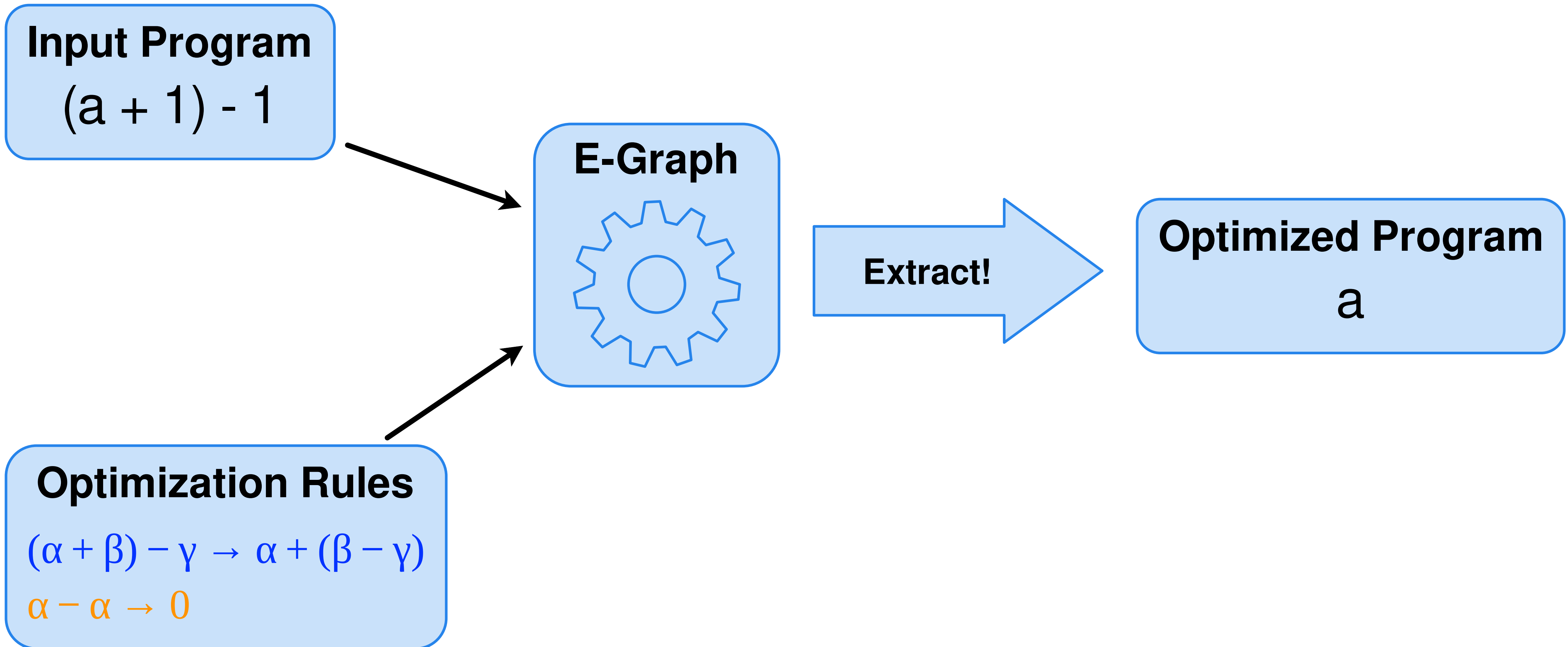
2

3

4

Example Effective E-Graph

E-Graphs Explore a Big Space

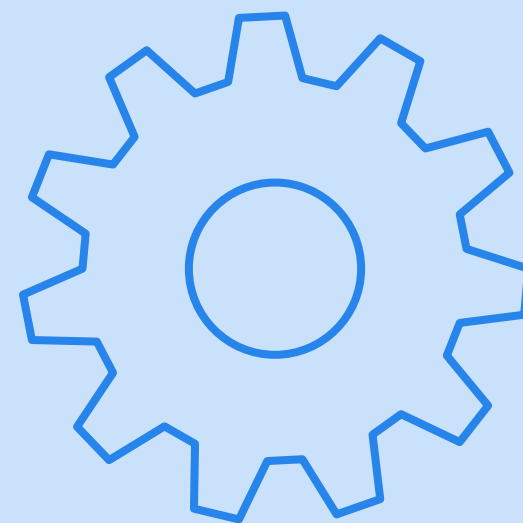


E-Graphs Explore a Big Space

Input Program

```
v1 = update(a, x)
v2 = update(a, x+1)
```

E-Graph



Extract!

Optimized Program

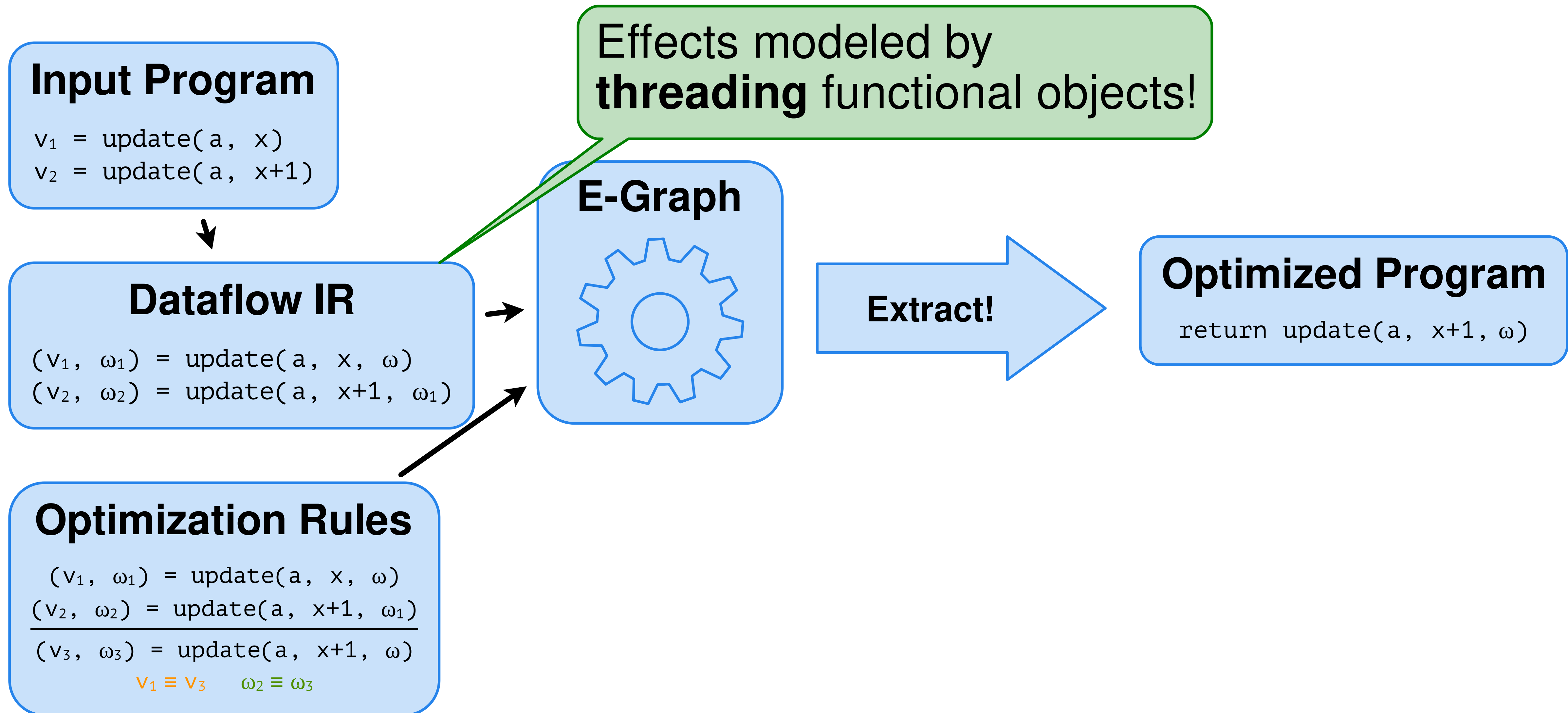
```
return update(a, x+1, ω)
```

Optimization Rules

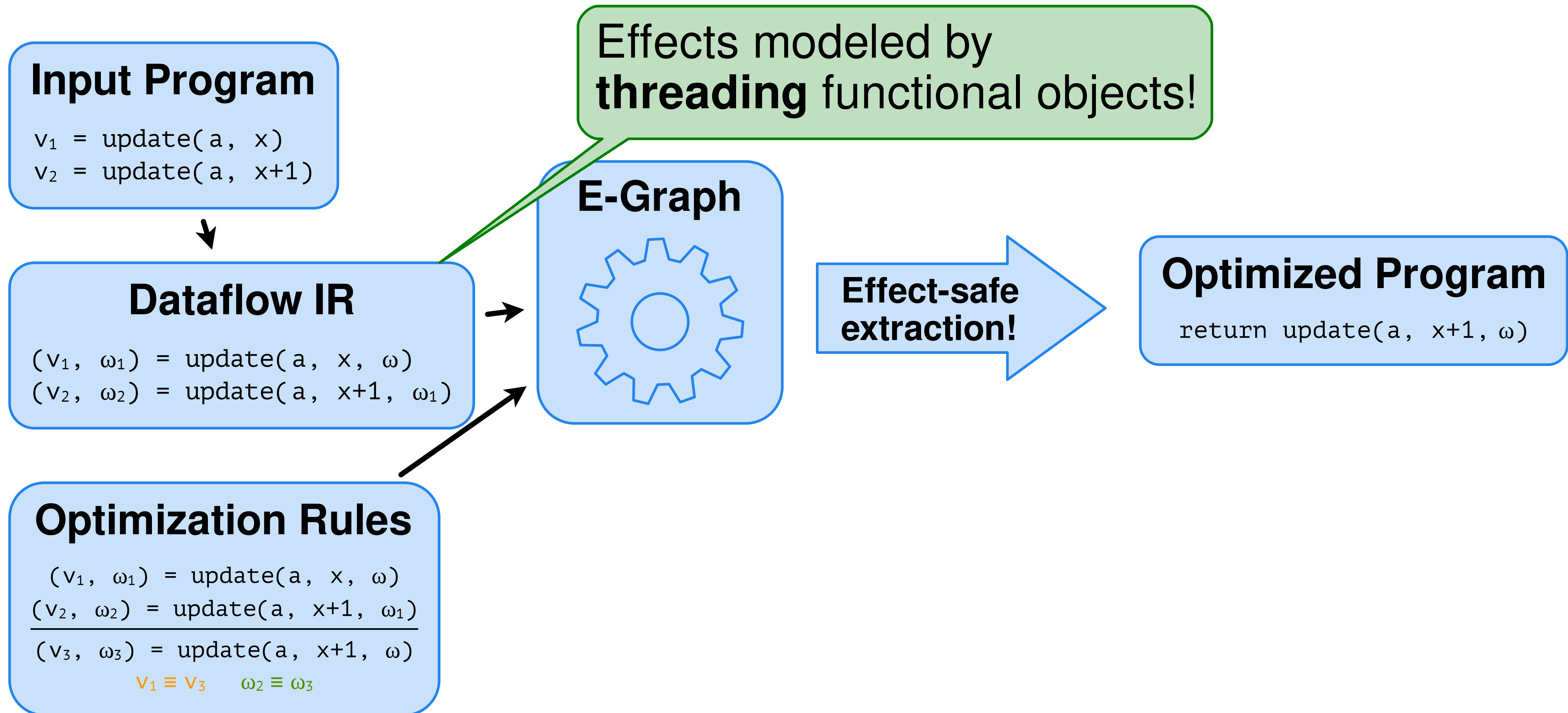
$$\frac{\begin{array}{l} (v_1, \omega_1) = \text{update}(a, x, \omega) \\ (v_2, \omega_2) = \text{update}(a, x+1, \omega_1) \end{array}}{(v_3, \omega_3) = \text{update}(a, x+1, \omega)}$$

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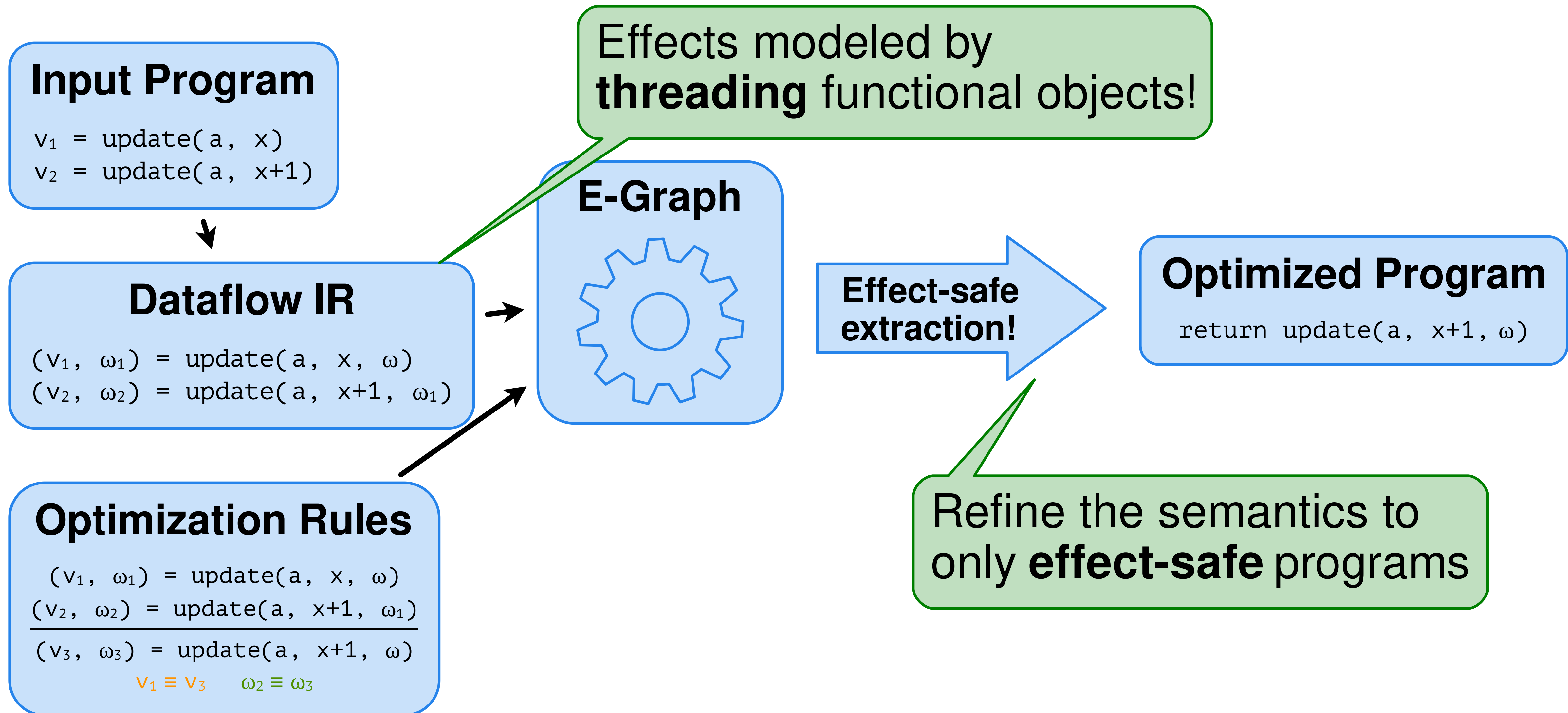
E-Graphs Explore a Big Space



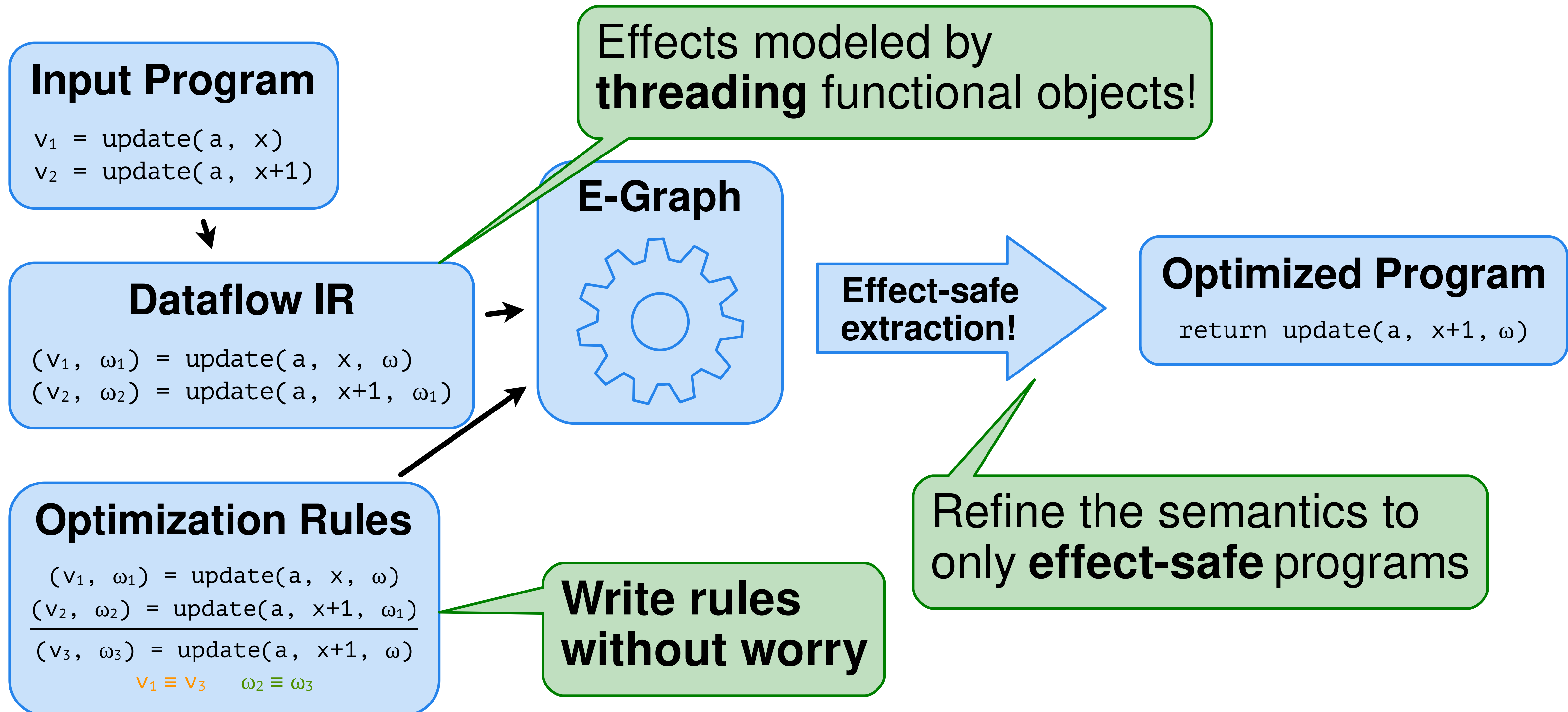
E-Graphs Explore a Big Space

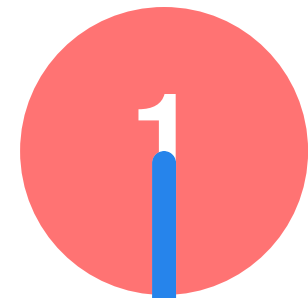


E-Graphs Explore a Big Space

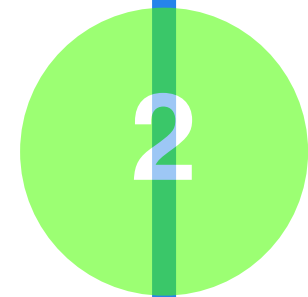


E-Graphs Explore a Big Space





Motivate Effectful E-Graphs



Example Effectful E-Graph



Effect-Safe Extraction



It Works!



Effect-Safe Extraction

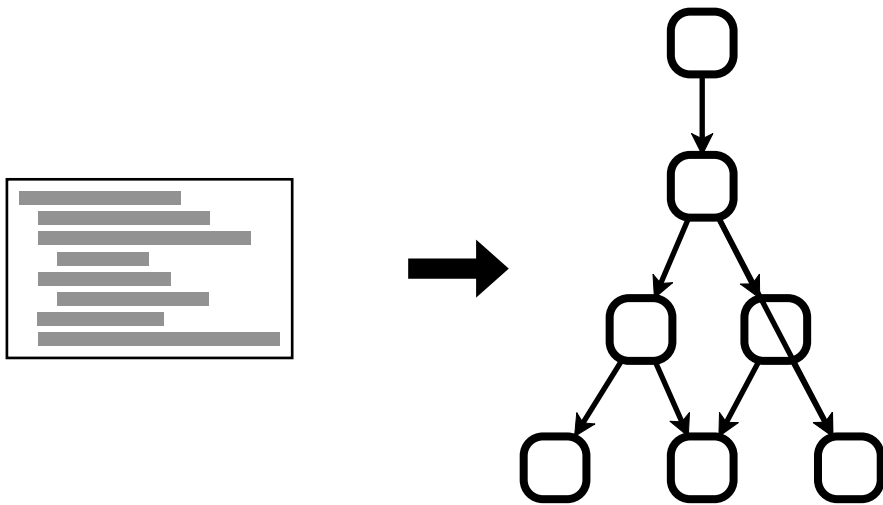
1

2

3

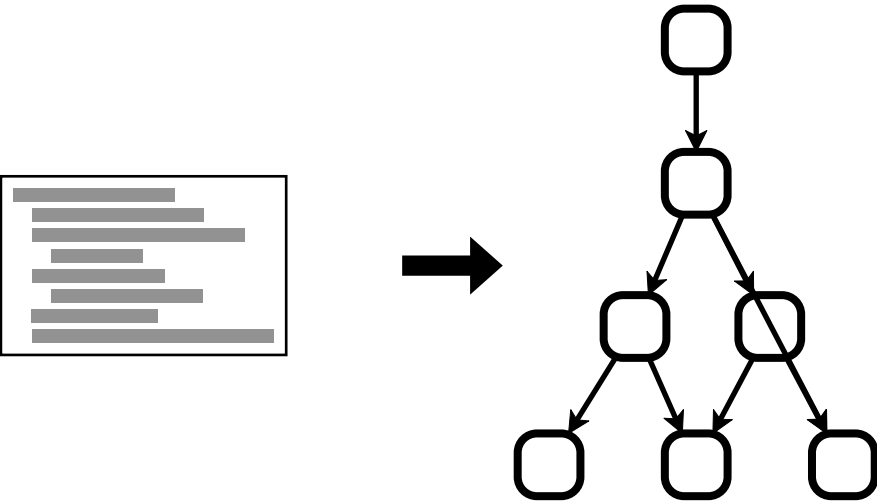
4

Effect-Safe Extraction is NP-Hard



$$\phi = \neg x_0 \wedge (\neg x_1 \vee x_0) \wedge (x_2 \vee x_1 \vee \neg x_2)$$

Effect-Safe Extraction is NP-Hard



$$\phi = \neg X_0 \wedge (\neg X_1 \vee X_0) \wedge (X_2 \vee X_1 \vee \neg X_2)$$

X_0

$\neg X_0$

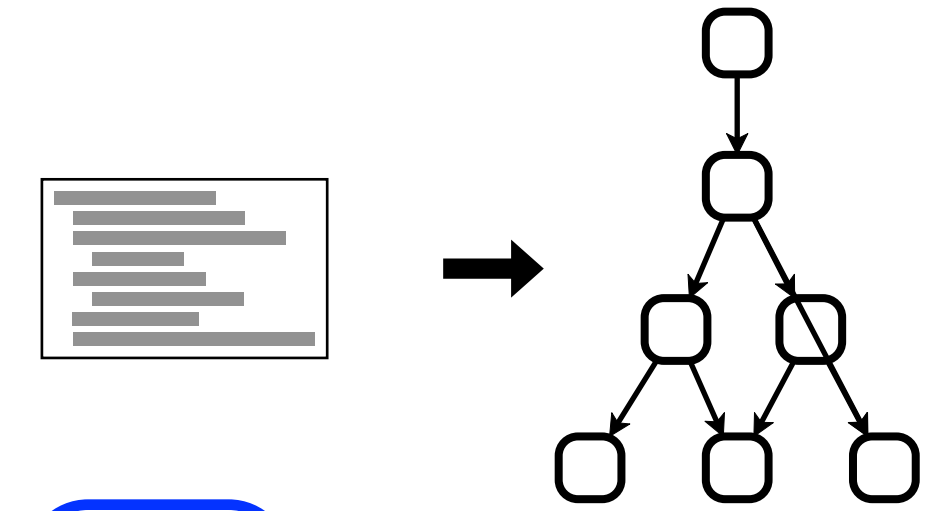
X_1

$\neg X_1$

X_2

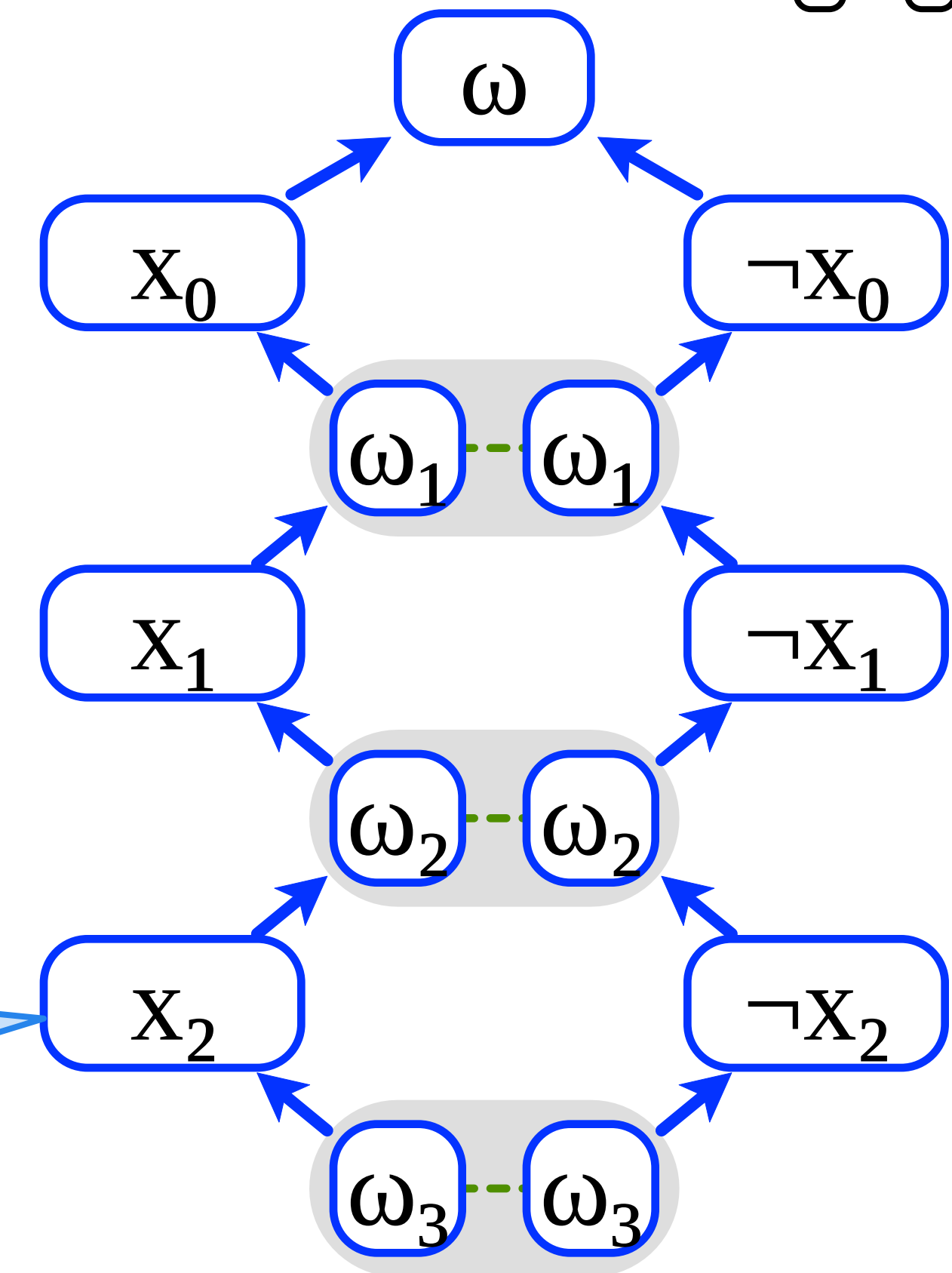
$\neg X_2$

Effect-Safe Extraction is NP-Hard

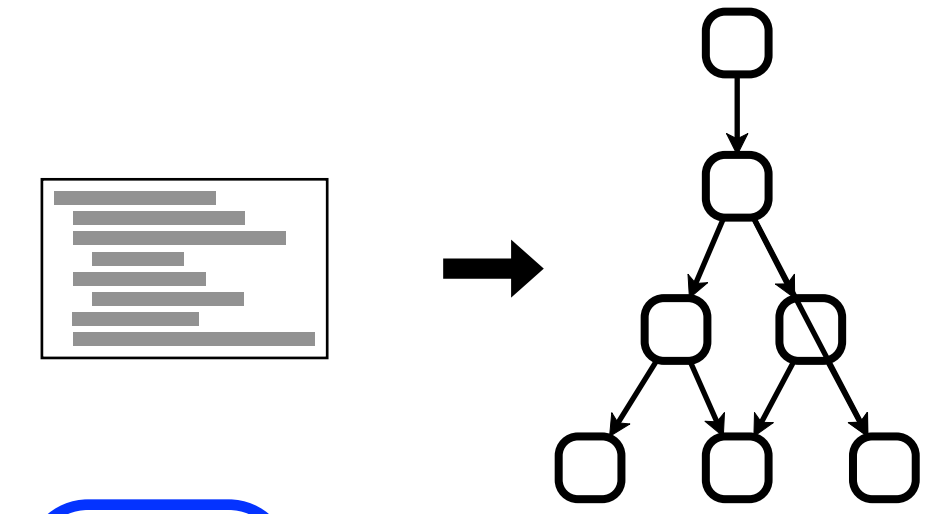


$$\phi = \neg X_0 \wedge (\neg X_1 \vee X_0) \wedge (X_2 \vee X_1 \vee \neg X_2)$$

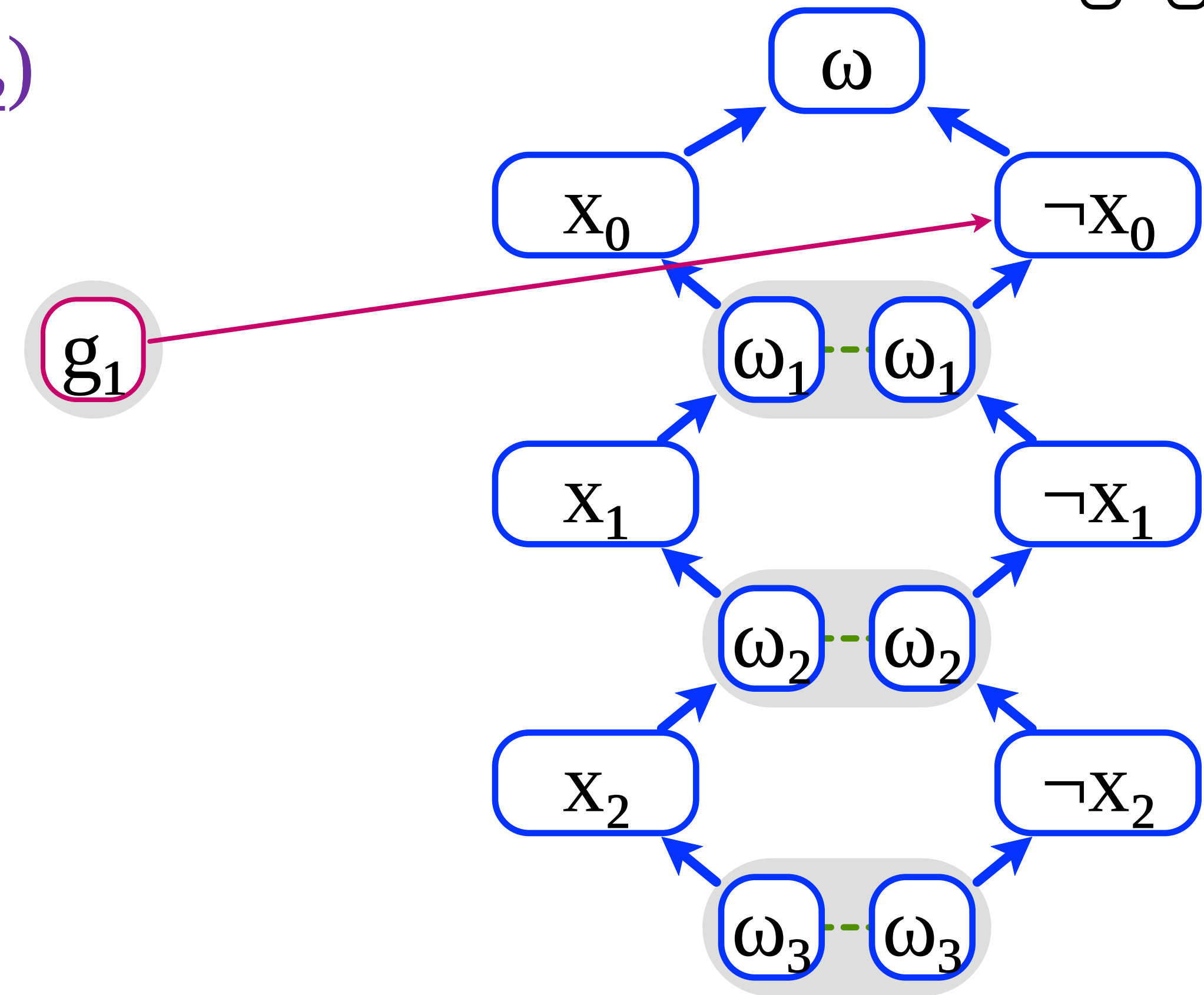
state walks encode assignments



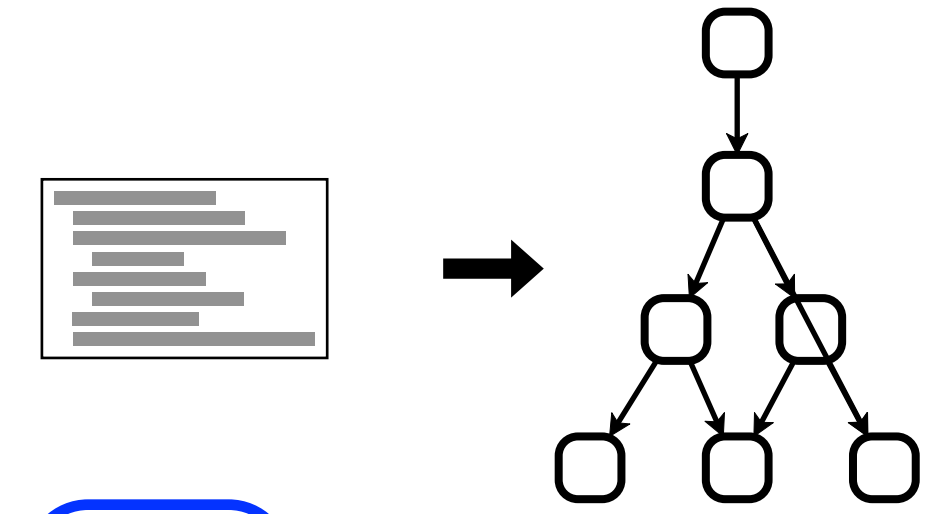
Effect-Safe Extraction is NP-Hard



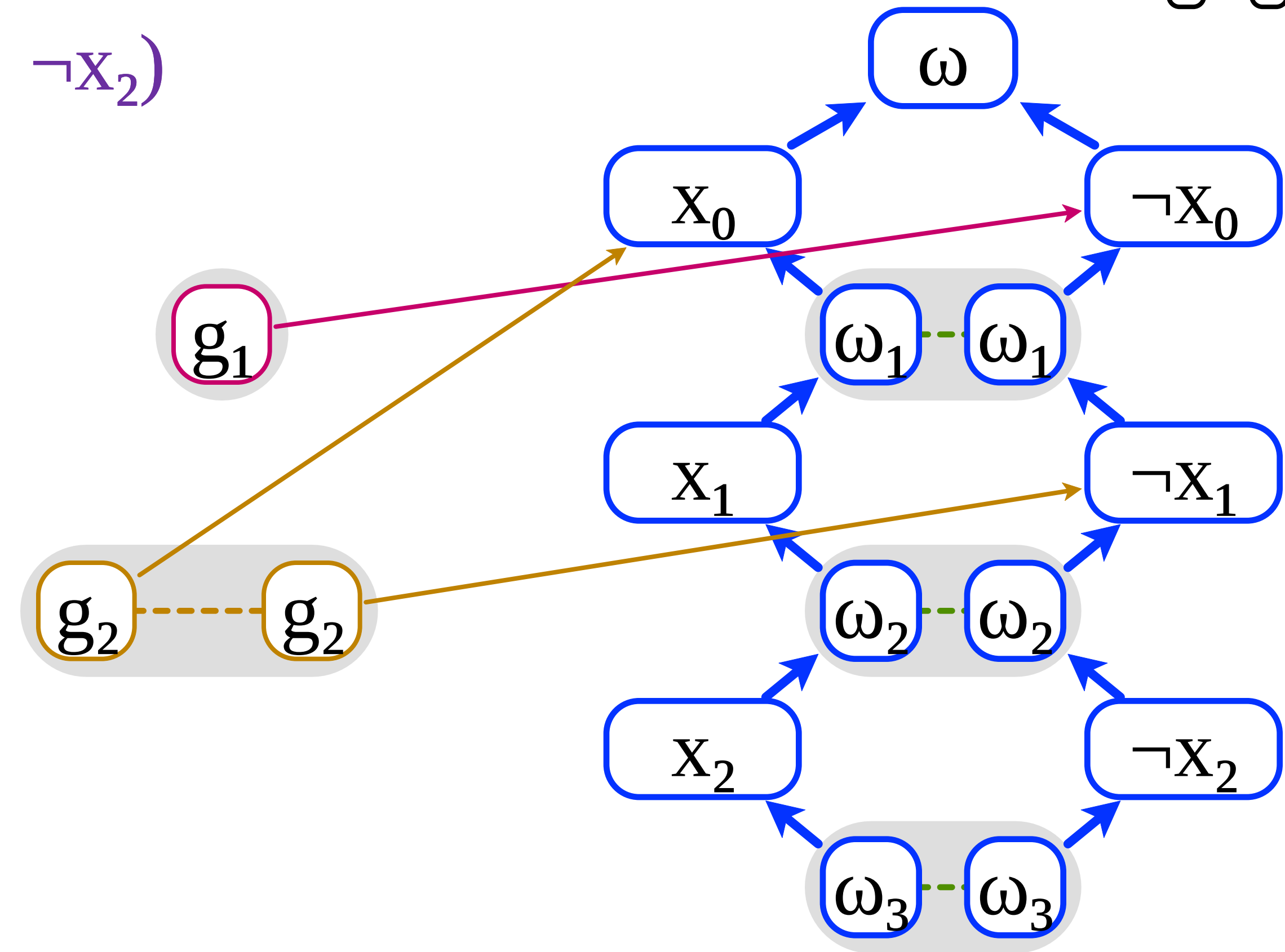
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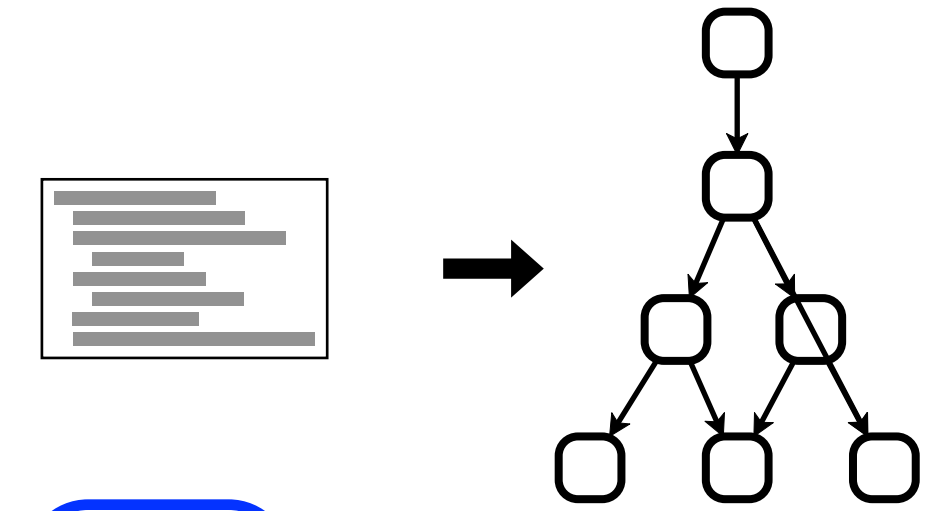
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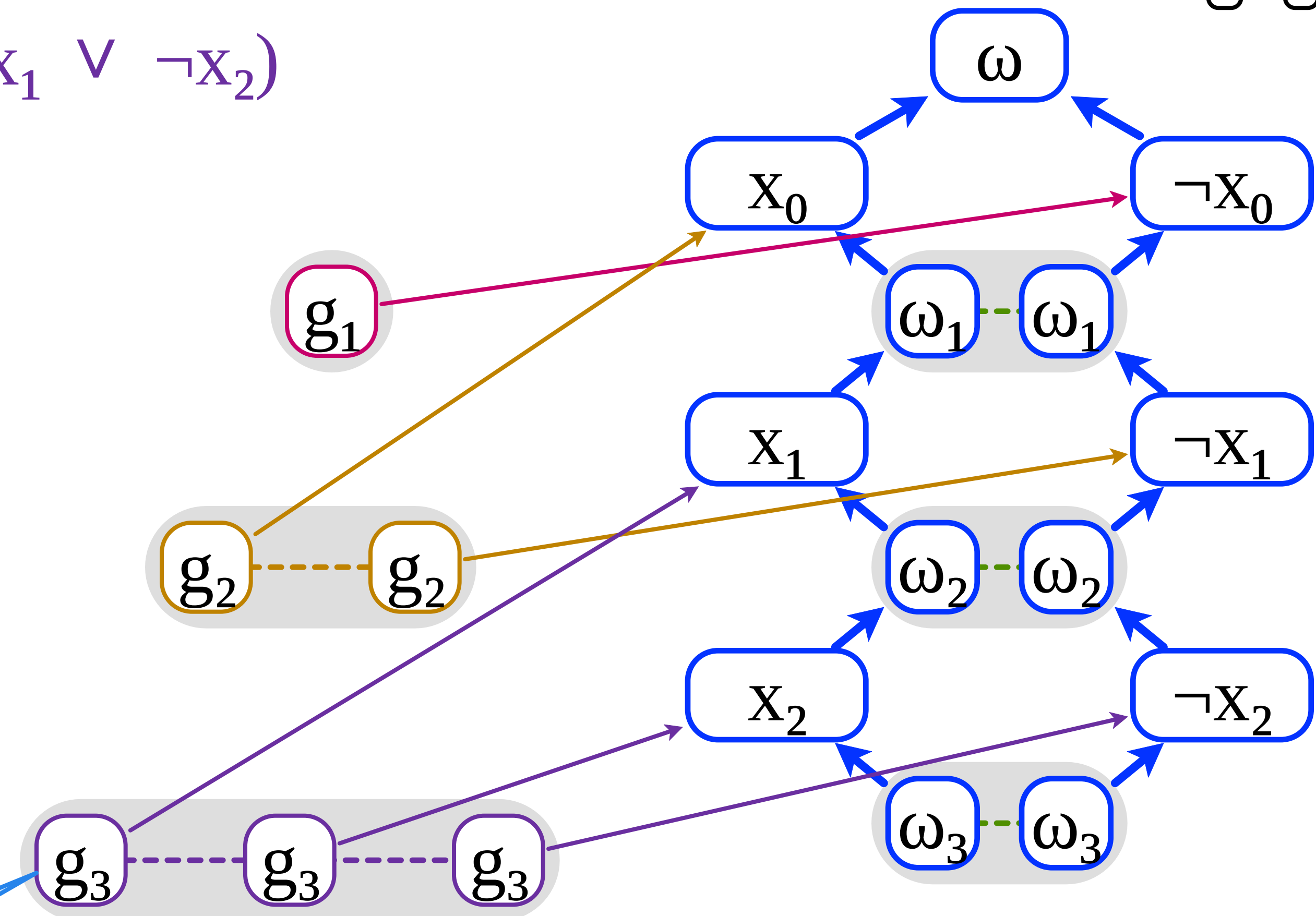
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Effect-Safe Extraction is NP-Hard

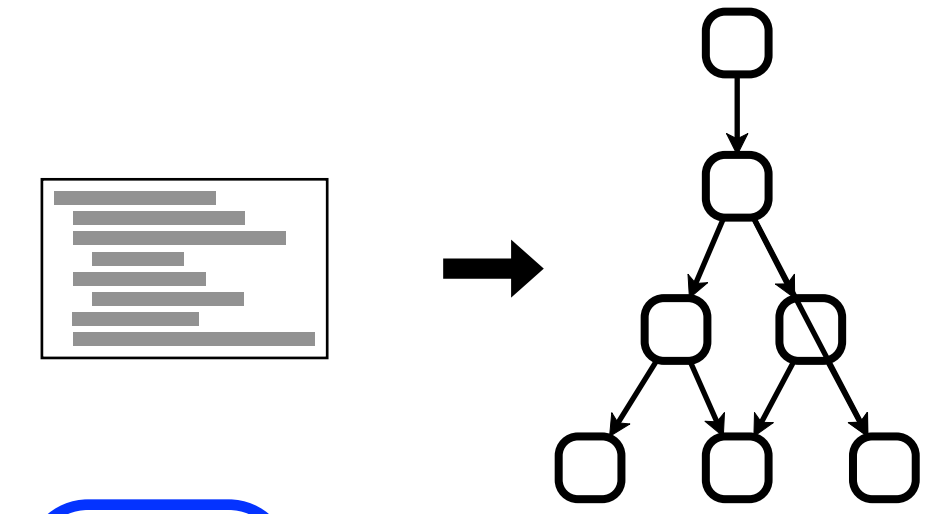


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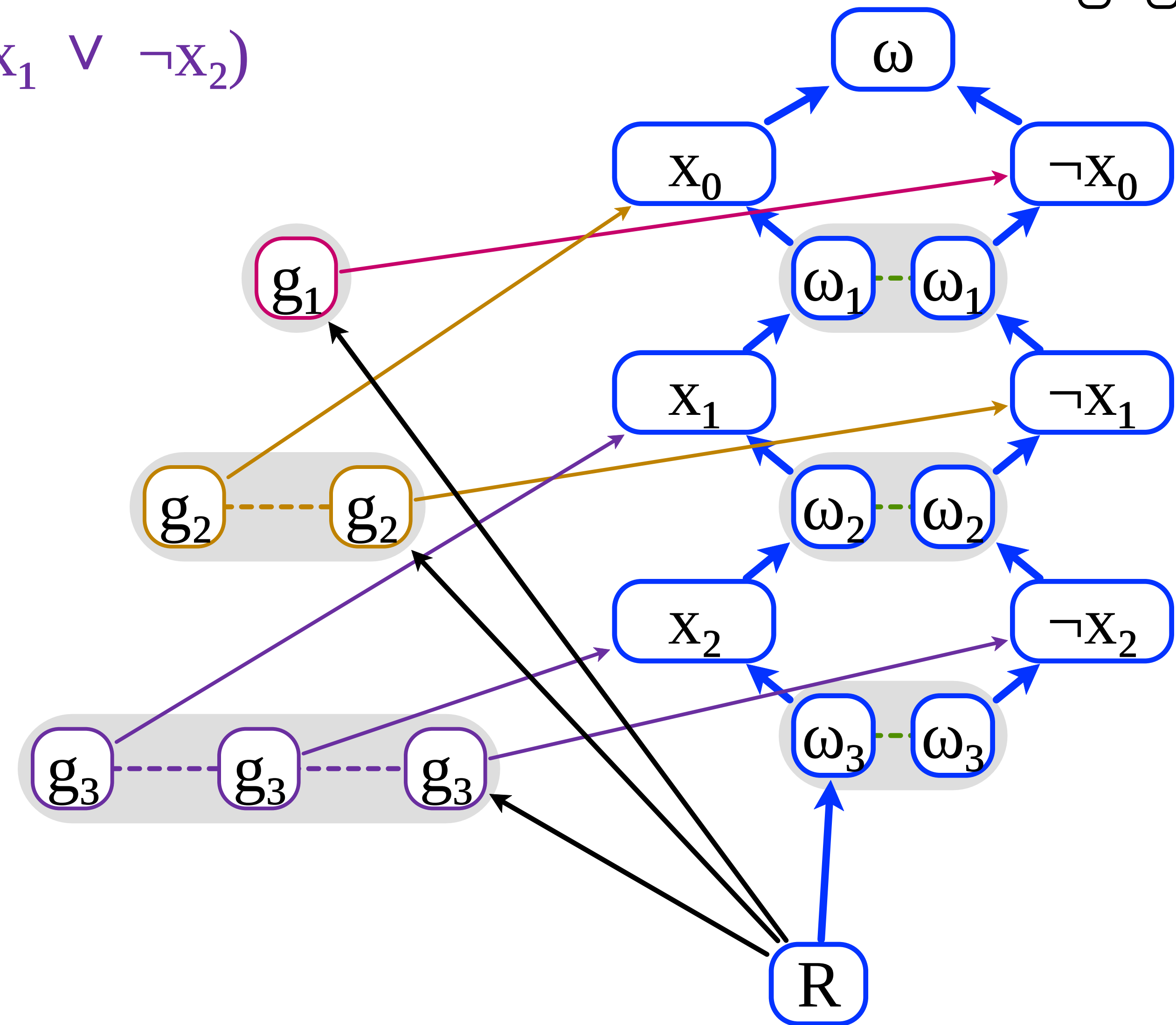


e-classes encode clauses

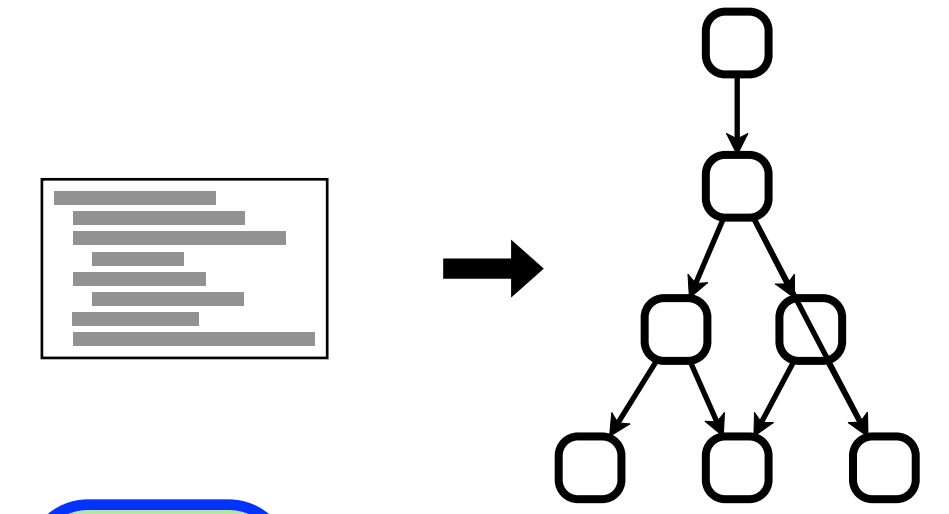
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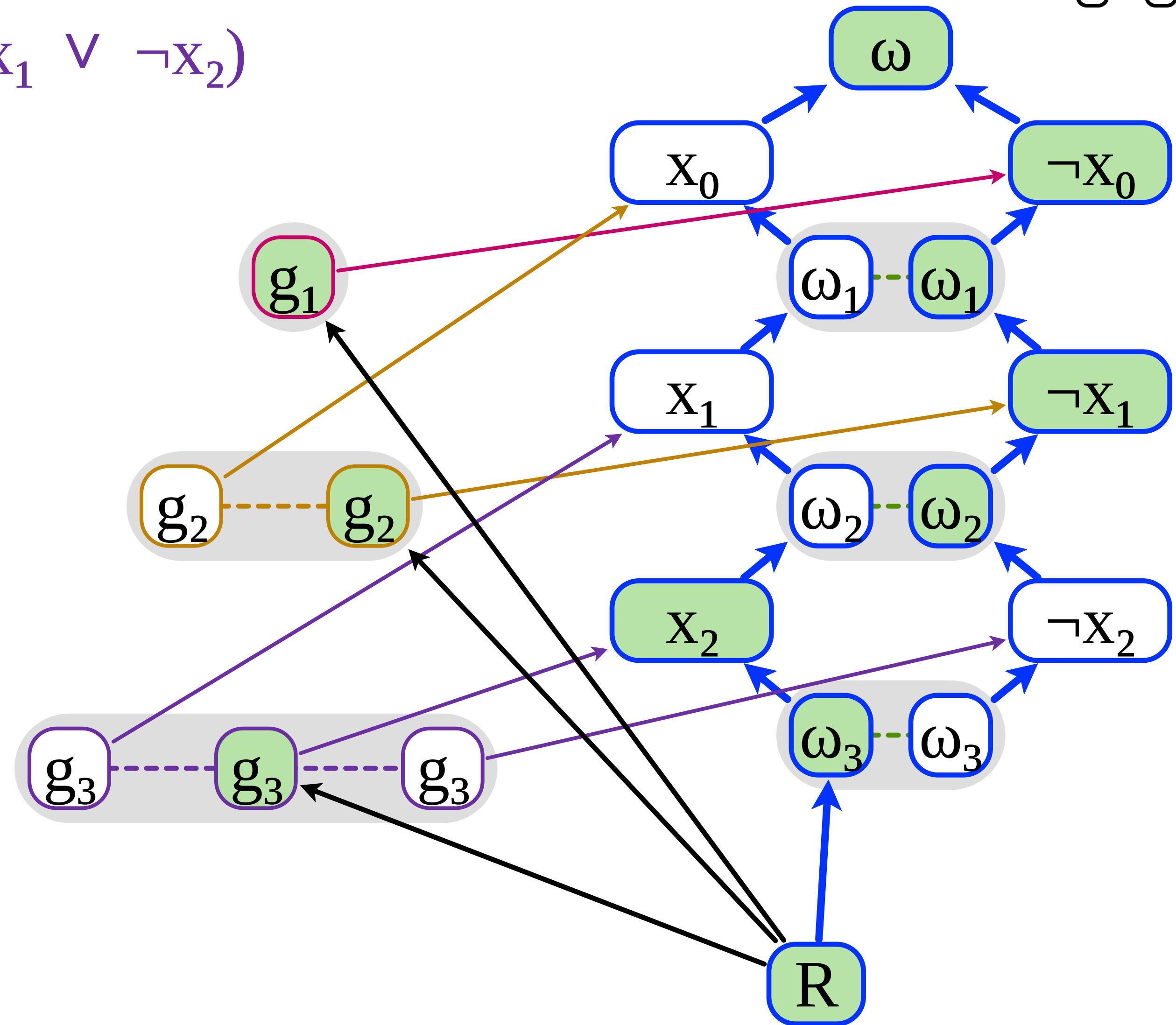
Effect-Safe Extraction is NP-Hard



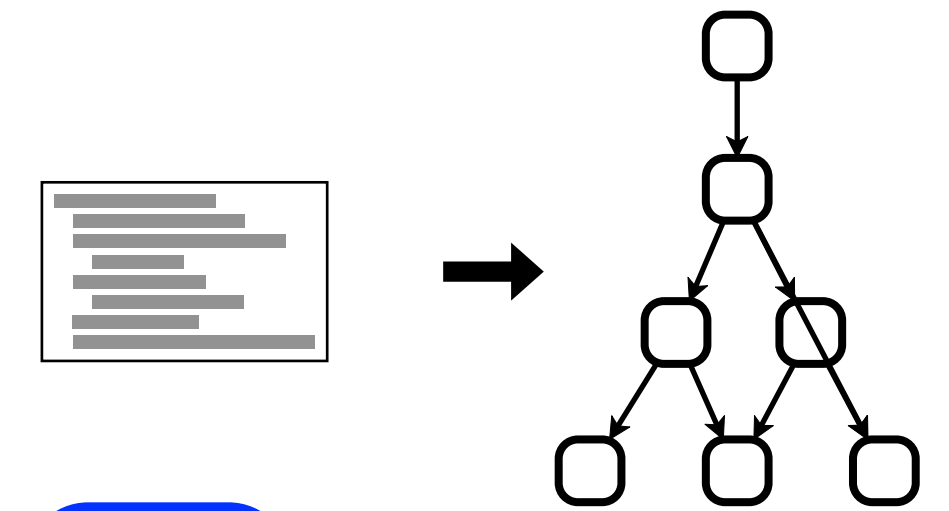
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Satisfying assignment:

$$X_0 = \perp \quad X_1 = \perp \quad X_2 = \top$$



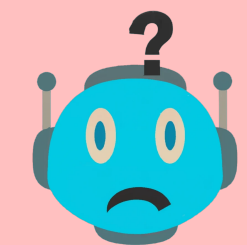
Effect-Safe Extraction is NP-Hard



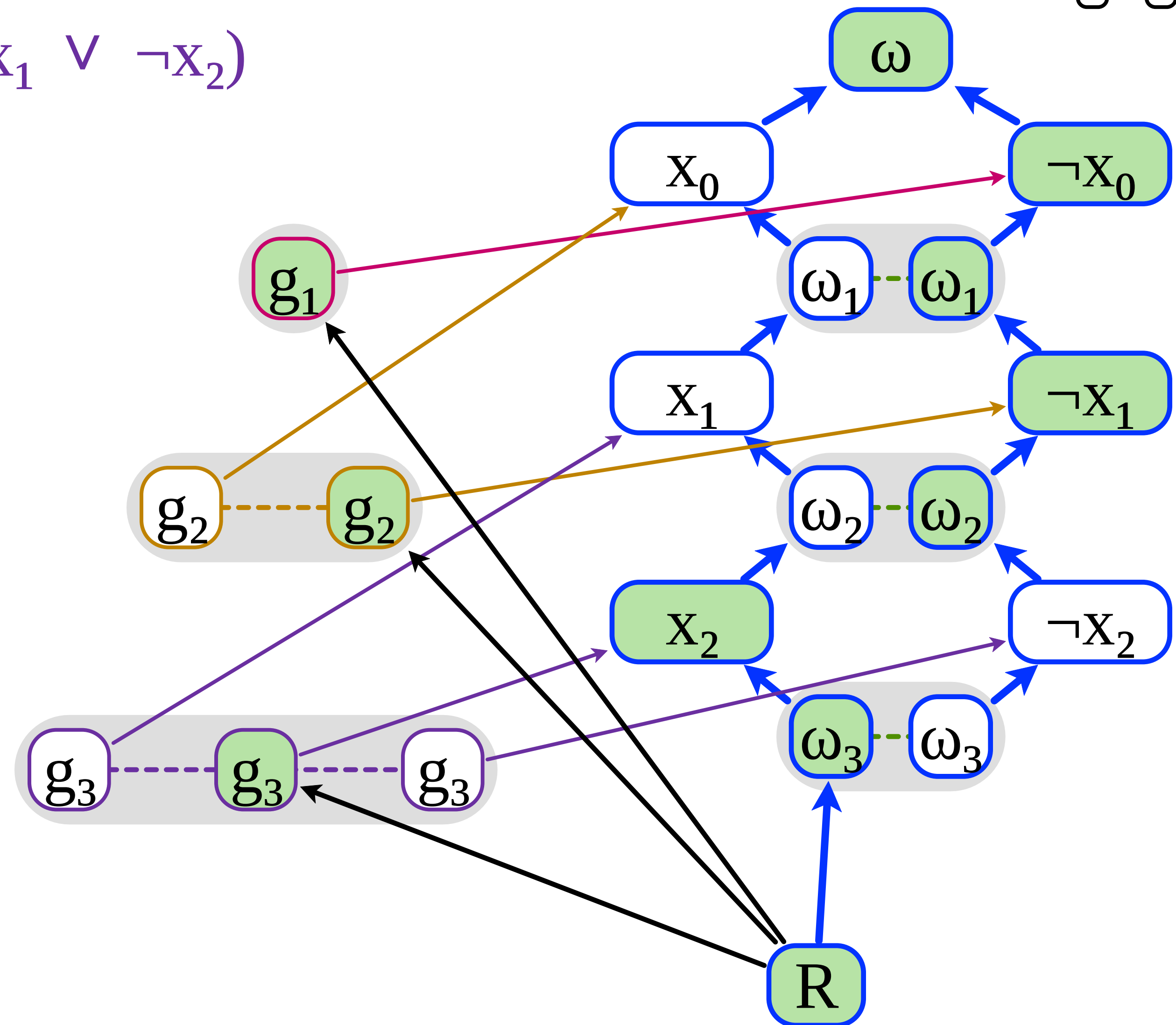
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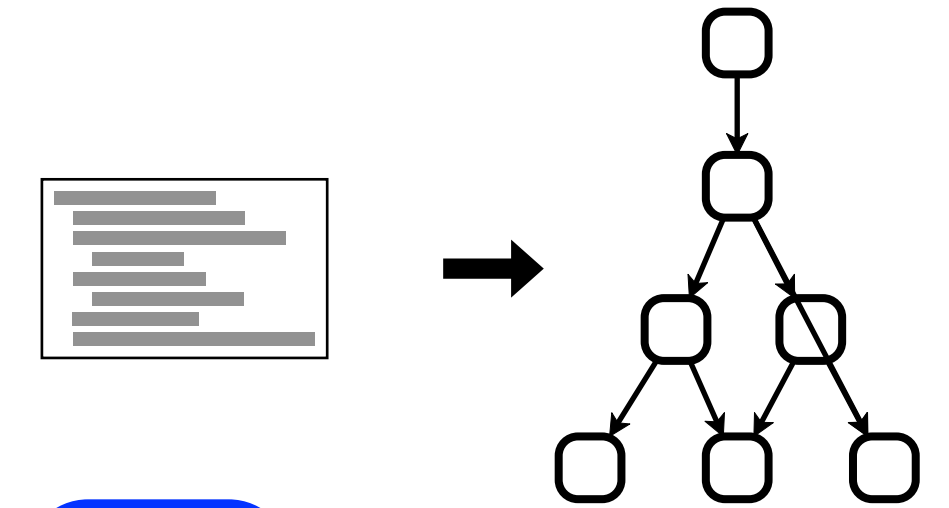
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Problem is NP-hard!
Give up??



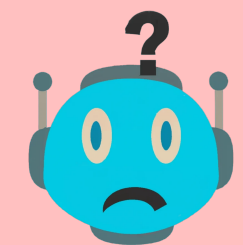
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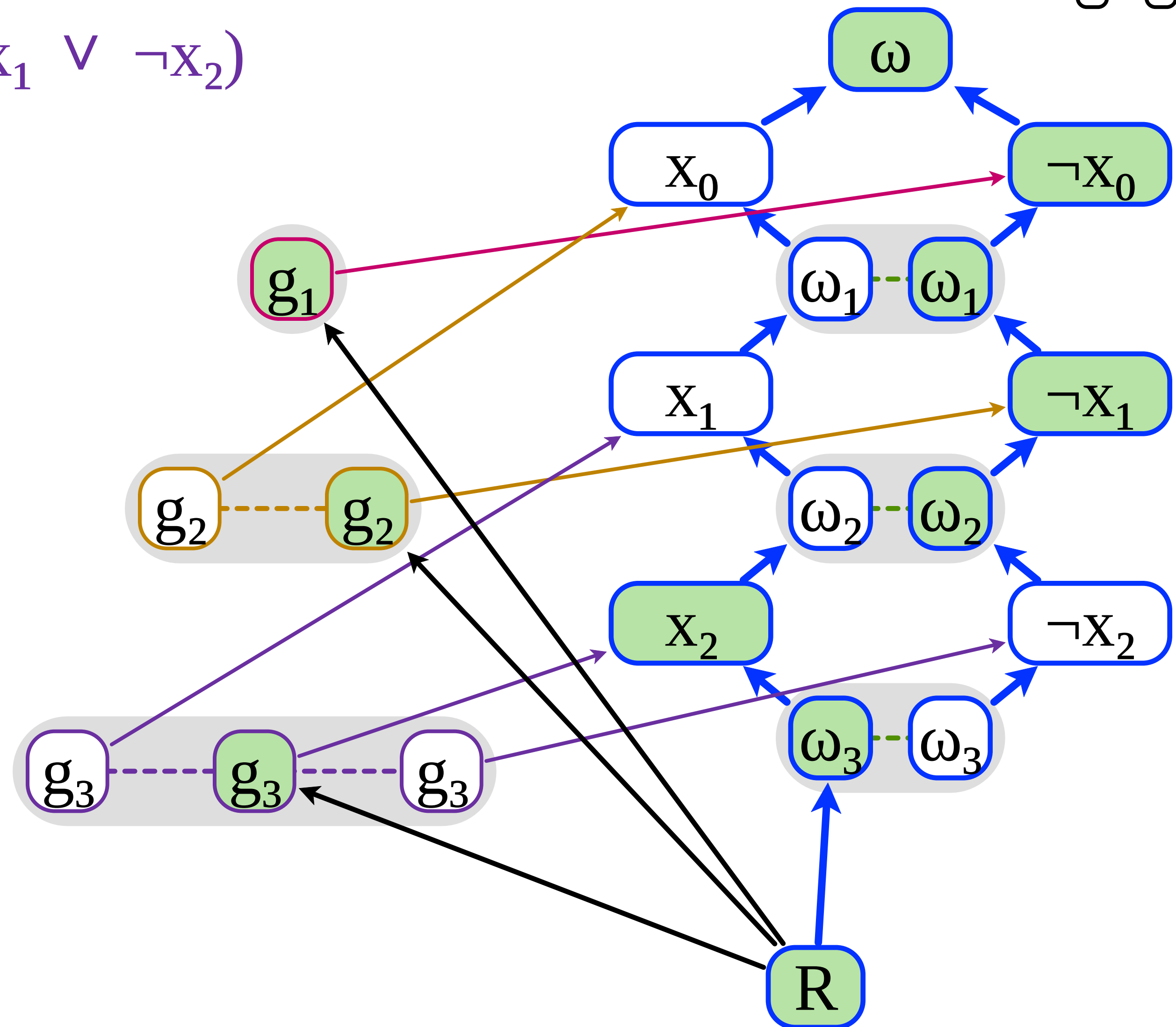
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Problem is NP-hard!
Give up??

**No! Real-world e-graphs
are easier, measured
with a *parameter***



The Statewalk DP Algorithm

```
fun foo(a, x, ω):
```

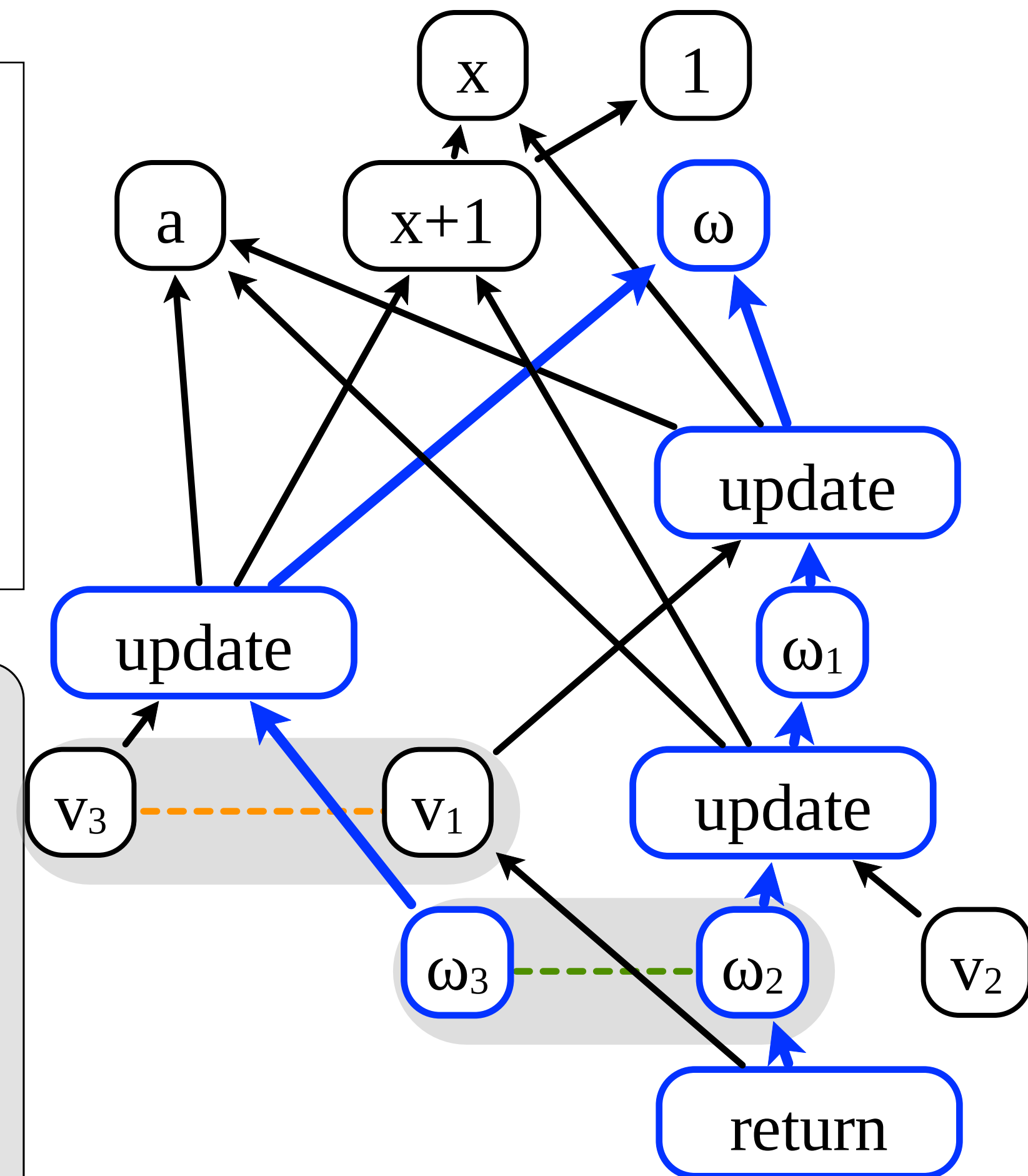
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```

```
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```

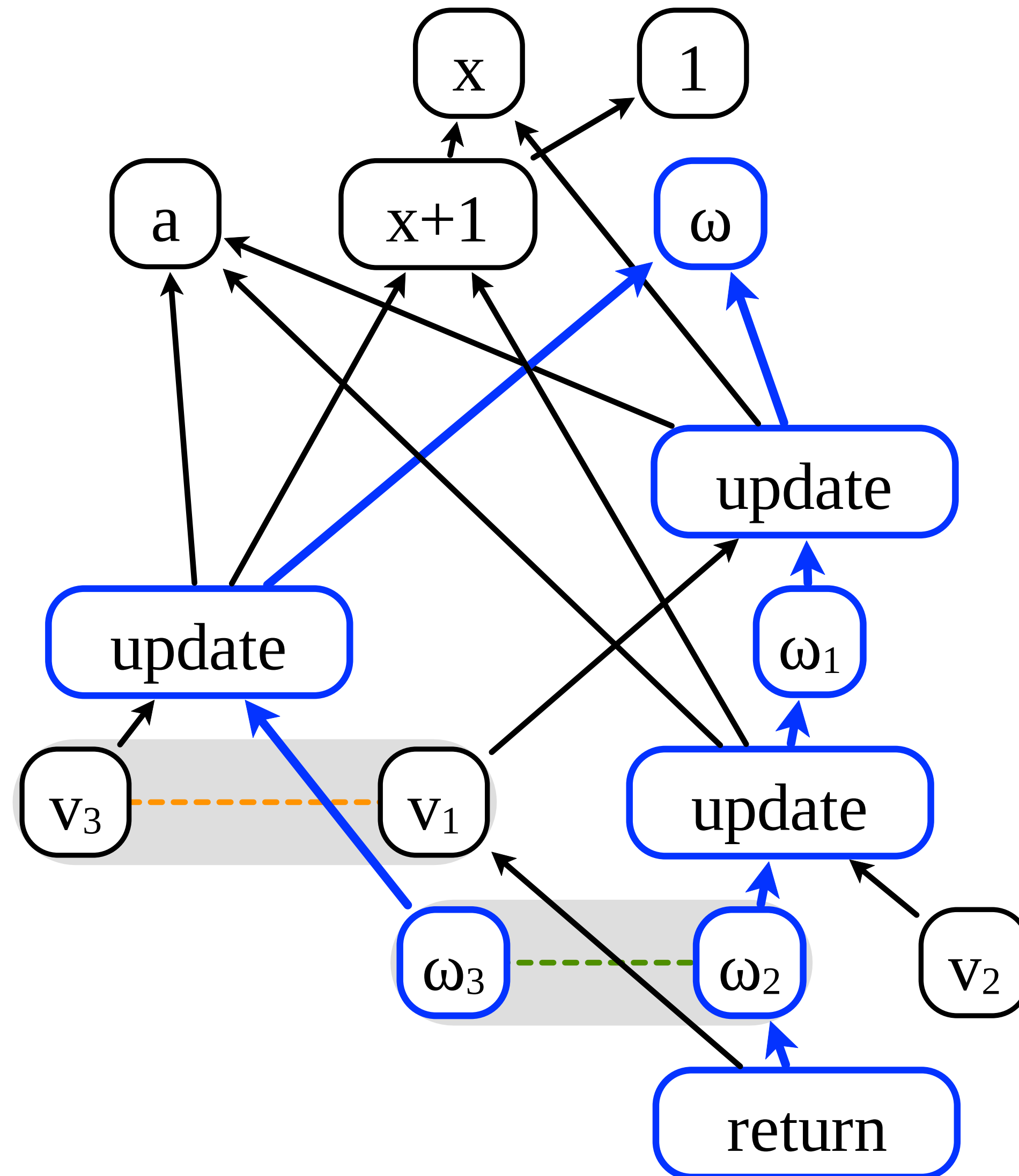
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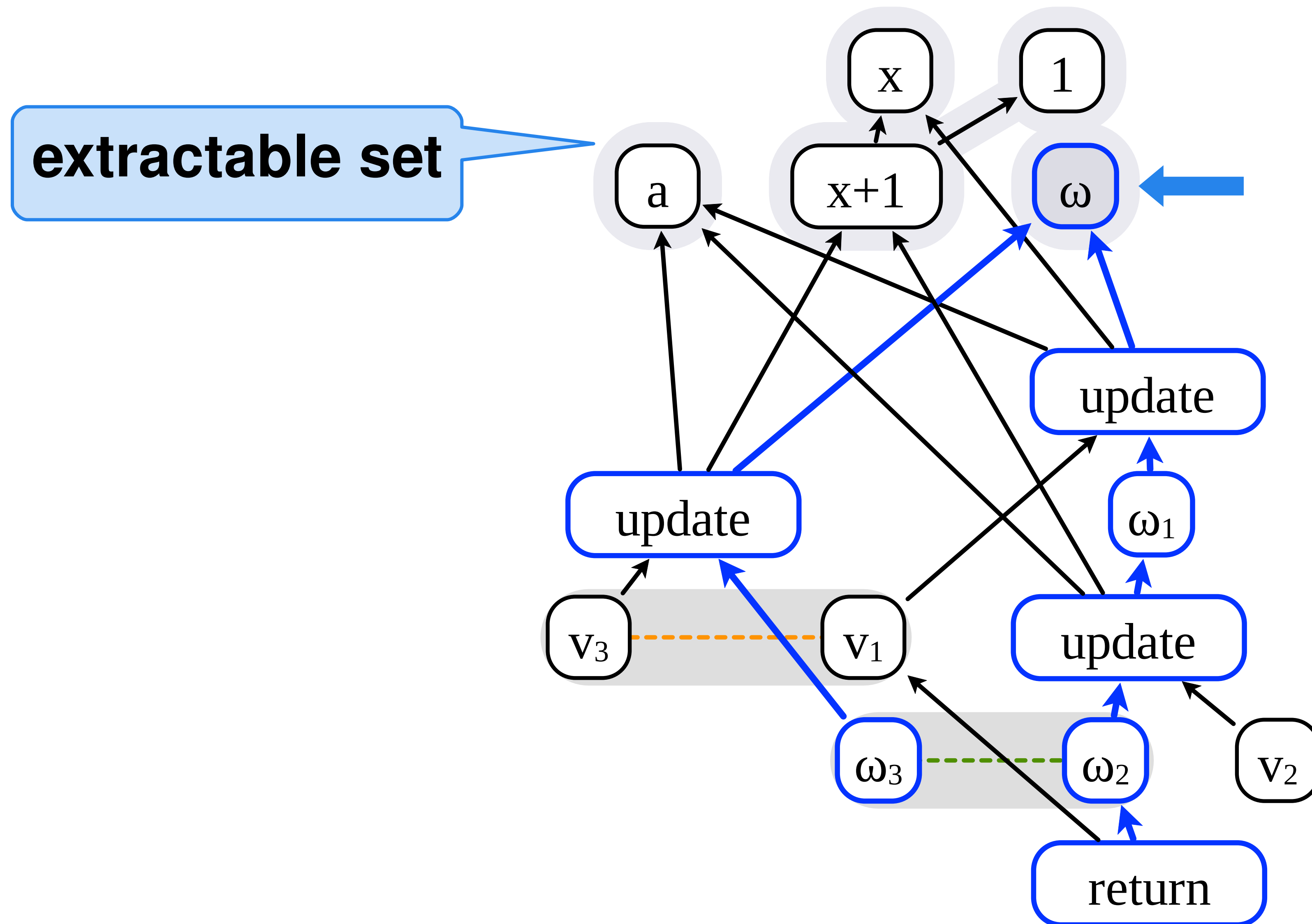
$v_1 \equiv v_3$ $\omega_2 \equiv \omega_3$



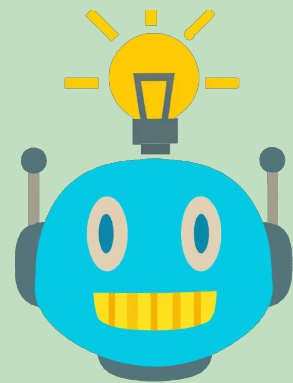
The Statewalk DP Algorithm



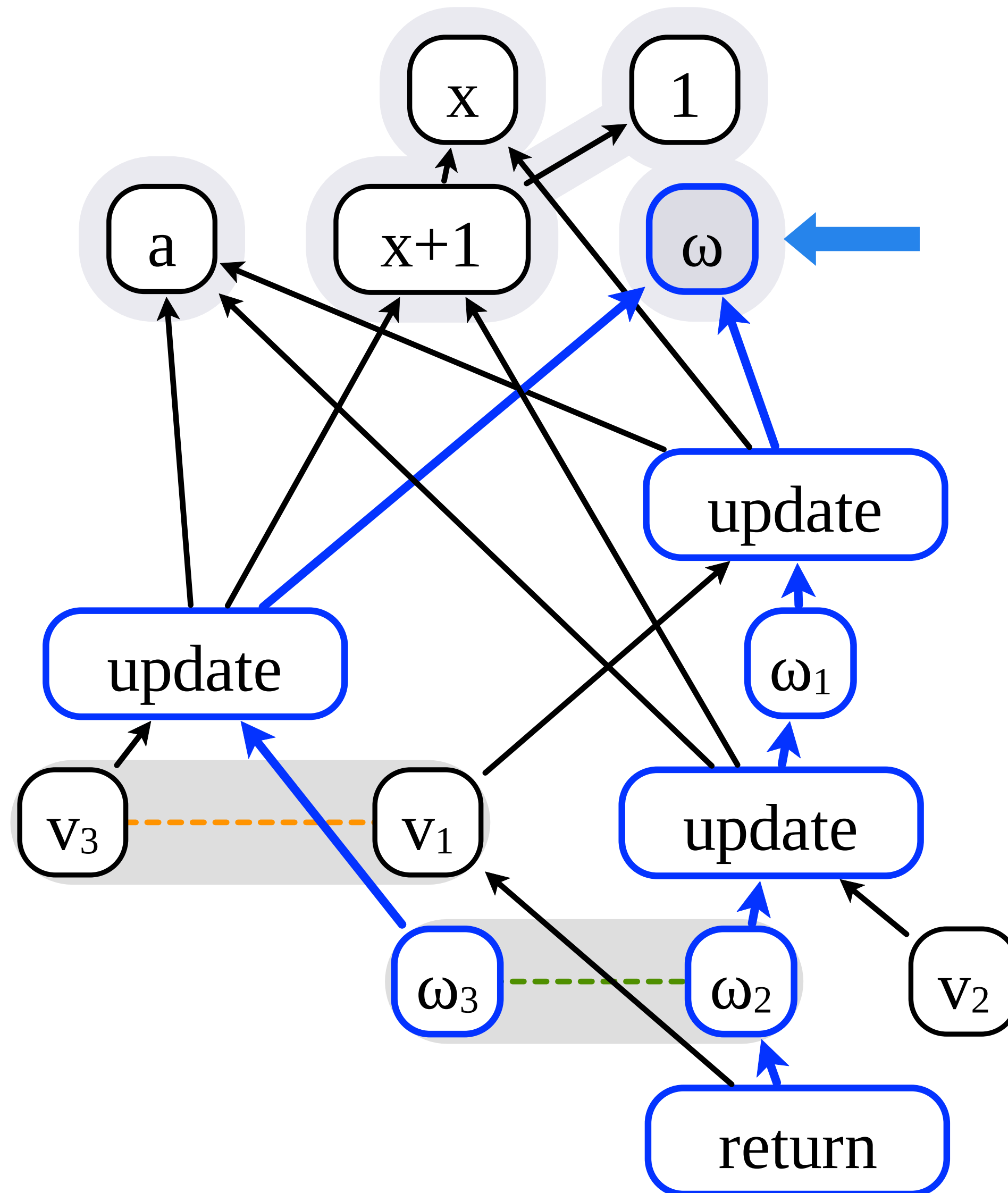
The Statewalk DP Algorithm



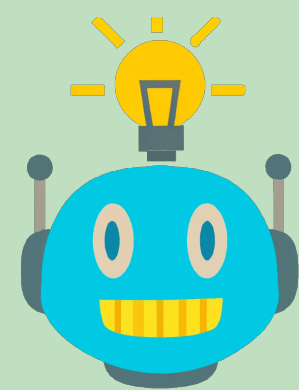
The Statewalk DP Algorithm



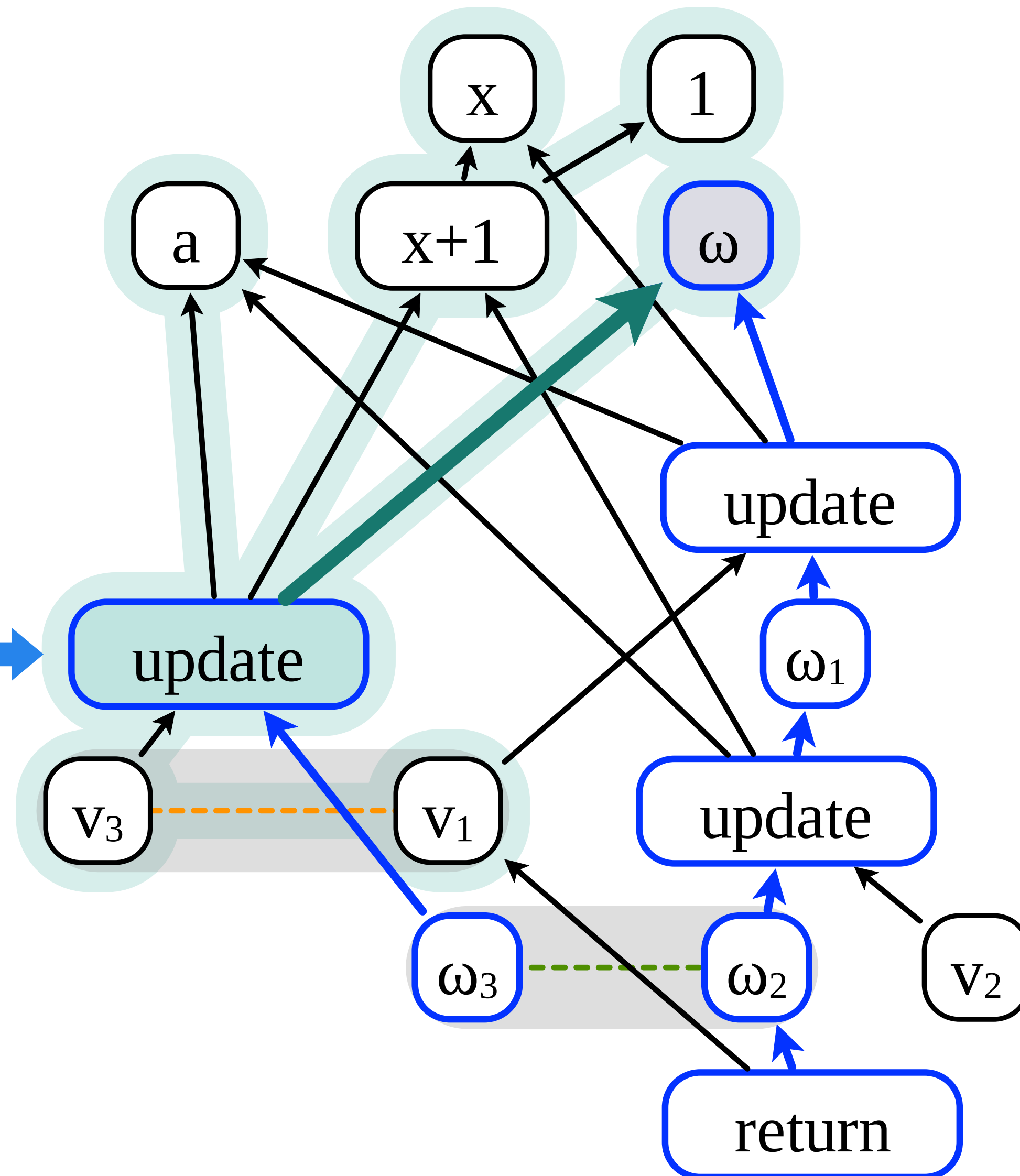
Key idea:
extend the
statewalk,
keep one per
extractable set



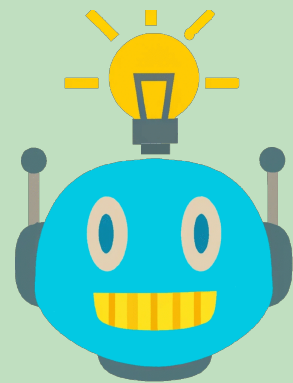
The Statewalk DP Algorithm



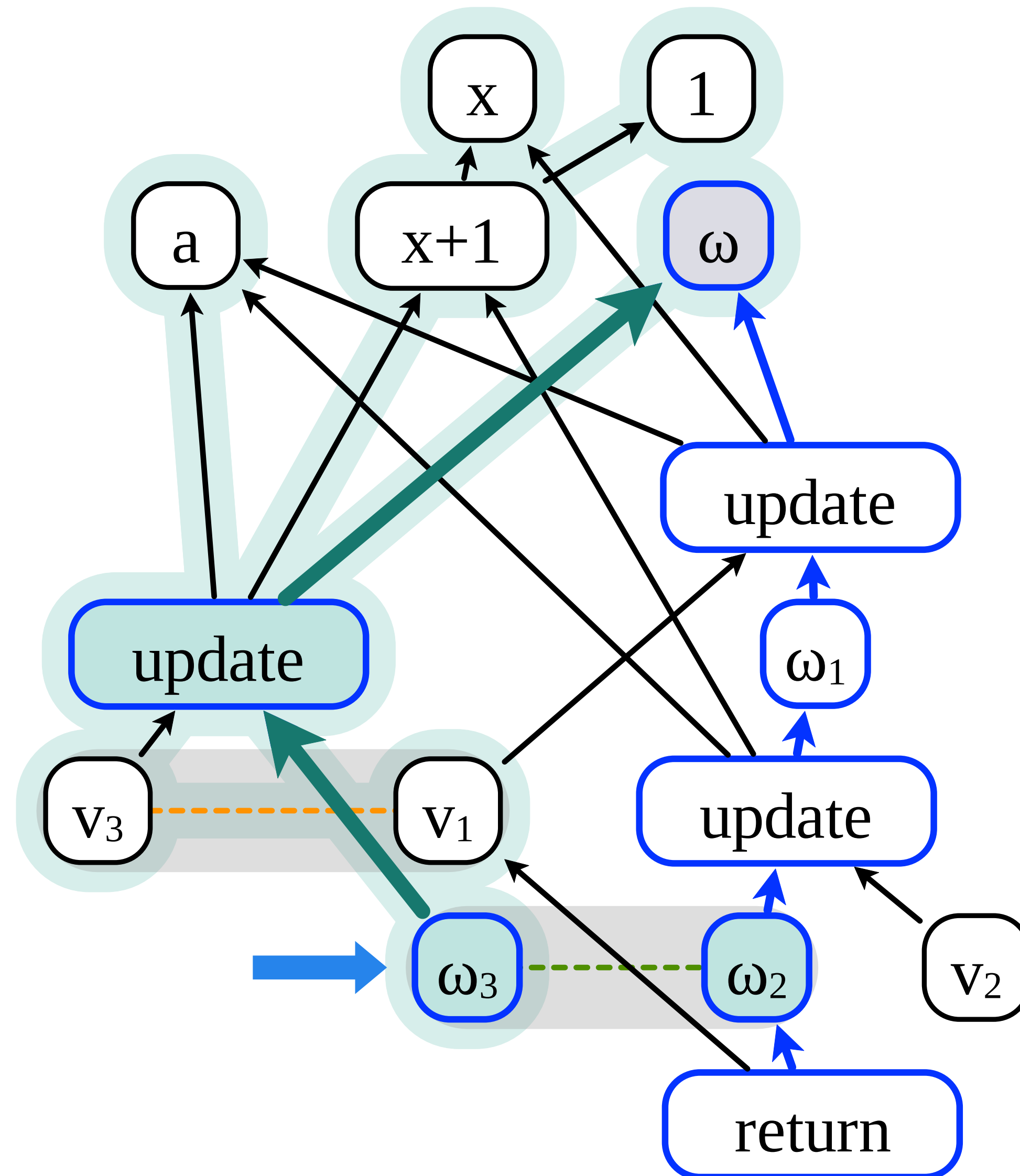
Key idea:
extend the
statewalk,
keep one per
extractable set



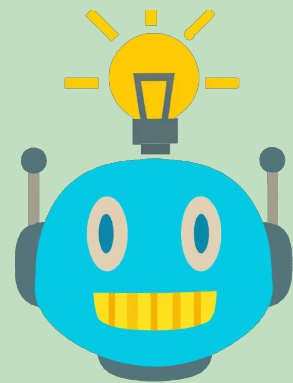
The Statewalk DP Algorithm



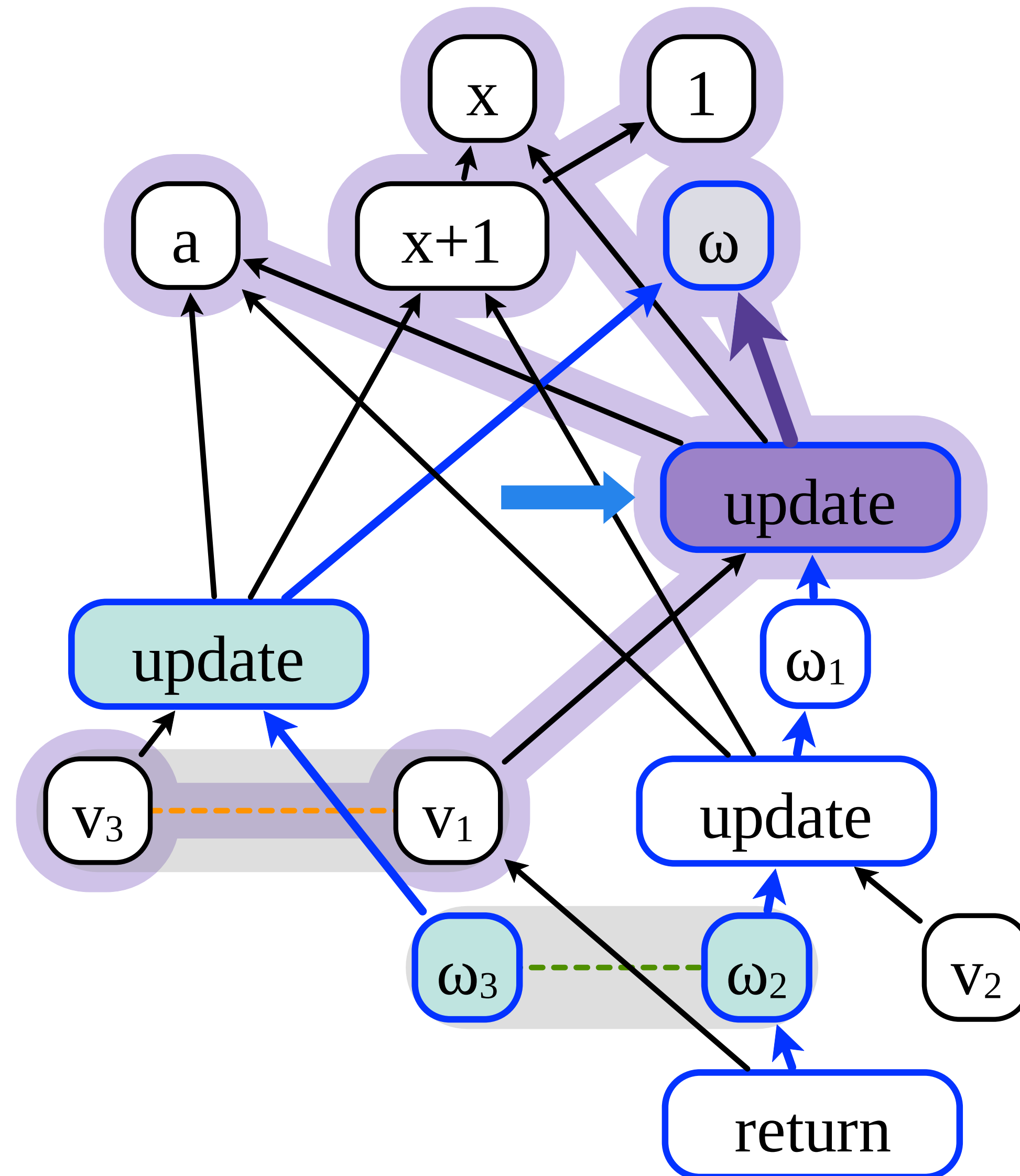
Key idea:
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statewalk,
keep one per
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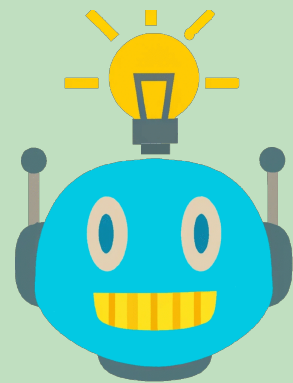
The Statewalk DP Algorithm



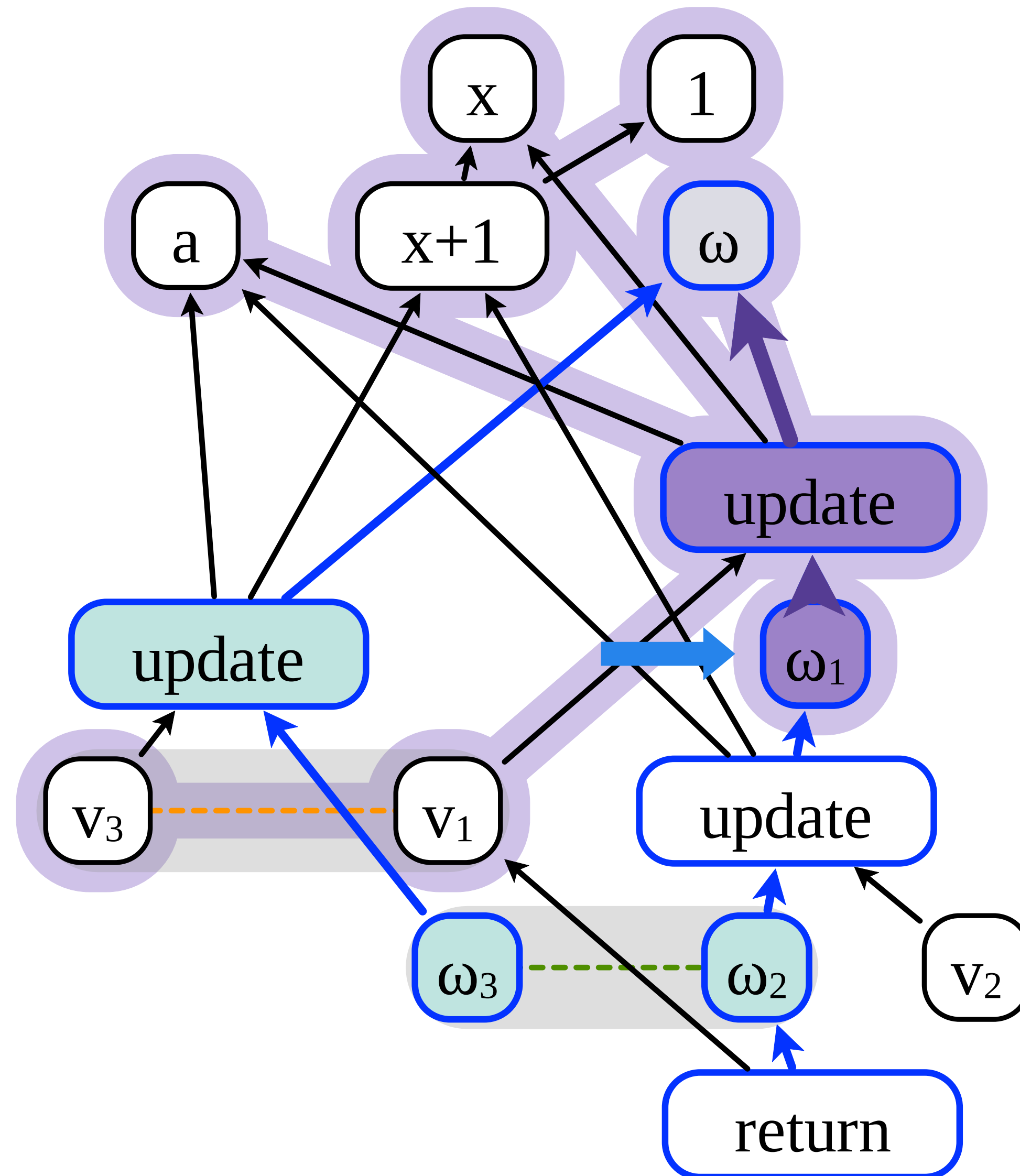
Key idea:
extend the
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keep one per
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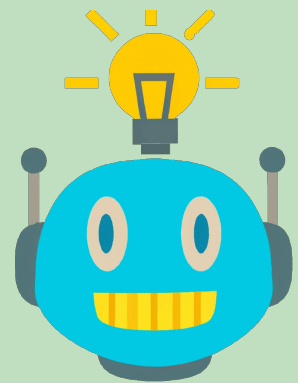
The Statewalk DP Algorithm



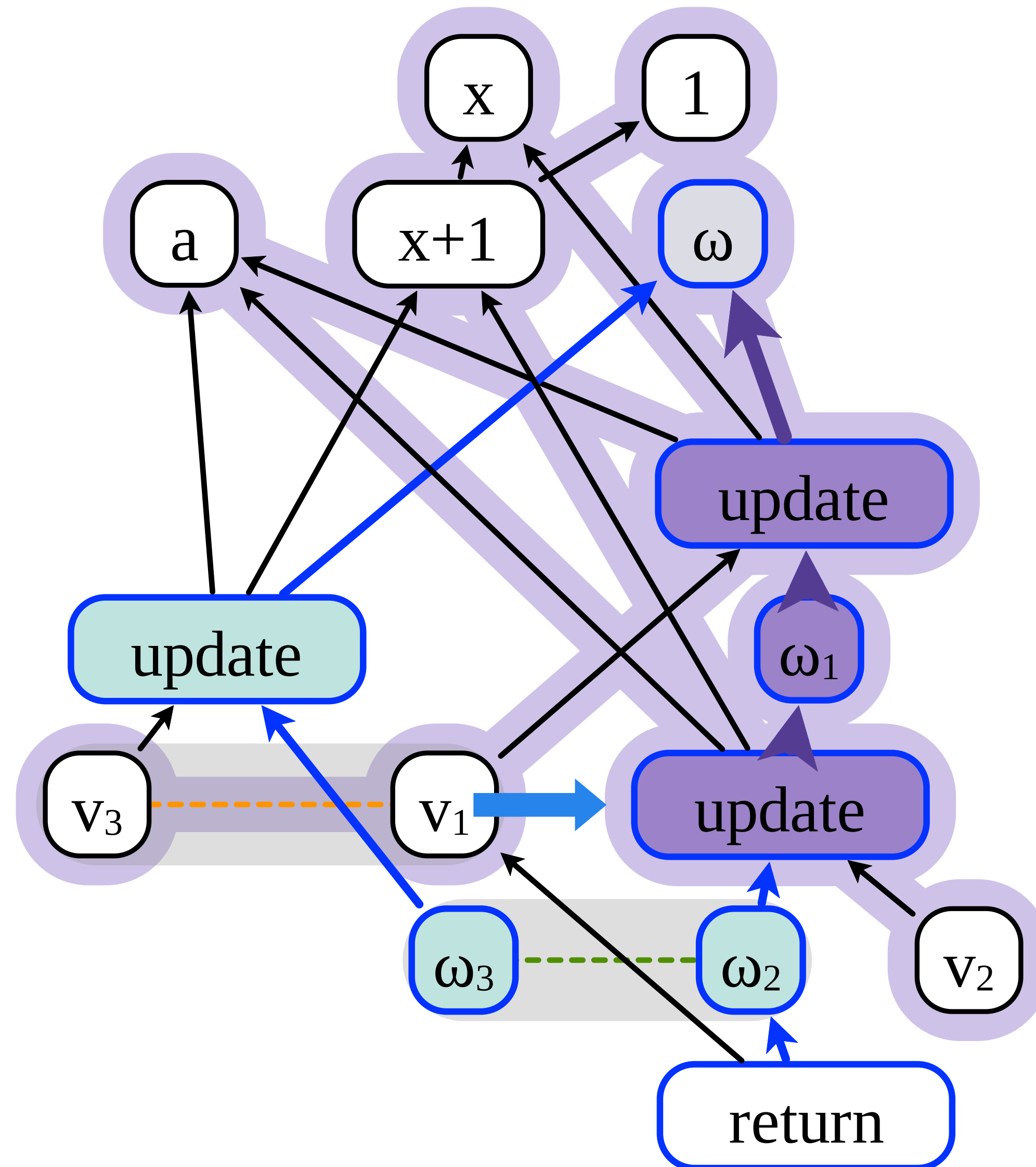
Key idea:
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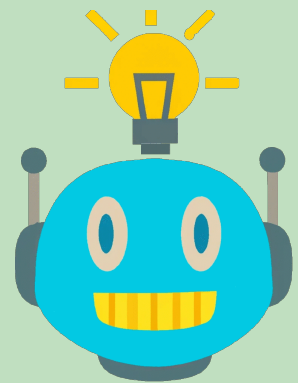
The Statewalk DP Algorithm



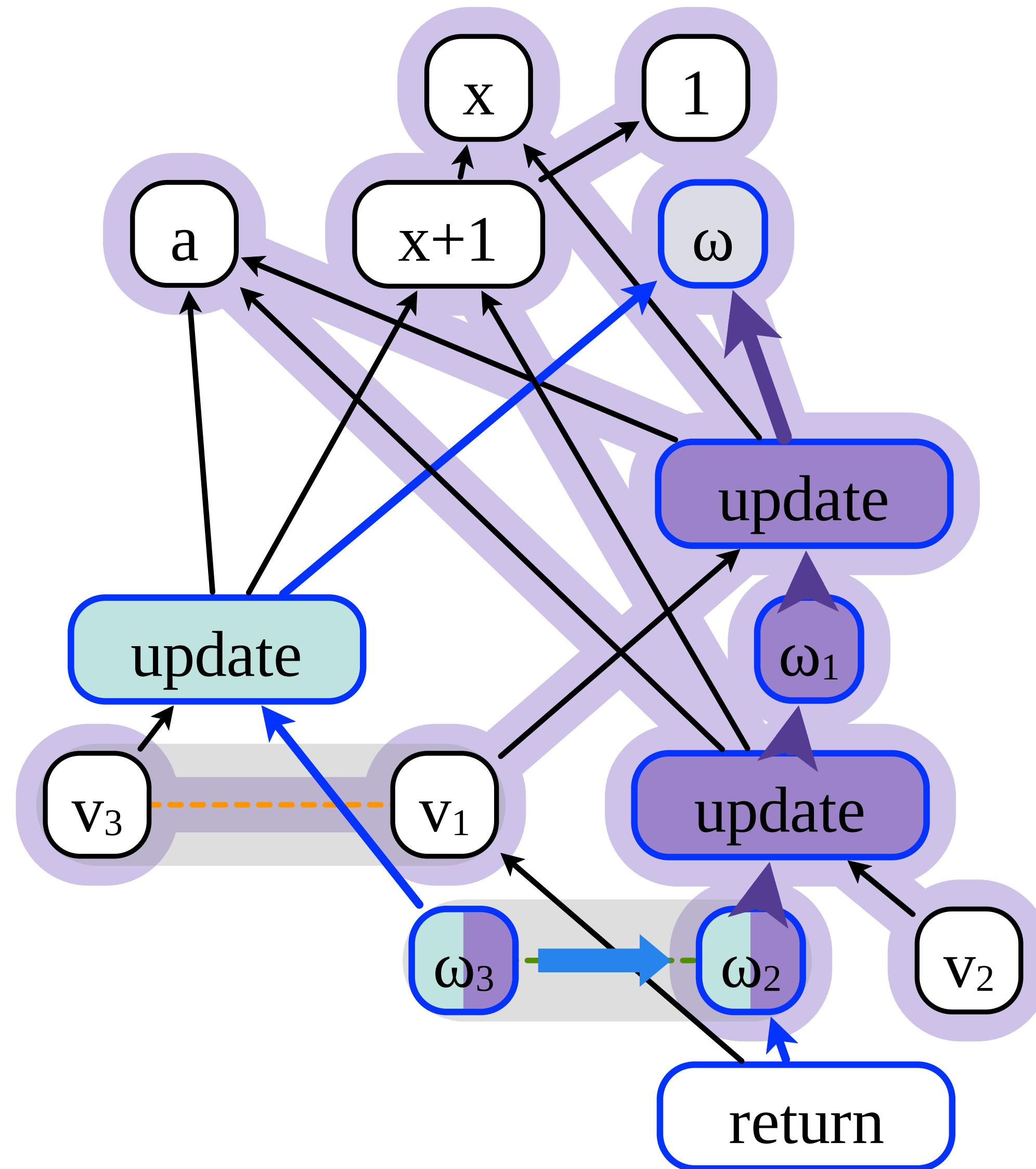
Key idea:
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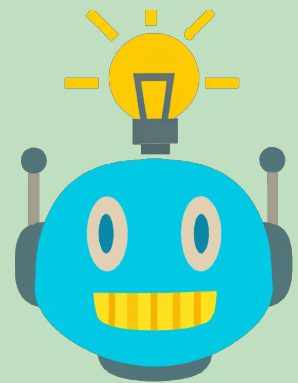
The Statewalk DP Algorithm



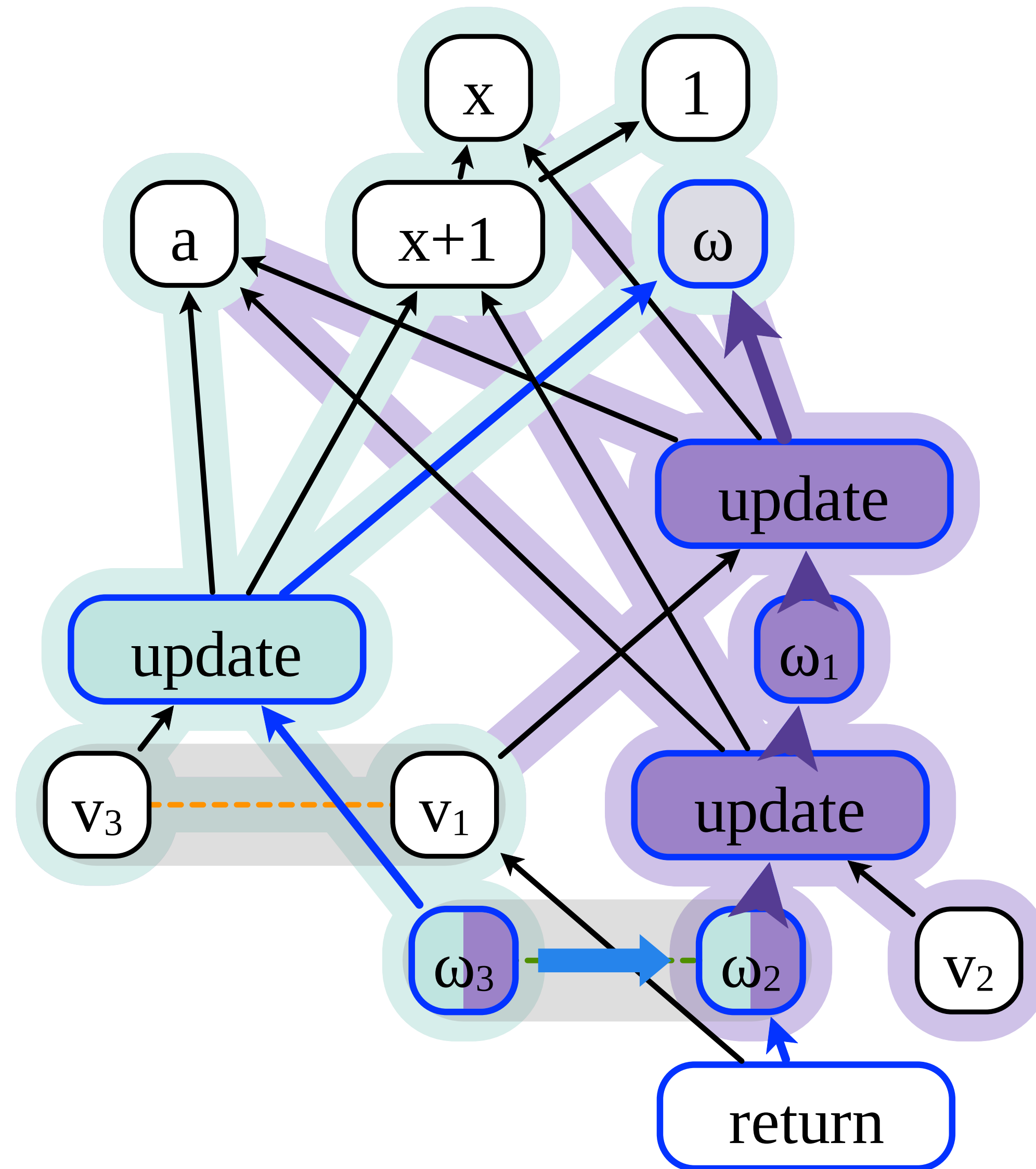
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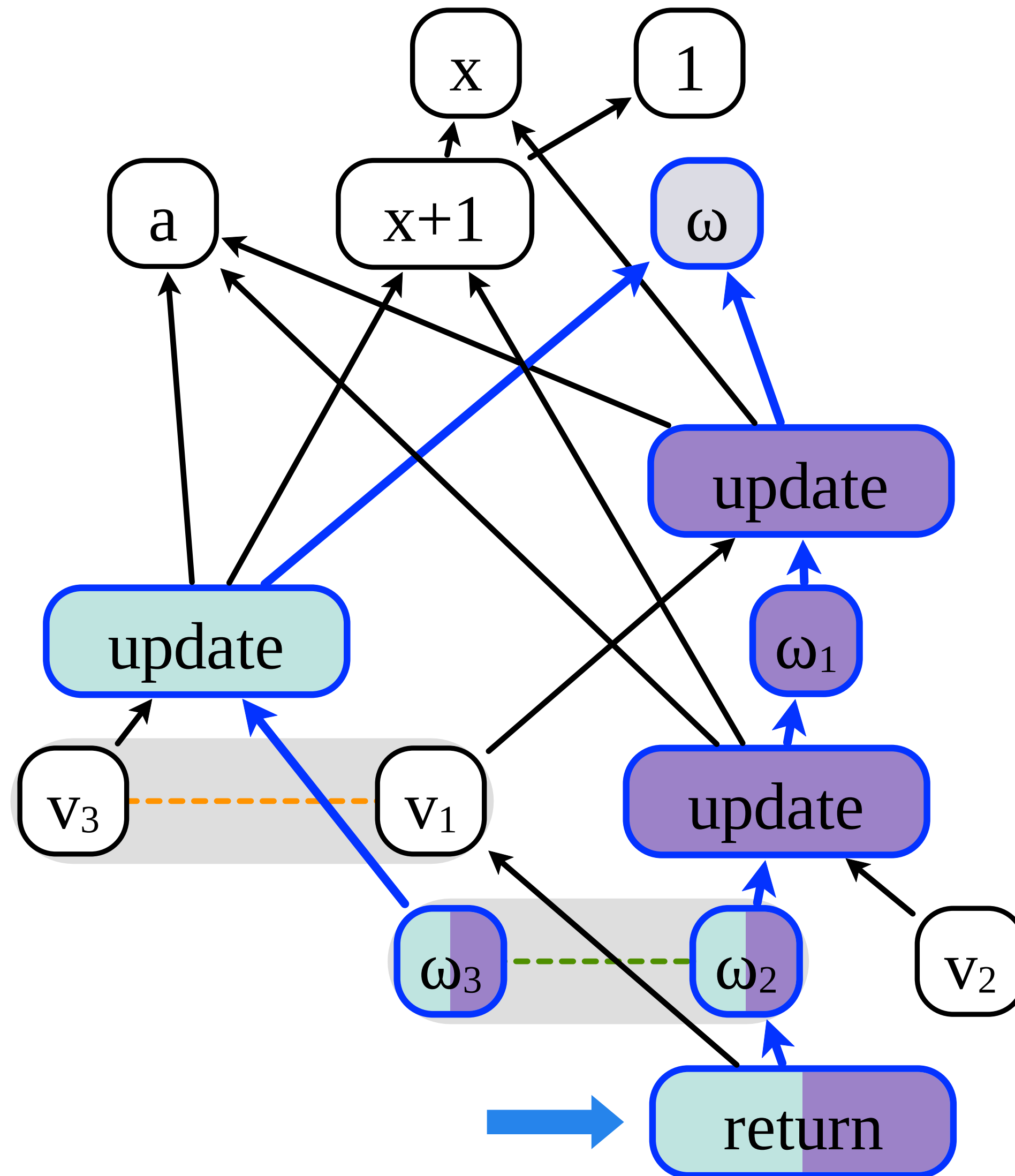
The Statewalk DP Algorithm



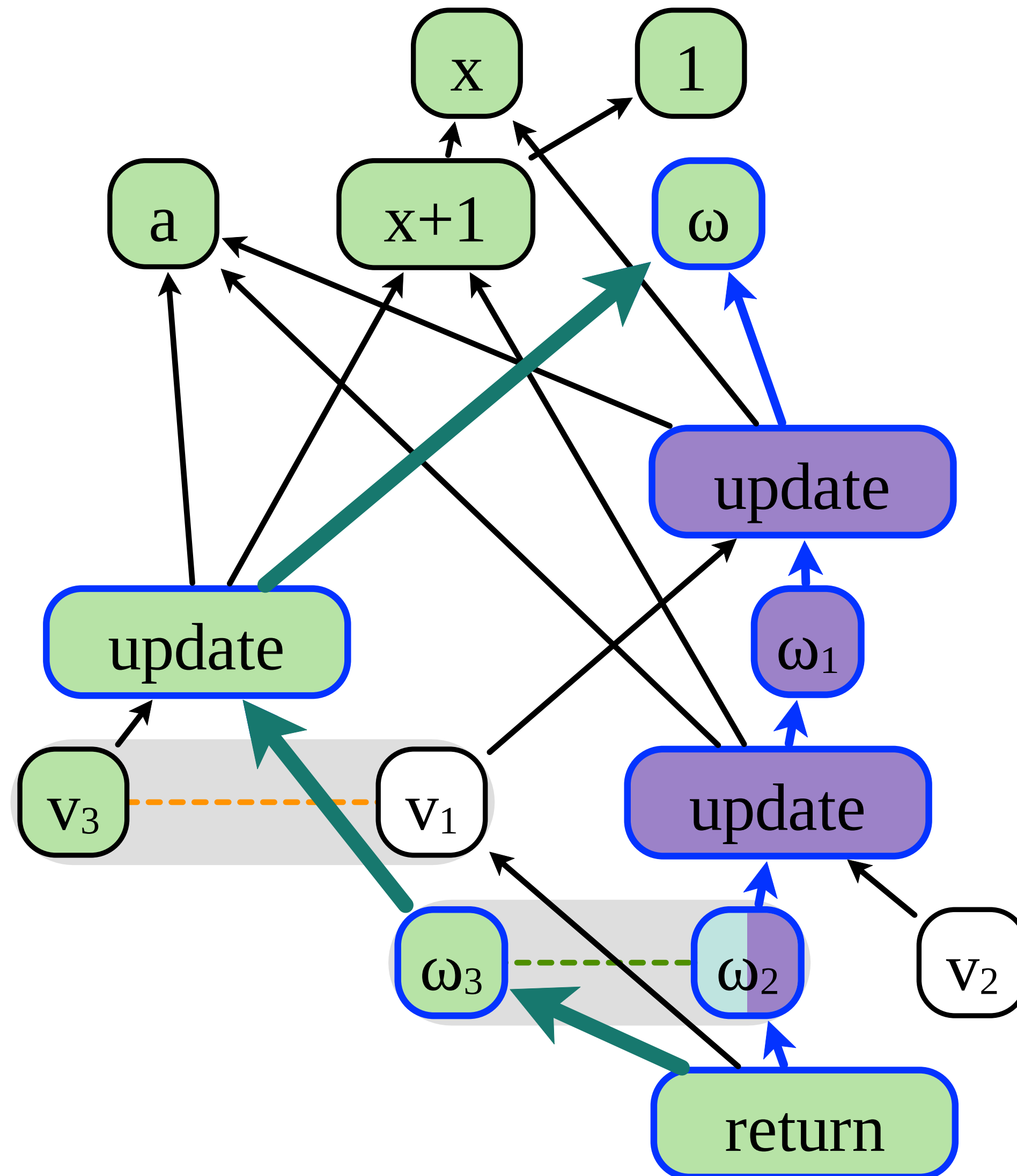
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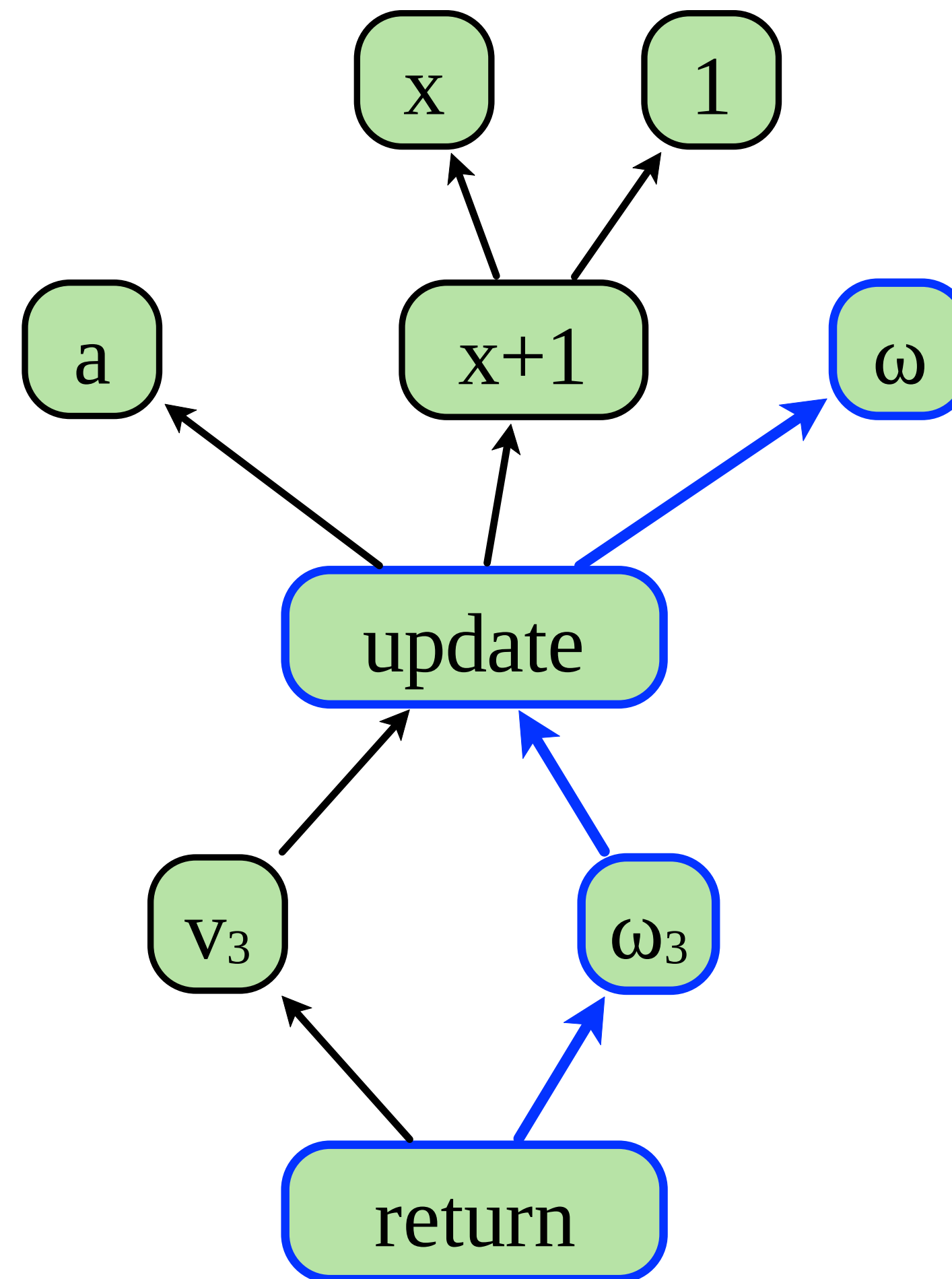
The Statewalk DP Algorithm



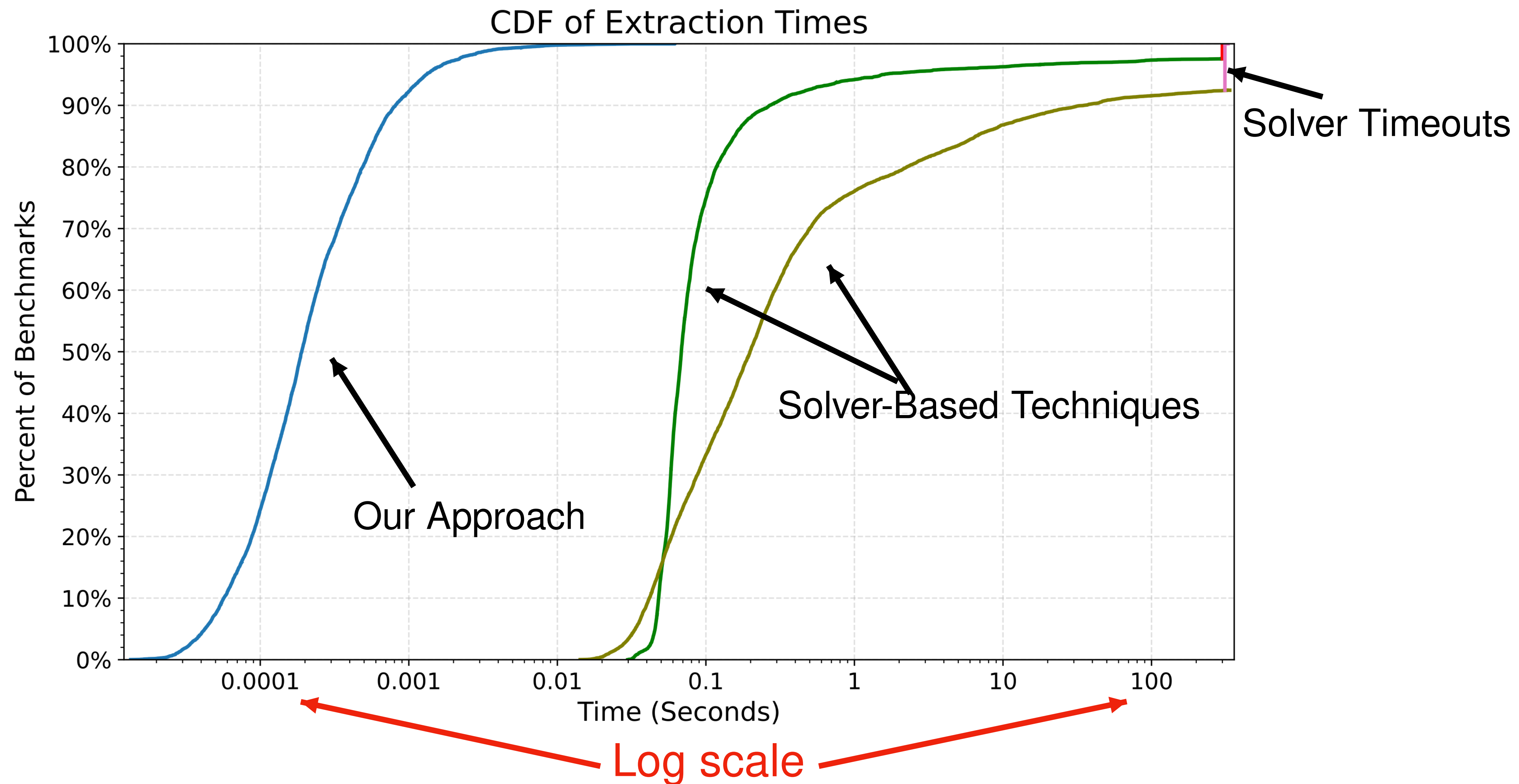
The Statewalk DP Algorithm



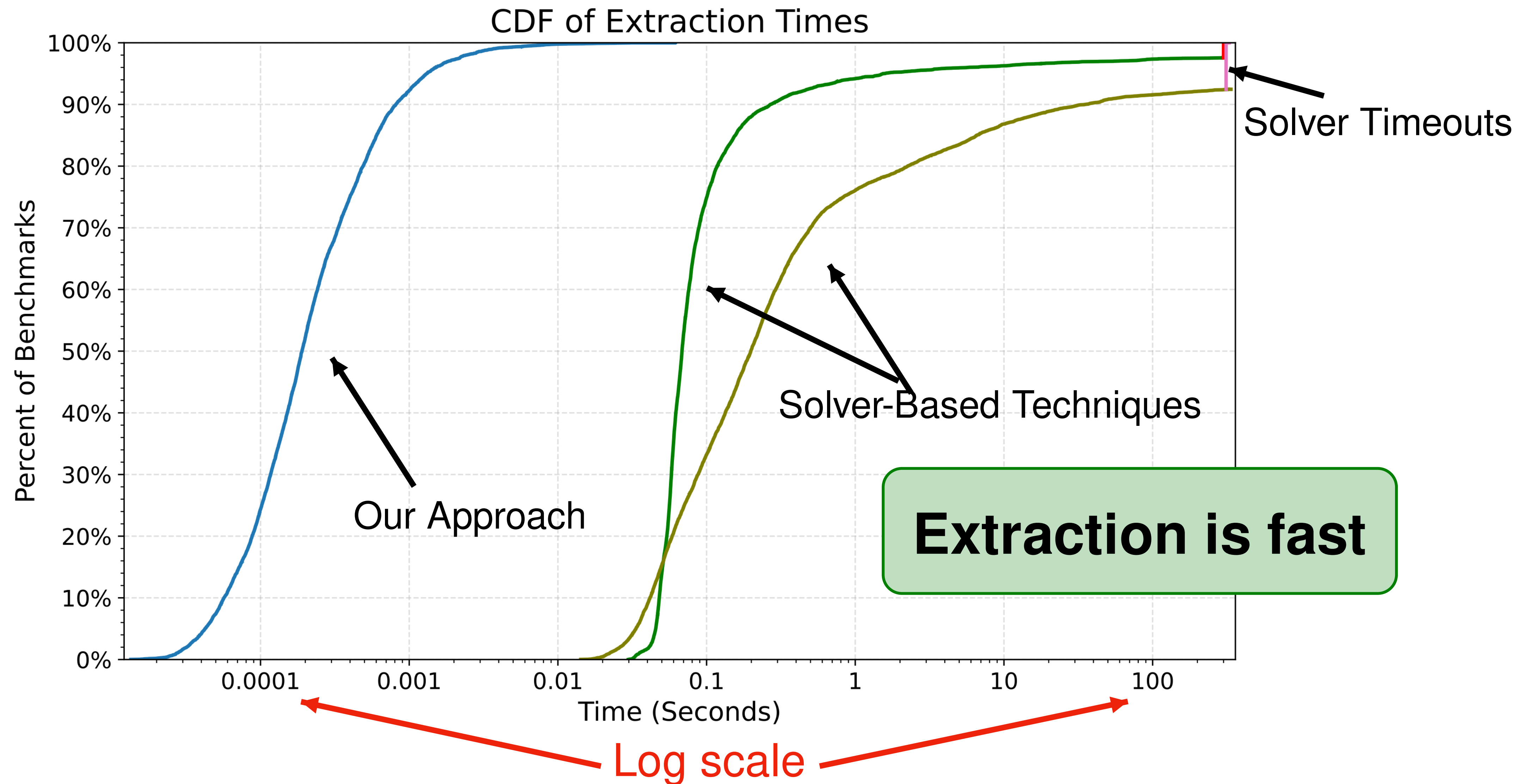
The Statewalk DP Algorithm

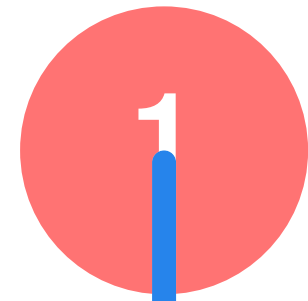


Fast Effect-safe Extraction in Practice

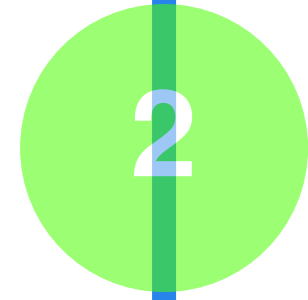


Fast Effect-safe Extraction in Practice

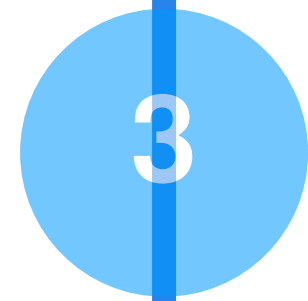




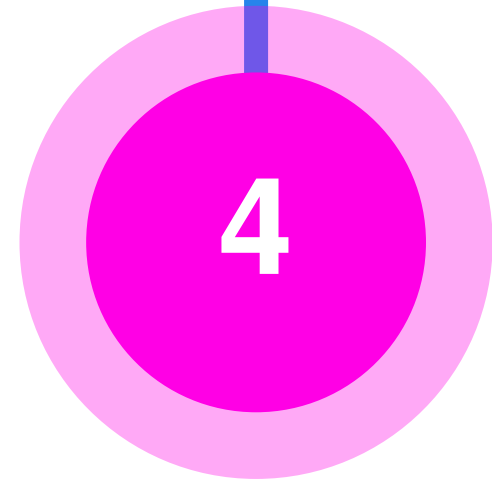
Motivate Effectful E-Graphs



Example Effectful E-Graph



Effect-Safe Extraction



It Works!



It Works!

1

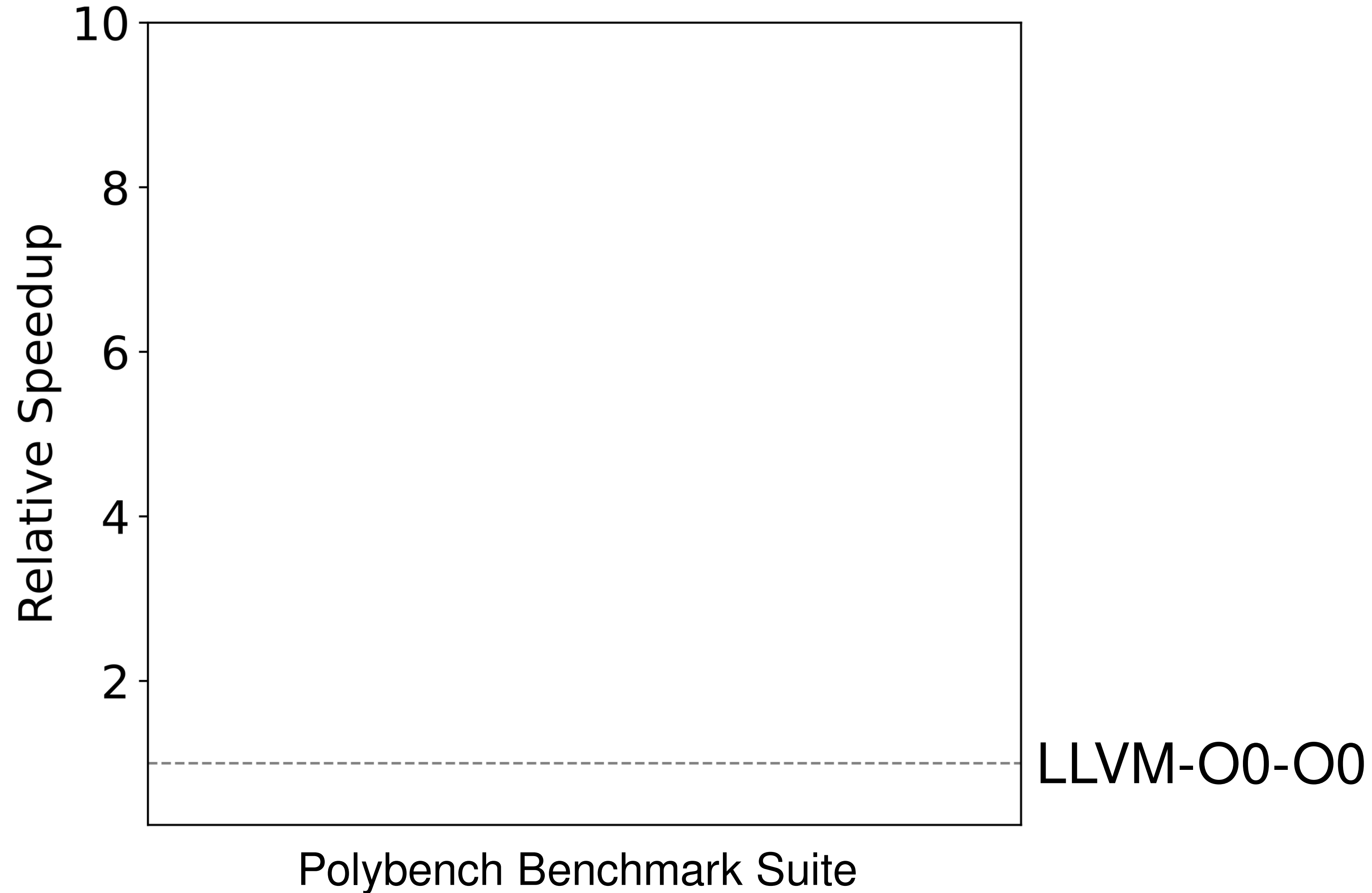
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3

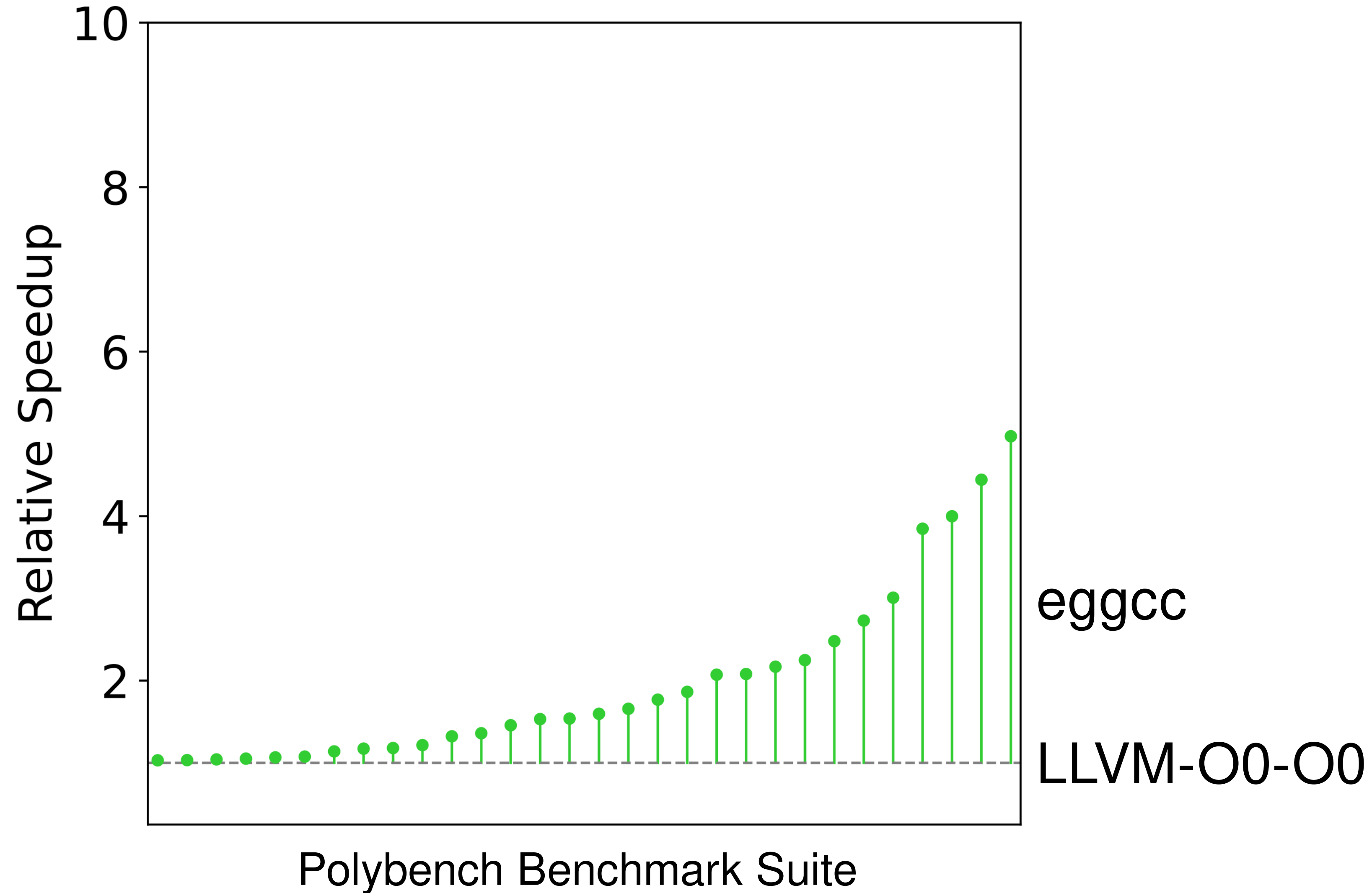
4

It Works!

Fast Effect-safe Extraction in Practice



Fast Effect-safe Extraction in Practice



1

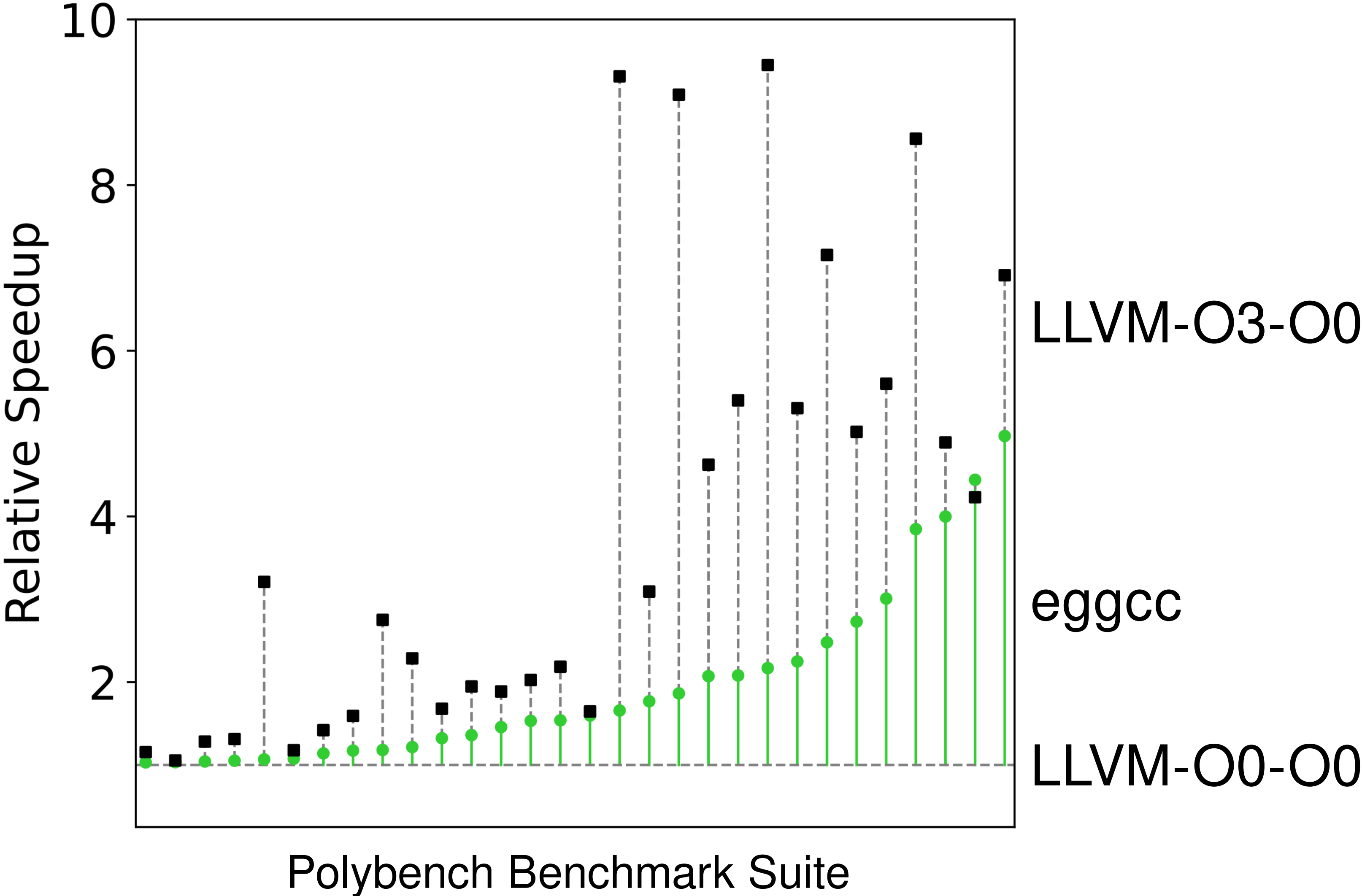
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3

4

It Works!

Fast Effect-safe Extraction in Practice



1

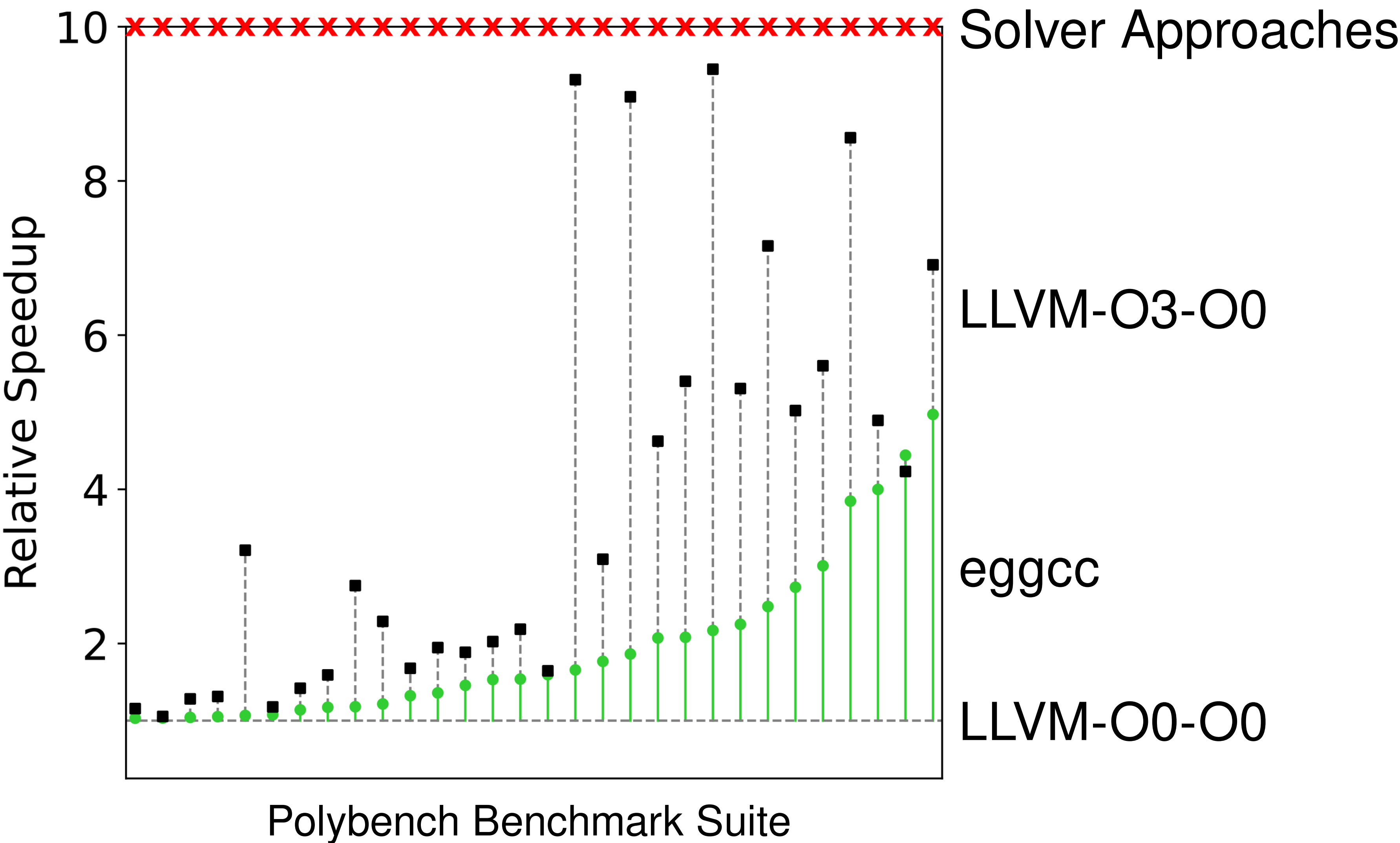
2

3

4

It Works!

Fast Effect-safe Extraction in Practice



1

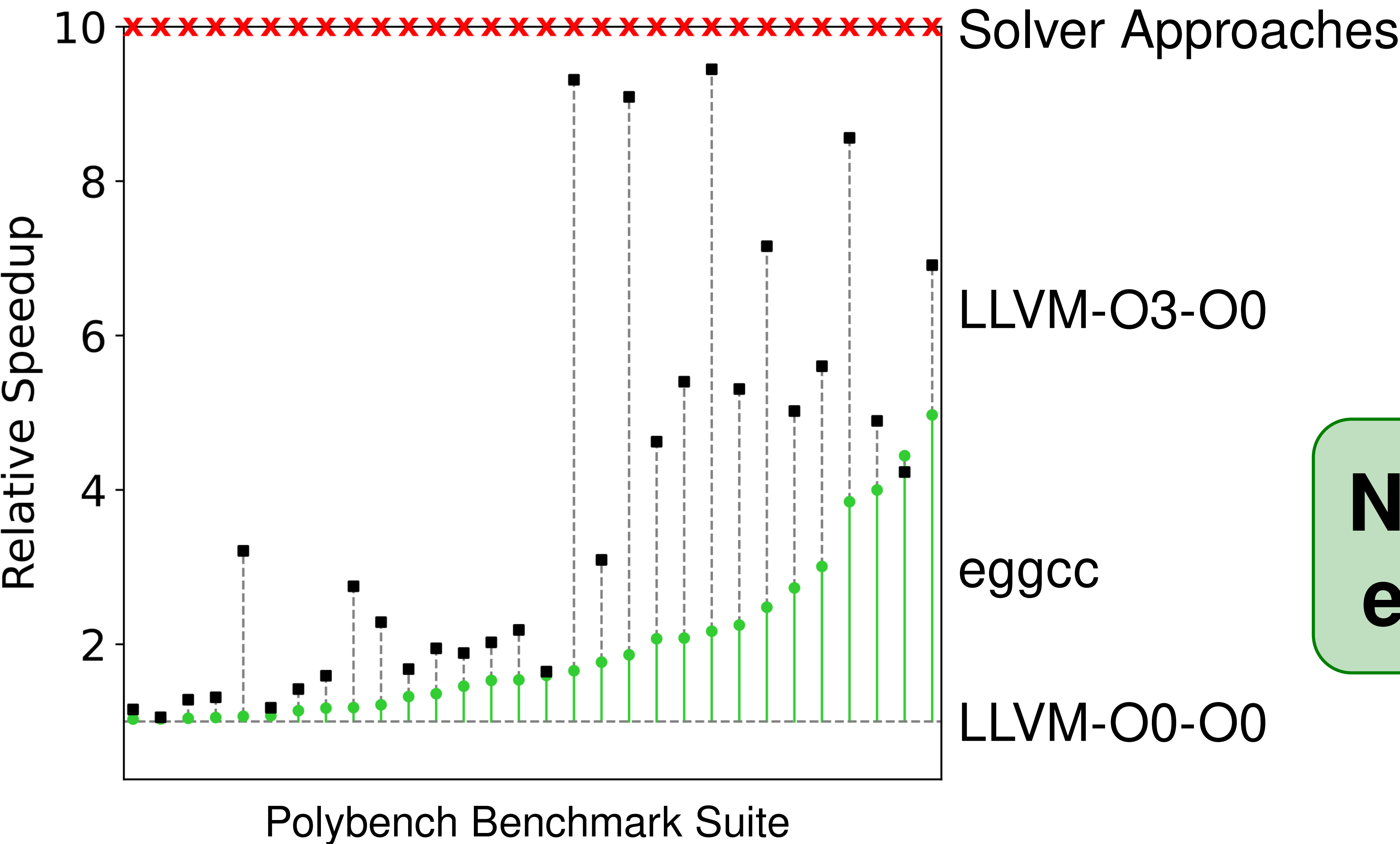
2

3

4

It Works!

Fast Effect-safe Extraction in Practice



**Non-trivial
e-graphs!**